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IIM A
Product Mindset

(ProdMan Prep Book)
2025-26

THE PRODMAN
CLUB OF IIM AHMEDABAD



ACKNOWLEDGEMENT

There has been a steep increase in the number of firms offering product management roles in the recent past. This is not only maneuvered by an increase in the number of firms visiting the graduate students, but also by expansion in their scope to associate product manager roles for undergrads. In spite of this, there exists a gap in terms of comprehensive preparation material suited for the Indian audience. Filling this gap was an uphill task, which couldn't have been possible without the stakeholders who were a part of the journey at different points in time.

First and foremost, we extend our deepest gratitude to the entire **ProdMan Club (2025-26)** and the team of collaborators. They dedicated countless hours to the rigorous process of ideating, curating, and refining the content within these pages. We are also incredibly thankful to the wider student and alumni community, whose valuable efforts in the past have greatly enriched this work.

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Finally, we thank you, **the reader**. By engaging with this material, you become a vital part of this initiative. We hope you find it valuable, and we welcome any feedback that will help us strive for continuous improvement.

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The ProdMan Club
IIM Ahmedabad

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Section 1

PM Concepts

What Is Product Management

Throughout business history, innovators have kindled ideas that solve real problems. They built them, tested them, sometimes failed, and iterated with one goal: to create and deliver meaningful solutions. Today, that end-to-end process is embodied in the role of the Product Manager.

A common definition is that a **product manager is the CEO of a product**. PMs shoulder broad responsibility for vision, execution, and outcomes. They **guide teams to discover, develop, and ship the right product**, while bridging gaps among stakeholders.

What does a PM do?

At its heart, product management is a **strategic function at the intersection of business, technology, and user experience.** A few key traits of a PM include:

- **Customer empathy**: Uncovering both explicit and latent needs is essential. The challenge lies in detecting unstated problems, going beyond what customers say to what they actually require. Sometimes people don't know what they require unless someone shows it to them. PMs listen to customers to define problems but take ownership in shaping the solution.
- **Solution Ownership**: PMs decide *how to solve a problem*. The quality of the solution, and its fit with customer context, sets apart great PMs from average ones.
- **Prioritization of Decisions**: A PM constantly evaluates competing ideas. By weighing metrics like feasibility, impact, effort, and business value, PMs ensure teams focus on what matters most so that faster delivery of critical features is achieved while preventing wasted effort.
- **Excellent Communication:** PMs must adapt their message for different audiences, executives, engineers, designers, or marketers, using the right vocabulary and context to align everyone toward the same product vision.

Key Responsibilities of a Product Manager

While responsibilities vary by company size, industry, and maturity, your journey as a PM generally include:

- **Market Research & Insight**: Conducting customer and competitive analysis to guide decisions
- **Vision & Strategy**: Crafting a clear product vision linked to business goals and translating it into a roadmap
- **Idea Generation & Prioritization**: Conducting meetings to address customer needs and prioritizing features using criteria like impact, feasibility, and value
- **Design & Development Collaboration**: Shaping wireframes, writing user stories, facilitating development with engineering teams, and supporting testing and iteration
- **Go-to-Market & Launch**: Coordinating marketing strategy, aligning with launch timelines, and ensuring support readiness
- **Ownership of Success Metrics**: Tracking performance via customer satisfaction, adoption, retention, revenue etc.

Performance measurement of PMs

Your performance as a PM will be measured using activities associated with product success, customers, and the overall process of product design and launch.

KPIs commonly used are:

- Monthly Revenue – Month-wise revenue generated by the product
- Delivery-On-Time - Delivery of product or new feature on time
- Product Quality - Number of deviation from standard quality practices
- Customer Counts - Increase in customer over a specified duration
- Stickiness Ratio - Daily active user/ daily monthly user (DAU/DMU)
- Customer Retention - Number of customers staying with the company over a specified period
- Customer Satisfaction Score Measures overall satisfaction of customers with the product or new features

Product Design for PMs

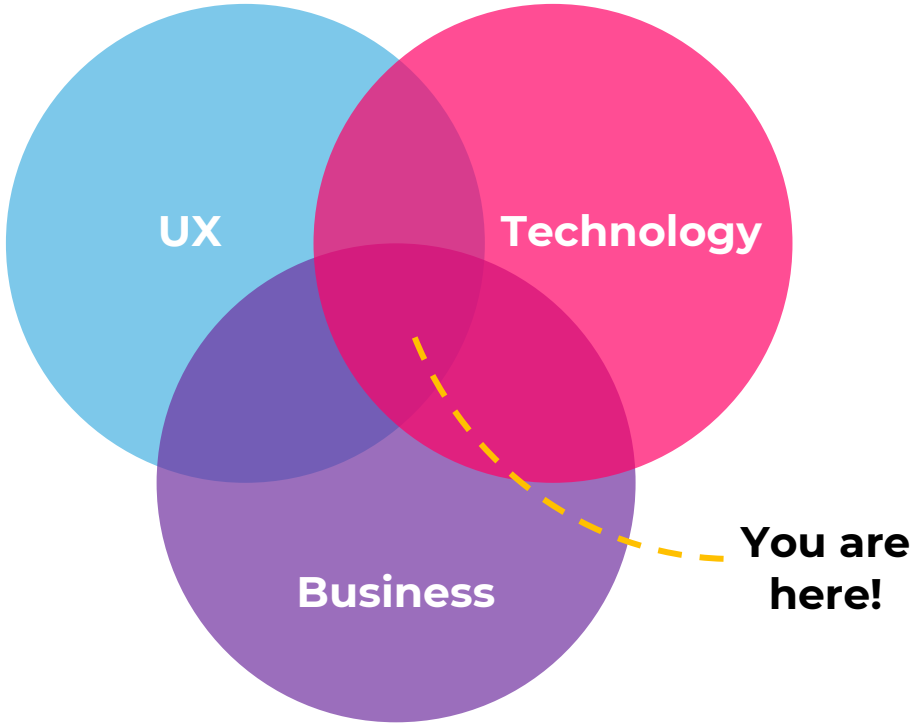
Why Product Design Matters for PMs

A Product Manager bridges customer needs, business goals, and technical delivery. You're not just deciding what to build, you're making sure it solves the right problem in a way that is valuable, usable, and feasible.

Good product design ensures:

- The **right problem** is solved (not just what's easy to build)
- Solutions are **feasible** for engineering and **viable** for the business
- Users can achieve their goals **smoothly and intuitively**

As a PM, you don't need to be the lead designer but must guide design decisions, so the product delivers value consistently.



Reference: https://www.aela.io/en/blog/all/product-manager-business-technology-and-user-experience

How PMs Should Approach Product Design

1. **Start with the user**: Deeply understand their pain points, workflows, and environment.
2. **Collaborate early**: Involve design and engineering in ideation early, not just delivery.
3. **Prioritize trade-offs**: Every design choice impacts cost, time, and user experience.
4. **Iterate fast**: Get early versions into users' hands, test, and refine. Use prototypes, MVPs, and metrics to validate before scaling.

Key Concepts in Product Design

PLC (Product Life Cycle)

What it is:
The stages a product goes through in the market: Introduction → Growth → Maturity → Decline.

1. **Introduction:** You've just launched. Focus is on **adoption** and finding your first true fans (early adopters). You're validating your core hypothesis with a real-world audience.

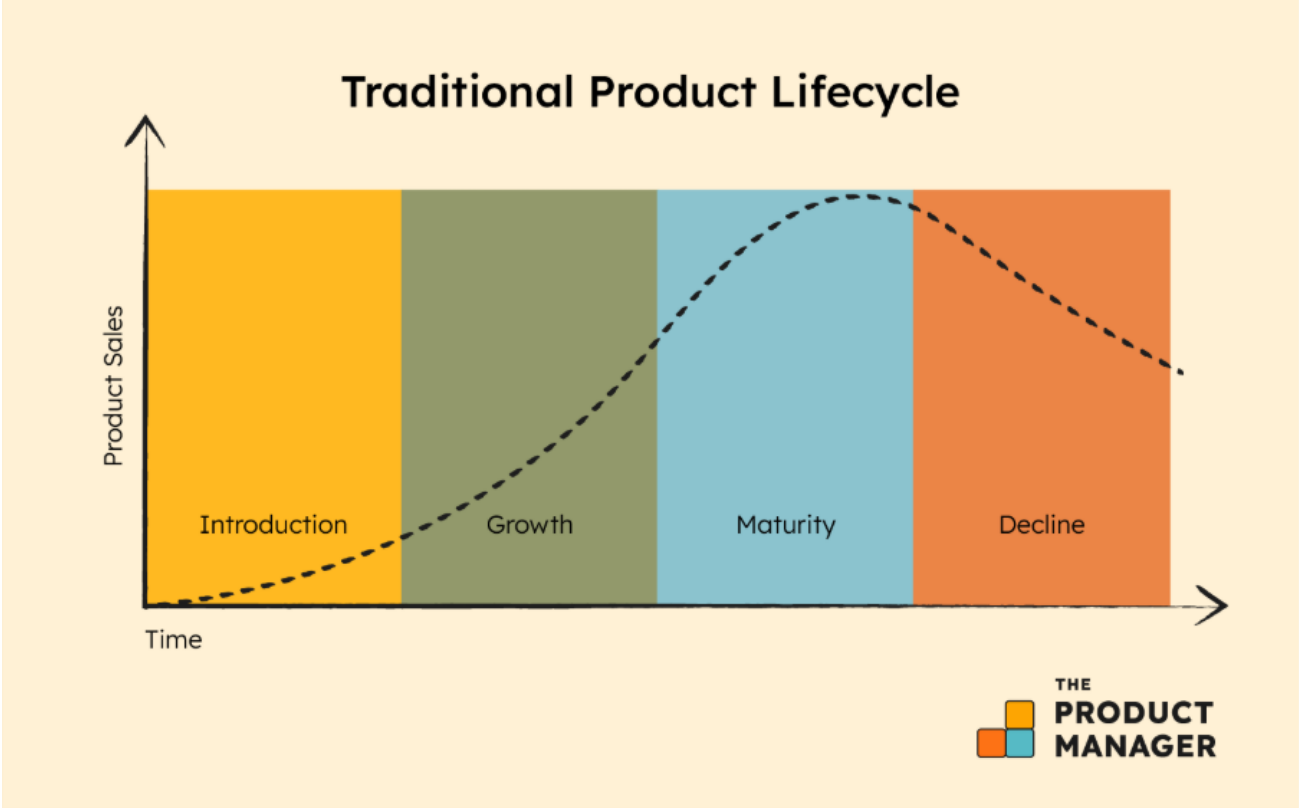
2. **Growth:** You have product-market fit! Focus shifts to **scaling**. You're rapidly adding features, entering new markets, and acquiring users quickly to build a defensible market position.

3. **Maturity: Growth is slowing down.** Focus is now on **optimization and retention**. You're fending off competitors, increasing efficiency, and finding new ways to get more value from your existing user base.

4. **Decline:** The market is shrinking, or a superior technology has emerged. Focus is on **profitability**. You may be "sunsetting" the product, minimizing costs, or milking the remaining cash flow before retiring it.

Why PMs should care:

- Different stages need different strategies (e.g., heavy marketing in Introduction, cost optimization in Maturity).
- Helps decide whether to invest, sustain, pivot, or sunset a product.



Reference: https://theproductmanager.com/topics/product-life-cycle-management/

Example:

Instagram Reels in 2020 was in Introduction with rapid feature pushes and heavy promotion. In 2025, it's in Maturity, focusing on refining UX and increasing ad monetization.

User Personas

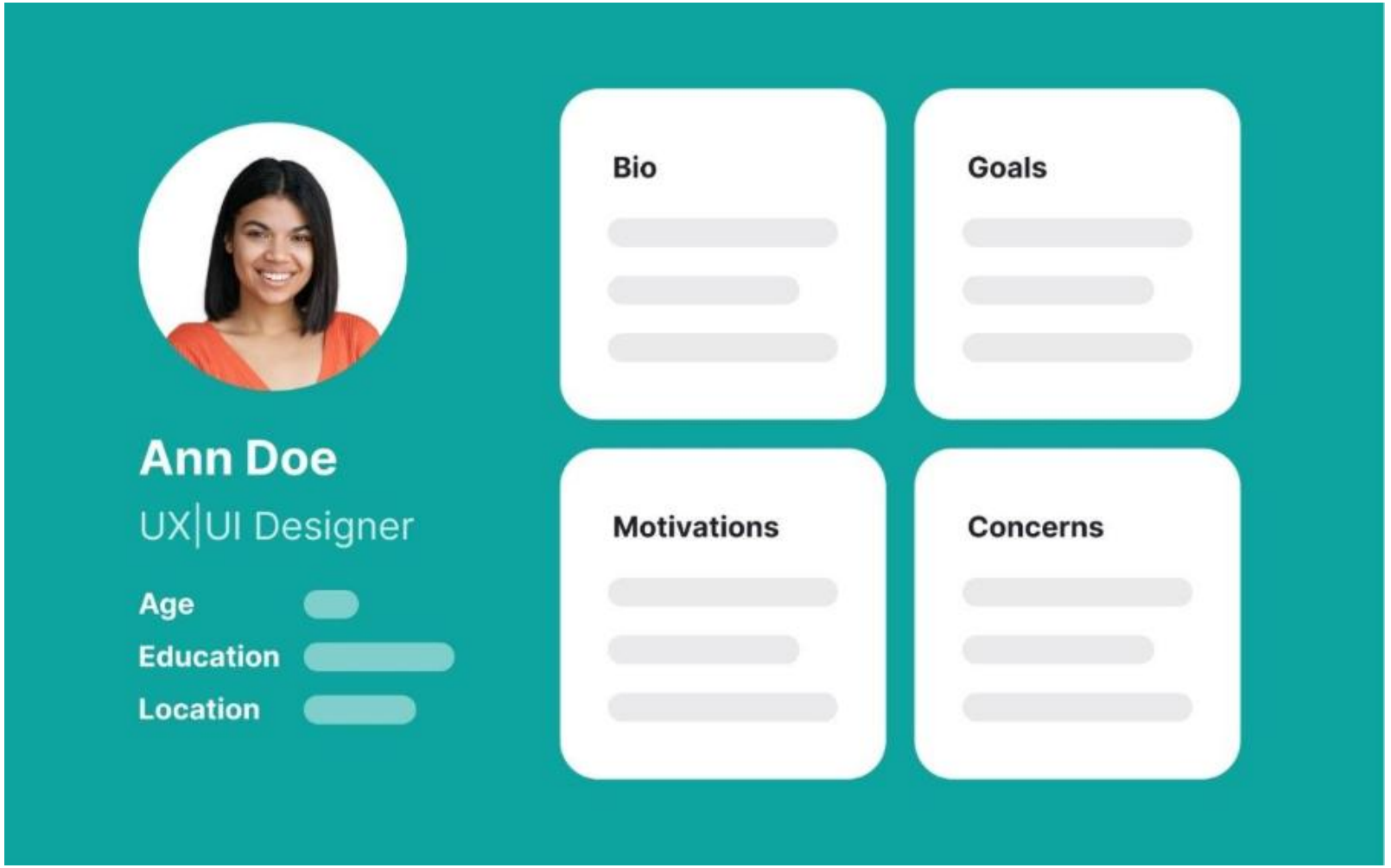
What it is:
Fictional, research-backed representations of your target customers, including demographics, product usage behaviors, goals, and challenges. This helps in achieving a product-market fit. It is recommended to come up with around 3-4 user personas.

Why PMs should care:

- Personas anchor design choices in real user needs rather than internal opinions.
- They help align the team (designers, engineers, marketing) on who you're building for (even prioritize some personas over others)
- In roadmap discussions, you can use personas to defend priorities ("Feature X matters because Persona A struggles with Y problem.")

Example:
For a budgeting app, your primary persona might be "Ravi, 28, urban professional, wants to track expenses easily on mobile, hates manual entry." That means you might prioritize **bank sync** over advanced analytics.

Reference: https://app.uxcel.com/courses/design-foundations/ux-deliverables-002/user-persona-7788


Ann Doe
UX|UI Designer
Age
Education
Location
Bio
Goals
Motivations
Concerns

User Stories

What it is:
Short, structured statements of a feature from a user's perspective:
"As a [user type], I want [goal] so that [benefit]."

Why PMs should care:

- Keeps requirements user-focused rather than tech-focused.
- Makes it easier for developers to understand the context and purpose of a feature.
- Helps you slice work into smaller, deliverable pieces for faster iteration.

Example:
"As Ravi, I want to scan receipts so that I don't have to type expenses manually." This drives a feature decision to add an OCR-powered upload option.

UX Design

What it is:
User Experience (UX) is how a user interacts with and feels about your product and its ease of use, clarity, flow, and satisfaction.

Why PMs should care:

- Poor UX kills adoption even if the feature is powerful.
- PMs must advocate for usability testing, accessibility, and intuitive workflows.
- Good UX can reduce customer support costs and increase adoption and retention.

Example:
In a food delivery app, if users drop off at the address entry, simplifying it with “use current location” can improve order completion rates.

PRD (Product Requirements Document)

What it is:
A detailed, formal document that specifies how customers will interact with a product, the functionality needed to fulfil all use cases, and the requirements for engineering and other teams. It often includes wireframes, user stories, and technical constraints. The PRD serves as a central reference point for engineering, quality assurance, customer service, sales, and product marketing, ensuring everyone understands what is to be built and why.

Key Sections:

- **Problem Statement**: The customer pain points the product aims to solve, backed by market research or customer discovery.
- **Goals/Objectives**: Desired user and business outcomes, aligned with the product vision and company strategy.
- **User Personas & Stories**: Detailed personas and narratives describing how representative users will accomplish key tasks.
- **Features & Scope**: Clear definition of in-scope and out-of-scope features for the release, with prioritization.
- **Requirements & Use Cases**: Comprehensive description of all the ways the product will be used, including functional and non-functional requirements.
- **Wireframes/Prototypes**: Visual layouts or interactive models to guide engineering and design.
- **Success Metrics**: Quantitative and qualitative measures of success (e.g., adoption rates, engagement metrics, conversion rates).

Why PMs should care:

- Provides necessary detail for engineering to build without ambiguity, reducing back-and-forth.
- Acts as the single source of truth for all stakeholders, minimizing misalignment.
- Enables cross-functional teams to translate product goals into actionable work plans.
- Reduces the risk of scope creep by clarifying priorities and constraints upfront.

Example:
Before launching a “Dark Mode” feature:

- **Problem**: Users experience eye strain in low-light conditions.
- **Personas & Story**: “Melody, a working mother in her 40s, uses the app late at night after putting her kids to bed. She finds the bright interface uncomfortable and wants a softer display option.”
- **Scope**: Apply dark mode to all core screens; exclude legacy admin dashboard in v1.
- **Requirements**: Toggle switch in settings, persist mode preference, ensure color contrast meets accessibility standards.
- **Success Metric**: ≥ 40% of nightly active users enable dark mode within one month of release.

MVP (Minimum Viable Product)

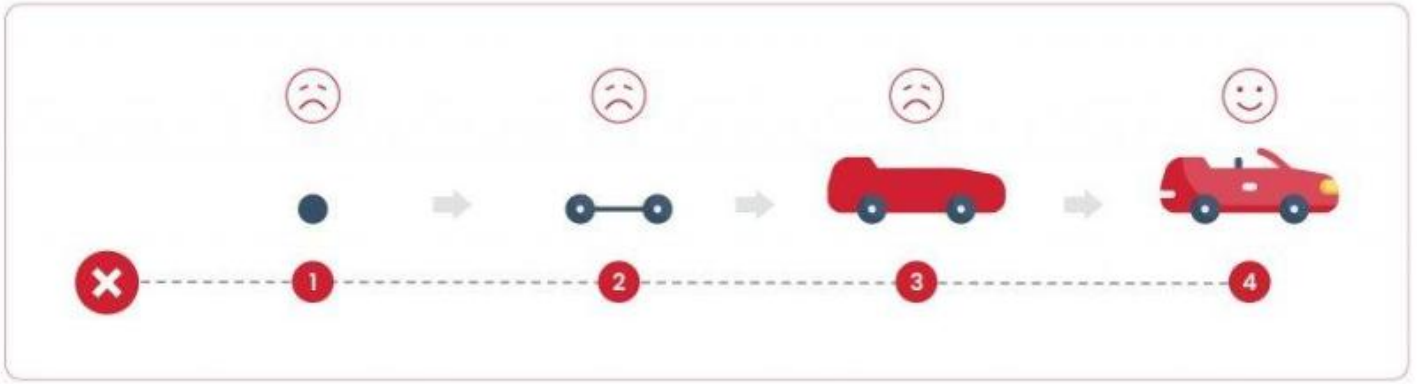
What it is:
An MVP is the most misunderstood concept in product management. It is **not** a half-finished, buggy product. It is the version of a new product that allows you to collect the **maximum amount of validated learning about your customers with the least amount of effort.**

Why PMs should care:

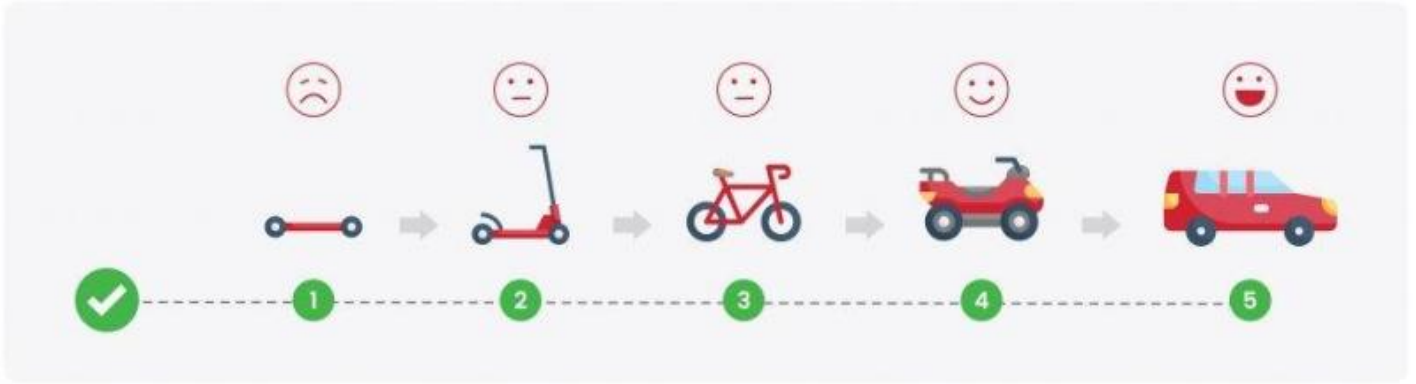
- Prevents overbuilding and wasting resources.
- Lets you test assumptions early and pivot quickly if needed.
- Gives real-world feedback to inform future iterations.

Example:
Instead of building a complete AI résumé review platform, launch a simple "upload PDF, get instant score" tool to see if job seekers are interested.

Reference: https://shiftasia.com/column/mvp-development-steps-and-benefits-explained/



MVP = MINIMUM VIABLE PRODUCT



CIRCLES Framework

What it is:
A problem-solving framework for design questions

- **C**omprehend the Situation: Ask clarifying questions to understand the goal.
- **I**dentify the Customer: Define your target user personas.
- **R**eport the Customer's Needs: State their problems and use cases (via user stories).
- **C**ut Through Prioritization: Prioritize which use case to solve first.
- **L**ist Solutions: Brainstorm multiple potential solutions.
- **E**valuate Trade-offs: Analyze the pros and cons of your proposed solutions.
- **S**ummarize Your Recommendation: State your final recommendation

Why PMs should care:

- Useful in structured thinking during design discussions.
- Helps avoid jumping to solutions without fully understanding the problem.

HOOK Model

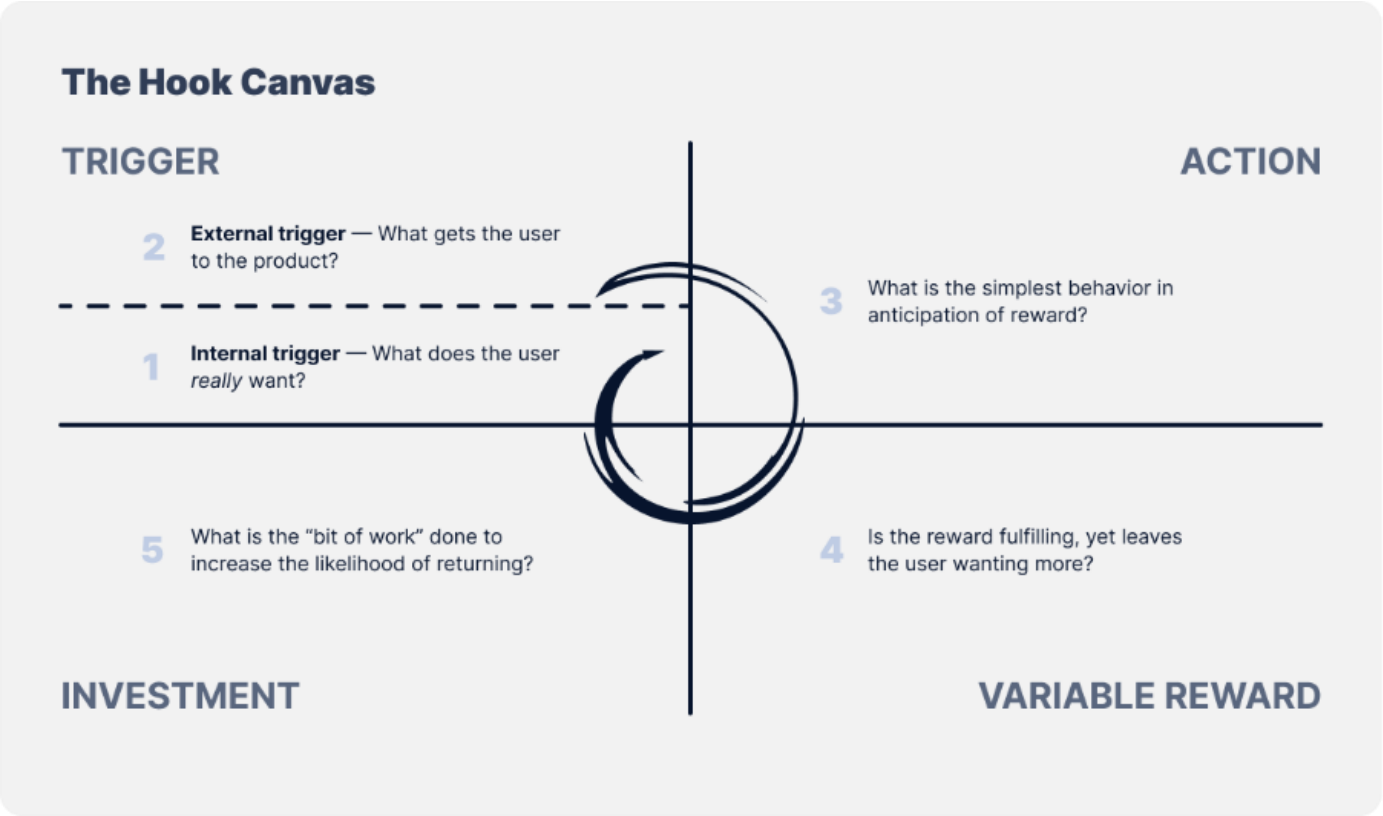
What it is:
A behavioral design loop for building habit-forming products:

- **Trigger**: The cue that initiates a behavior. This can be external (like a notification) or internal (like the feeling of boredom).
- **Action**: The simplest action a user can take in anticipation of a reward (e.g., scrolling a feed, clicking a play button).
- **Variable Reward**: The reward that satisfies the user's need but is delivered unpredictably. This is what creates craving (e.g., the thrill of seeing what's next in your social feed).
- **Investment**: A small bit of work the user puts into the product, which loads the next trigger and increases the likelihood of them returning (e.g., following a user, creating a playlist, sending a message). This work could bring returns from the user (e.g., user spending money/inviting other people to the platform/providing data, etc.)

Why PMs should care:

- Helps create products users naturally return to.
- Encourages thinking about long-term engagement rather than just acquisition.

Reference: https://artkai.io/blog/the-hooked-model



Example: Duolingo

- Trigger: Daily push notification.
- Action: Complete a lesson.
- Variable Reward: Fun streak animations.
- Investment: User progresses in a skill tree, making them return tomorrow.

Why Feature Prioritization Matters for PMs

As a PM, you will *always* have more feature requests than you can build. Your job is to decide **what to work on first**, balancing:

- **User Value**: How much pain does this solve? How much delight does it create?
- **Business Value**: Does this drive revenue, increase market share, improve retention, or reduce costs?
- **Strategic Fit**: Does this feature move us closer to our long-term product vision?
- **Effort & Feasibility**: What will it take to build this, and do we have the resources and dependencies in place?

Poor prioritization wastes engineering effort; delays impact and frustrates stakeholders.
Good prioritization **maximizes ROI** and keeps the team focused.

How PMs Should Approach Feature Prioritization

Start with Your North Star: Always begin with your company's strategic goals and your product's vision. If a feature doesn't align with these, it should be heavily questioned, no matter how exciting it seems.

Gather & Centralize Inputs: Create a single repository for all ideas. This includes user feedback from support tickets, requests from the sales team, insights from market research, new technology opportunities, and ideas from your internal team.

Choose the Right Framework: No single framework is perfect for all situations. Select a tool that fits the stage of your product, the data you have available, and the decision you need to make (more on this below).

Socialize & Align: Prioritization is not a solo activity done in a dark room. Share your framework and reasoning with key stakeholders (engineering leads, design, marketing, leadership). This transparency builds trust and alignment, even when you have to say "no".

Iterate and Revisit: Priorities are not set in stone. The market changes, you get new data, a competitor makes a move. Revisit and adjust your roadmap regularly, typically every quarter.

[Refer to the Metrics section to understand what a **North Star** is.]

Key Frameworks for Prioritization

RICE

What it is:
A scoring formula: **(Reach × Impact × Confidence) / Effort**

- **Reach**: How many users will this feature affect over a specific period?
  - e.g., 5,000 users/month

- **Impact**: How much will this move the needle for those users?
  - Use a scale:
  - 3 for massive impact,
  - 2 for high,
  - 1 for medium,
  - 0.5 for low.

- **Confidence**: How sure are you about your Reach and Impact scores?
  - (100% for high confidence, 80% for medium, 50% for low).
  - This is crucial for de-biasing.

- **Effort**: How many "person-months" will this take from the entire team (product, design, engineering)?

Why PMs should care:

- Gives a quantitative way to compare ideas.
- Reduces bias towards “loudest voice in the room”.

Example:
You have two ideas:

- Add “One-Click Checkout” (Reach: 50K users, Impact: High, Confidence: 80%, Effort: 3 months).
- Add “Gift Wrapping Option” (Reach: 10K, Impact: Medium, Confidence: 90%, Effort: 1 month).

The RICE score will likely show that “One-Click Checkout” has a higher impact overall, even though it’s harder to build.

Reference: https://whatfix.com/blog/rice-scoring-model/

R.I.C.E Prioritizations Method

Priority	Project	Reach	Impact	Confidence	Effort	R.I.C.E Score
1	Biometric Authentication	81	●●●●●	75%	●●○○○	152
2	Stripe Pay integration	70	●●●○○	50%	●●●○○	35
3	Shopify e-commerce cart	58	●●●●○	50%	●●●●○	29
4	API Dashboard	43	●●○○○	50%	●●●○○	14

whatfix

Value-Effort Matrix

What it is:
A 2×2 chart:

- **High Value / Low Effort →** Quick Wins, they provide immediate impact and build momentum.
- **High Value / High Effort →** Major Projects, these are your strategic initiatives. They need to be broken down into smaller phases and planned carefully
- **Low Value / Low Effort →** Fill-ins, only do these if you have free time, which you rarely will. They don't move the needle
- **Low Value / High Effort →** Avoid, at all costs.

What brings "Value"?

- **User Value (Customer Impact):** Improves satisfaction, solves pain points, or creates delight for users.
- **Business Value (Commercial Impact):** Drives revenue, accelerates acquisition and growth, boosts retention, or reduces costs.
- **Strategic Value (Future Impact):** Strengthens competitive advantage, supports long-term goals, and aligns with the company's vision.

What demands "Effort"?

- **Development Effort:** Time and complexity required for design, coding, and testing.
- **Resource Cost:** Financial investment, team bandwidth, and opportunity cost.
- **Risk & Uncertainty:** Technical challenges, dependency bottlenecks, or potential market failure.

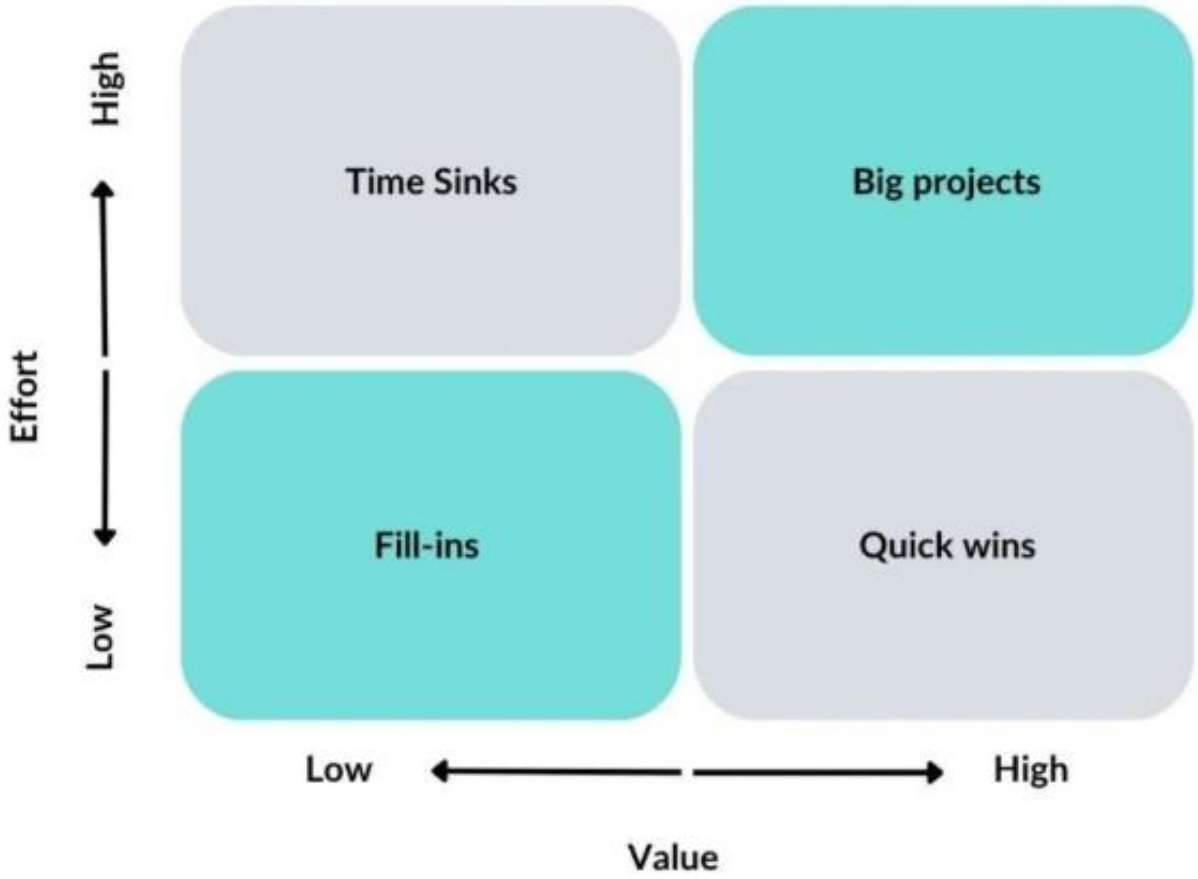
Why PMs should care:

- Simple visual tool for quick decisions.
- Useful for early-stage or small-team prioritization.

Example:
A "Search Bar Improvement" that boosts retention but takes 2 days to code? → Quick Win → Build now.
A "New AI Recommendation Engine" that takes 6 months with unclear ROI? → Likely avoid for now.

Reference: https://www.userxd.co/post/how-to-understand-value-vs-effort-in-ux-a-comprehensive-guide

The Value vs. Effort Matrix



Kano Model

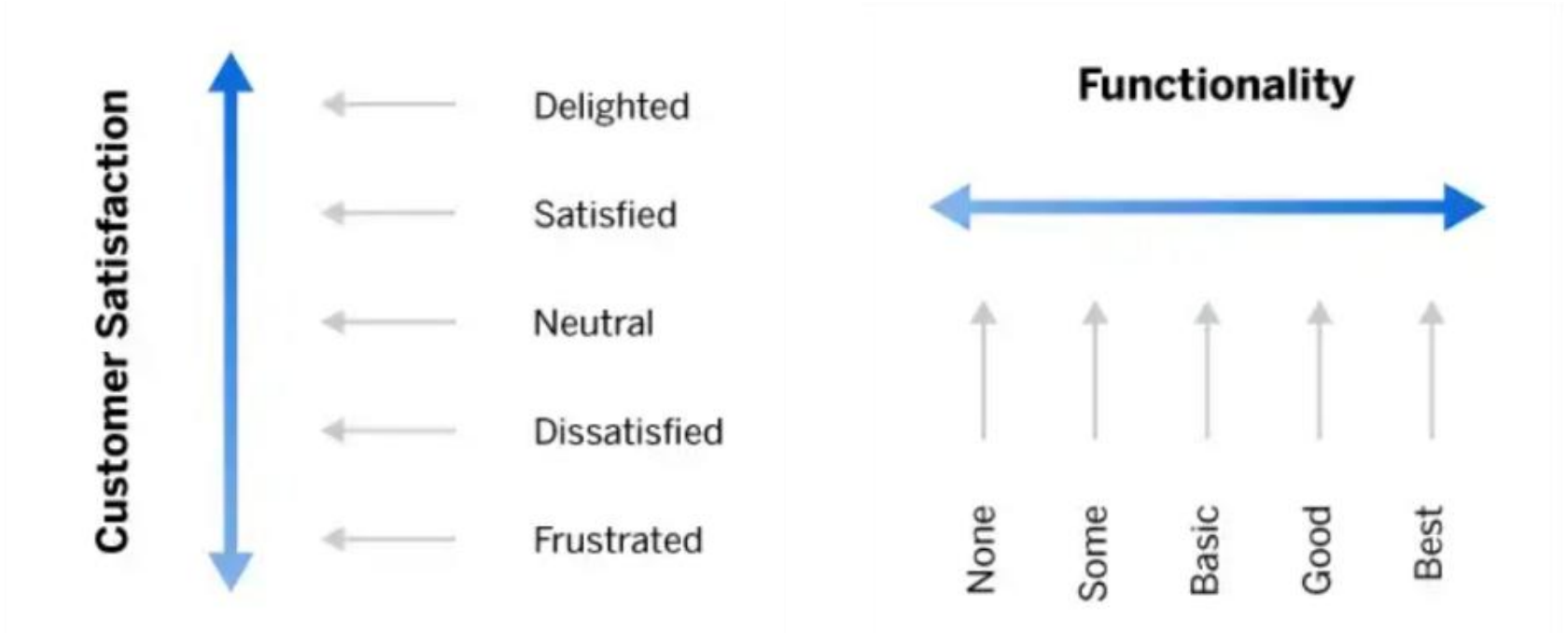
What it is:
The Kano Model shifts your thinking from "what features should we build?" to "how will this feature affect user satisfaction?" It classifies features into:

- **Must-haves (Basic Expectations)**: Users expect these and will be dissatisfied if they are missing, but they won't be delighted if they are present.
  - (e.g., A hotel room must have a bed).

- **Performance Features (Satisfiers):** The more you provide, the more satisfied users become.
  - (e.g., Faster Wi-Fi, more storage space).

- **Delighters (Exciters):** Unexpected features that create delight. Their absence doesn't cause dissatisfaction, but their presence can create loyal fans.
  - (e.g., a free welcome gift in your hotel room).

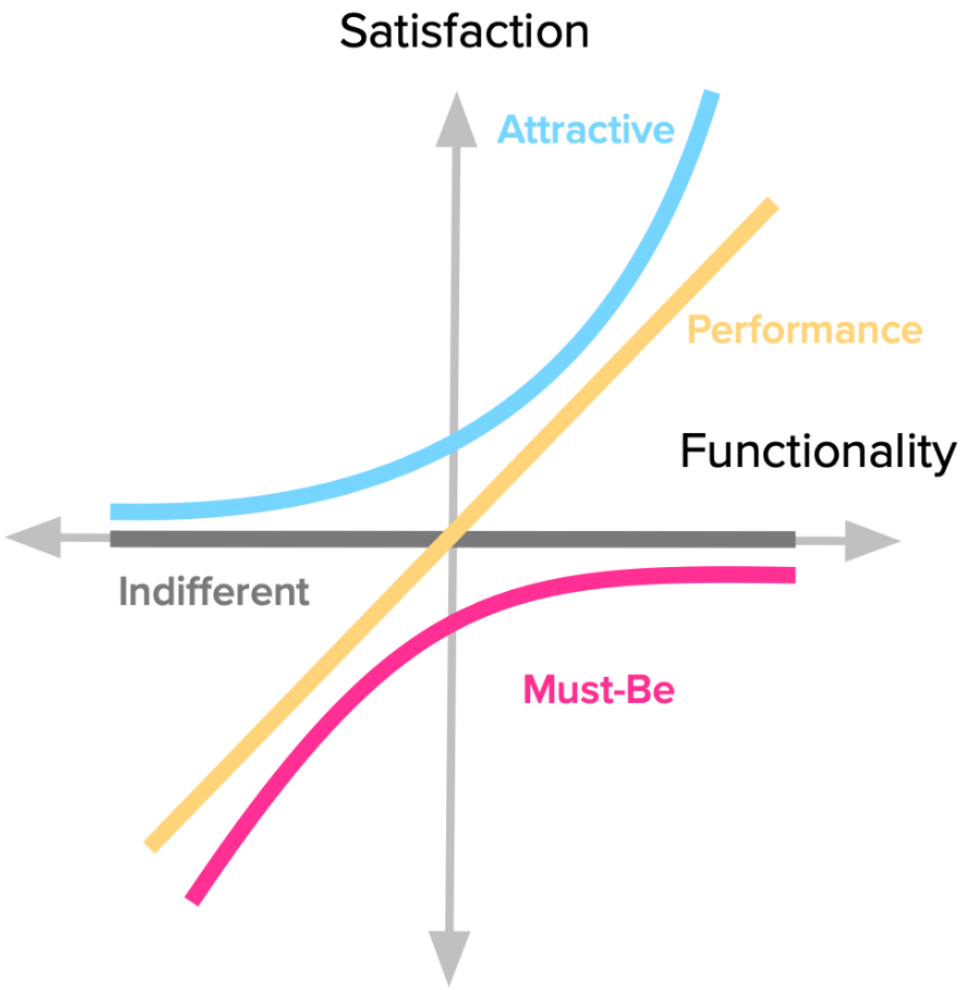
- **Indifferent**: Features that customers think are neither good nor bad. They can be unimportant features to the customer.



Why PMs should care:

- Helps avoid wasting effort on over-optimizing basic features when you could add delighters.
- Keeps user satisfaction in mind, not just business metrics.
- Today's "Delighter" is tomorrow's "Must-have." Features naturally migrate down the Kano curve over time as user expectations change.

Reference: https://www.qualtrics.com/en-au/experience-management/research/kano-analysis/#:~:text=The%20Kano%20Analysis%20model%20(pronounced,can%20be%20measured%20and%20explored.



Example:
For a hotel booking app:
Must-have: Accurate room availability
Performance: Lower prices, better filters
Delighter: Free late checkout for loyal customers

Reference: https://foldingburritos.com/blog/kano-model/

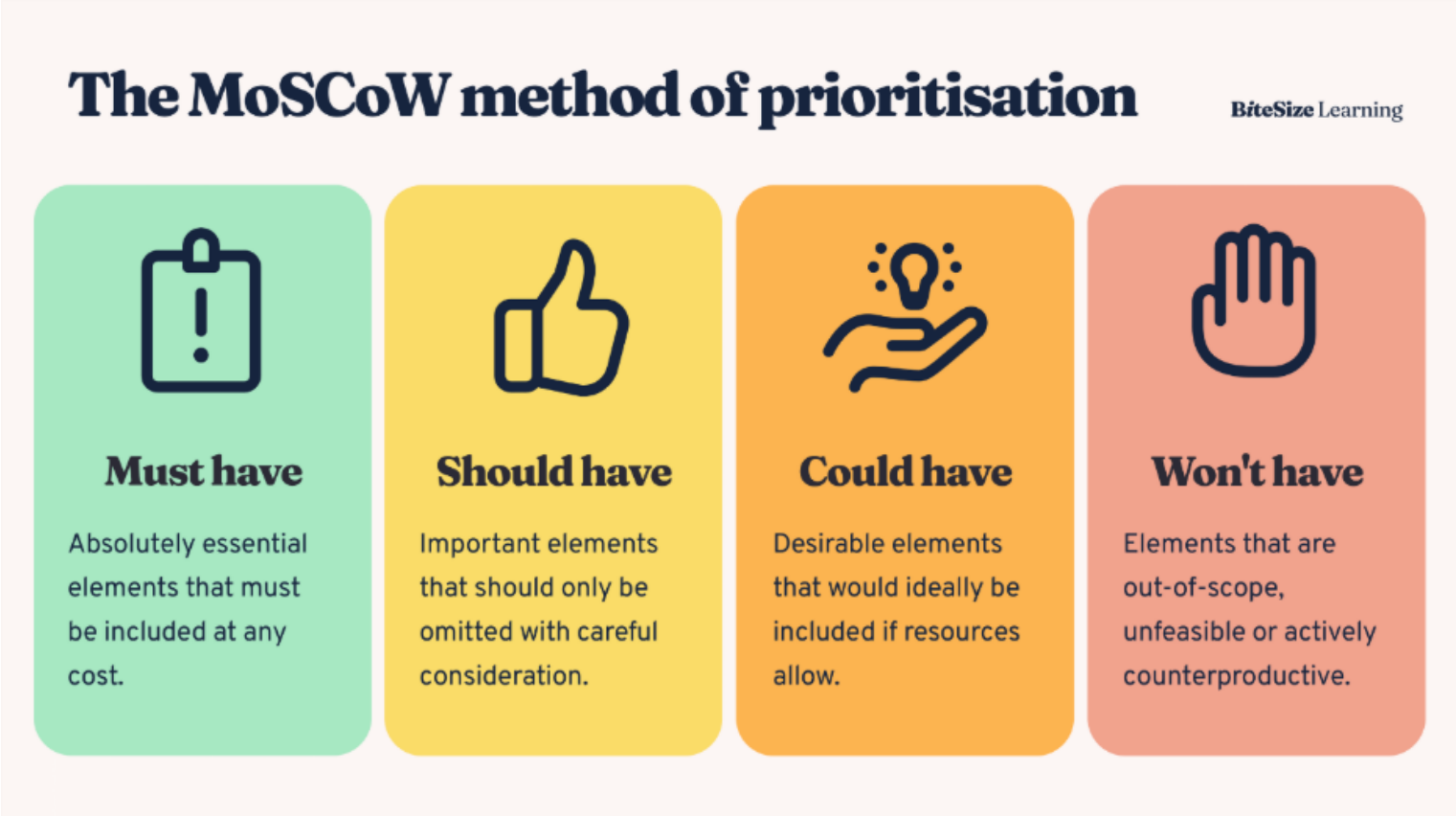
MoSCoW

What it is:
MoSCoW is less about prioritizing a backlog and more about defining what must be included in a specific release or MVP.

- **Must Have**: Non-negotiable features for the release to be viable. If even one of these is missing, the launch is a failure.
- **Should Have**: Important but not vital. The release is still valuable without them, but they would be painful to leave out.
- **Could Have**: Desirable but not necessary. These are nice-to-haves that will be included only if time and resources permit.
- **Won't Have (This Time)**: Explicitly out of scope for the current release.

Why PMs should care:

- Works well in Agile for release planning.
- Helps manage stakeholder expectations when resources are limited.



Example:
For a conference app MVP:

- Must Have: Agenda & Speaker Info
- Should Have: In-app Q&A
- Could Have: AR Navigation
- Won't Have: Virtual Reality Tour (this year)

Reference: https://www.bitesizelearning.co.uk/resources/moscow-prioritisation-model

ICE

What it is:
Score = **Impact × Confidence × Ease** (simpler cousin of RICE, without Reach).

- **Impact**: How much will this positively affect the user or business? (Scale of 1-10)
- **Confidence**: How sure are you about the impact score? (Scale of 1-10)
- **Ease**: How easy is this to build? (Scale of 1-10, where 10 is very easy)

Why PMs should care:

- Fast to apply for backlog grooming.
- Ideal when you don't have detailed user reach data yet.
- However, ICE can bias you towards features that deeply impact a small number of users.

Example:
You're deciding between "Dark Mode" vs "Custom Themes" in a beta app:

- Dark Mode: Impact 8, Confidence 9, Ease 7 → Score = 504
- Custom Themes: Impact 5, Confidence 7, Ease 5 → Score = 175
- → Build Dark Mode first.

Example ICE Prioritization

Your Growth Experiments:	Impact	x	Confidence	x	Ease	=	Total Score
Redesign homepage hero	8		7		5		280
Add exit-intent popup on Blog	6		8		9		432
Launch referral program test	9		6		4		216

Reference: https://growwithward.com/ice-prioritization-framework/

Why Metrics Matter for PMs

Without metrics, you're flying blind. PMs need **measurable indicators** to:

- **Objective Decision-Making**: Replace gut feelings and the "loudest voice in the room" with evidence-backed choices.
- **Diagnosing Problems:** Identify precisely where your product is succeeding or failing, so you can focus your efforts where they matter most.
- **Driving Alignment:** Give your team and stakeholders a shared definition of success and a straightforward way to track progress toward it.
- **Telling a Story:** Use data to craft a compelling narrative about your product's journey, its challenges, and its victories.

A PM's success is often judged by impact, and **impact** is demonstrated through **clear, relevant metrics**, i.e. tracking the *right things*.

How PMs Should Approach Metrics

1. **Start from Your Objective**: What is the single most important goal for your product right now? Is it acquiring new users? Increasing engagement? Driving revenue? Your primary objective dictates your primary metric.

2. **Build a Metric Tree**: Don't just pick a single metric. Map out how your primary goal (the North Star) breaks down into smaller, actionable drivers. This helps your team see how their daily work contributes to the bigger picture.

3. **Choose Leading and Lagging Indicators:**

    a. **Lagging Indicators** confirm long-term success (e.g., monthly revenue, customer retention). They are easy to measure but hard to influence directly.

    b. **Leading Indicators** predict future success (e.g., daily active users, number of items added to a cart). They show early signs of progress and are more actionable.

4. **Identify Counter-Metrics:** For every metric you try to improve, you must track a counter-metric to ensure you're not causing harm elsewhere. If you're trying to increase ad clicks (your goal), you must also monitor session time or user complaints (your counter-metric) to make sure you're not just annoying users.

5. **Avoid Vanity Metrics:** A vanity metric is a number that looks good on paper but doesn't actually help you make decisions (e.g., total number of sign-ups since launch). Focus on actionable metrics that change your behavior.

Key Frameworks & Models for Metrics

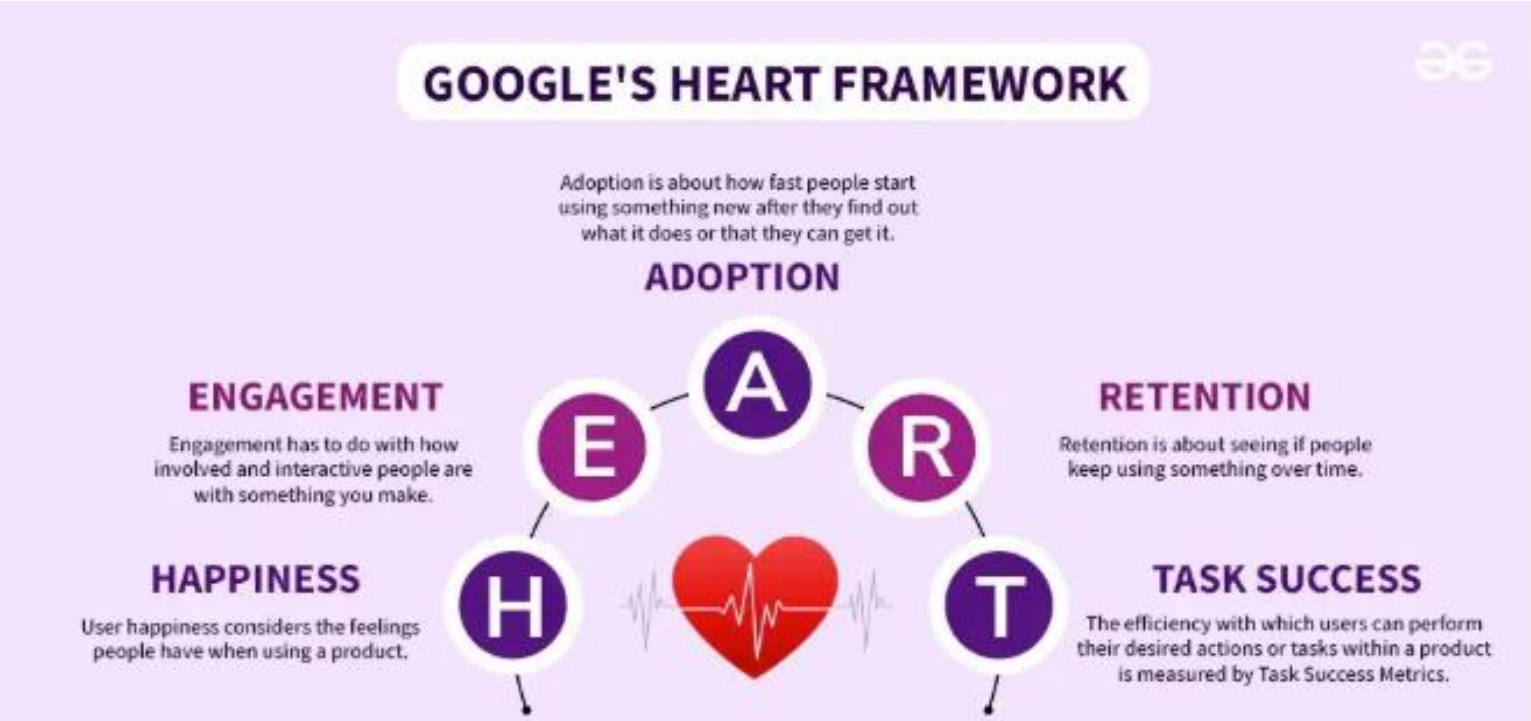
HEART Framework

What it is:
A UX measurement framework from Google, focused on:

- **Happiness**: How do users feel about your product?
  - (Measured via surveys like NPS, CSAT, or app store ratings).
- **Engagement**: How often and how deeply do users interact with your product?
  - (e.g., sessions per user per week, photos uploaded per day).
- **Adoption**: How many new users are trying your product or a specific feature?
  - (e.g., new accounts created, % of users who use a new feature in its first 7 days).
- **Retention**: What percentage of users are returning over time?
  - (e.g., daily/monthly active users, churn rate).
- **Task Success**: Can users accomplish their goals quickly and easily?
  - (e.g., time to complete a key task, success rate of a search query).

Why PMs should care:

- Gives a holistic view of product experience beyond pure numbers. Helps understand the "why" behind the numbers.
- Helps balance short-term engagement with long-term satisfaction.



Example:
For a project management app:

- Happiness: NPS = +40
- Engagement: Avg. 3 logins/week
- Adoption: +15% new sign-ups MoM
- Retention: 70% of teams are active after 3 months
- Task Success: 90% of tasks created are completed within deadlines

If Retention is dropping while Happiness is high, maybe users like the app but don't find it essential daily.

Reference: https://www.geeksforgeeks.org/websites-apps/googles-heart-framework-for-measuring-ux/

AARRR Framework (Pirate Metrics)

What it is:
A funnel-based model to track growth:

- **Acquisition**: How do users find you? (e.g., visitors from organic search, cost per install from ads).
- **Activation**: Do users have a great first experience (their "aha!" moment)? (e.g., % of users who complete onboarding, % who use a key feature in their first session).
- **Retention**: Are users returning? (e.g., 30-day active user rate, churn rate).
- **Referral**: Are users telling others about your product? (e.g., viral coefficient, number of invites sent).
- **Revenue**: Are you making money? (e.g., customer lifetime value (LTV), average revenue per user (ARPU)).

Why PMs should care:

- Perfect for mapping the user journey, especially new products, startups, etc.
- Pinpoints where the funnel is leaking.

Validated learning with pirate metrics (AARRR)



Example:
For a music streaming app:

- Acquisition: 50K installs from ads last month
- Activation: 60% play a song in the first session
- Retention: 30% listen weekly after 1 month
- Referral: 5% invite a friend
- Revenue: ₹150 average monthly subscription per paying user

If Retention is weak, you focus on playlists, personalized recommendations, and offline mode.

Reference: https://www.gartner.com/en/digital-markets/insights/growth-loops-vs-aarrr-funnel

North Star Metric (NSM)

What it is:
A single, guiding metric that reflects the core value you deliver to users.

The Litmus Test for a Good NSM:

- It expresses user value.
- It represents your product strategy and measures progress.
- It's a leading indicator of revenue.

Why PMs should care:

- Creates alignment across teams, i.e., your entire company.
- Prevents teams from chasing random metrics that don’t drive long-term value.

Example:

Airbnb's NSM:
Nights Booked. Because it reflects both guest satisfaction (they're booking stays) and host success (earning income).

Duolingo's NSM:
Daily Active Users (DAU) who complete at least one lesson. Just tracking DAU would be a vanity metric. By adding the qualifier "complete at least one lesson," they ensure they measure engaged learning, which is the core of their value proposition.

Uber's NSM:
Rides Completed. Because it reflects value for both sides, i.e. the rider and the driver. Further, it acts as a leading indicator of revenue.



Reference: https://www.slingshotapp.io/blog/north-star-metric

Why Testing Matters for PMs

Even the best product ideas are just hypotheses until proven in the real world.

Testing helps PMs:

- **De-risk Decisions**: Would you bet a million dollars on a coin flip? Building a feature without testing is the business equivalent. Testing allows you to validate your assumptions on a small scale before committing significant engineering resources.
- **Allocate Capital Intelligently**: Your team's time is your company's capital. Testing ensures you invest in features that demonstrably improve the user experience and drive business metrics.
- **End "Opinion-Based" Debates**: Testing provides an objective referee in arguments about design or functionality. Instead of debating which button color is better, you can say, "Let's test it and let the data decide." This builds a culture of evidence over authority.
- **Learn Faster**: Every test, whether it succeeds or fails, provides a valuable insight into your users' behavior. The faster you learn, the faster you can iterate and outpace the competition

How PMs Should Approach Testing

- **Formulate a Crystal-Clear Hypothesis:** This is the most critical step. A strong hypothesis is specific, testable, and connects an action to an expected outcome.
  - Weak Hypothesis: "Changing the button will improve clicks."
  - Strong Hypothesis: "If we change the primary call-to-action button on the homepage from blue to orange (the change), we will increase the click-through rate (the metric) by 10% (the expected impact) because the new color has higher contrast and creates more urgency."

- **Define Your Success Metric (and Counter-Metric):**
  - **Primary Metric:** Choose the one key metric that will determine if the test is a success (e.g., conversion rate, user retention).
  - **Counter-Metric:** Identify a metric that could be negatively affected. If you're testing a change to increase ad clicks, your counter-metric might be session duration or user block rate to ensure you're not just annoying your users.

- **Ensure Statistical Significance:** You need to run the test long enough and on a large enough group of users to be confident that your results aren't due to random chance. Use a sample size calculator to determine this beforehand.

- **Analyze and Act:** Once the test concludes, analyze the results. There are three possible outcomes:
  - **Success**: Your hypothesis was correct. Roll out the change to 100% of users.
  - **Failure**: The change had no effect or an adverse effect. Scrap the idea and, most importantly, document what you learned from the failure.
  - **Inconclusive/Interesting:** The primary metric didn't move, but a secondary metric did. This is an opportunity to dig deeper and form a new hypothesis.

Key Method: A/B Testing

A/B Testing

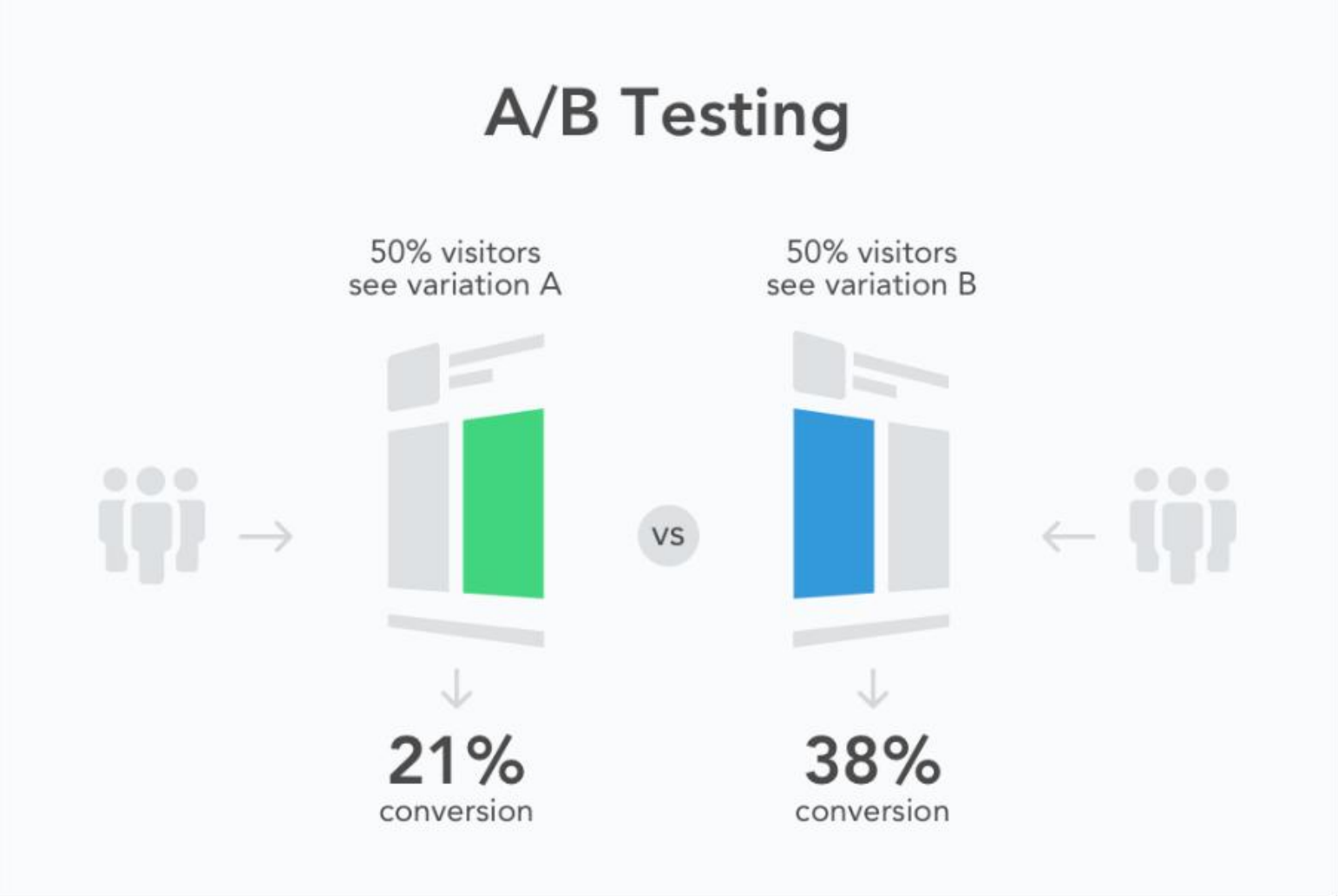
What it is:
A controlled experiment where you compare **two versions** (A = control, B = variation) to see which performs better on a specific metric.

Why PMs should care:

- Avoids “design by opinion” and lets data decide.
- Can be used for UI changes, pricing models, onboarding flows, or feature variations, but just one test variable at a time.
- Minimizes risk by testing on a subset of users before a full rollout.

Example:
You want to improve checkout completion on your e-commerce site.

- **Hypothesis**: A “Buy Now” button in the cart will reduce drop-offs by 10%.
- **Test**:
  - Group A (Control): Regular checkout process.
  - Group B (Variant): Cart page with “Buy Now” button.
- **Result**: Group B has a 14% higher completion rate → Roll it out to all users.



Reference: https://devopedia.org/a-b-testing

Section 2

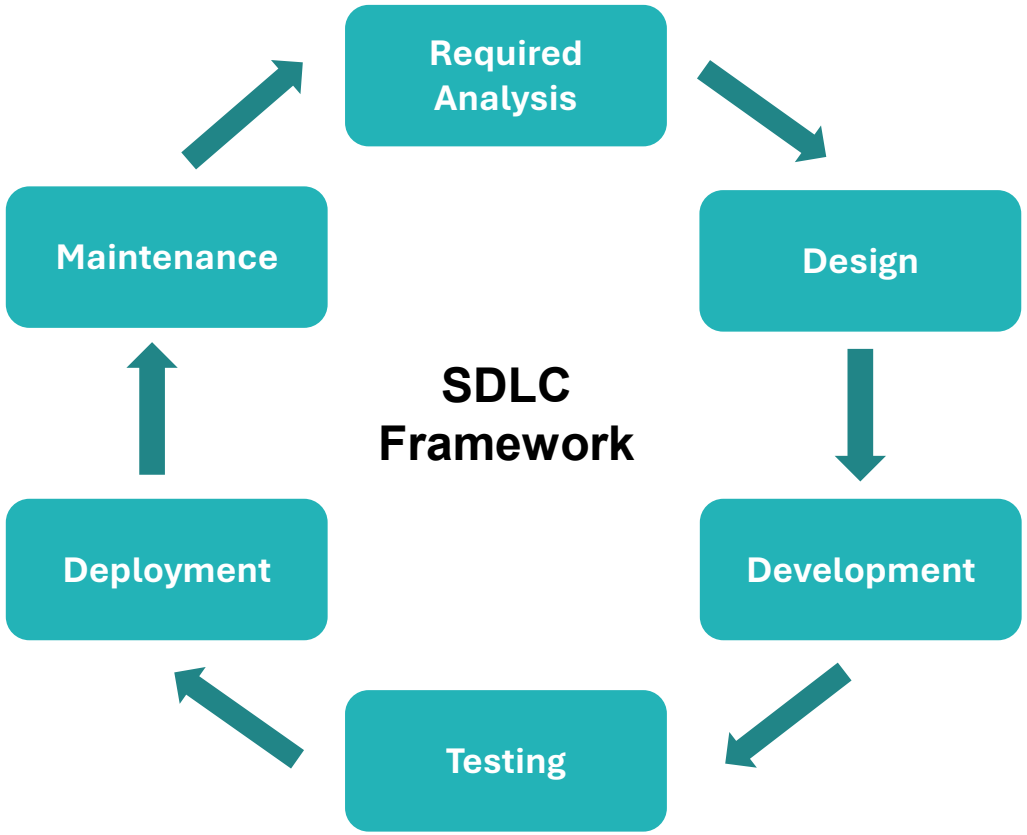
Tech Concepts

SDLC Framework – Why PMs should care

- SDLC keeps the product journey predictable and efficient.
- Understanding each stage helps you time the decisions. You know when to involve which stakeholders, how to manage timelines, and how to prevent costly rework.
- As a PM, you'll often decide when a feature is ready to move to the next phase and balance speed vs. quality.

Example: If you're building a new checkout flow, you lock requirements early so developers aren't guessing mid-coding, which avoids costly rework.

There are two common methodologies for SDLC- **Waterfall** and **Agile**



Waterfall Model

The Waterfall model follows a linear, sequential process where each phase begins only after the previous one is complete, with no overlap. Best suited for small, well-defined projects with minimal expected changes.

Linear flow: Requirement Analysis → Design → Implementation → Testing → Maintenance



Limitations

- **Low adaptability**: Late-stage defects (e.g., in testing) require extensive rework and fallback to previous stages.
- **Delayed delivery**: Sequential phases can lead to late releases, reducing competitive advantage.
- **Defect risk**: Defects are identified during testing and it occurs at the end of SLDC, thus the risk of project failure is concentrated at the end.

Agile Methodology

Agile

Agile is an iterative, time-boxed approach that delivers software in small increments, enabling rapid adaptation to change. It prioritizes:

- Working software over comprehensive documentation
- Responding to change over following a fixed plan

Core Principles

- Deliver valuable software early and continuously.
- Embrace changing requirements, even late in development, to maintain competitive edge.
- Release working software frequently (every 2–8 weeks) to get rapid feedback and reduce defects.
- Focus on technical excellence and good design to enhance agility.
- Maximize work not done—avoid non-value-adding activities.
- Conduct regular retrospectives for continuous improvement.

Agile Methodology

Agile Development

Scrum

It is built around **self-organizing, cross-functional teams** working in short, fixed-length cycles called **sprints** (typically 1–4 weeks) while delivering incremental developments.

Roles: Product Owner (defines priorities), Development Team (executes), Scrum Master (facilitates process)

Events: Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective.

Kanban

Visualises task on a Kanban Board with columns like “To Do”, ”In Progress” and “Completed”.

Facilities progress tracking, identifies bottlenecks, and adapt to changing requirements.

Ideal for environments with frequently shifting priorities.

Waterfall vs Agile

Aspect	Waterfall	Agile
Process Flow	Linear, sequential phases	Iterative, incremental sprints
Flexibility	Low – changes are costly and disruptive	High – welcomes changes even late in development
Delivery	Single final release after all phases	Frequent, small releases (every 2–8 weeks)
Customer Involvement	Primarily at start and end	Continuous feedback throughout
Risk Management	High risk at end (defects found late)	Risks addressed incrementally
Best Suited For	Small, well-defined projects	Dynamic, evolving requirements
Testing	Performed after implementation is complete	Continuous testing within each sprint
Team Collaboration	Limited between business and developers	Daily collaboration, cross-functional teams
Documentation	Heavy, detailed documentation	Lightweight, just enough for clarity
Examples of Frameworks	N/A (Waterfall is itself a framework)	Scrum, Kanban, XP

User Experience Design

User Experience Design is the process of fabricating products that provide purposeful and relevant experiences to users.

To achieve this, designers follow the process of design thinking.

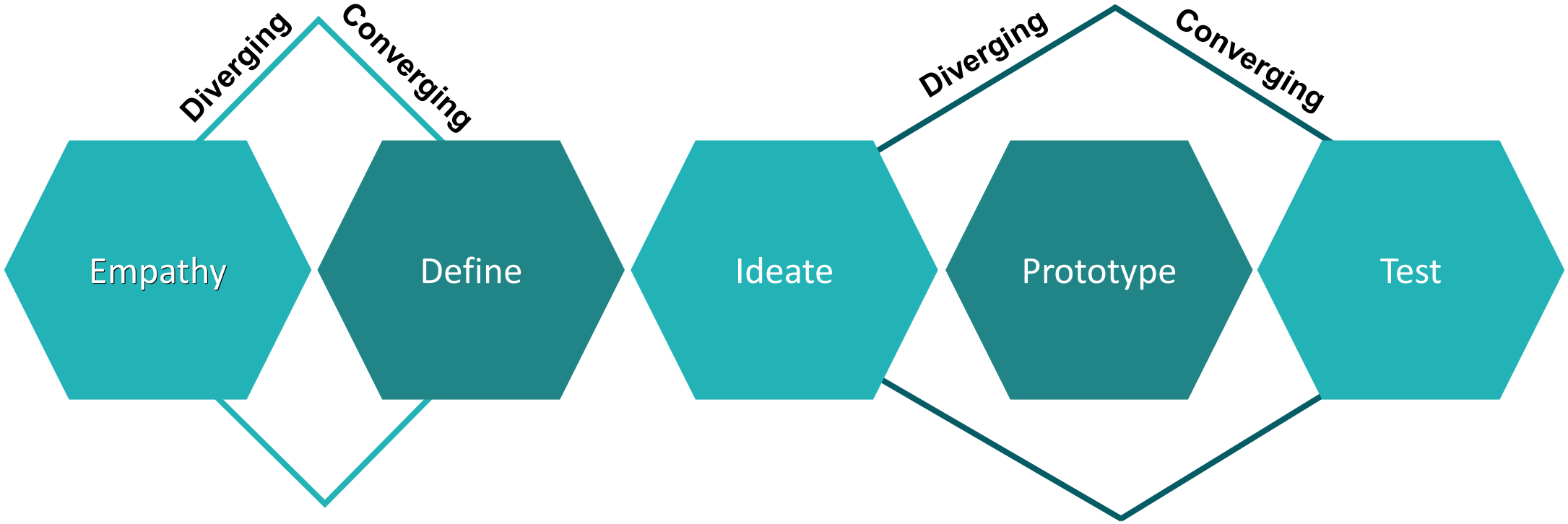
Design Thinking Process

Design Thinking is an **iterative, user-centric problem-solving approach** to create products with meaningful and relevant experiences. It focuses on understanding user needs, redefining problems, and exploring alternative solutions.

The following are the five stages of Design Thinking Process:

1. Empathy
2. Define
3. Ideate
4. Prototype
5. Test

Five Stages (EDIPT)



Deeply understand the person for whom you are designing for

Circumscribe the problem based on emerging institutions

Brainstorming on ideas and possible solutions

Create a tangible representation of the possible solution

Come back to the people you are designed for, to show prototype, test ideas and gather feedback

Step 1: Empathize

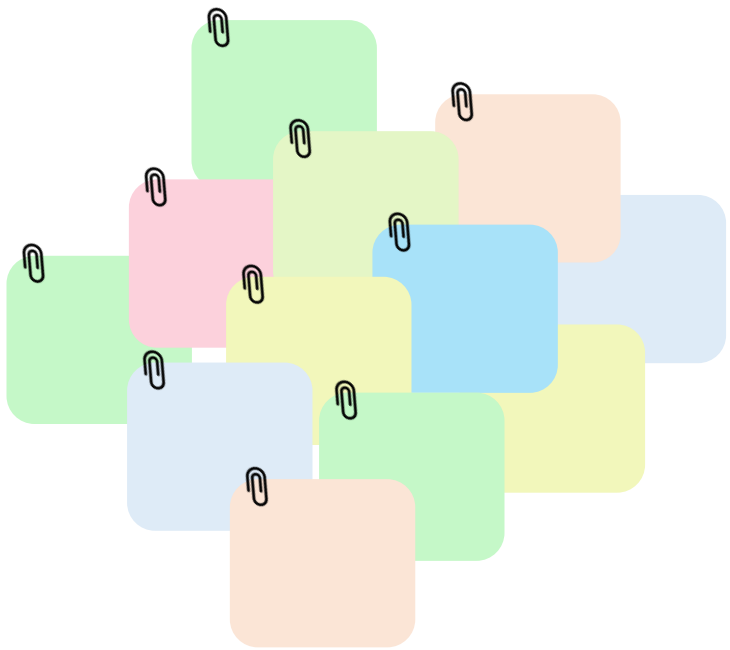
- Conduct secondary research to understand context and current solutions.
- Perform competitive analysis to find market gaps.
- Gather primary data using ethnographic studies, focus groups, diary studies, cultural probes, surveys, interviews and expert consultations

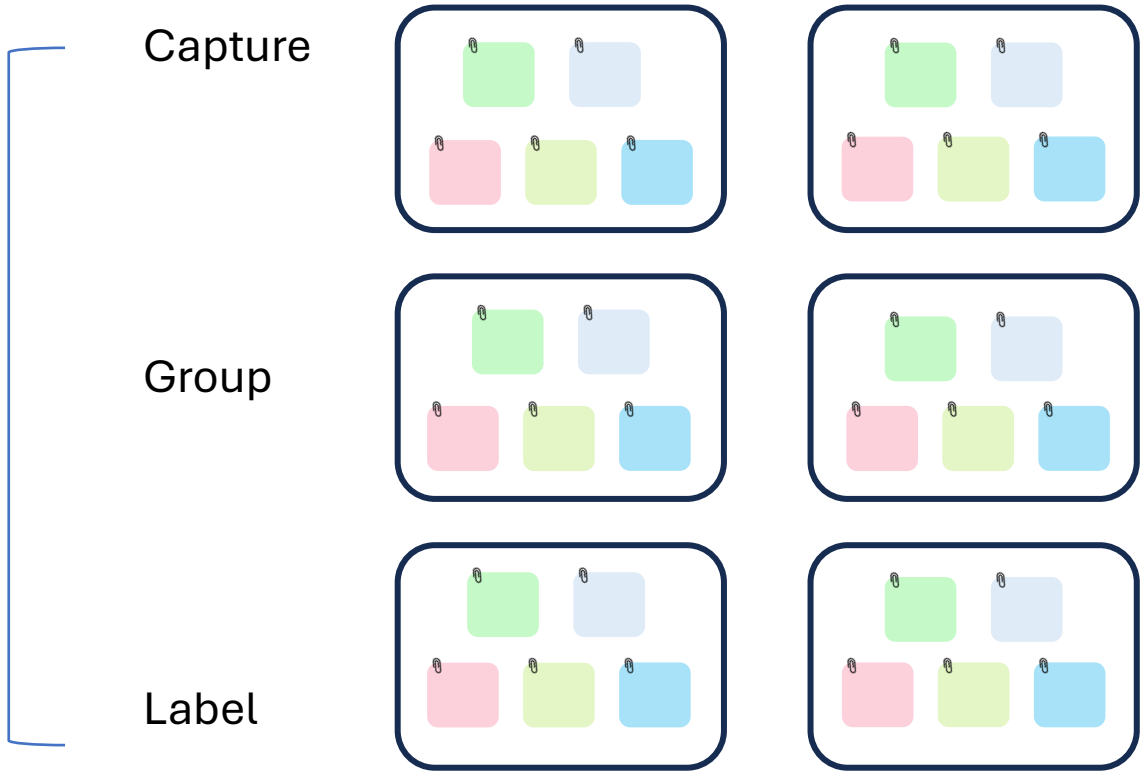
Step 2: Define

- Use affinity diagramming (KJ method) to organize research insights, identify pain points, and uncover patterns.
- Entire team participates in this and the aim is to identify the range of problems, and uncover similarities among problems from multiple stakeholders.

The KJ Method and Affinity Mapping

The KJ Method, developed by Jiro Kawakita, is a structured approach to affinity mapping.





User Experience Design

Step 2: Define

Develop user personas—archetypes representing user goals and behaviours—based on behavioural variables (activities, attitudes, skills, motivations).

Persona types:

- **Primary**: Main target user
- **Secondary**: Minor additional needs
- **Supplemental**: Needs met via a combination of primary and secondary personas
- **Customer**: Buyer, not necessarily the user
- **Served**: Indirectly affected individuals
- **Negative**: Users explicitly excluded
- Formulate a problem statement that captures user pain points and business impact.

E.g. “Company X’s customer satisfaction ratings are low and market share has diminished by 10% over the past year because users don’t have adequate tools to perform A, B, and C tasks that would help them meet their goal of G.”

Step 3: Ideate

- Diverge to generate many solutions without judgment (brainstorming, brainwriting, SCAMPER, Crazy 8s).
- Converge to select promising ideas using dot-voting, impact/effort matrices, or feasibility–viability–desirability checks.

Step 4: Prototype

- **Start simple**: Use sketches, paper models, or clickable wireframes before moving to more detailed designs.
- **Show the main features**: Focus on the key parts of the product that matter most to the user’s experience.
- **Try different versions**: Make several options to compare and learn from.
- **Keep it quick**: The goal is to learn fast, not to make it perfect.
- Aim of prototyping is to get something real in front of users so you can see what works and what doesn’t before spending too much time or money.

Step 5: Test

The aim is to see how people use the prototype and learn what needs to change. You watch, listen, and ask questions while they interact with it.



- Use A/B or multivariate tests when appropriate.
- Conduct heuristic evaluations using Nielsen's 10 usability heuristics (used for testing usability of online products).

Visibility of System Status – Keep users informed with timely feedback (e.g., loading indicators).
Match Between System & Real World – Use familiar terms, icons, and mental models.
User Control & Freedom – If users get into an unwanted scenario by mistake, they should have an emergency exit. Allow undo/redo and easy exits.
Consistency & Standards – Maintain consistent terminology and design patterns. E.g. "add to cart" and "add to basket" should not be used interchangeably.

Error Prevention – Avoid error-prone situations; confirm critical actions.
Recognition Over Recall – Reduce cognitive load with visible options (e.g., autocomplete).
Flexibility & Efficiency – Support both novice and expert users (e.g., shortcuts).
Aesthetic & Minimalist Design – Show only relevant information upfront.
Error Recovery Support – Provide clear, actionable error messages.
Help & Documentation – Offer concise, searchable, task-focused guidance within the interface.

User Experience Design

PM Takeaways

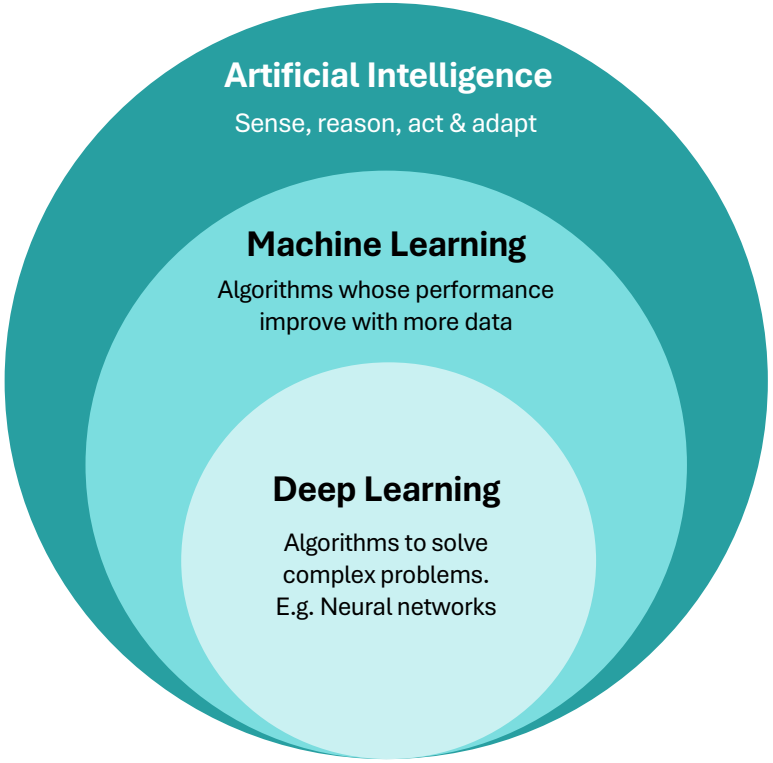
Empathy and research form the foundation of effective product design

Personas and problem statements align teams and ensure business–user fit.

Usability heuristics serve as a quick evaluation checklist before launch

Successful PMs integrate user needs, technical feasibility, and business goals throughout the design cycle

Machine Learning & Artificial Intelligence

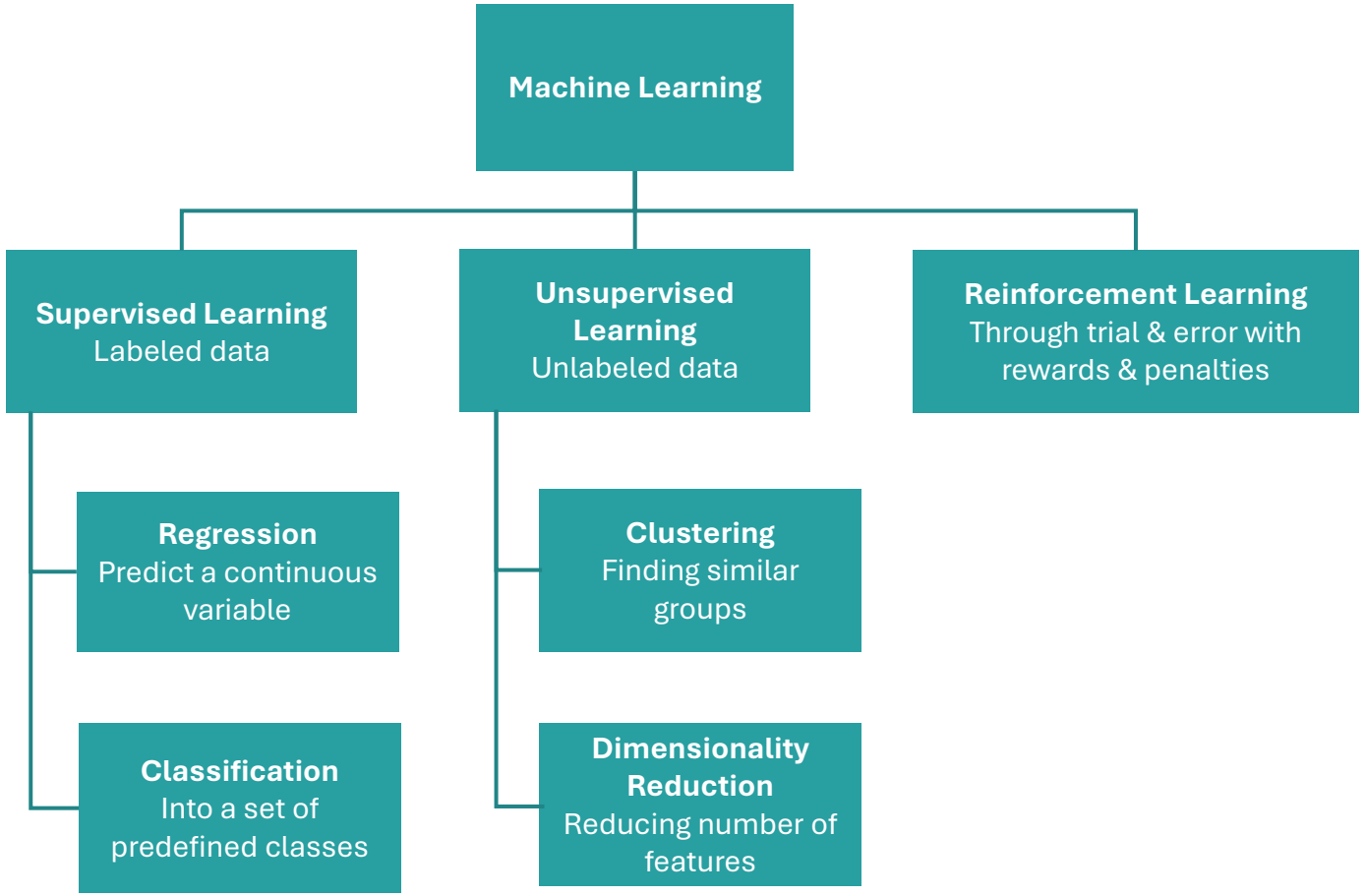


Terminologies

1. **Artificial Intelligence (AI):** Broad field aiming to create machines that can *sense, reason, act, and adapt* like humans.
2. **Machine Learning (ML):** A subset of AI where algorithms learn patterns from data to make predictions or decisions without explicit rule-based programming.
3. **Deep Learning (DL):** A subset of ML using multi-layered neural networks to handle complex tasks like image recognition, speech processing, and natural language understanding.

Why it matters for PMs

1. Enables data-driven product features (recommendations, personalization, automation).
2. Shapes user experience and competitive advantage.
3. Requires balancing performance, data needs, cost, and ethics.

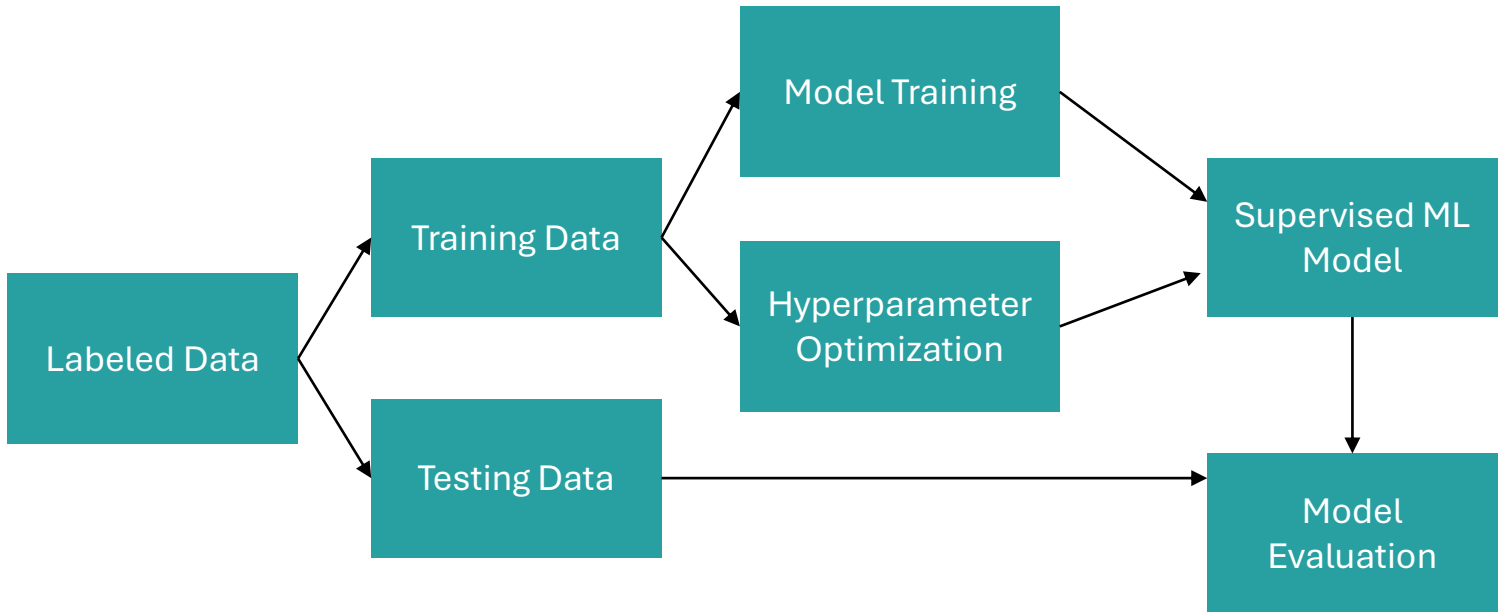


Types of Machine Learning

1. Supervised Learning: Learn from labelled data.
a) ***Regression*** → Predict continuous values (e.g., stock prices, weather).
b) ***Classification*** → Assign to categories (e.g., fraud detection, spam filter).

2. Unsupervised Learning: Find patterns in unlabelled data.
a) ***Clustering*** → Group similar data (e.g., customer segments).
b) ***Dimensionality Reduction*** → Simplify features while keeping patterns.

3. Reinforcement Learning: Learn by trial-and-error to maximize rewards (e.g., game AI, robotics).



Model training basics

1. **Labelled Data** is split into:
   - **Training Set** → Used to teach the model patterns in the data and adjust **hyperparameters** (settings that control learning).
   - **Testing Set** → Used after training to measure accuracy and check for overfitting (model memorizing rather than generalizing).
2. A well-trained model can then make predictions on **unseen data**:
   - **Classification** → Predict discrete categories.
   - **Regression** → Predict continuous values.

Once a model is trained, it can be fed unseen data to predict outcomes.

Recommender Systems

Core to products like Netflix, Amazon, YouTube, Spotify that help users discover things they didn't know they wanted.

1. **Content-Based** → Recommend items similar in attributes to what the user liked.
   - ***Pros*:** Simple, uses item features.
   - ***Cons*:** Misses items without shared attributes.

2. **Knowledge-Based** → Ask users for preferences and recommend accordingly (e.g., Netflix asking for favourite shows).

3. **Collaborative Filtering** → Use ratings/behavior of similar users to make recommendations.
   - ***Memory-Based*:** Find similar users/items and vote.
   - ***Model-Based*** *(e.g., Matrix Factorization with ALS)*: Learn hidden patterns ("latent factors") from partially filled rating matrices to predict missing ratings. Combines user preferences and item attributes.

Why it matters for PMs

Drives personalization → engagement → retention.

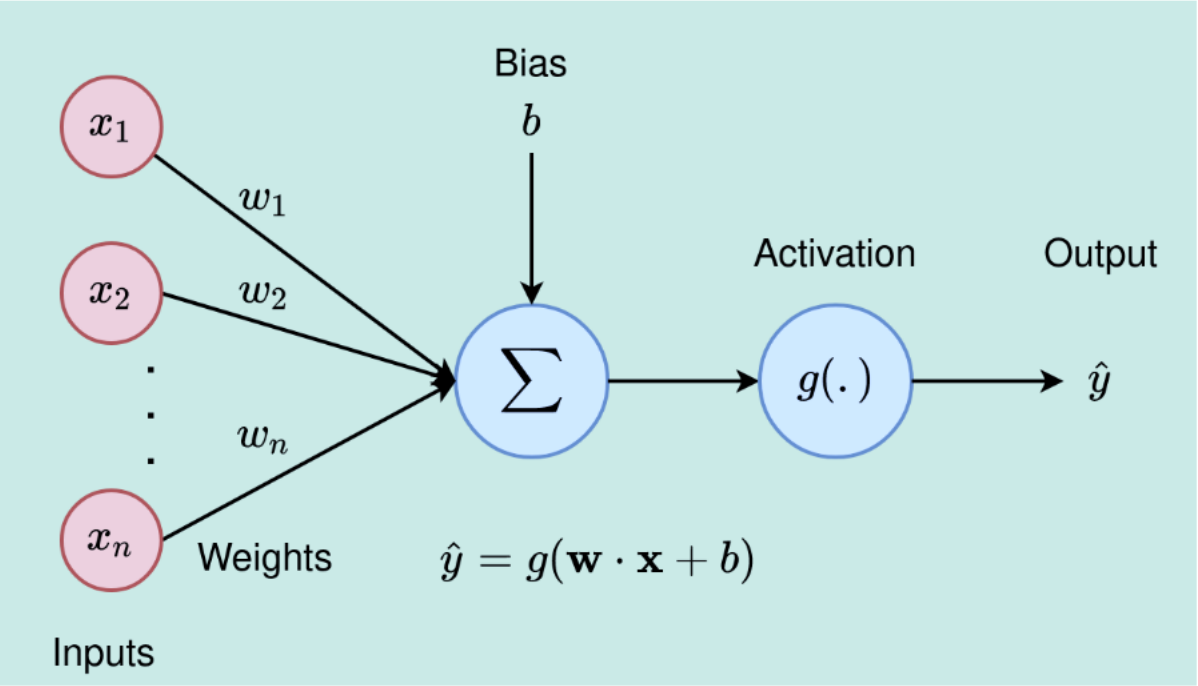
Choice of algorithm impacts user satisfaction, data needs, and scalability.

Neural Networks

Inspired by how the human brain works, neural networks take inputs, process them through connected "neurons" with weights (importance factors), and produce outputs.

1. **Biological Analogy**: In the brain, a neuron fires when the input exceeds a threshold, sending signals through axons. Similarly, in an artificial neuron, inputs are weighted, summed, passed through an **activation function** (e.g., tanh), and turned into outputs.

2. **Feature Extraction**:
   - ***Classical ML*:** Humans manually choose features from data.
   - ***Deep Learning (DL)*:** Networks with many **hidden layers** automatically learn and extract complex, high-level features.



Why it matters for PMs

DL powers cutting-edge applications (vision, speech, NLP) and reduces the need for manual feature engineering, but requires large datasets and strong compute resources.

AI/ML — Advantages for PMs

1. **Personalization at Scale** → Tailor experiences for millions of users automatically and improve customer service using virtual assistants.
2. **Automation** → Reduce manual work, speed up processes.
3. **Better Decision-Making** → Data-driven insights and predictions.
   - Helps in risk management and security
   - Identify and target new customer segments
4. **Competitive Edge** → Differentiates product features in crowded markets.

AI/ML — Disadvantages/Challenges

1. **Data Dependence** → Needs large, high-quality datasets.
2. **High Costs** → Computing, storage, and engineering talent are expensive.
3. **Complexity** → Harder to build, test, and explain than rule-based systems.
4. **Bias & Ethics Risks** → Can reinforce unfair patterns or invade privacy.

Algorithm Family	Great For	Ask Your Team	Watch Out For (The PM's Concern)
Linear Models	Establishing a simple baseline, predictable outcomes.	"Is the relationship between our inputs and output likely linear?"	Oversimplification. May miss complex patterns in user behavior.
Tree-Based Models	Explaining why a decision was made (high interpretability).	"How can we visualize the decision rules for the business team?"	Overfitting. The model might learn the training data too well and fail on new data.
Neural Networks	Finding complex, non-linear patterns in massive datasets (e.g., image or speech).	"Do we have enough data and computing power to train this effectively?"	The "Black Box" Problem. It's very difficult to explain why the model made a specific prediction, which is a risk for user trust and debugging.

A Framework for Applying AI/ML

Before embarking on an AI/ML project, ask yourself and your team these critical questions:

1. **Product Goal:**
   - What user problem are we solving?
   - How does this ML feature align with the overall product vision and goals?

2. **Integration:**
   - How will the ML component be integrated into the existing product?

3. **Starting Point:**
   - What is the minimum viable product (MVP) for this ML feature?
   - What data do we have, and what data do we need?
   - Is this data representative of all our users?

4. **Benchmarking:**
   - How will you measure success?
   - What are the key performance indicators (KPIs) and benchmarks?
   - What is our tolerance for false positives vs. false negatives, and what is the user experience impact of each?

5. **User Interaction:**
   - How will users interact with and control the AI-driven features?
   - How will we know if the model's performance is degrading over time?
   - What is our rollback plan?
   - What happens when the AI is wrong?

Augmented Reality & Virtual Reality

Augmented Reality (AR) and Virtual Reality (VR) are both technological experiences that change how digital technology interacts with the physical world.

Augmented Reality

- AR refers to observing the real-life environment with a digital augmentation overlaid. AR embeds video or images onto a display of the physical world, typically through a smartphone.
- The overlaid image/video creates an interaction between the user, the digital, and the physical worlds, allowing them to make connections or have new experiences. E.g. Pokémon Go

Virtual Reality

- In VR, the user interacts with an entirely virtual world using a headset and a controller.
- It creates a sensory experience by stimulating the sensory organs into interacting with the virtual world as they would with the physical world.

Why it matters for PMs

- For product managers, AR and VR offer powerful tools to create immersive user experiences, from virtual product demos and interactive training to new forms of data collection and rapid prototyping.
- However, PMs must evaluate platform compatibility, user readiness, and content strategy to design impactful, accessible, and future-ready products.

Augmented Reality & Virtual Reality

Augmented Reality Properties	Virtual Reality Properties
Visualisation: AR enables users to see complex details with clarity by overlaying digital imagery onto the real world. For instance, it can project the exact location of human veins to assist in blood draws or display how components fit together in intricate mechanical assemblies.	**Interaction:** AR changes the way interaction between humans and machines occurs by bypassing the need for physical controls in the future with virtual control panels.
Instruction: AR transforms learning and training by delivering real-time, context-aware guidance. It can overlay step-by-step instructions, diagrams, or videos directly onto the user's field of view, allowing them to follow procedures more accurately and efficiently.	**Interactivity:** VR environments respond dynamically to user inputs such as head movements, hand gestures, or controller actions. This real-time responsiveness allows users to explore, manipulate, and engage with objects or scenarios within the virtual world, increasing engagement and realism.
Interaction: AR changes the way interaction between humans and machines occurs by bypassing the need for physical controls in the future with virtual control panels.	**Sensory Feedback:** VR stimulates multiple senses to enhance realism. Visual and audio cues are standard, while advanced systems add haptic feedback (vibrations, touch sensations) or even motion platforms. This multisensory approach strengthens the feeling of presence and helps users better interpret the virtual environment.

Augmented Reality Working

1. Computer Vision:

The system analyses the live camera feed to understand the surrounding environment. It identifies objects, surfaces, and spatial context, creating a basic map of the real world from the user's perspective. This can be marker-based (identifying pre-programmed objects) or marker-less (recognising items on the go).

2. Rendering:

Based on this understanding, the system selects relevant digital content and overlays it precisely onto the real-world view. This content is aligned in size, position, and perspective to blend seamlessly with the physical environment, enhancing the user's experience.

Virtual Reality Working

1. Content Source:

The VR headset receives video input either from a connected PC via HDMI, a smartphone placed inside the headset, or its own built-in graphics processor (in standalone headsets).

2. Display Output:

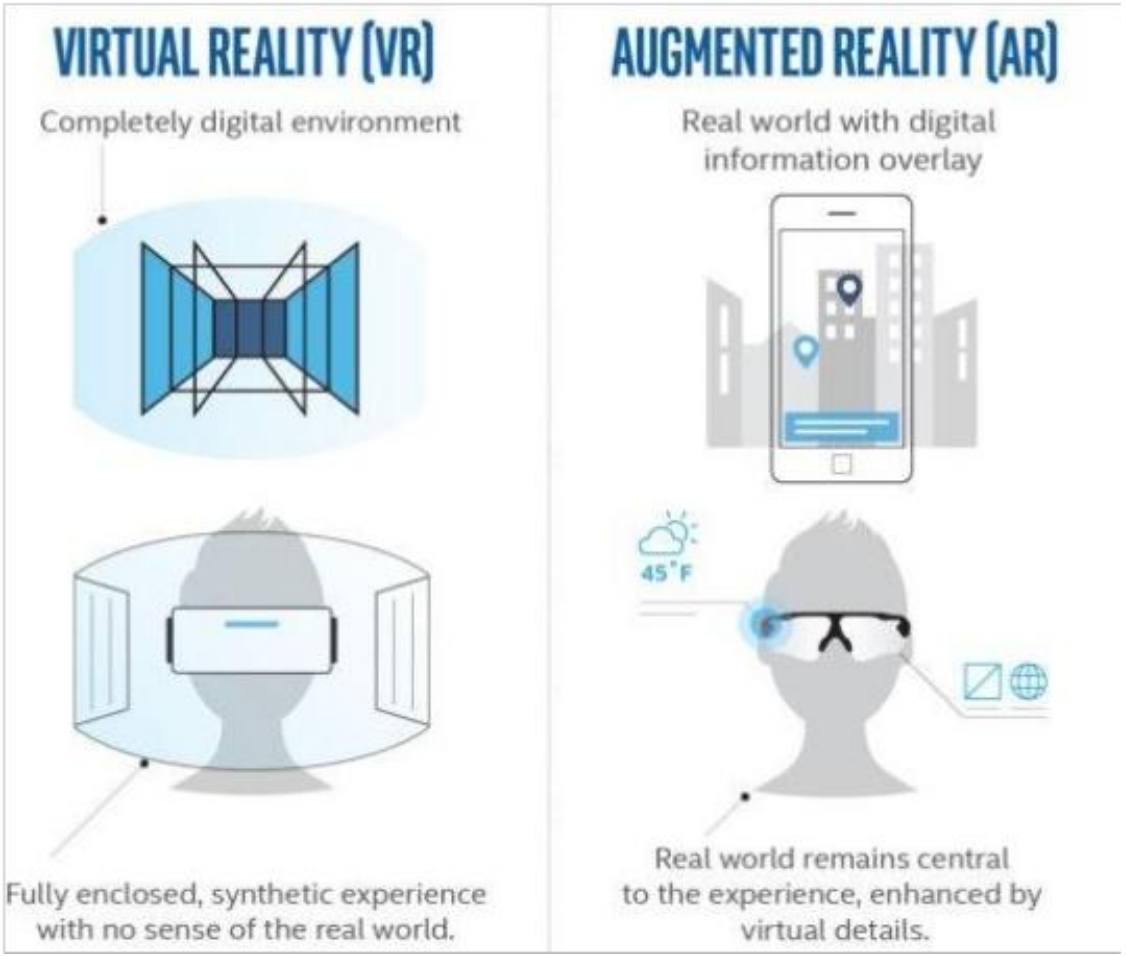
The VR headset receives video input either from a connected PC via HDMI, a smartphone placed inside the headset, or its own built-in graphics processor (in standalone headsets).

Feature	Augmented Reality	Virtual Reality
Goal	To enhance the real world with digital overlays.	To replace the real world with a simulated one
Environment	A mix of real and virtual elements	Fully digital and computer-generated
Hardware	Primarily smartphones, tablets, and smart glasses	VR headsets and controllers (e.g., Meta Quest).

3. Optical Processing:

Lenses positioned between the eyes and the screens adjust and distort the images so they align correctly for each eye. This creates a stereoscopic 3D effect, giving the user a realistic sense of depth and immersion.

Differences Between Augmented Reality & Virtual Reality



Applications of Augmented Reality

Smart Glasses

Wearable, transparent devices like Google Glass Enterprise and Microsoft HoloLens overlay digital information directly onto the user's field of view, enhancing real-world scenes with 3D visuals, notifications, and contextual data.

Healthcare

AR is used in healthcare for advanced medical imaging and surgical planning, enabling doctors to visualise internal anatomy (like tumours) in a 3D format before and during procedures, improving precision and patient safety.

Lifestyle

Smartphone apps such as measurement tools and decor visualizers use AR to overlay digital content on real-life objects, enabling users to measure dimensions or preview how furniture will look in their homes.

Entertainment

AR enhances experiences in gaming and social media through playful overlays like Snapchat filters, lens effects, and Pokémon GO-style interactive content that blends into the user's real-world surroundings



AR use in navigation

Applications of Virtual Reality

Entertainment

VR creates ways to access movies, places, and video games in 360 degrees. It allows users to look where they want in virtual environments, creating a new experience.

Healthcare

VR allows surgeons and those training in the medical field to watch surgery simulations, learn about tools, and experience patient dynamics in a virtual setting before surgery.

Training

VR can help with training for new drivers as the virtual environment is safer and more accessible than learning to drive in the physical world.

Education

VR brings students into other environments to get a clear picture of what they are learning about, making it more exciting. Additionally, it could make remote learning easier as students could participate in a virtual classroom.



AR use in navigation

Know more:

1. https://medium.com/xrpractices/generative-xr-space-aware-ai-bringing-new-era-of-immersion-9e5e863d1a1c
2. https://www.coursera.org/articles/augmented-reality-vs-virtual-reality
3. https://www.geeksforgeeks.org/blogs/top-modern-day-applications-of-augmented-reality-ar/

Cloud Computing

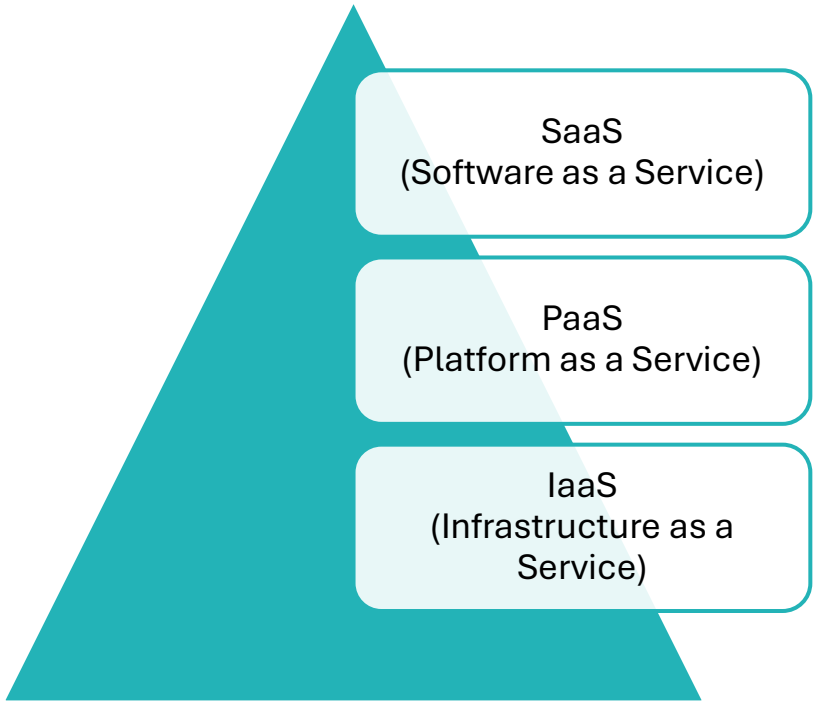
What is Could Computing?

At its simplest, **cloud computing** is the on-demand delivery of computing services—servers, storage, databases, networking, software, and advanced analytics—over the internet. Instead of owning and maintaining costly data centres, you rent resources from a provider like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).

Why is it important?

The shift to the cloud is the most significant technological change since the client-server model. It fundamentally alters business agility, financial models, and the very architecture of products.

Core Services Model



Cloud Computing

Core Services Model	IaaS (Infrastructure as a Service)	PaaS (Platform as a Service)	SaaS (Software as a Service)
What it is	Renting the fundamental building blocks of computing: servers, storage, and networking. It's like leasing a plot of land where you have maximum control but also maximum responsibility to build everything.	A platform for developers to build, deploy, and manage applications without worrying about the underlying infrastructure. It's like renting a fully equipped workshop with all the tools provided.	Ready-to-use software delivered over the internet on a subscription basis. The provider manages everything.
The PM's view	Use IaaS for maximum flexibility and control over the tech stack, especially for complex or highly custom applications. It requires significant engineering overhead.	Ideal for increasing developer velocity. It abstracts away infrastructure complexity, allowing the team to focus purely on application code and features. This is the sweet spot for many agile product teams.	As a PM, you might be building a SaaS product for your customers or integrating with other SaaS products (like Salesforce or Slack) to enhance your own offering.
Examples	Amazon EC2, Google Compute Engine	AWS Elastic Beanstalk, Google App Engine, Heroku.	Salesforce, Google Workspace, Dropbox.

Benefits of Cloud Computing	Drawbacks of Cloud Computing
Cost Efficiency – Pay-as-you-go model, reduced CapEx, more flexible OpEx model.	**Security & Privacy Risks** – Data on third-party servers; misconfigurations can cause breaches.
Scalability & Elasticity – Instantly scale resources up/down based on demand. Avoid cost of idle infrastructure	**Vendor Lock-in** – Hard to switch providers without high cost & complexity.
Speed & Agility – Provision servers in minutes, accelerating development & deployment.	**Cost Spikes** – Poor monitoring can lead to unexpectedly high bills due to Pay-as-you-go models
Global Reach – Deploy in multiple regions for lower latency & better UX.	**Downtime Risks** – Provider outages = your downtime.
Reliability & Disaster Recovery – Geo-distributed backups and failovers.	**Compliance Challenges** – Varying regional data laws add complexity.

When should a PM Consider Cloud Solutions?

Scenario	Why Cloud Helps
New Product from Scratch	Avoids high upfront costs; easy to scale while finding product–market fit.
Unpredictable / Spiky Traffic	Elasticity handles sudden demand surges without overprovisioning.
Need Faster Development	PaaS frees engineers from infra management to focus on features.
Global Expansion	Deploy apps close to users worldwide for low latency.
Big Data / AI / ML Workloads	On-demand, powerful compute & analytics without in-house build.
Aging On-Prem Infrastructure	Opportunity to migrate instead of reinvesting in costly hardware.

Public Cloud
Shared infrastructure, owned by third-party (AWS, Azure), accessed over the internet.

- ✅ Scalable & cost-effective
- ❌ Less control & potential compliance issues

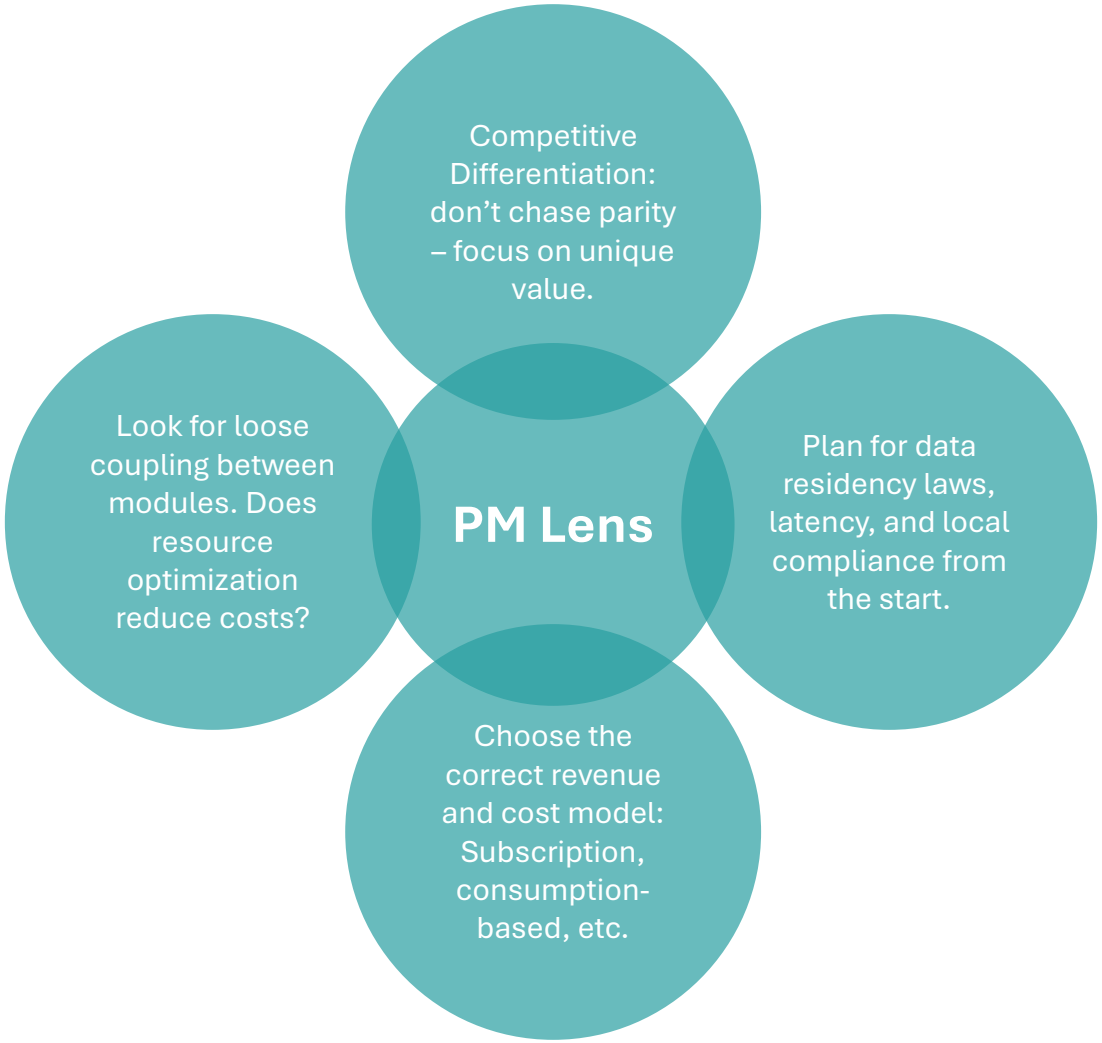
Hybrid Cloud
Mix of private + public cloud; data/apps move between them.

- ✅ Security + scalability
- ❌ More complex to manage

Private Cloud
Exclusive infrastructure for one org, on-prem or hosted.

- ✅ High security & control
- ❌ Expensive & less scalable

Cost Pricing Models	How It Works	PM's View
Pay-As-You-Go (On-Demand)	Pay per hour/second, no commitments.	Max flexibility for unpredictable workloads & dev/test; highest per-unit cost.
Reserved Instances	Commit for 1–3 years for big discounts (up to 75%).	Best for steady, predictable traffic; cost-optimization lever.
Spot Instances	Buy unused capacity at steep discounts (up to 90%); can be reclaimed anytime.	Great for fault-tolerant, non-critical jobs (batch/data processing); not for core user-facing services.
Freemium / Free Tier	Limited free usage each month.	Strong user acquisition strategy; lets customers try before buying.
Tiered Pricing	Multiple pricing tiers (Basic, Pro, Enterprise).	Aligns pricing with features/usage for different customer segments.


Competitive Differentiation: don't chase parity – focus on unique value.
Look for loose coupling between modules. Does resource optimization reduce costs?
PM Lens
Plan for data residency laws, latency, and local compliance from the start.
Choose the correct revenue and cost model: Subscription, consumption-based, etc.

BlockChain

Introduction:

- A blockchain is a distributed database or ledger shared across a computer network's nodes.
- They maintain a secure and decentralized record of transactions but are not limited to cryptocurrency uses.
- In an industry, blockchains can be used to make data industry immutable, i.e., it cannot be altered.
- It is a form of a mathematical data structure and is programmed to be immune to manipulation.

Properties:

1. **Open** → *Distributed ledger technology* - All network participants can access the distributed ledger. It records transactions only once, eliminating the duplication effort typical of traditional business networks.
2. **Immutable** → *Immutable records* - No participant can tamper with a transaction after it's been recorded in the shared ledger. If a transaction record includes an error, a new transaction must be added to reverse the error, and both transactions are then visible.
3. **Secure** → *Public key cryptography* - It secures transactions and data on the blockchain by leveraging two cryptographic keys: a public key and a private key. The private key holder can authorize transactions, provide security and verify ownership, while the public key allows others to send data to the correct address.

Working:

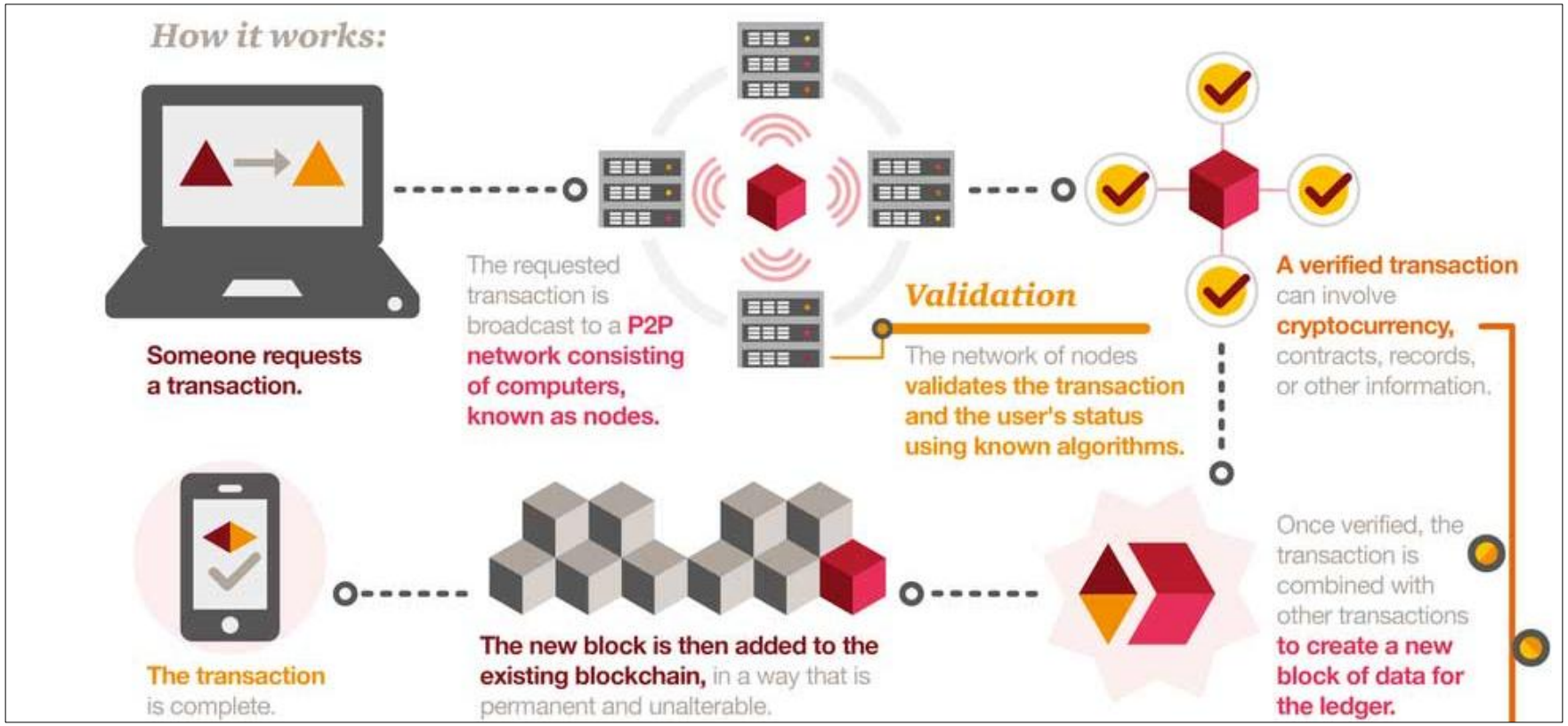
Each transaction is stored as a "block" containing specific details along with a timestamp to preserve chronological order.

Each block is connected to the previous and subsequent blocks via cryptographic hashes, making alterations virtually impossible without changing all the subsequent blocks.

All validated blocks are added to the chain permanently using consensus algorithms like Proof of Work (PoW) or Proof of Stake (PoS), creating a tamper-proof ledger.

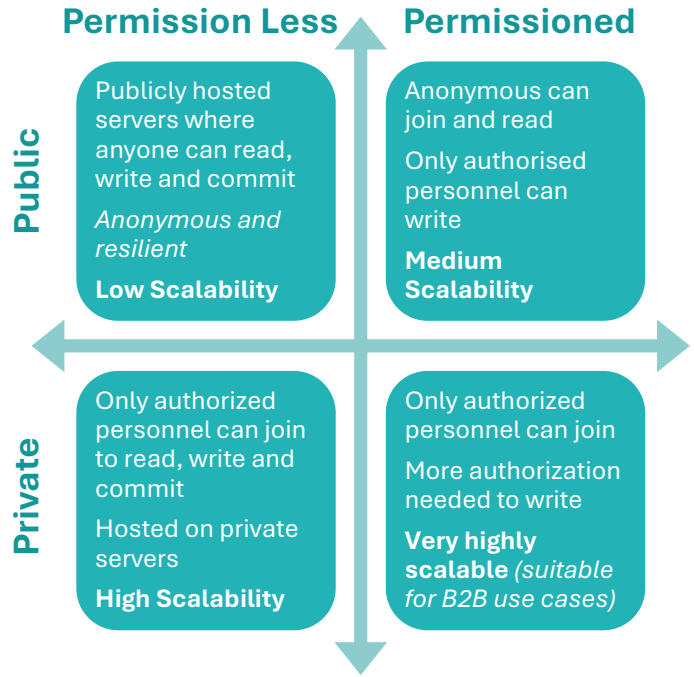
The chain's immutability builds a trusted, transparent record accessible to all network participants, preventing fraud and ensuring accuracy.

Working of Blockchain Technology



Types

1. **Public Blockchain** → A fully open network where anyone can join, view, and validate transactions (e.g., Bitcoin, Ethereum). It offers maximum decentralization but often has slower transaction speeds, higher energy use, and limited privacy.
2. **Private Blockchain** → A decentralized peer-to-peer network controlled by a single organization. The governing entity decides who can join, run the consensus process, and maintain the ledger.



Further types based on permission

1. **Public Permission-less** → Anyone can join the network, read data, and perform transactions without authorization. It operates on public servers and ensures anonymity and decentralization, making it resilient and censorship-resistant. However, due to its consensus-heavy design, it suffers from low scalability and slower transaction speeds. *E.g., Bitcoin, Litecoin and Polkadot*.
2. **Public Permissioned** → Anyone can access and read the blockchain data, but only selected, authorized participants can write or validate transactions. It balances openness with control, offering medium scalability. It suits use cases where transparency is key, but data integrity, such as government registries or regulated public services, must be protected. *E.g., Ripple (XRP Ledger)*
3. **Private Permission-less** → Only authorized individuals can join the network, and all participants can read, write, and commit data freely within the network. Hosted on private servers, it ensures tighter access control and better performance than public systems. It offers high scalability, which is ideal for internal enterprise applications requiring speed, privacy, and moderate decentralization. *E.g., MultiChain (configured in open-write mode)*
4. **Private Permissioned** → Most restrictive and enterprise-friendly model where only pre-approved participants can join, and even among them, writing rights are further limited to select users. Hosted privately and tightly governed, it ensures maximum control, security, and compliance. With very high scalability, it's best suited for B2B settings like supply chains, banking, and healthcare, where trust, speed, and confidentiality are critical. *E.g., Hyperledger Fabric and R3 Corda*

Business Value:

1. **Lower risk & compliance costs** → Shared, blockchain-based KYC lets banks verify a customer once and reuse the verification securely across institutions, cutting costs and improving onboarding speed.
   For e.g., In Oct 2017, OCBC, HSBC, and MUFG, with Singapore's IMDA, completed Southeast Asia's first blockchain-based KYC POC. KPMG estimated 25–50% cost savings by removing repeated checks and adding a clear audit trail.
2. **Faster & cheaper transactions** → Digitizing trade documents on a blockchain eliminates paper handling and manual checks, speeding up cross-border deals.
   For e.g., in May 2018, HSBC and ING settled a soybean shipment from Argentina to Malaysia in 24 hours (vs. 5–10 days), the first live commercial trade finance transaction using blockchain.
3. **Automated, secure contract fulfilment** → Smart contracts execute automatically when conditions are met, reducing human errors and delays.
   For e.g., in 2018, Walmart and IBM's Food Trust traced mango shipments in 2.2 seconds (vs. 7 days), improving food safety, recall speed, and transparency.

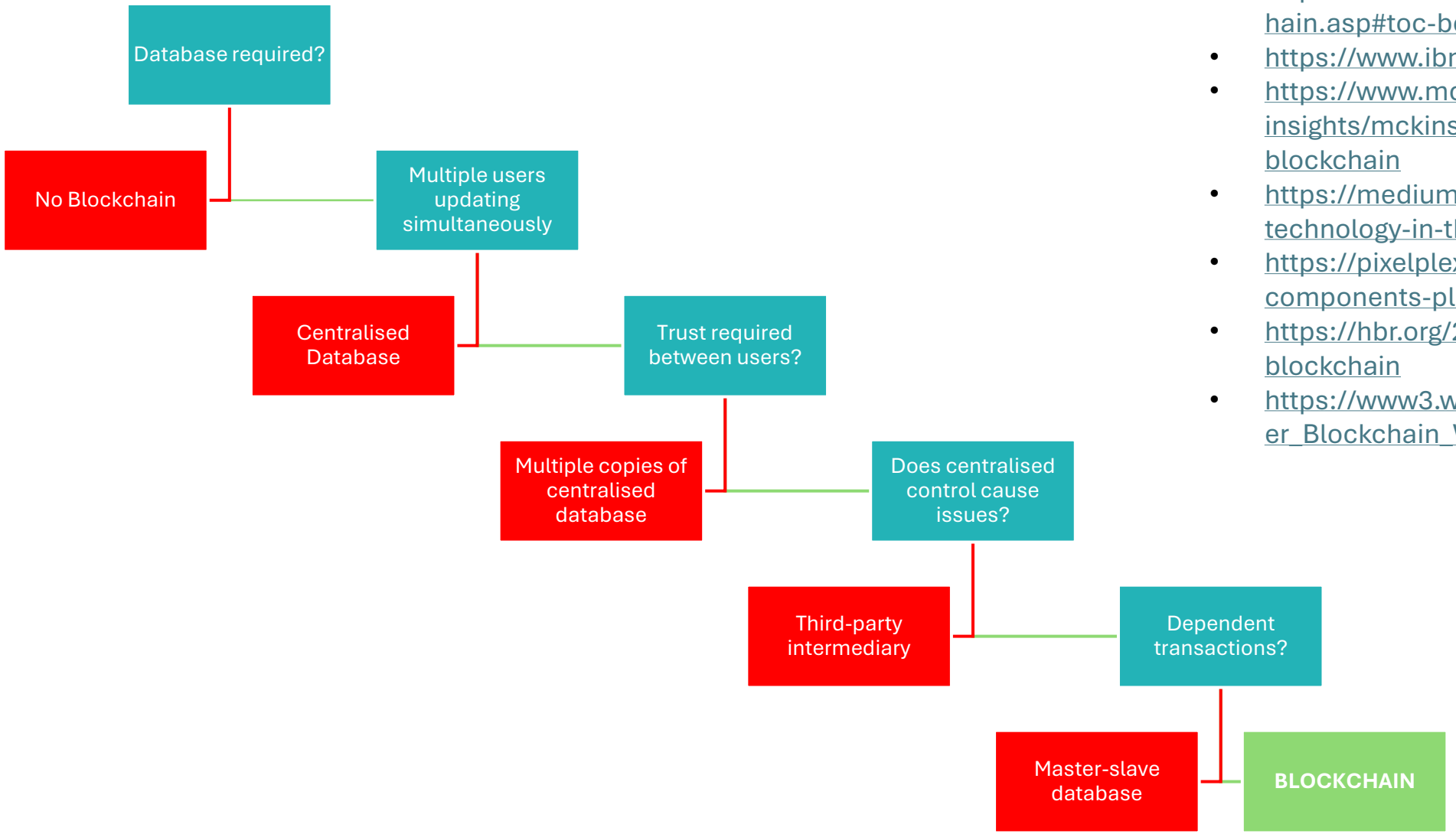
Shortcomings:

1. **High Energy consumption** → Computational power needed for mining results in enormous energy consumption. This environmental cost poses a barrier to sustainability and green innovation.
2. **User adoption due to complexity** → Despite its potential, blockchain is not user-friendly. Wallets, private keys, and smart contracts are complicated for the average user to navigate. Poor UX and technical jargon limit mainstream adoption, especially outside tech-savvy communities.
3. **Regulatory & legal uncertainty** → Blockchain faces inconsistent global regulations that hinder adoption.

Applications:

1. **Cryptocurrency** - Blockchain enables secure, peer-to-peer digital money transfers without banks or intermediaries.
2. **Supply Chain Tracking** - It records every step of a product's journey from source to customer in a tamper-proof ledger. This increases transparency and helps verify authenticity.
3. **Smart Contracts** - Blockchain can run coded agreements that automatically execute when conditions are met. This removes the need for middlemen and speeds up transactions.
4. **Digital Identity** - It offers individuals and organizations secure, verifiable digital IDs. This helps reduce fraud and simplifies online verification.
5. **Voting Systems** - Blockchain creates transparent, auditable voting records that are hard to alter. This improves trust and reduces the risk of election fraud.

How PMs decide when to use Blockchain



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- https://www.ibm.com/think/topics/blockchain
- https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-blockchain
- https://medium.com/@lvooO/blockchain-technology-in-the-food-industry-ddf2d34cced6
- https://pixelplex.io/blog/blockchain-explained-components-platforms-use-cases/
- https://hbr.org/2017/01/the-truth-about-blockchain
- https://www3.weforum.org/docs/48423_Whether_Blockchain_WP.pdf

Internet of Things (IoT)

Introduction

1. The Internet of Things (IoT) is a network of physical devices embedded with sensors, software, and connectivity, enabling them to collect, exchange, and act on data. These “smart” devices range from consumer gadgets like fitness trackers to large-scale systems like smart cities, which result in improved efficiency, convenience, and decision-making.

2. IoT will enable PMs to offer personalized experiences through real-time monitoring. It can further be coupled with AI to gather actionable insights and perform predictive analytics, and additional sources of revenue can be unlocked, thus providing a competitive edge.

Properties

1. **Connectivity** → IoT devices use wired or wireless connections to exchange data with other devices or platforms. This typically includes Wi-Fi for home automation, Bluetooth for wearables, Zigbee for low-power smart devices and 5G for high-speed industrial IoT.

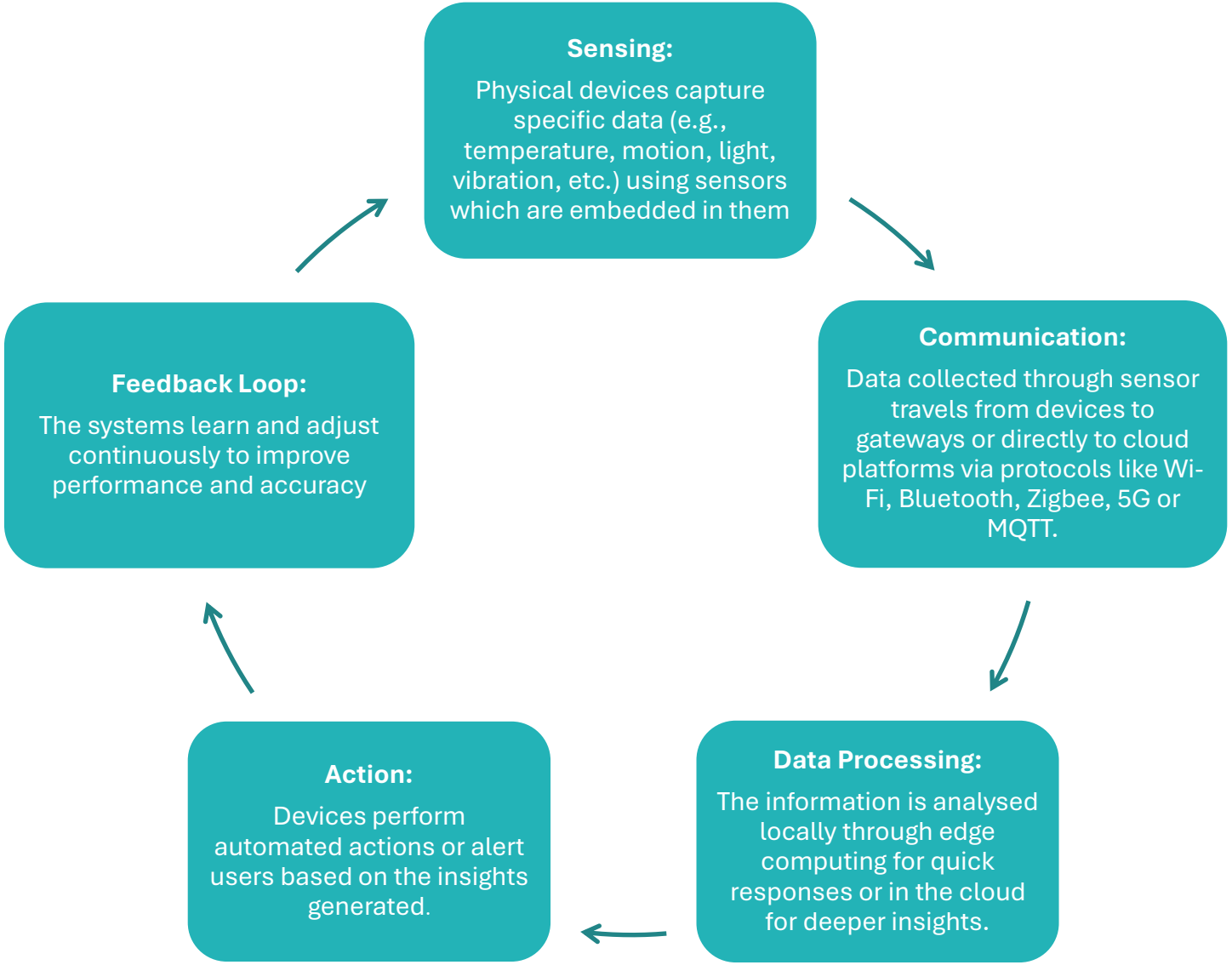
2. **Automation** → Devices can execute actions automatically when triggered by certain conditions, reducing manual intervention.

3. **Scalability** → IoT platforms are designed to handle thousands to millions of connected devices simultaneously, with the ability to add more without degrading performance.

4. **Interoperability** → IoT ensures devices from different manufacturers can communicate and work together using standard protocols or APIs.

5. **Real-Time Insights** → IoT allows for continuous data capture and analytics for quick decisions and immediate action.

Working


Sensing:
Physical devices capture specific data (e.g., temperature, motion, light, vibration, etc.) using sensors which are embedded in them
Communication:
Data collected through sensor travels from devices to gateways or directly to cloud platforms via protocols like Wi-Fi, Bluetooth, Zigbee, 5G or MQTT.
Data Processing:
The information is analysed locally through edge computing for quick responses or in the cloud for deeper insights.
Action:
Devices perform automated actions or alert users based on the insights generated.
Feedback Loop:
The systems learn and adjust continuously to improve performance and accuracy

Internet of Things (IoT)

Types

Type	Consumer IoT	Commercial IoT	Industrial IoT	Infrastructure IoT
Focus	• Convenience • Lifestyle enhancement	• High security standards (sensitive data) • Customer experience	• Predictive maintenance • Safety	• Sustainability • Safety
Sector	• Home • Fitness • Entertainment	• Healthcare • Retail • Hospitality	• Manufacturing • Logistics • Energy • Infrastructure	• Smart cities • Utilities • Transportation
Applications	• Smart homes (voice assistants, smart lighting, connected appliances) • Connected health devices (smart glucose monitors)	• Smart retail (personalized in-store offers) • Asset tracking in warehouses	• Predictive maintenance for machinery • Remote equipment monitoring in manufacturing plants	• Smart grids for energy distribution • Smart water management (leak detection, quality monitoring)

Internet of Things (IoT)

Business Value of IoT

Operational Efficiency

IoT streamlines processes by automating tasks, reducing delays, and optimizing resource allocation. This is especially valuable in industries where downtime is expensive.

For example, Amazon warehouses use IoT-enabled robots to locate and transport items, speed up order fulfilment, and reduce labor costs.

Data-Driven Decision Making

Real-time data collection enables organizations to respond quickly to changing conditions and improve decision quality.

For example, Starbucks uses IoT-connected coffee machines to monitor drink quality and adjust recipes remotely based on performance analytics.

Enhanced Customer Experience

IoT enables personalization, convenience, and faster service, increasing customer loyalty and satisfaction.

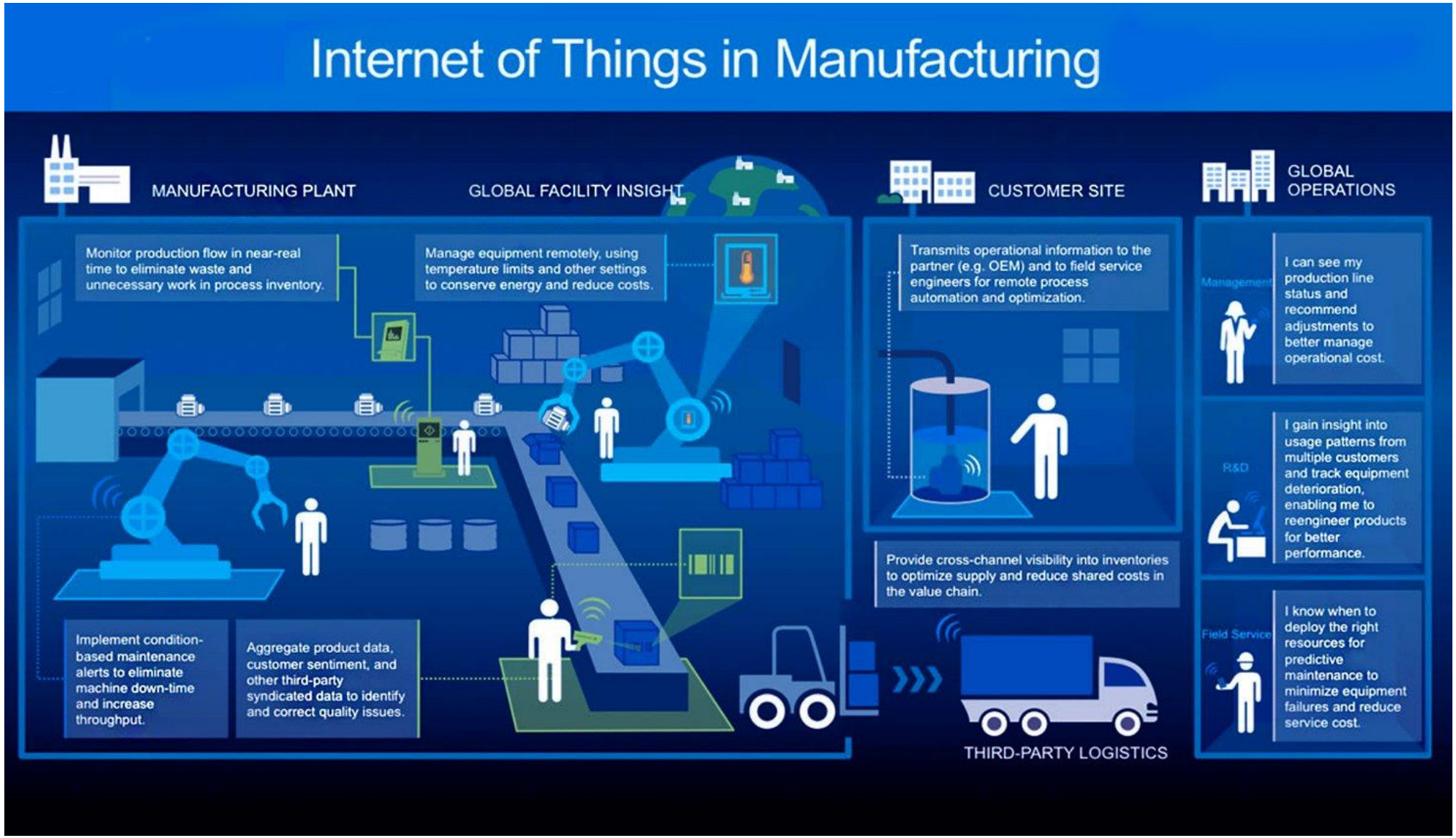
For example, Disney's MagicBand allows park visitors to seamlessly unlock hotel rooms, enter rides, and make cashless payments.

New Revenue Models

IoT provides subscription-based services, pay-per-use models, and data monetization opportunities.

For example, John Deere offers precision farming subscriptions that use IoT data to optimize planting and harvesting schedules.

Internet of Things (IoT)



Economic Features of Digital Platforms – Differentiability

Sustainable differentiation lives in the **customer's perception** (not just internal process improvements). For example, if Uber creates a new algorithm that increases cab availability for customers, but the customers do not perceive any significant difference between Ola and Uber, then Uber has not differentiated itself. There are two kinds of differentiation – Uber has not differentiated itself. There are two kinds of differentiation –

- **Horizontal differentiation:** Different tastes/identities (e.g., Reddit communities, niche job boards, gamers vs professionals).
- **Vertical differentiation:** Objectively "better" experience (lower search cost, higher trust/safety, better matches, faster delivery).

Platforms typically win on vertical (quality/governance/trust) and cement their position with horizontal community/identity over time.

Two levers dominate in differentiation -

- **Cost of convergence:** Copying the core product is not difficult in digital platforms. Hence, businesses need to find ways to have a competitive edge over competitors. Some ways include data moat, exclusive deals/partnerships, developing highly trusted and dense networks, and eventually creating a strong and memorable brand.
- **Switching costs:** For platforms, the aim is to raise the *real* cost of leaving (time, money, convenience, lost benefits) without resorting to dark patterns. Practical levers include:
  - Reputation and history (ratings, reviews) that users don't want to abandon
  - Deep product/workflow integrations (APIs, catalogues, payments, logistics) that would be painful to rebuild elsewhere
  - Loyalty and financial hooks (subscriptions, fast payouts/financing) that accrue over time
  - Social and community graphs (followers, groups, messaging) that users value
  - Personalisation and saved data (preferences, lists, models) that make your experience meaningfully better
  - Fair contractual/verification frictions (KYC, guarantees, insurance) that improve trust while making switching non-trivial.

Takeaway: Late entrants win by differentiating rather than fighting a pure scale war.

Economic Features of Digital Platforms – Network Effects

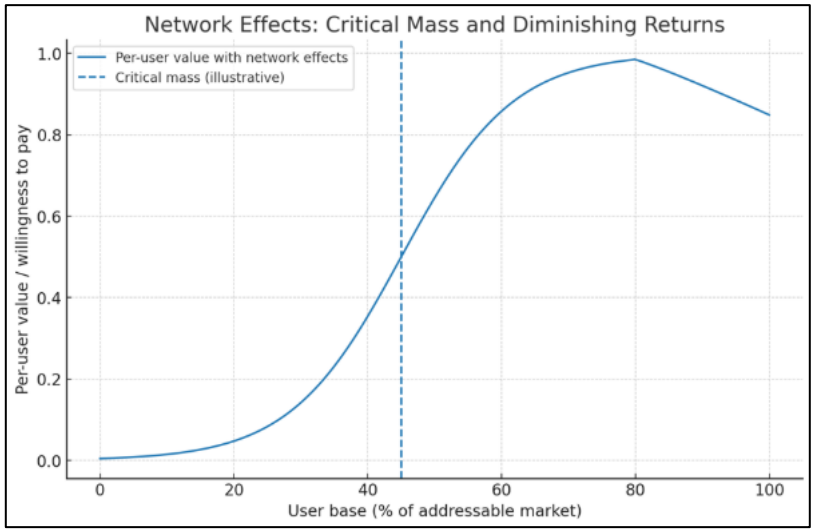
A network effect exists when a user's value increases as more users join. There are various ways in which Network Effects can kick in.

Side of network effects:

- ***Same-side:*** Value rises as more peers join. For example, Instagram is more useful if your friends are on the platform.
- ***Cross-side:*** Value rises across sides. More Airbnb hosts incentivise more guests to book on the platform.

Scale of network effects:

- ***Local:*** Effects confined to geographies/segments. Generally, these are more competitive given the increased investments required to scale the platform across geographies. For example, Uber competes with local rivals in almost every geography it plays in.
- ***Global:*** Inventory or interactions travel globally. For example, the more hosts there are on Airbnb, the more attractive the platform becomes worldwide.



Network Effects S-curve

Pricing and Competition in Multi-Sided Markets

Two-Sided Price Structure

Platforms set asymmetric prices to internalise cross-side externalities. Typically, they **subsidise the more price-sensitive or network-generating side** (e.g., free usage for consumers) and **monetise the other side** (e.g., advertisers, merchants). The split evolves over the lifecycle—heavy subsidies at launch to cross the cold start, then gradually tapering.

Competition, Multihoming, and Bottlenecks

- **Multihoming:** If users/suppliers easily switch between multiple platforms, rivalry remains active, and mark-ups stay in check.
- **Competitive bottlenecks:** When one side is locked in (exclusivity, high switching costs), the platform with privileged access can capture outsized value. Example here is Apple App Store. Given the heavy lock in of users on the platform, Apple can charge 30% commissions to developers, which is difficult to imagine if there were multiple App stores.
- **Expectations & tipping:** User expectations about future participation can stall launches or accelerate dominance once momentum builds.

Strategic Issues in Platform Management

- **Launching problem:** Overcome cold start via subsidies, exclusive/supply seeding, or a single-player utility that's useful before a network forms. Expect cash burn until critical mass. overcome cold start via subsidies, exclusive/supply seeding, or single-player utility that's useful before a network forms. Expect cash burn until critical mass.

- **Marketplace vs Reseller:** Marketplaces scale breadth and take rates with low capital intensity; reseller/hybrid models offer tighter quality control and margin on selected lines.

- **Platform leakage:** Prevent off-platform transactions by making on-platform strictly safer/better (payments, escrow/insurance, dispute resolution, guarantees).

Demand – Supply, Surplus, and Value Capture

In competitive markets, consumer and producer surplus are split at equilibrium. Platforms frequently deploy **price discrimination** (auctions, personalized ads, tiered fees), enabling them to capture more surplus. Strong network effects also steepen effective demand (higher willingness to pay), increasing total surplus—of which a well-governed, powerful platform can claim a large share.

Conclusion

Digital platforms combine powerful **economies of scale**, **network effects**, and **differentiation** to create self-reinforcing advantages. These forces explain both rapid growth and a tendency towards concentration. Strategy thus revolves around crossing critical mass, choosing sound price structures and governance, and sustaining defensible differentiation—while policy safeguards competition and user welfare without dulling innovation.

LLMs

- LLM refers to a special type of AI model called a Large Language Model. LLMs, as the name suggests, are trained with extensive text data.
- With an input of text, they can respond with probabilistically possible next set of words or text based on the statistical patterns it had learnt about the natural language of humans from the training data.

How LLMs *Really* Work: A Probabilistic Approach

- At its core, a Large Language Model is a sophisticated **probabilistic engine**.
- When you give an LLM a prompt like *"The first person on the moon was..."*, it doesn't "know" the answer. Instead, it calculates the statistical probability of every word in its vocabulary that could come next.
  - "Neil" might have a 75% probability, "Buzz" a 15%, and "Mickey" a 0.001%.
- It chooses a word based on these probabilities and then repeats the process, predicting the next word after "Neil," and so on, to generate a complete response.

Under the Hood: Encoding, Transformers, and Decoding

- To predict the next word, a model needs to understand the input you provide. This happens through a three-step process powered by architecture called the **Transformer**.
- **Encoding: Turning Words into Meaningful Numbers-** Models don't read words, they process numbers. The first step, **encoding**, converts your prompt into a series of numbers called "tokens." Each token represents a word or a piece of a word. These numerical representations (vectors) capture the word's semantic meaning and relationship to other words.
- **The Transformer: Understanding Context-** This is the engine room. The **Transformer architecture**, introduced by Google in 2017, was a revolution because of its "attention mechanism." This allows the model to weigh the importance of different words in the input sequence. When processing the sentence "The robot picked up the heavy box because **it** was strong," the attention mechanism helps the model understand that "**it**" refers to the "robot," not the "box."
- **Decoding: Turning Numbers Back into Words-** After the Transformer processes the encoded input, the **decoding** phase begins. The model generates a new token, a numerical representation of the most probable next word. This token is then converted back into human-readable text. This process repeats, with each newly generated word being fed back into the model to help predict the next one, until the full response is complete.
- This entire architecture is what the "GPT" in models like GPT-4 stands for: **G**enerative **P**re-trained **T**ransformer. It's **G**enerative because it creates new text, is **p**re-trained on a massive dataset, and is built on the **T**ransformer architecture.

Controlling the Output: Key Parameters

- You can influence an LLM's probabilistic choices using specific parameters. While the prompt is your primary tool, the standard technical parameters controlling the model's creativity are Temperature, Top-P and Top-K.
- **Temperature (Creativity vs. Coherence):** Temperature controls the randomness of the output. A **low temperature** (e.g., 0.2) makes the model more deterministic. It will try to choose the word with the highest probability. This is ideal for factual tasks like summarisation or data extraction where you need predictable, coherent answers. On the other hand, a **high temperature** (e.g., 0.9) increases randomness. The model might pick a less likely word, leading to more creative, diverse, or unexpected outputs. This is useful for brainstorming, creative writing, or generating multiple different options.
- **Top-K Sampling:** This parameter instructs the model to create a shortlist of the 'K' most likely next words and then choose only from that list. A **low K** (e.g., K=1) makes the model very safe and deterministic.
- **Top-P (Nucleus Sampling):** Top-P is another method to control randomness. It tells the model to consider only the most probable words that comprise a certain cumulative percentage (the "nucleus").
- For most applications, you'll use a low temperature for predictable outputs. For creative tasks, you might increase it.

The AI Model Landscape: LLMs, SLMs, and Reasoning Models

The term "LLM" is often used as a catch-all, but the landscape is more nuanced. The following table captures the different models that are in use today –

Model Type	Key Characteristics	PM Use Case
LLM	Large, general-purpose text models (GPT-4, Gemini). Versatile but costly	Chatbots, content generation, research assistance
SLMs (Small Language)	Small, efficient, task-specific models	Ticket classification, quick text summarisation
Reasoning Model	Optimized for multi-step problem-solving	Strategic analysis, complex decision support
Multimodal Model	Handle text, images, audio, and video together	Generate UI from sketches, read & summarize reports.
Image Gen Model	Create images from text prompts (DALL·E, Midjourney)	Marketing creatives, product mock-ups
Speech Model	Convert speech ↔ text, add voice to apps	Voice assistants, meeting transcription
Domain Models	Trained for one field (finance, law, medicine)	Contract review, medical advice, market risk
AI Agents	Use multiple models/tools to act autonomously	Automated research, workflow execution

Prompt Engineering

Large language models use text prompts (sometimes with image prompts) to predict the output. **Prompt Engineering** is the practice of strategically designing your input to guide a model toward a specific, high-quality, and relevant output. If an LLM is a brilliant, improvisational actor, the prompt is your script and direction. Your word choice, style and tone, structure, context, training data, model configurations, and the model itself all have a role in how effective your prompt is. Some basic and popular prompt engineering methods are -

Technique	Explanation	Use Case	Example
Zero / One / Few-Shot	Give the model zero, one, or a few examples so it learns the pattern instantly. Zero-shot = just instructions, few-shot = multiple examples.	Classify user feedback into **Bug**, **Feature Request**, or **UI/UX Issue**	"Here are 3 labelled reviews. Now, label these 20 new ones the same way."
Role Prompting	Tell the model to act in a specific role so tone and style match that persona.	Create different outputs for engineers vs. leadership	"Act as a Product Manager. Summarise this idea for the CEO."
Chain-of-Thought (CoT)	Ask the model to think step-by-step before answering, ensuring logical reasoning.	Break down a market entry decision logically	"Should we launch in Southeast Asia? Think step-by-step: market size → competition → costs."
ReAct (Reason + Act)	Alternate between reasoning and using tools (e.g., web search) to get fresh info before final answer.	Gather competitor data and prepare analysis	"Search for current iPhone prices in India, compare with Samsung, then suggest positioning."

Other vital guidelines to craft an effective prompt include

- **Start Simple** – Give a short, explicit instruction first. Add details only if needed.
- **Be Clear** – Say exactly what format, length, and tone you want.
- **Use Positive Language** – Tell the model what to do, not what to avoid.
- **Show Examples** – Give correct, well-formatted samples to copy.
- **Test & Save** – Try your prompt, note what works, and keep a record.

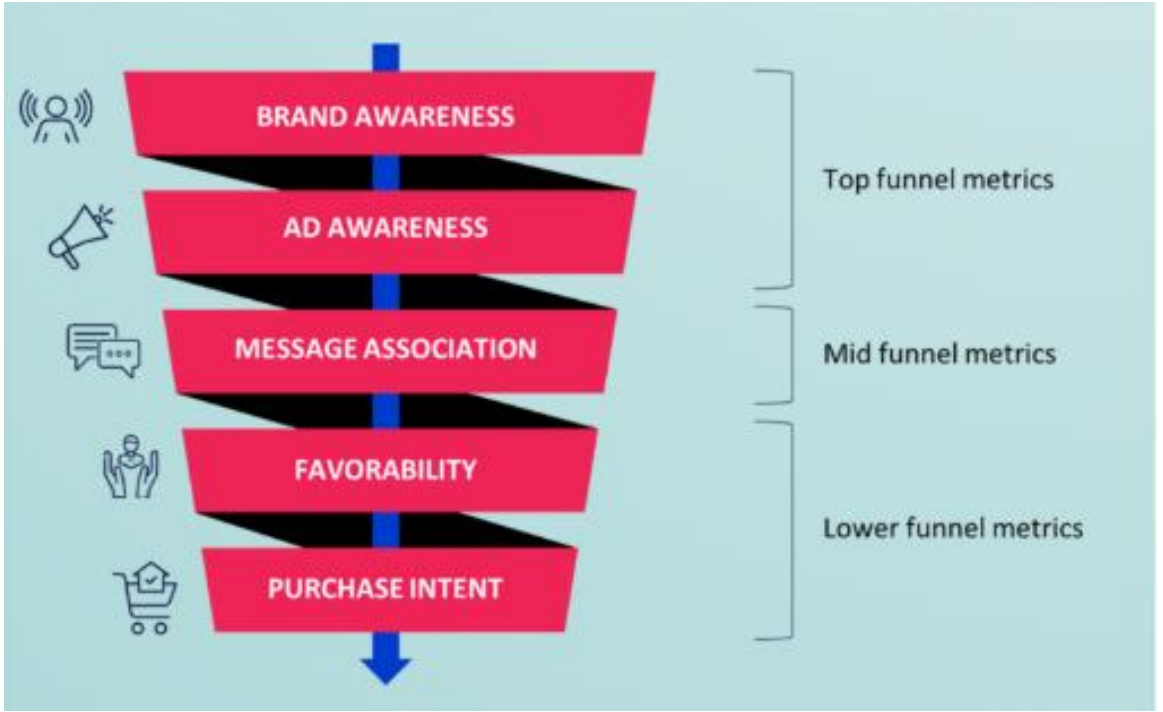
Reference

https://www.kaggle.com/whitepaper-prompt-engineering

Customer Funnel & Brand Lift

Introduction

As a product manager, understanding the impact of your marketing efforts on a feature improvement or new product development is crucial. A **brand lift study** is a powerful tool to measure how well your campaign has changed people's **awareness** and **perception** of your product. Awareness improves conversion at the top end of the customer funnel, while perception helps with the bottom end.



A brand lift study helps you measure the effectiveness of your marketing campaigns in improving how people **perceive** and are **aware** of your new feature or product. It's like taking a before-and-after snapshot to see how your campaign has influenced people's opinions and behavior.

Brand Lift Studies probe brand parameters amongst control and exposed audience sets, to measure the impact (delta) of any feature improvement. It is similar to the **RCT (Randomized Controlled Trials) method**.

Customer Funnel & Brand Lift

What questions can you see to get answers to?	
Awareness	Are more people aware of the new feature or product after the campaign?
Perception	Do people have a better opinion of the new feature or product?
Intent	Are more people interested in using or buying the new feature or product?
Engagement	Are people more engaged with the new feature or product (e.g., increased usage, positive feedback)?

Benefits of a Brand Lift Study

Understand Impact

Observe how well your marketing efforts for the new feature or product are working.

Make Data Driven Decisions

Use the insights to improve future campaigns and product strategies.

Optimize Budget

Allocate resources more effectively by knowing what works best.

Prove ROI

Demonstrate the return on investment (ROI) of your marketing spend to stakeholders.

Customer Funnel & Brand Lift

How does a Brand Lift Study work?

1

Setting Up the Study
Determine what you want to measure, such as awareness, favorability, or usage intent for the new feature or product. Develop questions to ask people about their awareness and perceptions of the feature or product before and after the campaign.

2

Collecting Data
Split your audience into two groups. One group sees the marketing campaign (exposed group), and the other doesn't (control group). Ask both groups about their awareness and perception of the feature or product before the campaign starts.

3

Running the Campaign
Promote the new feature or product as planned.

4

Post-Campaign Survey
After the campaign ends, ask both groups the same questions again to see if there are any changes in their awareness and perception.

5

Analyzing Results
Look at the differences in responses between the control and exposed groups before and after the campaign. Calculate the "lift," which is the improvement in awareness, perception, or intent caused by the campaign. For example, if 20% of people in the control group knew about the feature before the campaign and 40% in the exposed group knew about it after, the brand lift in awareness would be 20%.

Real-world problem statements

YouTube

We're *rolling out YouTube Shorts globally. Are our campaigns making people aware that Shorts exists as a competitor to TikTok, and do they see it as fun and engaging enough to use? They would track* ***awareness & perception*** *here.*

WhatsApp

We introduced disappearing messages and new encryption features. Do users feel WhatsApp is safer after our awareness campaign, and are they more likely to trust us with private conversations? This would focus on ***perception lift (safety, trust)*** *and* ***intent*** *(are people using WhatsApp more for sensitive chats?), especially after Facebook's history with privacy concerns.*

AirBnB

We introduced 'I'm Flexible' search to help users explore new destinations and dates. Did our ads and in-app nudges make travelers aware of this tool, and has it changed their perception of Airbnb from being just a booking site to a discovery platform? This tracks engagement and awareness.

What is RAG?

Retrieval Augmented Generation (RAG) is a feature in modern LLMs that allows them to fetch updated information on questions asked by the users. It allows LLM models to access external knowledge to update its output, rather than being dependent only on the knowledge present in its training data. This has made models more useful for general QnA, as LLMs before RAG (like the initial versions of ChatGPT) were restricted in their knowledge base to the timestamp of its training data, which was till 2021.

What are these external sources?

External sources of data may be results generated by search engines or 3rd party knowledge graphs. They also may be updated company documents for internal LLMs used for reporting or summarization purposes.

Applications

Search Queries: For search queries on LLMs like ChatGPT related to very recent instances, RAG may be used to search for updated web pages covering the instance and using their knowledge to improve the LLM output

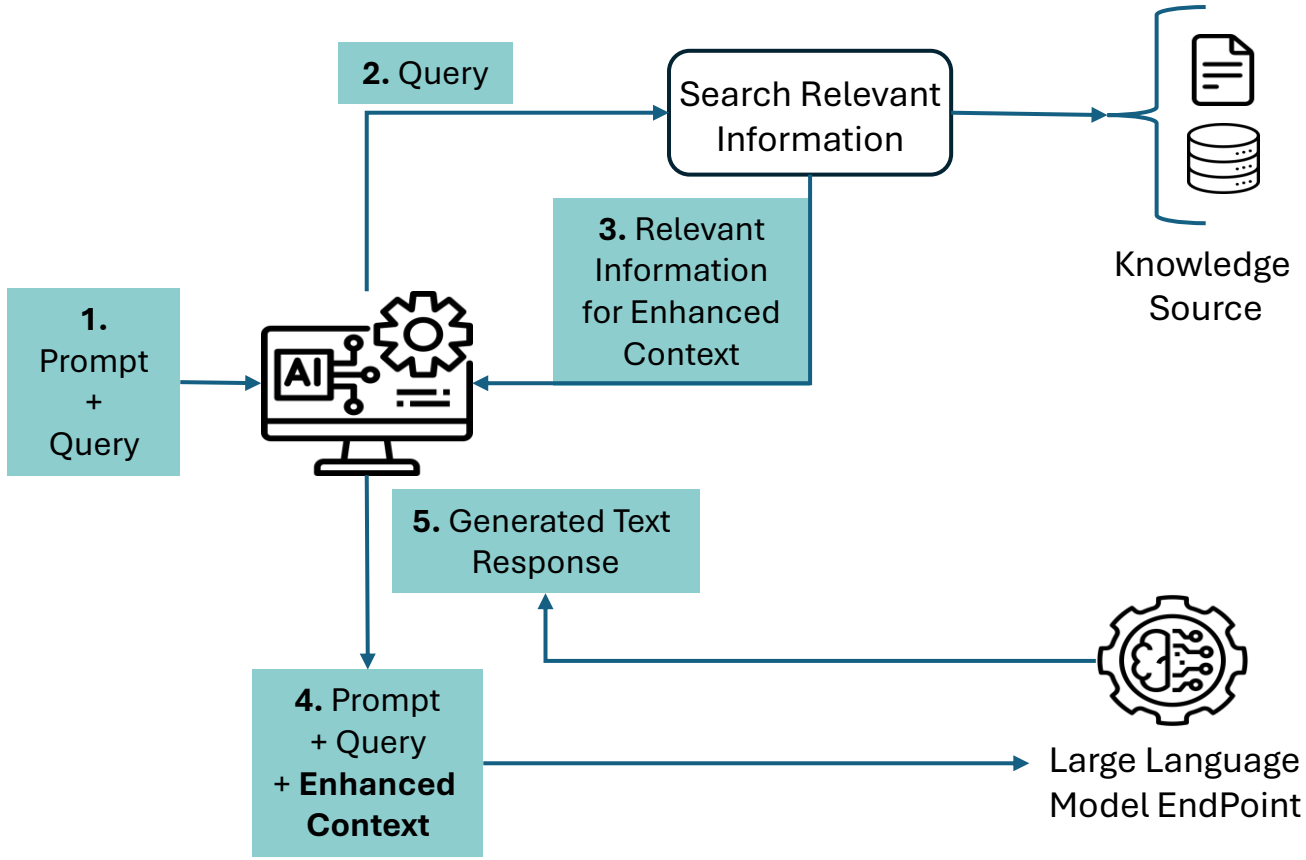
Business Intelligence: For LLMs being used to summarize daily competitor analysis reports and daily updates of internal company performance, RAG can be used to access the updated reports and generate summaries in an instance

Personalized Recommendations: In recommendation systems looking to generate product lists for users, RAG can be used to access user's search and purchase history, their comments on other products etc. to improve the recommendations and provide personalized results

How does RAG improve an LLM System?

Base LLMs, disconnected from external sources of information, can suffer from

- **Hallucination**: generating false information with confidence
- **Generic Responses**: LLMs provide generic answers to specific questions due to lack of knowledge
- RAG helps update LLM knowledge without re-training the whole model, which provides a cost-effective alternative and faster way of using LLMs for a variety of purposes.



What is MCP?

- Model Context Protocol (MCP) is an open standard developed by Anthropic that standardizes how applications provide information to LLM models.
- This enables the building of agents and workflows on top of LLMs without worrying about data transfer formats and focusing only on functionality.
- MCP is not just restricted to data retrieval but can also expose endpoints for automating functionality in external software through LLM-assisted agentic interventions.

What happened before MCP?

- Before MCP, all actions in an external service required a unique API and customized logic for implementation.
- Through MCP integration, AI agents can now send read/write/modify requests to external sources without needing to know the specifics of their implementation.

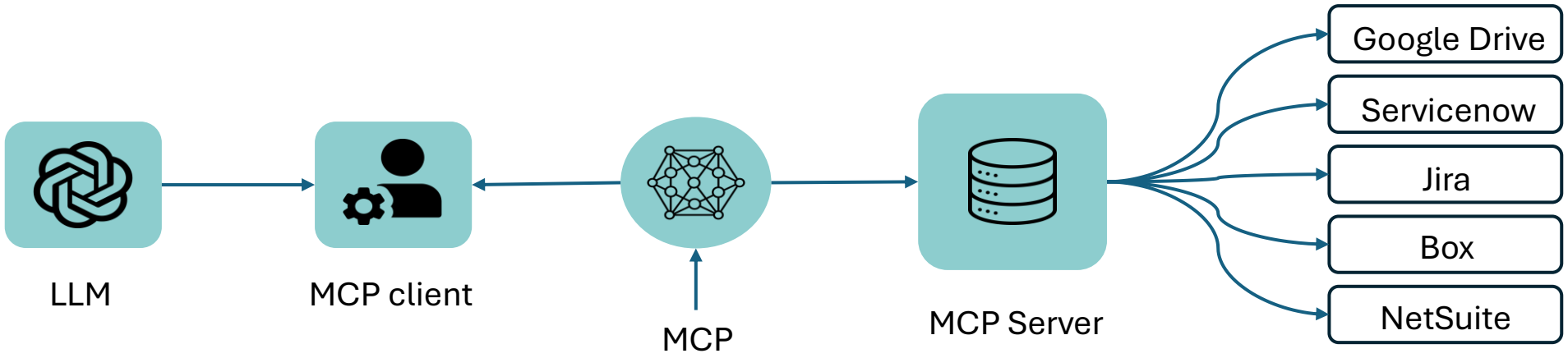
MCP Architecture

An MCP architecture includes

MCP Client: usually residing inside an AI assistant (-like Claude, ChatGPT), the MCP client is used by the AI to communicate with external tools and data sources. When The AI wants to use a tool, it goes through the MCP client to talk to the matching server

MCP Server: an MCP server is the translating entity in the architecture. It takes a text-based request from the MCP client (e.g.: modify the latest excel file with today's data) and translates it into commands that the external tool/data source understands.

MCP Protocol: it is the communication protocol that defines how the client and server communicate. It defines how the messages look like, how actions are described and how results are shown.



RAG vs MCP

Feature	RAG (Retrieval-Augmented Generation)	MCP (Model Context Protocol)
Purpose	To enhance the knowledge and accuracy of an LLM by providing external data.	To extend the capabilities of an LLM to perform actions in other systems.
Function	Retrieves information from documents, databases, or APIs to inform a response.	Connects to tools and APIs to execute a task or workflow.
Use Case Type	Best for enterprise AI search, chatbots that answer questions, or summarizing documents.	Best for "agentic AI" use cases where the AI needs to take a specific action.
How it works	Retrieves relevant documents, augments the LLM prompt with that data, and then generates an output.	Uses a client-server protocol to access and use "tools" that expose data or functionality from other applications.

In the world of AI, RAG and MCP are not competitors but are often seen as complementary technologies that can work together to create more sophisticated AI systems.[5] For example, an MCP-powered system could use RAG as a tool to first retrieve information and then use other MCP tools to act on that information.

Agentic AI

Agentic AI refers to autonomous AI systems that can act towards predetermined goals without human intervention. They differ from existing ML models in that they are adaptive to different use cases and can exhibit autonomy in actions.

How is Agentic AI different from Gen AI

GenAI refers to recent AI capabilities for generating content. Given a certain input by a human user, GenAI uses LLM models to generate output. It requires human intervention for manual input and cannot function independently. However, an AI agent can act independently and orchestrate with other GenAI platforms to achieve pre-set goals.

Feature	Generative AI	Agentic AI
Core Function	Responds to specific, human-given prompts for generating an output	Autonomously pursues pre-defined goals.
Initiative	**Reactive.** It waits for instructions.	**Proactive.** It plans and initiates actions on its own
Process	One-shot execution (Input -> Output)	Continuous loop (Perceive -> Reason -> Act -> Feedback -> Learn)
Example	If a human asks Google Gemini to “generate a meal recipe using tomatoes”, it will generate a recipe in text-format using pre-determined knowledge.	However, if given an AI agent with the goal of managing your food inventory and meal planning, it will check for the available food-stuffs in your inventory and then automatically suggest you a recipe for the day, alongside creating a to-do list of refreshing your food inventory (or maybe even requesting an order on an online food app if it has permission to do so).

AI Agent Characteristics

Proactive: Agents can function on complex inputs and take actions on their own to achieve pre-determined goals. They don't need to be pre-set for different conditions and can take actions **without human approval**.

Adaptable: Agents learn from their experiences and take feedback on its actions and responses, much like traditional ML systems. Multiagent systems can over time increase the scope of their coverage through proper feedback loops

Specialized: While there are hopes of achieving AGI, most agents perform specialized tasks. They are given a restricted set of goals to achieve and the permissions to execute tasks to move toward that goal. Multi-agent architectures can specialize in a range of tasks; however, at their core, they constitute individual agent systems that work toward a defined goal.

Acting: The ability to take actions is the core identity of an agentic system. This includes interacting with other systems and/or generating data to achieve a desired outcome.

AI Agents & Automated workflows

- Workflows are rule-based, pre-determined sequential actions that are triggered given a certain stimulus. They follow a restricted set of actions and cannot function outside their programmed domain. For example: receiving a notification whenever your e-mail inbox size exceeds 10,000 is an automated workflow.
- An agent combines the decision-taking capability of workflows with complex logic understanding to take actions that may not have been coded before.
- For example, if a workflow is triggered to send a notification when inbox size exceeds 10,000, an agent may further analyze the inbox and segregate the e-mails on basis of relevance, alongside sending you a summary of all the mails.
- Since it is impossible to pre-define workflows for every scenario, AI agents will help in such cases of **open-ended problems**.

When should AI agents be used?

Is the task open-ended and complex?: Agents increase latency and cost whereas close-ended or simpler tasks can be cheaply automated using rule-based alternatives

Is the cost of being wrong low?: Autonomous systems can make mistakes, especially while learning. If the cost of the model being wrong is high (such as in financial or healthcare related tasks) it is best to use LLMs in an advisory capacity rather than giving it decision making powers

Is there a clear feedback loop?: There must be a feedback mechanism to improve agent's actions. For example: using AI agents in medical diagnosis can have huge repercussions if the diagnosis is wrong. Having human intervention is best advised in these scenarios. Rather, developing workflows to summarize patient history would be a more suitable use case.

AI agents may be used in cases of documents handling. For example: maintaining file systems. When a new file is added to a file system, the agent can determine where it should best be stored. In this case, the cost of being wrong is lower and the agent can learn from human intervention of moving the file to a more suitable location.

Increasing complexity: AI Agents are massively complex systems, and designing them involved defining proper goal alignment, otherwise there are risks of the agent going "off the rails". E.g. A vaguely defined goal like "maximize sales" could lead an agent to spam customers with discounts, damaging brand reputation.

Increased Latency time: Since agents interact with LLM systems for context understanding, they can take time in deciding actions and may not be suited for low latency situations.

Importance of Agentic AI for a PM

Modernizing application functionality

- With the growing demand for intelligent functionality in all applications, it is imperative for product managers to understand how AI Agents work and how they can be integrated into their current software. Knowledge of AI agents can help a product manager ideate
- How existing workflows in the customer journey can be automated to improve customer satisfaction and help them explore product functionalities
- Develop interfaces to allow their application to communicate with other 3rd party apps without development of customized protocols for inter-app communication
- For example, Myntra can develop an AI agent that updates a user's buying list residing in another application (like Google Keep) whenever their wishlisted product comes back in stock
- Explore new product categories: AI agents and their smart functionality opens avenues for multi-step, cross-tool tasks without constant user input (e.g.: booking applications that automate travel planning and recommendations through constant updating of user's current journey)

Improving their PM workflow

PMs must communicate with multiple stakeholders every day and interact with a variety of applications to ensure product success. Needless to say, intelligent automation of many such tasks will reduce their workload

For example, AI agents can

- Automatically update their calendars based on new invites and past user interactions with similar events
- Changes made in one document can automatically be updated across all related documents without manual intervention to keep all stakeholders updated
- Agents can provide recommendations on improving product KPIs through updated news articles, company documents etc., providing A/B test recommendations, new product functionality, integration strategies etc.

Concept Note: Low-Code & The AI-Powered Development Stack

Imagine you need to build a house. You could spend years learning to make bricks, mix mortar, and frame walls from scratch.

OR you could use a set of high-quality, pre-made LEGO blocks to build a great-looking house in a single afternoon.

This is the core idea behind Low-Code/No-Code (LCNC) platforms. They provide the building blocks for software, allowing you to create functional applications visually, drastically reducing the need for traditional, line-by-line coding. For a Product Manager, it's a strategic lever for speed.

The New Development Stack: A PM's Toolkit

The following table summarizes the commonly used tools by PMs, each suited for a different job. Choosing which tool to pick is a key skill.

Category	Analogy	Use Case	Key Tools
No-Code	Using a pre-made LEGO kit with instructions	You need to track user feedback. You use **Airtable** to create a form and a database. Then, you use **n8n** or **Zapier** to automatically send a Slack message to your team every time a new piece of feedback is submitted	Bubble, Airtable, n8n, Shopify
Low-Code	Using a LEGO kit, but also having custom 3D-printed pieces for special parts	Your sales team needs a custom dashboard to see which customers use a new feature. You use **Retool** to build a dashboard that pulls live data from the company's internal database, something a simple spreadsheet can't do	Retool, OutSystems, MS Power Apps
AI-Assisted Pro-Code	Having a master builder robot that builds what you describe	Your engineering team is building a new, complex feature. They use **GitHub Copilot**, an AI "pair programmer," writing routine code for them.	GitHub Copilot, Replit AI

English is the New Programming Language

The evolution of this trend is the direct use of generative AI, where your instructions in plain English become the code itself.

Conversational Coding (Vibe Coding)

Coined by AI researcher Andrej Karpathy, it describes a workflow where a PM can guide an AI to build an application through a conversation.

How it Works in Practice:

1. **You start with a simple prompt** → "I need a Python script that scrapes Twitter for mentions of our competitor."
2. **The AI (like Claude or ChatGPT) generates the code.**
3. **You test and refine with more English** → "That's good but modify it to analyze the sentiment of each tweet and save the negative ones to a Google Sheet." This iterative, conversational process allows to create powerful tools and prototypes without writing a single line of code.

AI Agent & Workflow Builders

The newest frontier is platforms that allow you to build and deploy autonomous AI agents.

- **Tools in Action** → You could use a platform like **Lovable.dev** or **Bolt.ai** to build a sophisticated customer support agent that can access a knowledge base, interact with APIs to check order statuses, and escalate issues to a human—all configured through a visual, workflow-based interface.

Analysis for using these tools

Adopting these platforms is a strategic decision with significant trade-offs. While they improve productivity, they also have some challenges, as discussed in the table below.

As Andrew Ng states,

"Engineers are 10x faster. Product managers haven't sped up at the same rate. Now they're the bottleneck."

As engineering execution accelerates, the critical constraint shifts to the speed and quality of product strategy and decision-making.

So, a new challenge for PMs is using these AI and LCNC tools to accelerate their workflows, from user research synthesis to prototyping.

Reference:

Watch Andrew Ng discuss building faster with AI

Advantages	Potential Risks & Limitations
Accelerated Time-to-Market: Drastically reduce development cycles for MVPs, internal tools, and AI-powered features	**Vendor Lock-In:** Migrating an application off a proprietary platform can be difficult and costly, sometimes requiring a complete rebuild
Reduced Development Costs: Lower reliance on engineering talent for many typical applications and automation	**Scalability & Performance Ceilings:** Pure no-code platforms may struggle with the demands of large-scale, enterprise-grade applications
Increased Agility & Iteration: PMs can build and modify prototypes and internal tools directly	**"Black Box" & Debugging Issues:** When AI-generated code fails, debugging can be complex if the team doesn't fully understand the underlying logic
Democratized Innovation: Business users can solve their problems, freeing up core engineering teams	**Security & Governance Risks:** The proliferation of "shadow IT" built by citizen developers can create security vulnerabilities

Building Trust in Products

The New Currency of Trust

In today's world, products compete not just on features or price, but on something deeper: **trust**. A decade ago, consumers often accepted trade-offs; convenience at the cost of privacy, or speed at the cost of reliability. Today, **the balance has shifted**. **Users** are more aware, **regulators** are more watchful, and **competitive** landscapes are unforgiving. Security, resilience, and governance are no longer "back-end concerns." They are the front-line tests of whether a product deserves to exist.

Building Blocks

Security	Resilience	Governance

The PMs Takeaway

Security, resilience, and governance may sound like technical or abstract terms, but at their heart, they are human. They answer the question: *Can I trust this product with my time, my data, my money, or even my beliefs?* For aspiring product managers, especially in interviews, it helps to see these themes not as checkboxes but as lenses. They show how to think about product decisions beyond features — **to ask not just "will it work?" but "will it be safe, will it endure, and will it do right by the user?"**

Building Trust in Products

Security

When users share data, make a payment, or sign in with their identity, they extend trust. **Breaches are not just technical failures; they are broken promises**.

For product managers, thinking about security means asking:

"What are we promising our users when they interact with us, and can we keep that promise under stress? "

A secure product is not only about encryption or compliance; it is about **designing experiences that make users feel safe without friction**.

The PM's role is to champion this balance — building in safeguards while preserving usability.

Resilience

If security is about keeping promises, **resilience** is about keeping them consistently. In a world of sudden traffic spikes, malicious attacks, and unpredictable failures, resilience becomes the silent foundation of trust.

Users rarely notice when systems are resilient — but they immediately feel when they are not. Latency spikes, outages, or fraud attacks don't just frustrate; they erode credibility.

For PMs, resilience means asking:

"Can my product absorb shocks and still deliver its core value?"

It also means prioritizing fault-tolerance and recovery as much as shiny new features.

Governance

As products grow powerful, especially with AI, new questions emerge:

"Should we build this? Who could be harmed?"

Governance is about guiding products with a moral compass, ensuring decisions don't just chase growth but respect fairness, transparency, and accountability. This is not a far-removed concern for lawyers or ethicists; it sits squarely with PMs, who decide what gets built, how it is measured, and how it is explained to the world.

Good governance is not about saying "no" to innovation; it is about ensuring that innovation sustains trust over time.

As a Product Manager, you won't be expected to write the code, but you will be expected to work with the team that does. **System design** is fundamentally about creating robust, scalable, and reliable applications that can handle millions of users and massive amounts of data.

Imagine you're the product manager for **Zomato**. Your job is to ensure that when millions of people across India open the app, they can browse menus, place orders, and get their food on time, all without the system crashing.

Database – The Digital Storehouse

Think of the database as Zomato's massive storehouse of information, restaurant details, menus, prices, user profiles, ratings, and order history.

Just like warehouses store goods in an organized way, a database stores data so that it can be quickly retrieved and updated. When you search "Biryani," the database is where Zomato pulls all restaurants serving Biryani in your city.

API – The Waiter Who Takes Your Order

The API (Application Programming Interface) is like the waiter who listens to your order and takes it to the kitchen. It's the middleman between the user (you) and the backend system.

When you tap "Order Now," the API takes that request to the correct part of the system and brings the confirmation back to you. We will discuss APIs more in later sections.

Microservices - Specialised Kitchens

Instead of one giant kitchen cooking everything, Zomato has different "kitchens" for specific tasks — payments, restaurant search, order tracking, and delivery partner matching.

Each kitchen (microservice) operates independently, forming the complete service together. The payment microservice processes your UPI transaction while the delivery microservice finds a rider.

Message Queues - The Order Dispatch System

Sometimes, tasks can't be handled immediately and need to be queued. Like a restaurant's dispatch counter holds completed orders until delivery boys pick them up, message queues hold tasks until the next service is ready. Once the payment is processed, the order request sits in a queue until a delivery partner is available.

Security - The ID Check

Before giving access, Zomato checks who you are and what you can do — just like a gated community guard checking IDs.

Key Security Measures include

- **Authentication** → Verifying user identity
- **Authorization** → Checking user permissions
- **Encryption** → HTTPS/TLS for data in transit, encryption for data at rest
- **DDoS Protection** → Preventing malicious traffic overloads
- **Rate Limiting** → Preventing API abuse

Relational Databases (SQL)

- Organize data into tables
- Use SQL (Structured Query Language) for data manipulation
- Known for strong consistency, transactions, and integrity
- Examples: MySQL, PostgreSQL

ACID Properties

- **A**tomicity → Transactions are "all or nothing"; they either complete entirely or don't happen at all
- **C**onsistency → Ensures that after a transaction, the database remains in a valid state, enforcing defined rules and constraints
- **I**solation → Concurrent transactions do not interfere with each other
- **D**urability → Once a transaction is committed, the data is permanently stored and will survive system failures

Non-Relational Databases (NoSQL)

- Designed for flexibility and scalability, often sacrificing some consistency
- Schema-less → Do not enforce rigid schemas or foreign key relationships
- Types include key-value stores (e.g., Redis), document stores (e.g., MongoDB), and graph databases (e.g., Neo4j)
- Ideal for unstructured or semi-structured data and quick iteration

In- Memory Databases

Store data primarily in RAM for fast retrieval, often used for caching and session storage (e.g., Redis, Memcached).

PM Tip

Choosing the right database affects development speed, costs, and user experience.

SQL databases are great for structured data and complex relationships, while NoSQL databases excel at rapid iteration and handling diverse data types.

Scaling and Performance

When an application gains more users, a single server might become overwhelmed. To handle increased demand, systems need to scale.

- **Vertical Scaling (Scaling Up)** → Increase the resources of a single server, such as adding more RAM or upgrading the CPU. It's simple to implement but has limited potential because there's a maximum capacity for a single machine.
- **Horizontal Scaling (Scaling Out)** → Add more servers (replicas) to handle a subset of requests, distributing the load across multiple machines. This approach is more powerful because it allows almost infinite scalability and doesn't require high-end individual machines. It also inherently provides redundancy and fault tolerance, as other servers can continue to fulfil requests if one goes down, eliminating a single point of failure.

Load Balancers - Smart Traffic Managers

Returning to the Zomato analogy, imagine multiple kitchens inside a food court. A load balancer decides which kitchen should take the next order to keep things moving efficiently. If one server is overloaded, the load balancer sends the request to a less busy one.

Load Balancers are essential components (often acting as reverse proxies) that direct incoming requests to the appropriate server in a pool. Their primary goal is to even out traffic, maintain a smooth user experience, and prevent bottlenecks.

Content Delivery Networks (CDNs)

CDNs are a specific type of reverse proxy network consisting of servers distributed worldwide. They are primarily used to serve static files like images, videos, HTML, CSS, and JavaScript.

CDNs copy content from your origin server onto their globally distributed servers. When a user requests content, the CDN delivers it from the nearest CDN server, significantly reducing latency.

Networking Fundamentals

IP Address → A unique identifier assigned to each device on a network, allowing devices to locate each other. IPv4 and IPv6 are different versions of IP addresses.

TCP/IP (Internet Protocol Suite) → These rules define how data is sentover the Internet. Data is broken into packets, sent across the network, and reassembled at the destination.

- **TCP (Transmission Control Protocol)** → Ensures reliable communication. It guarantees the complete and correct delivery of data packets, managing connection setup and retransmitting missing packets. Many other protocols, like HTTP and WebSockets, are built on TCP.
- **UDP (User Datagram Protocol)** → Faster but less reliable than TCP. It doesn't establish a connection or guarantee delivery/order, making it suitable for time-sensitive communications where some data loss is acceptable (e.g., video calls).

DNS (Domain Name System) → Acts as the Internet's "phone book," translating human-friendly domain names (e.g., neetcode.io) into their corresponding IP addresses.
Ports → Within a device, specific processes or services are identified by ports (e.g., port 80 for HTTP).
Firewalls → Monitor and control network traffic to protect networks.

Other Important Terms

- **Availability** → Measures how much time a system is up and running when users need it, often expressed as a percentage *(e.g., "five nines" or 99.999% availability)*.
- Reliability: Ensures a system works correctly and consistently.
- **Fault Tolerance** → How a system handles unexpected failures or attacks.
- **Redundancy** → Having backups to ensure continued operation if one part fails.
- **Throughput** → Measures how much data a system can handle over time *(e.g., requests per second, queries per second, bytes per second)*
- **Latency** → Measures how long it takes to handle a single request *(time for a request to get a response)*

API Patterns and Protocols

HTTP (Hypertext Transfer Protocol)

Built on TCP/IP, it's the standard for web browsing, following a client-server model.

- **Request/Response** → A client initiates a request (with header and body), and the server sends a response (with header and body)
- **Stateless** → Each HTTP interaction is separate, with no memory of past requests
- **Status Codes** → Provide feedback on the result of a request (e.g., 200 for success, 400 for client error, 500 for server error)
- **Methods** → Common methods include GET (fetching data), POST (creating data), PUT/PATCH (updating), and DELETE (removing)

REST (Representational State Transfer)

A popular API pattern that standardizes HTTP APIs.

- It's stateless and uses standard HTTP methods
- Often uses JSON for data exchange
- Can sometimes lead to over-fetching or under-fetching data

REST APIs are widely understood and easy to debug, making them suitable for most applications.

gRPC (Google Remote Procedure Call)

A framework released by Google, considered an improvement over REST APIs, is mainly used for server-to-server communication.

- It leverages HTTP/2 and Protocol Buffers (a binary format for serializing data)
- Very efficient in terms of bandwidth and resources
- Less human-readable than JSON

WebSockets

They provide a two-way (bi-directional) communication channel over a single, long-lived connection.

- Crucial for real-time updates, unlike HTTP's one-way request/response model
- Commonly used in chat applications, for real-time features like live chat, collaborative editing, or live updates

The Layered View of a Modern Software Product

Layer	Core question	Typical components	Tools /Tech
Experience	How do users interact?	Web & mobile UI, API clients	React/Next.js, Swift/Kotlin
Application logic	What business rules run?	Services, controllers, workflows	Spring Boot, Express/Fastify, Django
Integration	How do parts talk cleanly?	REST/GraphQL/gRPC APIs, gateways	OpenAPI/Swagger, Postman, Kong, Apigee (Swagger, OpenAPI Initiative)
Data	Where is truth stored and read fast?	SQL/NoSQL DBs, caches, search	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
Async & real-time	How are spikes and events handled?	Queues, streams, WebSockets	Kafka, RabbitMQ, SQS; Socket.IO
Delivery & runtime	Where does code run and scale?	Containers, functions, autoscale	Docker, Kubernetes, AWS Lambda; CNCF ecosystem (CNCF, Cloud Native Now)
Reliability	How is health measured and protected?	SLOs/SLIs, error budgets, alerts	Google SRE practices, Prometheus, Grafana, Datadog (Google SRE)
Security & access	Who can do what?	AuthN/Z, secrets, WAF, IAM	OAuth 2.0, JWT, Vault, Cloudflare, AWS IAM
Edge & delivery	How is content made fast/globally?	CDN, edge compute, caching	Cloudflare/Akamai, Fastly; Redis

Section 3

Know Your Sector



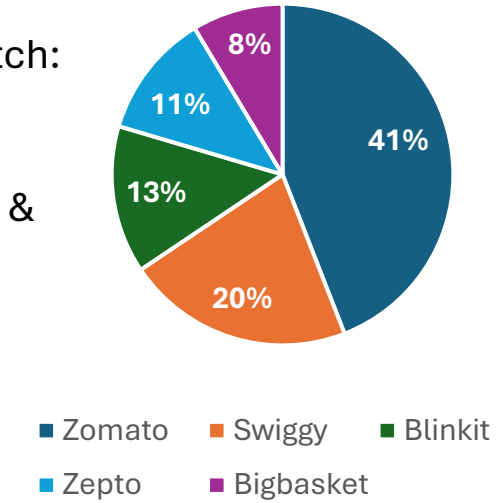
Quick Commerce

Key Players

India: Blinkit (Zomato), Swiggy Instamart, Zepto; (legacy/on-watch: Dunzo)

Global: Gopuff (US; retail media & owned dark stores), Getir (exited US/UK/EU; focusing on Turkey), Flink (Germany; multi-category grocery products)

Reference: Tracxn



Common Characteristics

Dark stores within 1-3 km of dense demand; 4-8k SKUs; micro-fulfilment pick/pack in 2-6 minutes

Promise windows: 10-20 minute "instant" or 30-45 minute rapid delivery; surge/slotless

Assortment mix: everyday groceries + "urgent" items (ice cream, beverages, snacks), growing into beauty, home, pharma-lite, and seasonal

Use Cases

- **Top-up and Emergencies:** Missed ingredients, baby products, Over-the-Counter (OTC) products
- **Occasion-driven:** Festivals, party supplies, live-sports spikes
- **Segments:** Urban singles/roommates, young families, premium convenience seekers, office micro-kitchens top-up, students in coaching hubs

Reference: https://redseer.com/articles/quick-commerce-indias-retail-darling-or-profit-mirage/

Quick Commerce

Business Model Canvas

Value Proposition:

- **Consumers:** instant convenience, reliable ETAs, curated essentials, late-night availability
- **Brands:** high-visibility placements on platform; rapid trials & promos based on data, benefits from impulse-led sales
- **Last-Mile Delivery Partners:** predictable zones of dense demand

Revenue Streams: fees (platform, delivery, surge), subscription, listed on-site ads, product margins

Costs: Dark store and rentals, technology, rider payouts, pick/pack, spoilage/shrinkage, discounts, marketing and CAC

Channels: Mobile apps, super-app entry points (Zomato → Blinkit; Swiggy → Instamart), web, affiliate links (via offers in Pluxee, Cred), CRM (push notifications), brand collaborations (seasonal campaigns)

User Journey

- **Consumer:** Browse → add to cart → promo/fees visibility → pay → live ETA → doorstep handoff → rate/support
- **Dark Store (Ops):**
  - **Supply-Chain:** Forecast → receive → storage → inventory control and cycle counting → quality control → replenishment
  - **Order Management:** Order prioritization → locate and pick → quality checks and pack → dispatch
- **Delivery Partner:** Accept → navigate → deliver → issue resolution
- **Brand:** Onboard → list SKU data (cataloguing of products) → run promos/ads → track metrics like return on ad spends, price demand elasticities

Quick Commerce

Key PM Metrics

- **Growth:** Total Orders, Average Order Value (AOV), Monthly Active Users (MAU), retail-media revenue % (ads on app), membership subscriptions/renewals
- **Ops and Quality:** On-time %, speed promise accuracy, fill rate (% orders fulfilled without stockouts), cancellations and returns %
- **Unit Economics:** Contribution margin per order, Customer Acquisition Cost (CAC), payback period (CAC recovery via margins)
- **Trust:** Complaint rate, cold-chain breach rate, fraud/promo abuse %.
- **Investor Lens:** Gross Order Value (GOV) growth vs. adjusted EBITDA

Macro Trends and Opportunities

- **Emerging and Futuristic Trends:** Drones for last-mile delivery, warehouse robotics, eco-friendly alternatives like electric vehicles for delivery
- **New Business Models and Experiences:** Subscription-based ordering, gamified rewards (scratch card, push lever to win cashbacks), better privacy control (hiding sensitive order history with user consent), ONDC apps, nudge economics (opting users for eco-friendly packaging, donation to charities), move towards profitability
- **Category and Geography Expansion / Consolidation:** non-grocery items (electronics, gold), festival specific offerings, cooked meals delivered in 10-15 minutes (Zepto Cafe); Tier-2 and Tier-3 cities expansion in India, Global players retrenching to home markets (Getir → Turkey)
- **AI and Hyper-Personalization:** personalized discounts, data-driven recommendations, customer support
- **Retail Media:** On-site ads and sponsored brand and SKU placements on the platform

Quick Commerce

Common Product Pitfalls

- **Unit Economics Myopia**: Deep discounts lead to unsustainable burn
- **Scaling Too Wide, Too Fast**: Expanding into low-density geographies dilutes utilization (orders/hour)
- **Overpromising ETAs:** Optimistic promises → missed SLAs → churn
- **Ops Complexity:** Mismanaged dark store inventory, cold-chain breaks, packing errors, address/entry issues
- **Limited Assortment Trap**: Ignoring adjacent high-margin categories (beauty, home, private label) limits profitability
- **Ad Monetization Pitfalls**: Retail media promises high margins, but poor targeting or cluttered UX can alienate users and reduce purchase intent

Interview Tips

- **Dissect unit economics live (whiteboarding):** Given AOV ₹500, delivery cost ₹70, pick/pack ₹20, discounts 8%, ads revenue ₹5/order - what is the contribution margin?
- **Design for density:** Choose where to open the next five dark stores; trade off coverage vs. cannibalization
- **Raise AOV without killing speed:** Smart bundles ("chips + soda" during cricket matches), free-delivery thresholds, cross-sell at checkout
- **Reliability over raw speed:** Promise accuracy, dynamic SLAs, proactive delay communications
- **Measure what matters:** On-time %, fill rate, delivery utilization, retail-media Return on Ad Spends (ROAS)
- **Ethics and safety:** Cold-chain quality checks, fair fees and wages, transparent surge, customer support availability Recruiters look for balanced thinking, not just optimization

Sector Evolution Timeline

Pre-2016

Digital Wallets

2016-20

UPI & Demonetisation Boom

2021-23

Formation of Super Apps & BNPL craze

2024-25

Embedded Finance, Regulations & Focus on profitability

B2C

Focus on frictionless UX, building engagement loops, and establishing trust.

- **Digital Payments (UPI & Wallets)** → PhonePe, Google Pay, Paytm, CRED UPI. The NPCI is a key enabler with its UPI infrastructure.
- **Wealthtech & Broking** → Zerodha, Groww, Upstox.
- **Lending & Credit (incl. BNPL – Buy Now and Pay Later)**: CRED Cash, LendingKart, DMI Finance, Simpl, LazyPay.
- **InsurTech** → PolicyBazaar, Acko, Digit Insurance.
- **Neobanks & Personal Finance** → Digital-only banks that offer modern, user-friendly banking experiences. E.g. Jupiter, Fi, Niyo.

B2B

Focus on robust APIs, developer experience, reliability (SLAs), & compliance.

- **Payment Gateways & Merchant Solutions** → Razorpay, PayU, CCAvenue, Pine Labs
- **SME Lending & Supply Chain Finance** → KredX, M1xchange, CredAble.
- **Banking & Financial Infrastructure (Fintech SaaS)** → They provide APIs and software for tasks like KYC, fraud detection, and more. E.g. Perfios, Zeta, M2P Fintech, Setu

Common Characteristics

- **Frictionless UX** → Fast, intuitive onboarding (under 2-3 mins) to minimize user drop-offs.
- **Trust & Compliance** → Visible security cues and RBI/SEBI-aligned messaging to build user trust.
- **Engagement Loops** → Gamification and rewards to drive DAU/MAU and create habit-forming usage.
- **Tier-2/3 Inclusivity** → Multilingual UX and low-data optimization for the broader Indian market.

- **Integration Speed** → API-first, plug-and-play onboarding for rapid enterprise client integration.
- **SLA & Reliability** → Guaranteed high reliability with Service Level Agreements (SLAs) promising 99.95%+ uptime.
- **Compliance-by-Design** → Built-in modules for KYC, AML, and PCI-DSS for out-of-the-box regulatory compliance.
- **Data Security & Privacy** → Enterprise-grade encryption, comprehensive audit trails, and SOC2 readiness are standard.

Business Model Canvas

Value Propositions → Instant low-cost payments, seamless access to credit, hyper-personalization, 24/7 access, and richer data for underwriting.

Channels → Direct B2C apps (PLG), B2B enterprise sales, embedded finance partnerships (e.g., on e-commerce sites), and API platforms for developers.

Revenue Streams →

- **Payments** → Merchant Discount Rate (MDR), transaction fees.
- **Lending** → Net Interest Margin (NIM), processing fees.
- **Wealth** → Brokerage fees, subscription fees (SaaS).
- **Infrastructure** → API call fees, SaaS licensing.

Key Partnerships → Banks & NBFCs (for credit lines, accounts), card networks (Visa, Mastercard), data providers, and channel partners (e-commerce platforms).

Cost Structure → Dominated by technology & talent costs, followed by Customer Acquisition Cost (CAC), compliance & legal overhead, and fraud losses.

Use Case & Pain Points

Use Case	Pain Point	Opportunity
Retail Consumers (Payments, Lending, Investing)	Complex onboarding/KYC, transaction failures, hidden fees, fear of fraud.	Build a sub-2-minute onboarding flow, provide clear failure-state messaging, and design transparent, trust-building user interfaces.
SMBs / Merchants (Payments, Working Capital)	Slow settlement times (T+2), high integration complexity, managing chargebacks and disputes.	Offer faster settlements (e.g., T+1), build low-code/no-code integration plugins, and provide an automated dispute resolution dashboard.
Developers (using Fintech APIs)	Poor documentation, inconsistent API performance, high costs, complex compliance requirements.	Create best-in-class developer documentation, offer a reliable sandbox environment, and provide compliance-as-a-service features.
NRIs / Expats (Remittances, Global Investing)	High foreign exchange (forex) markups, opaque transfer fees, fear of fraud during cross-border transfers, and complex regulatory hurdles for investing abroad.	Build a secure, low-cost remittance platform with real-time tracking and a simplified, compliant interface for accessing global investment products.

User Journey

Payments (Customer)

App download → Account creation → KYC → Account linking → First transaction → Issue resolution

Account creation → High competition and security fears create user mistrust.

- *Opportunity* → Emphasize security to build immediate trust.

KYC verification → Complex, lengthy KYC causes high user drop-offs.

- Opportunity → Aim for a sub-2-minute KYC using India Stack APIs to reduce friction.

Account linking → Technical failures and confusing errors cause frustration.

- *Opportunity* → Provide clear error messages & guided flows to ensure a high success rate.

First transaction → User inertia and no clear incentive to transact.

- *Opportunity* → Drive activation with gamified rewards for the first payment.

Issue resolution → Opaque processes and fear of losing money break user trust.

- *Opportunity* → Design a transparent, self-serve resolution center with real-time tracking

Payments (Merchant)

Account creation → KYC → Technical integration → Configuration → Go-live → Operations

KYC verification → Slow, manual verification delays the merchant's ability to accept payments.

- **Opportunity** → Automate document analysis & send reminders to speed up activation.

Technical integration → Inconsistent APIs and poor documentation create integration challenges.

- **Opportunity** → Provide best-in-class developer documentation and reliable testing tools.

Platform configuration → Complex dashboards are often confusing for business owners to use.

- **Opportunity** → Create intuitive UI with guided setups and smart defaults.

Go-live → Anxiety about potential failures impacting customers and revenue.

- **Opportunity** → Offer a "Go-Live" checklist and proactive support for a smooth launch.

Compliance & reporting → High overhead of managing chargebacks and complex reports.

- **Opportunity** → Build an automated dispute dashboard and analytics tools.

Key PM Metrics

Growth & Adoption →

- **MAU/WAU** → Monthly/Weekly Active Users
- **TPV/GTV** → Total/Gross Transaction Volume

Unit Economics →

- **LTV/CAC Ratio** → Target >3
- **Contribution Margin** → Revenue per transaction minus variable costs.

Risk & Quality →

- **Authorization Rate** → % of transactions successfully approved. Target >95%.
- **Chargeback Rate** → % of transactions disputed by customers.
- **Platform Uptime** → Target >99.95%.

Engagement & Trust →

- **Onboarding Drop-off Rate** → % of users who start but don't complete KYC.
- **NPS** → Net Promoter Score.

Macro Trends and Opportunities

Credit on UPI → Unifying payments and credit into a single, seamless flow.

- *Opportunity* → Build innovative credit products (e.g., small-ticket loans) delivered instantly at the point of sale via UPI.

Open Finance (Account Aggregator) → Shift towards consent-based data sharing for a holistic financial view.

- *Opportunity* → Create automated personal finance managers or build superior underwriting models using real-time cash flow data.

AI Agents → Moving from simple chatbots to autonomous agents for financial management.

- *Opportunity* → Design AI agents that can monitor finances, detect fraud, and execute investment strategies on behalf of the user.

Embedded Finance → Financial services are becoming invisible infrastructure, forming superapps.

- *Opportunity* → Develop vertical-specific B2B2C solutions, like "Lending-as-a-Service" for SaaS platforms.

Common Product Pitfalls

- **Onboarding Friction** → A lengthy or confusing KYC process leading to high user drop-offs.
- **Ignoring the "Money Flow"** → Not deeply understanding the entire transaction lifecycle, leading to settlement and reconciliation failures.
- **Compliance as an Afterthought** → Treating RBI/SEBI regulations as a final check-box, leading to product rollbacks or fines.
- **Shallow Fraud Detection** → Relying on simple rules instead of a multi-layered fraud defense & data security (AI models, velocity checks, manual review), leading to high losses.
- **Negative Unit Economics** → Burning cash on unsustainable rewards or discounts without a clear path to profitability (LTV<CAC).
- **Feature Bloat** → Cluttering the interface of a simple payments app with too many non-core services, harming the primary user experience.

Interview Tips

- **Think Compliance-First** → Start your product design by addressing the regulatory guardrails (RBI, SEBI, NPCI). It shows maturity.
- **Anchor on Trust, Not Just Growth** → For every feature, explain how you would build and measure user trust, even if it means sacrificing short-term metrics.
- **Map the Full Money Flow** → Be able to whiteboard the end-to-end journey of a transaction—from user action to settlement and reconciliation.
- **Know Your Revenue Levers** → Speak the language of finance. Discuss impact in terms of basis points (BPS) on MDR or Net Interest Margin (NIM).
- **Design for Failure** → What is the fallback flow when a UPI payment fails? How do you communicate errors? A great PM plans for exceptions.

Travel Tech

Key Players

Online Travel Agencies (OTAs) / Digital Marketplaces

- Global: Booking Holdings (Booking.com, Priceline, Agoda), Expedia Group (Expedia, Hotels.com), Trip.com (Ctrip)
- India: MakeMyTrip, Yatra, Cleartrip (Flipkart-owned), EaseMyTrip

Discovery and Reviews → Google Flights, Skyscanner, Trivago

New Models → Airbnb (home-sharing), Hopper (AI-powered price prediction and multi-category bookings), Zostel (hostels and curated group travel packages), PickYourTrail (personalized Holiday packages)

Common Characteristics

Marketplace / aggregator model → Connects and partners with travel suppliers (airlines, hotels, tours) and consumers

Low-margin, high-volume nature → Monetization via commissions & ancillary (cab pickup, holiday packages, forex card) sales, dynamic pricing

Fragmented supply → Especially hotels in India → aggregation adds value. Heavy reliance on trust, reviews (user generated content), and seamless UX

Seasonal & cyclical demand → Holidays, events

Use Cases

Leisure Travelers → Families, couples, solo trips	**Business Travelers** → Corporate booking via TMCs (Travel Management Companies)	**Budget vs Premium and Experiential** → Price-sensitive backpackers or students vs high-end curated stays	**Hosts/Hotels** → Listing on OTAs for reach

Travel Tech

Business Model Canvas

Value Proposition → Convenience, price comparison, discovery, reviews, flexible booking/cancellations, loyalty perks

Revenue Streams →

- Commissions
- Service fees, ads, white-label SaaS (example: 'powered by Expedia')
- Fintech add-ons: insurance, buy-now-pay-later, cancellation protection

Costs → Tech and infra, supplier integrations (example: via APIs), customer acquisition (marketing, discounts), support/refunds operations, loyalty & rewards

Channels → Mobile apps, websites, super-app tie-ins (Paytm, Flipkart), affiliate sites, meta-search (Google, Kayak)

User Journey

- **Traveler**
  Discover → compare → book → pay → travel → review
- **Supplier (hotel/airline)**
  Onboard → manage inventory → set pricing → receive bookings → service traveler → settlement/payout
- **OTA**
  Acquire user → aggregate supply → optimize search & pricing → enable booking & payment → upsell ancillaries → resolve support

Travel Tech

Key PM Metrics

- **Growth** → Monthly Active Users (MAU), search-to-book conversion %, Average Order Value (AOV) per booking, repeat percentage / stickiness
- **Ops and Quality** → cancellation rates, refund turnaround time (TAT), review ratings
- **Unit Economics** → commissions retained by OTA, customer acquisition cost (CAC), payback period, contribution margin
- **Trust** → fraud rate/chargebacks, fake review detection, on-platform dispute resolution
- **Investor Lens** → Gross Booking Value (GBV) v/s adjusted EBITDA

Macro Trends and Opportunities

- **AI-assisted Bookings and Insights** → Bookings shift from apps to chat-first platforms (Google Gemini), personalized itineraries, consumer trends prediction (example: shift towards colder destinations as a result of climate change)
- **Real-Time Translation** → AI translation removes language barriers, opening remote markets
- **Bleisure and Experiential Travel** → Work trips increasingly extend into leisure; Event Travel (Taylor Swift Eras Tour, Wimbledon)
- **Digital Detox and Well-being Tourism** → Rising demand for wellness retreats and unplugged experiences
- **Self-Guided Tours, Augmented Reality (AR)** → Self-guided audio tours, AR layer for immersive experience (you are standing at the Colosseum → AR shows gladiators fighting)
- **Fintech Layer** → Pay-later, insurance, cancellation protection, FX wallets bundled with travel

Reference: https://www.statista.com/topics/13141/travel-and-tourism-trends-2025/#topicOverview

Travel Tech

Common Product Pitfalls

- **Commoditization** → OTAs fight in red ocean → differentiation is hard
- **Thin Margins:** Especially for flights; over-reliance on discounts
- **Refund/Support Nightmares** → Delays and cancellations result in poor customer experience if not timely resolved
- **Supplier Conflict** → Hotels/airlines push direct bookings resulting in reduced margins for OTAs
- **Trust Issues** → Fake reviews, hidden fees, bait-and-switch listings
- **Over-expansion** → Launching too many verticals (tours, activities, forex) without focus → diluted UX

Interview Tips

- **Traveler Journey Gaps** → Redesign booking flow to cut drop-offs; think search filters, payment friction, refund transparency
- **Supplier retention** → A hotel chain wants to give fewer OTA commissions - how would you retain them? (Ads, featured placement, dynamic pricing tools)
- **Measure what matters** → Stress on conversion %, measure NPS against industry benchmarks, fraud prevention and loss savings
- **Strategic Trade-offs** → Speed of market expansion v/s deep local supply integration (curated inventory); top skimming v/s mass penetration
- **Ethics angle** → Transparent fees, fair cancellations, reliable refunds handling

Food Tech

Key Players

- **Aggregators:** Swiggy, Zomato, Uber Eats, DoorDash
- **Cloud Kitchens:** Rebel Foods (Faasos, Behrouz Biryani), Swiggy Access, Ghost Kitchens International
- **Restaurant SaaS**: POSist, PetPooja, Toast
- **Meal Kits/D2C:** FreshMenu, HelloFresh, iD Fresh
- **Quick Commerce (adjacent):** Blinkit, Swiggy Instamart, Zepto, BigBasket

Common Characteristics

- **Two-sided or three-sided marketplace:** Restaurants (and/or cloud kitchens) ↔ diners; and at times a third side (couriers)
- **Thin margins, high frequency:** Profit comes from scale plus operational discipline
- **Local density and ETAs:** Unit economics hinge on short distances, batching, reliable prep times
- **Ops and trust heavy:** Food safety, order accuracy, complaint resolution and refund speed matter
- **Demand peaks:** Mealtimes, weekends, rain events → load balancing & incentives

Business Model Canvas

Value Proposition:

- **Diners:** Convenience, variety, reliable ETAs, offers, tracking
- **Restaurants:** Incremental demand, logistics, payments, discovery, insights
- **Last-Mile Delivery Partners:** Predictable zones of dense demand

Revenue Streams: Commissions on orders, delivery and surge fees, subscriptions (example: Swiggy One), **retail-media/ads and sponsored listings**, B2B supply margins (Zomato Hyperpure, Swiggy Access), SaaS/POS fees (POSist, PetPooja), network effects further help onboard more restaurants, riders, and consumers.

Costs: Delivery rider payouts, discounts/offers, restaurant commissions/subsidies, tech and infra, support/refunds, marketing and loyalty spends

Channels: Mobile apps, web, CRM (push/WhatsApp), brand tie-ups, corporate portals (Pluxee)

Use Cases

- **Diners:** Everyday meals, late-night orders, group/family dining, health/diet-specific needs, celebrations
- **Enterprises:** Corporate meal programs, office micro-kitchen supplies, meal vouchers, team events
- **Restaurants:** Independents (reach new customers via platforms), cloud kitchens (online-only brands, virtual menus)
- **Delivery Partners:** Flexible earning, area-based routing, peak-time incentives

User Journey

Diner: Discover → customize → checkout/pay → track → receive → rate → re-order

Restaurant: Onboard → menu/catalogue & pricing → accept → prep → handoff → settlement → analytics/ads

Delivery Partner: Log in → accept/stack batch (multiple orders in one trip, or next order queued before finishing current) → pick up → navigate → deliver → earnings/feedback

Platform: Acquire users/merchants → match & dispatch → collect payments → support/refunds → monetize (commission, fees, ads) → retain (loyalty/CRM)

Macro Trends and Opportunities

Profitability over Blitz-scaling: Tighter discounts, fees rationalization, ops efficiency, ads as margin lever

Retail-media Rise: Brands move ad spend into food apps (sponsored listings, search ads) for high-intent visibility.

Cloud-kitchens: Fewer but better-utilized kitchens, virtual brands based on data

Regulation and Workforce: Transparency on fees, gig-worker standards, hygiene compliance

AI Everywhere: Demand forecasting, dynamic ETAs, menu intelligence, fraud detection, delivery routing

Key PM Metrics

Growth & Mix:	Ops and Quality:	Unit Economics:	Trust and Safety:	Investor Lens:
Active users, total orders, Average Order Value (AOV), repeat vs new users, membership uptake	On-time delivery, accurate ETAs, prep/fulfillment speed, cancellations/returns, order accuracy, delivery efficiency (utilization & batching)	Commission rate (take rate), delivery cost per order, contribution margin per order, Customer Acquisition Cost (CAC), ad/retail-media share	Complaint rate, food safety issues, fraud/promo abuse	Gross Order Value (GOV) growth, adjusted EBITDA, customer retention

Common Product Pitfalls

- **Discount Addiction →** Bad unit economics
- **Prep-time Blindness:** Great dispatch but slow kitchens = missed ETAs
- **Low Density Expansion:** Long trips, poor batching, high cost/order
- **Menu/Catalog Debt:** Bad images, tags, or availability → drop-offs & cancellations
- **Support Black Holes:** Slow refunds and canned replies kill trust
- **Ad Overload:** Retail-media that hurts discovery → conversion drops
- **Quality/safety lapses:** Food incidents outweigh months of growth

Interview Tips

Dissect unit economics live (whiteboarding): Given AOV ₹400, commission 20%, delivery ₹65, payment 1.5%, ads + ₹5/order, discounts 8% - what's contribution? Which levers move fastest (AOV, batching, distance, ads, fees)?

Design for density: Choose next 5 zones; trade off demand hotspots vs cannibalization; target orders/hr and average trip distance

Speed vs reliability: Dynamic ETAs, prep-time SLAs with restaurants, proactive delay credits

Raise AOV without slowing: Smart bundles, fee-free thresholds, time-of-day upsell, occasion packs

Reduce cancellations: Real-time inventory sync, auto-reroute to nearest branch, smart throttling

Measure what matters: On-time %, prep time p95, cancellation/return %, refund TAT, cohort retention, ads ROAS.

E-Commerce

Key Players

Global



- Alibaba
- Amazon
- Pinduoduo
- JD.com
- Coupang
- Rakuten

India



- Flipkart-Myntra (48%)
- Amazon (31%)
- Meesho (8%)
- Jiomart-Ajio (8%)

Common Characteristics

- **Supply chain:** centralized warehouses (hub and spoke model); logistics follows Inbound to FC → Mid Mile (Line haul, Sortation, Cross-dock) → Last mile hub → reverse logistics
- **Promise windows:** 1-3 days/ scheduled slots/ same day delivery timelines; faster delivery for premium plan subscribers (e.g. Amazon Prime); free reverse logistics
- **Assortment mix:** 50K-100M SKUs covering most categories
- **Trust Signals:** product reviews and ratings driving customer confidence

E-Commerce

Business Model Canvas

Value Proposition:

- **Consumers:** Wide product assortment, reliable delivery ETAs, personalized recommendations, attractive discounts & cashbacks, hassle-free refunds/returns.
- **Brands/Sellers:** National/global reach; high-visibility ad placements; targeted recommendations to buyers; data-driven trials and promotions; uplift from impulse-led sales; cost-efficient last-mile delivery.

Key Partners: Payment gateways (stripe, razorpay), Delivery partners (Delhivery, Ecom Express, DHL)

Revenue Streams: Marketplace commissions, advertisement charges, premium subscriptions, fulfilment/storage fees, and product margins (private labels)

Business Models: Inventory-led, Marketplace

Delivery Models: Platform fulfilled, Merchant fulfilled, Drop shipping**:** Warehouse lease, platform maintenance & upgrade, discounts, delivery partner payouts, unsold inventory, customer support, payment gateway fees, packaging, returns handling.

Use Cases

Urban mass-market consumers: convenience driven shoppers seeking competitive prices and home delivery, brand-loyal consumers.

Deal/Discount driven: Shoppers responding to festive offers, trending products, or seasonal events (e.g., Prime Day, Big Billion Day).
Tier 2/Emerging city shoppers: price sensitive consumers, first-time users, prioritizing reliable delivery
Marketplace sellers: D2C, SMBs seeking wider market reach.

User Journey

Consumer: browse → add to cart → promo/fees visibility → pay → track order → doorstep delivery → rate/support

Delivery partner: Assigned to route → pick up from last mile hub → deliver to customer → report issues / handle exceptions

Brand: onboard → list SKUs → update online catalogues → run promos/ads → fulfill orders → track performance metrics (return on ad spends, gross monthly sales, return rates)

Key PM Metrics

- **Growth & Mix:** Gross merchandise value (GMV), Average order value (AOV), Cart abandonment rate (CAR), Category mix
  Ops & Quality: OTD rate, Perfect order rate, Order picking accuracy, Return rate, Cancellation rate, Sell-Through rate
- **Unit Economics:** Contribution margin per order, Take rate, Customer lifetime value (CLV), Average cost per order
  Ad Performance: Click Through Rate (CTR), Cost per Click (CPC), Return on Ad Spends (ROAS), Ad conversion rate
- **Trust:** Complaint rate, Fraud/promo abuse %, Counterfeit complaint rate
- **Investor Lens:** Marketplace penetration, Adjusted EBITDA

Macro Trends and Opportunities

Futuristic technology trends:

- Drones for last-mile delivery
- Warehouse robotics
- Augmented reality for product visualization and trial
- Voice shopping using Alexa/Google Assistant.

New Consumer Experiences:

- Social commerce (Integration of shopping directly within social media platforms. Ex - Instagram, TikTok)
- Livestream shopping (Real-time product demonstrations by influencers/hosts with instant buying)
- Buy now pay later (Flexible credit at checkout, allowing deferred or split payments)
- Micro subscriptions (Small recurring payments for curated/niche products)

Emerging Business Models:

- Vertical marketplaces (Nykaa)
- Super apps (Grab, Tata Neu, GoTo)
- Cross-border sales (Amazon Global Sales)

Common Product Pitfalls

- **Listing Quality Gap:** Misclassified categories, inconsistent images, and broken variation mapping reduce product discoverability and conversions.
- **Price Blindspots:** Inefficient competitor tracking leads to uncompetitive pricing and lost transactions.
- **Trust Gaps:** Counterfeit/restricted/unsafe products and fake reviews erode customer confidence and lead to compliance issues.
- **Ad Waste:** Poor targeting and cluttered placements reduce ad ROI and purchase intent.
- **Supply Chain Inefficiencies:** Forecasting errors, incorrect stock displays, suboptimal routes, and inefficient order aggregation increase costs and delays

Interview Tips

- **Conversion Drop-offs:** If you diagnose a 30% cart abandonment in beauty– is it due to poor quality of product display page, trust gap, checkout friction, or poor recommendations? Propose fixes to lift conversion.
- **Returns at Scale:** Redesign the returns flow for apparel (E.g. 40% return rate). Balance customer satisfaction with lower reverse logistics cost.
- **Counterfeit Detection:** Build a proactive system to catch fake items before they reach customers – think workflows + tech levers.
- **Ad Slot Prioritization:** During peak events (Prime Day, Black Friday), design a system that allocates limited ad slots to maximize ROI without hurting UX.

Ground your answers in customer experience, articulate trade-offs clearly and show your awareness of unit economics.

Key Players

Global



- **Netflix (260M+ subscribers)**
- **Amazon Prime Video (200M+)**
- **Disney+ (149M)**
- **Tencent Video (115M+)**
- **iQIYI (101M+)**

India



National

- **JioHotstar**
- **Amazon Prime**
- **Netflix**
- **MX Player**
- **Zee5**
- **SonyLiv**

Regional

- **Sun NXT**
- **Hoichoi**
- **aha**

Common Characteristics

Digital supply chain: Content creation/Acquisition → Capture & Transcoding → Metadata entry (adding title, description) → Digital rights management (encrypting content, licensing, access controls) → Ingest (uploading content to platform) → Content distribution

Promise windows: Accessibility across devices, personalized recommendations, access to global content, buffer-free streaming, multi-language audio

Assortment mix: On-demand content, live streaming, music, Binge watching, D2C

Business Model Canvas

Value Proposition:

- **Consumers:** Content diversity, control over viewing experience, synchronization across devices, customized watchlists, price differentiation.
- **Content owners:** Low distribution costs, global audiences, viewer analytics insights, experimental content funding, flexible revenue options.

Revenue Streams: Subscription revenue, advertising revenue, rental/pay-per-view revenue, merchandise, licensing content

Business Models:

- SVOD (subscription video on demand | example - Netflix)
- AVOD (advertising video on demand | example - MX player)
- TVOD (transactional video on demand | example - Apple TV)
- Freemium (YouTube, Spotify)
- Others include: Studios & creators (licensing fees), app stores (platform fees), Smart TV OEMs (integration), payment gateways, cloud & CDN providers

Use Cases

Urban digital natives: Young professionals and students with demand for global content. Typically looking for single screen options.

Family households: Looking for bundled plans for the family. Favours regional content and popular local shows.

Sports fans: Key growth driver in the OTT segment. Subscribes to OTTs based on streaming rights of IPL/EPL/Olympics/World cup.

Content creators & owners: Sell licenses of movies, shows, streaming rights of sports for wider audience reach at low distribution costs. E.g. TVF, BCCI.

Stakeholder Journey

Growth & Mix: Subscriber acquisition rate, churn rate, monthly recurring revenue (MRR), content/genre mix

Engagement: Monthly active users (MAU), watch time, session duration, content completion rate, binge rate

Technical Performance: Buffering rate, video startup time (VST), streaming performance index (SPI), bitrate, latency & load time

Unit Economics: Average revenue per user (ARPU), Content spend efficiency, Content Cost per Hour/Title

Key PM Metrics

Consumer: subscription → content discovery → add to watchlist → view content → pause/replay → feedback/rating → renew subscription

Content creator or owner: Pitch for acquisition or production of originals → contract on licensing rights → content delivery → content promotion → analyze engagement data → monetization → contract renewal

Macro Trends and Opportunities

Emerging Content Formats:

Live watching sports/games (e.g Twitch), Short form videos, Non-linear storytelling (e.g. Black Mirror: Bandersnatch)

Futuristic Technology Trends:

Conversational search, Spatial Audio/Adaptive Sound (Netflix-Sennheiser partnership) , NFT-based collectibles and access tokens, VR watch parties, AR overlays

New Business Models:

Shoppable streaming/commerce-enabled OTT (Shop with Max), OTT bundling with telecom plans , FAST Channels, Gamified content models (Hotstar's cricket prediction contests)

Common Product Pitfalls

Content Discovery Gap: Poor metadata tagging and weak recommendations reduce engagement, while narrow personalization traps users in repetitive viewing loops, leading to content fatigue.

Subpar Streaming Quality: Poor video resolution, frequent buffering, latency issues in livestreaming and insufficient infrastructure to handle traffic spikes degrade the viewing experience.

Security Vulnerabilities: Poor DRM (Digital Rights Management) and anti-piracy measures can lead to content theft and revenue loss.

Ad Experience Breakdown: Irrelevant targeting, repetitive spots, and intrusive formats reduce user satisfaction and ad ROI.

Fragmented User Journeys: Lack of device sync and limited offline access disrupt continuity and lower customer satisfaction.

Interview Tips

In this sector, you can look out to the following:

Engagement Drop-offs: If new users stream <1 hour after week one, it signals churn risk. Diagnose causes and propose levers to boost early engagement.

Personalization Gaps: If users scroll endlessly but don't watch, refine recommendation engines to drive higher play conversion and watch time.

Ad Experience Design: For peak shows/events, optimize limited ad slots for yield while preserving immersion through smart placement and timing.

Content Navigation Design: A cluttered homepage reduces discovery. Redesign layout and categorization (genres, moods, trending, personalized rows) to improve relevance without overwhelming users.

Ground your answers in real user experiences and showcase your analytical strengths.

Ride Hailing Tech

Key Players

Global



- **Uber (28M rides daily),**
- **Didi Chuxing (30M rides)**
- **Grab (9M rides)**
- **Ola (4M)**
- **Lyft (2M)**

India



B2C

- **Ola**
- **Uber**
- **Namma Yatri**
- **Rapido**

B2B

- **Uber for Business**
- **Ola Corporate**
- **Shuttl**
- **Move in Sync**

Common Characteristics

Ride Workflow: Driver onboarding (verification, vehicle checks)--> User onboarding → Ride request → Matching & dynamic pricing → Navigation (route optimization) → Payment & settlements (in-app payments, driver payouts) → Feedback loop (ratings, complaints)

Corporate Ride-Hailing Workflow: Corporate onboarding → Employee onboarding (registration with company credentials) → Ride scheduling (on-demand or pre-scheduled) → Route optimization (multi-stop pickups/drops) → Ride execution → Payment & settlements → Feedback & analytics

Promise windows (passengers): Proximity-based matching, low wait times, transparent fares, safety features (SOS, ride tracking, verified drivers), personalization (saved location, preferred vehicles), well-maintained vehicles

Promise windows (drivers): Efficient vehicle utilization, incentive structures, use of real-time GPS data for navigation

Assortment mix: Multi-modal options (auto, bike, car, EV), carpooling, premium/luxury (e.g. Uber Black), scheduled rides, outstation trips, multi-stop trips

Ride Hailing Tech

Business Model Canvas

Value Proposition:

- **Passengers:** Seamless travel through easy booking, reliable availability, fair pricing, and trusted drivers
- **Drivers:** Higher earning potential through efficient matching, surge pricing benefits, seamless payment settlements

Revenue Streams: Trip-based commission, subscription fees, corporate contracts, in-app advertising, ancillary services (rentals)

Cost Structure: Driver payouts, platform development, customer support, driver incentives (e.g. underserved location, peak hour service), vehicle financing, insurance, mapping/navigation software purchase

Business Models: Commission-based model, Subscription/Membership model, B2B model (corporate partnerships)

Key Partnerships: Payment Gateways, map and technology providers, charging station networks, insurance providers (passenger & driver safety coverage)

Use Cases

Urban commuters: Office-goers, students, and city dwellers needing frequent/daily rides for commutes and social outings. Value cost efficiency, quick pickups, and minimal cancellations.

Tourists & leisure travellers: Travelers seeking airport transfers, sightseeing, and outstation rides. Prefer reliable service, comfortable or luxury cars, and transparent pricing.

Corporate users: Companies and business travelers using rides for employee commutes, client meetings, and airport transfers. Expect time-efficient aggregation, billing integration, and strong customer support.

Vehicle/Fleet owners: Drivers and operators leveraging the platform to maximize earnings with reduced idle time and fuel waste. Look for analytics-driven insights to optimize vehicle placement and quick payment cycles.

Ride Hailing Tech

Stakeholder Journey

Rider: Registration → ride search/request → driver matching & ETA → ride execution (pickup → trip → drop) → payment (in-app/cash) → feedback/rating → repeat usage

Driver/Fleet Partner: Onboarding & verification → vehicle check → receive ride request → navigate & complete ride → receive payment/payout → performance feedback → continued engagement / retention

Corporate Client: Company onboarding → policy setup (ride limits, billing rules) → employee onboarding → ride scheduling & tracking → consolidated billing & reporting → contract renewal

Key PM Metrics

Growth & Mix: Gross booking value (GBV), ride volume, active users, average daily rides, geographic coverage
Tech & Operations: Ride completion rate, ride cancellation rate, match time, driver utilization rate, driver acceptance rate, ETA accuracy
Trust: Safety incidents per million rides

Unit Economics: Contribution margin per trip, average revenue per user (ARPU), Take rate, driver earnings per hour, cost per ride

Macro Trends and Opportunities

Emerging and Futuristic Trends:

- Autonomous vehicles (Baidu Apollo Go partnership with Uber) - Self-driving fleets enabling cost-efficient and round-the-clock ride services.
- Sustainable mobility (Ola electric, Uber green) - EV adoption and green ride options reducing carbon footprint
- AI driven optimization (Didi's Gaia platform) **-** AI-based systems improving route planning, demand forecasting, and fleet utilization

New Business Models and Experiences:

- Mobility as a service (MaaS) - Integrated transport solutions combining ride-hailing with public and shared mobility
- Subscription rides (Lyft Pink, Uber One) - Membership models offering discounted fares and loyalty benefits
- Event & Tourism mobility (Uber Events) - Tailored rides for concerts, weddings, and tourist hotspots
- Micro-mobility (Rapido, Bounce) - Affordable short-distance travel via bikes, scooters, auto-rickshaws and self-drive options
- In-car Advertisement (GrabAds) - Showing targeted ads to passengers during rides
- Hyperlocalization (Namma Yatri in Bengaluru) - Platforms adapting to local cultures and pricing, often driver-cooperative owned

Ride Hailing Tech

Common Product Pitfalls

Real-time Matching Failure: Latency in the matching algorithm can result in slow driver-rider connections, leading to increased wait times and missed ride opportunities.

Inefficient Driver Dispatch: Weak forecasting and routing create driver shortages in some areas and oversupply in others, increasing wait times and reducing earnings.

Navigation & Routing Gaps: Inaccurate GPS, outdated maps, or weak route optimization increase ETAs, cause detours, and impact rider experience.

Driver Quality & Safety Concerns: Inconsistent driver onboarding, lack of background verification, or weak in-ride safety features erode user confidence.

Pricing Opacity & Surge Shock: Opaque surge pricing calculations erodes trust and drives rider churn.

Cancellation Handling Gaps: Unrestricted driver cancellations reduce reliability, delay trips, and frustrate passengers.

Interview Tips

Demand–Supply Imbalance: Rapid growth near Bangalore airport has increased rider demand, but driver reluctance to serve the area creates long wait times. Diagnose root causes and design levers. Think targeted incentives or pooled routing.

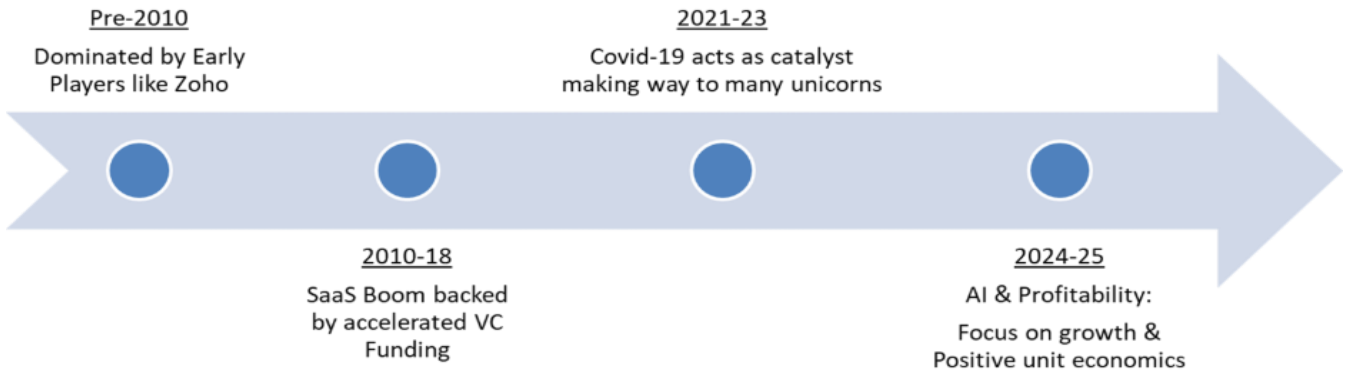
Driver Multi-App Switching: Surge-zone cancellations rise as drivers shift to rival apps offering better fares. Build loyalty mechanisms, earning parity, and reduce idle-time inefficiency to retain supply.

No-Show Costs: In corporate fleets, 20–30% employee no-shows inflate per-head trip cost and reduce utilization. Redesign with real-time confirmations, waitlist allocation, and automated rerouting.

Passenger Safety: Low SOS adoption and delayed response weaken trust. Reimagine the safety workflow with proactive nudges, live tracking, and faster escalation.

Keep your answers rooted in real rider and driver issues. Interviewers want to see clarity in identifying problems and actionable solutions.

Sector Evolution Timeline



The SaaS Stack & Key Players

L1: Horizontal SaaS (Function-Specific)

Focus on best-in-class user experience, deep integrations, and clear differentiation in crowded markets.

Key Players: Freshworks (Customer Support), Zoho (Business Suite), Darwinbox (HRTech), WebEngage (MarTech), Chargebee (Billing).

L2: Vertical SaaS (Industry-Specific)

Focus on deep domain expertise, industry-specific workflows, and becoming the "system of record" for a particular vertical.

Key Players: Lentra (BFSI), Innovaccer (Healthcare), Zenoti (Wellness), UrbanPiper (Restaurants).

L3: DevTools & Infrastructure SaaS

Focus on superior developer experience (DX), robust documentation, performance, and building a strong user community.

Key Players: Postman (API Platform), BrowserStack (Testing), Hasura (GraphQL).

Common Features

- **PLG (Product-Led-Growth) / Self-Serve Focused**
  - Frictionless Onboarding: Quick, self-serve sign-up and in-app guided tours to drive user activation.
  - Freemium/Trial Models: Offering a free tier or a time-bound trial is the default to lower the barrier to entry.
  - Community & Docs: Public documentation and a strong user community are key for support and engagement.
- **B2B / Enterprise-Focused**
  - Security & Compliance: SOC2, GDPR, and ISO certifications are non-negotiable table stakes.
  - Enterprise-Grade Features: Single Sign-On (SSO), role-based access control (RBAC), and audit logs are mandatory.
  - Integrations & APIs: A robust integration marketplace and well-documented APIs are critical for fitting into a customer's existing tech stack.

Business Model Canvas

- **Value Propositions**: Lower Total Cost of Ownership (TCO), faster time-to-value, scalability on demand, and continuous, automatic updates.
- **Channels**: Product-Led Growth (PLG) through freemium/trials, Sales-Led Growth (SLG) with inside and field sales teams, and content marketing.
- **Revenue Streams**:
  - Recurring Revenue: Annual/Monthly Recurring Revenue (ARR/MRR) through tiered subscriptions (based on features, users, or usage).
  - Other: One-time implementation fees and premium support packages.
- **Key Partnerships**: Cloud providers (AWS, Azure, GCP), technology partners for integrations, and channel resellers.
- **Cost Structure**: Dominated by Talent Costs (R&D) and Customer Acquisition Cost (S&M), followed by cloud infrastructure expenses.

User Cases and Pain Points

Use Case	Pain Point	Opportunity
SMBs (Self-Serve Users)	High costs of traditional software, lack of technical expertise, need for simple, all-in-one tools.	Offer affordable, easy-to-use products with a freemium model and excellent self-serve support.
Enterprise Buyers (Decision-Makers)	Complex procurement cycles, data security & compliance risks, unclear ROI, and integration challenges with legacy systems.	Provide clear ROI calculators, enterprise-grade security features, a robust integration marketplace, and dedicated support.
Developers (Using SaaS APIs)	Poor documentation, unreliable API performance, opaque pricing, and restrictive rate limits.	Create a superior developer experience (DX) with clear documentation, a reliable sandbox, and transparent, usage-based pricing.

User Journey

PLG User Journey
Discovery → Free Trial/Sign-up → Activation → Conversion → Expansion

- **Activation:** Steep learning curve and poor first-time user experience. → Design a seamless onboarding flow to a quick "win."
- **Conversion:** Unclear value proposition of paid plans. → Clearly gate high-value features and demonstrate ROI to encourage upgrades.
- **Expansion:** Product lacks features for growing needs. → Build modular features that users can adopt as they scale.

Enterprise Buyer Journey
Problem Awareness → Evaluation → Procurement & Legal → Implementation → Renewal

- **Evaluation:** Inability to properly test the product for security and integration. → Offer guided proof-of-concept (POC) trials and a robust sandbox.
- **Procurement & Legal:** Long negotiation cycles and inflexible contract terms. → Standardize contracts and empower sales teams with clear guidelines.
- **Implementation:** Difficult data migration and a lack of user training. → Provide dedicated implementation support and comprehensive onboarding for teams.

Key PM Metrics

Growth:

- ARR/MRR: Annual/Monthly Recurring Revenue

Unit Economics:

- LTV/CAC Ratio: Target > 3
- Payback Period: Target < 12-18 months

Retention:

- Net Revenue Retention (NRR): Target > 120% for best-in-class
- Gross Revenue Retention: Target > 90%

Engagement & Value:

- Activation Rate: % of users completing a key action
- NPS: Net Promoter Score

Macro Trends and Opportunities

AI as a Co-pilot: Generative AI is being embedded to automate tasks and act as an intelligent assistant within workflows.

- *Opportunity:* Move beyond simple features to create AI-native products that fundamentally change how work is done.

Vertical SaaS Deepening: Companies are building deep, industry-specific platforms that become a business's central operating system.

- *Opportunity:* Identify an underserved niche and build a "system of record" with tailored workflows and domain expertise.

Focus on Net Revenue Retention (NRR): The most efficient growth comes from the existing customer base.

- *Opportunity:* Design products for modular adoption, enabling clear paths for upselling and cross-selling.

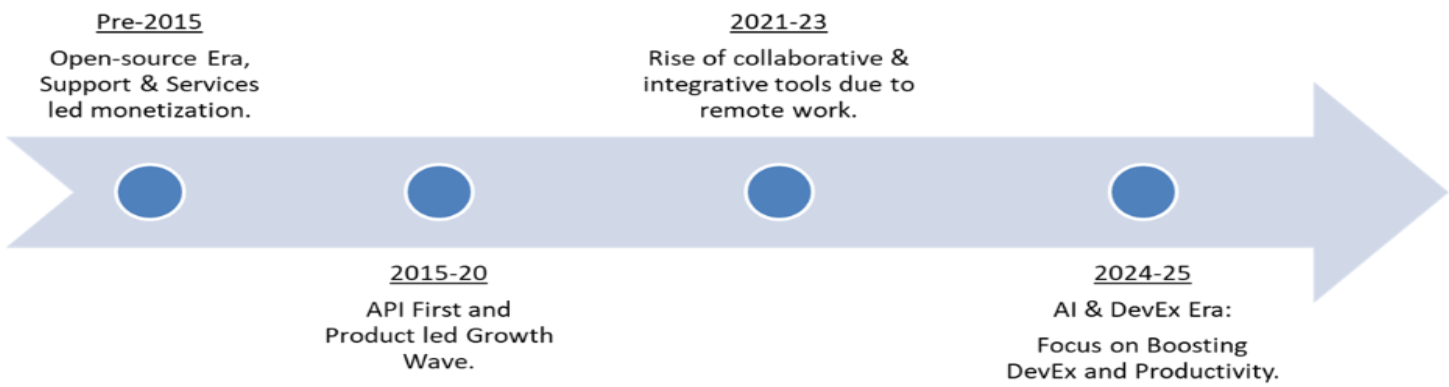
Common Product Pitfalls

- **Building a "Vitamin, not a Painkiller":** Creating a nice-to-have product that doesn't solve a critical business problem, leading to high churn.
- **Ignoring Go-To-Market (GTM) Fit**: Building a complex product but expecting a simple self-serve model to work, or vice versa.
- **Underestimating Enterprise Needs**: Trying to sell to large companies without mandatory features like SSO, audit logs, and SOC2 compliance.
- **Technical Debt**: Prioritizing new features over platform stability, resulting in a slow, buggy product.
- **Poor User Onboarding**: Failing to guide new users to the "aha!" moment quickly, leading to low activation and abandonment.

Interview Tips

- **Speak the Language of SaaS**: Always frame your answers using core metrics like ARR, NRR, LTV/CAC, and Churn.
- **Clarify the GTM Motion**: Start by asking, "Is this a Product-Led Growth (PLG) or Sales-Led Growth (SLG) model?"
- **Distinguish Between Personas**: Clarify if you are solving for the Buyer (focused on ROI) or the End User (focused on workflow efficiency).
- **Think in Workflows, Not Just Features**: Frame your ideas around improving a user's end-to-end workflow.
- **Anchor on Expansion Revenue**: When asked how to grow a product, focus on increasing Net Revenue Retention (NRR) through upselling and cross-selling.

Sector Evolution Timeline



The DevTools Stack & Key Players

L1: Code & Collaboration

Focus on performance, seamless collaboration features, and deep integrations into the developer's editor (IDE).
Key Players: GitHub, GitLab, Atlassian (Bitbucket), JetBrains.

L2: Build, Test & Deploy

Focus on speed, reliability, and providing insightful analytics to debug failures quickly.
Key Players (India-born): BrowserStack, LambdaTest.
Global Players: Jenkins (Open Source), CircleCI, Harness, Vercel.

L3: API Platforms & Infra

Focus on best-in-class documentation, API reliability, and a simple, predictable pricing model.
Key Players (India-born): Postman, Hasura.
Global Players: Stripe, Twilio, Docker, HashiCorp.

L4: Observability & Monitor

Focus on reducing noise, providing actionable alerts, and helping developers correlate data from different sources (logs, metrics, traces).
Key Players: Datadog, New Relic, Grafana (Open Source), Dynatrace.

Common Features

Developer Experience (DevEx) Focused

- **CLI & API Access:** A powerful Command-Line Interface (CLI) and a well-documented API are non-negotiable.
- **Stellar Documentation:** Documentation is treated as a core part of the product, not an afterthought.
- **Community is Key:** A thriving community (on Discord, Slack, etc.) for support and feedback is a critical feature.

Team & Enterprise Focused

- **Collaboration Features:** Real-time sharing, commenting, and multiplayer modes are becoming standard.
- **Governance & Security:** Role-based access control (RBAC), audit logs, and security features (DevSecOps) are mandatory for enterprise adoption.
- **Performance Analytics:** Dashboards showing build times, deployment frequency, and failure rates (DORA metrics) are crucial for engineering leaders.

Business Model Canvas

Value Propositions: Increased developer productivity, faster time-to-market for new features, improved software quality and reliability, and enhanced collaboration.

Channels: Product-Led Growth (PLG) driven by individual developers, open-source adoption, content marketing (technical blogs, tutorials), and a sales-led motion for enterprise accounts.

Revenue Streams:

- **Freemium & PLG:** A generous free tier for individuals and small teams is standard.
- **Usage-Based Pricing:** Charging based on consumption (e.g., build minutes, API calls, data storage).
- **Per-Seat Pricing:** Tiered subscriptions based on the number of developer seats.

Key Partnerships: Cloud providers (AWS, Azure, GCP), IDE makers (Microsoft for VS Code), and open-source foundations.

Cost Structure: Dominated by **Talent Costs (R&D)** for highly skilled engineers, followed by community management and cloud infrastructure costs.

User Cases and Pain Points

Use Case	Pain Point	Opportunity
Individual Developers	Time wasted on setting up local environments, repetitive boilerplate tasks, and a steep learning curve for new tools.	Offer a zero-config setup, build intelligent code-generation tools, and provide interactive tutorials.
Engineering Teams (Team Leads & Managers)	Slow CI/CD pipelines, inconsistent development environments across the team, and difficulty in reviewing and debugging complex code.	Build high-performance testing platforms, offer cloud-based development environments, and create powerful code review and collaboration tools.
Enterprise Buyers (VPs of Eng, CTOs)	Lack of visibility into developer productivity, ensuring security and compliance across the SDLC, and managing tool sprawl.	Provide comprehensive analytics dashboards (DORA metrics), build DevSecOps features, and offer an integrated platform solution.

User Journey

PLG Developer Journey
Discover (e.g., on Twitter/Blog) → Sign-up (often via GitHub) → Activation → Community Engagement → Expansion

- **Activation:** *The tool is complex to set up and the "aha!" moment is unclear.* → **Provide a "Getting Started" guide that leads to a meaningful outcome in under 5 minutes.**
- **Community Engagement:** *The developer gets stuck and has no one to ask for help.* → **Build an active Discord/Slack community for peer-to-peer support.**
- **Expansion:** *The developer loves the tool but can't convince their team to adopt it.* → **Build collaborative features that are easy to share and demonstrate value to the entire team.**

Enterprise Team Journey
Bottom-up Adoption → Team Evaluation (POC) → Security & Compliance Review → Procurement → Team-wide Rollout

- **Evaluation (POC):** *Difficult to prove the tool's value and ROI over existing solutions.* → **Offer a guided Proof-of-Concept with clear success metrics and dedicated support.**
- **Security & Compliance:** *The tool doesn't meet the company's strict security standards.* → **Achieve SOC2 compliance and build enterprise-grade features like SSO and audit logs.**
- **Rollout:** *Lack of training and integration makes team-wide adoption a challenge.* → **Provide detailed migration guides and offer onboarding sessions for the entire team.**

Key PM Metrics

Adoption & Growth:

- **Weekly Active Users/Teams:** Measures recurring usage.
- **Stars/Forks (for Open Source):** Indicates community interest.

Unit Economics:

- **LTV/CAC Ratio:** Target > 3, often higher due to low-touch sales.

Retention:

- **Net Revenue Retention (NRR):** Target > 130% for best-in-class DevTools.

Engagement & Value:

- **Activation Rate:** % of users who successfully complete a core workflow (e.g., run their first build).
- **Time to Value (TTV):** How quickly a new user finds the "aha!" moment.

Macro Trends and Opportunities

AI in the SDLC: AI is moving from being a code assistant to an autonomous agent that can write, test, and even deploy code.

- *Opportunity:* Build AI-native tools that automate entire workflows, not just individual tasks.

Platform Engineering: Enterprises are building internal developer platforms (IDPs) to standardize tools and improve DevEx.

- *Opportunity:* Create modular, API-first tools that can be easily integrated into these internal platforms.

DevSecOps is Mainstream: Security is shifting left and becoming an integral part of the developer's workflow.

- *Opportunity:* Build tools that embed security checks and vulnerability scanning directly into the CI/CD pipeline and the developer's IDE.

Common Product Pitfalls

- **Poor Developer Experience (DX):** A slow, buggy, or unintuitive tool will be quickly abandoned by developers.
- **Ignoring the Open-Source Community:** Treating the community as a marketing channel instead of a core part of the product development process.
- **Failing to Bridge the PLG-to-Enterprise Gap:** Not building the necessary security, compliance, and governance features required to sell to large companies.
- **"Leaky Bucket" Funnel:** Focusing on top-of-funnel sign-ups without a clear strategy for activating and retaining those users.
- **Solving a Niche Problem:** Building a tool for a workflow that only a very small subset of developers has, limiting the total addressable market.

Interview Tips

- **Be the User:** Demonstrate genuine empathy and a deep understanding of developer pain points, workflows, and the tools they use daily.
- **Think in Ecosystems:** Frame your answer around how your product would integrate with the broader DevTools ecosystem (e.g., GitHub, VS Code, Slack).
- **Community-First Mindset:** When asked about GTM, a great answer is to start with building a strong open-source or user community.
- **Developer Experience is Everything:** Anchor your product decisions on what will create the most seamless and delightful experience for the developer.
- **Articulate the PLG Flywheel:** Be able to explain how a great product with a strong community can lead to a powerful, low-cost acquisition and expansion loop.

Health Tech

Key Players

Global



Consultation - Teladoc Health, Amwell, Doctor on Demand
Hospital SaaS & EMR - Epic Systems, Oracle Cerner, Meditech
Fitness Tracking - MyFitnessPal, Strava, Fitbit

India



Consultation - Practo, Tata 1mg, MFine
Hospital SaaS & EMR - Practo Ray, HealthPlix, MediBuddy
Fitness Tracking - Cult.fit, HealthifyMe, Fittr

Common Characteristics

Promise windows (Patients/Users)

- Verified doctors, nutritionists and fitness coaches
- Remote consultations, workouts, home diagnostics
- Continuous health monitoring and progress tracking

Promise windows (Providers)

- Smart scheduling with integrated insurance, billing
- Wider patient pool through digital platforms
- Analytical insights from digitized health records

Assortment mix: Telehealth, enterprise solutions (EHR, hospital management), diagnostics and monitoring, medicine & nutrition delivery, wellness, lifestyle platforms

Key Bottlenecks

- Regulatory and compliance hurdles (HIPAA, GDPR, NDHM)
- Strict oversight of health claims & outcomes
- Tight controls on e-prescriptions & online medicine sales

Business Model Canvas

Value Proposition

- **Patients/Users:** Accessible healthcare & fitness, supported by expert consultants, healthcare services
- **Providers:** Simplified appointment, follow-up management, integrated patient/user records, expanded reach through discovery, add-on services

Revenue Streams

Consultation commissions, Diagnostic fees, Subscription plans (chronic care, wellness), SaaS licensing (EMR/EHR for hospitals/doctors), Corporate employee packages

Business Models

Pay-per-service (Practo, Tata 1mg), Freemium (HealthifyMe), Enterprise solutions (Practo Ray), Subscription models

Key Partnerships

Doctors, hospital networks, pharmaceutical companies, diagnostic labs, technology and cloud providers

Use Cases / Customer Segments

- **Care seekers:** Looking for quick and reliable access to doctors, medicines, and diagnostics. Value convenience, trust, and affordability.
- **Health & fitness enthusiasts:** Focused on fitness, diet, sleep, and stress management. Engage with trackers, workout videos, and habit-building programs.
- **Chronic condition patients:** Looking for continuous monitoring, specialist support, and personalized care plans.
- **Enterprise-linked users:** Access healthcare through corporate or insurance tie-ups. Seek preventive care, mental wellness, and subsidized consultations.
- **Providers:** Healthcare/wellness providers and hospitals adopting digital tools for practice management, teleconsultation, EMR/EHR, and patient engagement.

Stakeholder Journey

Patient: User onboarding → identify need → discover provider → book consultation/test → pay → consult/diagnosis → medicine delivery → follow-up → review

Provider: Registration and KYC → list offerings → setup profile and slots → deliver services (consult/therapy/training) → share plan/prescription/report → track performance/analytics

Healthcare SaaS subscribers: Organization onboarding → workflow setup (services, slots, billing) → record digitization and data management → billing & settlements → analytics & reporting → subscription renewal/upgrade

Chronic Care/Fitness: User onboarding → profile setup → need assessment → program discovery → subscription/payment → periodic consult/coaching → continuous monitoring → progress review

Metrics and KPIs

Growth & Mix: New user acquisition rate, Churn rate, Conversion rate (free to paid), CLV, CAC payback, GMV, ARPU

Ops & Engagement: Monthly active users (MAU), session frequency and duration, appointment wait time, appointment no show rate, diagnostic turnaround time
Functional Performance: Care plan adherence rate, clinical outcomes improvement, consultation completion rate

Technical Performance: Feature adoption rate, user feedback score, System downtime

Health Tech

Macro Trends

Consumer Experience:
- AI powered personalized health journeys
- VR/AR enabled healthcare training (e.g. Osso VR)
- Wearables + Healthcare Integration (e.g. Apple Health - aggregates health data with option to share with healthcare provider)

Futuristic Technology Trends:
- Gen AI in Diagnostics
- Blockchain for securing health data (e.g. BurstIQ)
- Remote patient monitoring (e.g. Medtronic)

New Business Models:
- Subscription based consultation
- Heath super apps (GrabHealth)
- Niche wellness platforms (Flo, Wysa)

Common Product Pitfalls

- **Subpar Device Connectivity:** Video drops, buffering, device pairing issues, and slow data updates undermine remote care and real-time monitoring.
- **Poor UX/UI Design -** Complex navigation, confusing workflows, and lack of intuitive design reduce user adoption and engagement, especially among elderly or less tech-savvy patients.
- **Privacy, Security & Compliance Risks:** Inadequate data protection, access control issues, and missing consent tracking put patient trust and regulatory compliance at risk.
- **Engagement Drop-off and Retention Issues:** Many users download wellness apps but show high churn after initial use due to lack of sustained motivation or personalization.

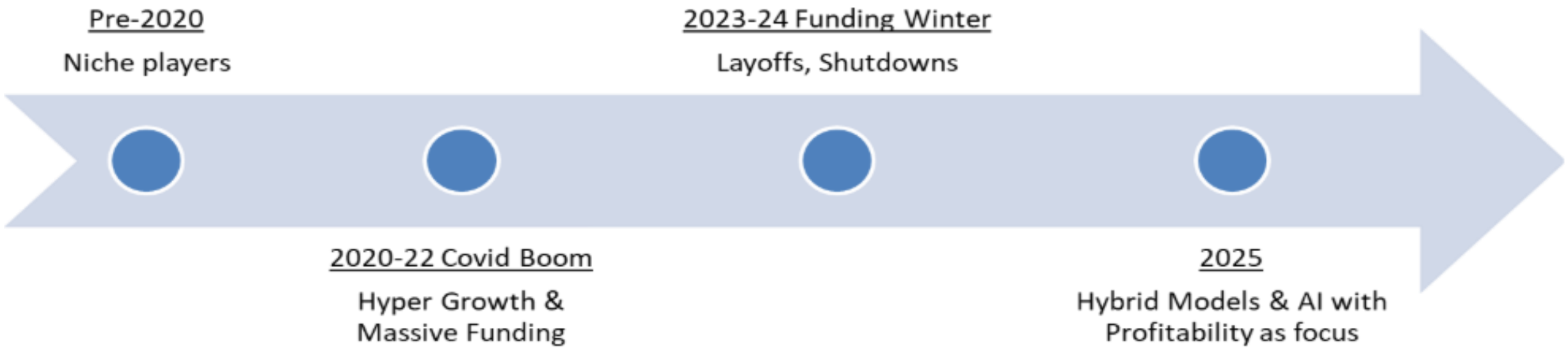
Interview Prep Tips

In this sector, you can look out to the following:

- **Engagement Drop-offs:** Wellness apps see steep usage decline after the first month. Diagnose friction points (complex flows, fatigue) and propose product levers. Think gamification and habit-building nudges.
- **Monetization Barriers:** High consult volumes on Practo aren't converting into medicine orders. Diagnose drop-off points and design levers to capture downstream revenue.
- **Elderly Adoption Gap:** Senior citizens are most in need of digital health support but often can't use the apps. Solve with simplified UI, voice support, and caregiver-linked accounts.
- **Data Fragmentation & Interoperability:** Patients juggle multiple apps (consults, fitness, chronic care) while providers manage siloed systems (labs, wearables, EHRs), leading to broken journeys and app fatigue. Diagnose integration gaps and propose unified health hub with interoperable standards to deliver seamless, continuous care.

Prioritize accessibility, reduce friction, and connect fragmented experiences to scale health tech products beyond a simple aggregator model.

Sector Evolution Timeline



Key Players

Global

1. Coursera
2. Udemy
3. Duolingo

India

1. PhysicsWallah (PW)
2. upGrad
3. Unacademy
4. Vedantu
5. Simplilearn

Common Features and Product Norms

- **Pricing Models:** Freemium, Subscription, One-time Cohort Pricing
- **Delivery Format: Hybrid** Learning is the norm (Live Classes + Recorded Content + Offline Centers).
- **Engagement Levers:** Gamification (leaderboards, points), Community (peer groups), and 24/7 Doubt-Solving (AI + Human TAs).
- **Tech Enablers:** AI Tutors, Adaptive Testing, Vernacular-First UI/UX and Scalable architecture.
- **Product Focus:** A shift from content delivery to **measurable learning outcomes** and career success

Business Model Canvas

- **Value Propositions:** Affordability for Tier-2/3, tangible career outcomes, AI-powered personalization, and trust via hybrid models.
- **Channels:** Digital marketing (SEO, YouTube), direct sales teams, offline centers, and organic referrals.
- **Revenue Streams:** B2C subscriptions & course fees, B2B SaaS licenses, and outcome-linked models (Income Share Agreements - students pay as a %age of their future income after they get a job).
- **Key Resources:** Brand & star educators, proprietary tech, content library, and physical offline network.
- **Key Partnerships:** Universities (for accreditation), corporates (for upskilling), and schools (for distribution).
- **Cost Structure:** Dominated by Customer Acquisition Cost (CAC), followed by employee salaries, content development, and tech infrastructure.

Key Use Cases

- **K-12 Supplementary Learning**: After-school tutoring and concept clarification.
- **Test Preparation**: Coaching for competitive exams (JEE, NEET, UPSC).
- **Professional Upskilling**: Acquiring new skills for career growth (e.g., Data Science, Digital Marketing).
- **Higher Education**: Pursuing accredited online degrees and certifications.

User Journey and Pain Points

User Journey: 1. Discovery → 2. Evaluation→ 3. Enrollment → 4. Learning → 5. Outcome

Journey	Pain Point	Opportunity
Discovery - Learner/parent sees ads, gets referrals, or searches online.	Information overload, aggressive marketing, and lack of trust.	Build a trustworthy brand via organic content and testimonials.
Evaluation - Compares platforms, attends demos, and checks pricing/reviews.	Unclear pricing, poor demos, and confusing refund policies.	Offer impactful demos and maintain simple, transparent policies.
Onboarding - Pays, gets platform access, and starts the first session.	Clunky payments and a poor first-time user experience (FTUE).	Design a seamless onboarding flow to a quick first "win."
Engagement - Attends classes, solves problems, and interacts with peers/mentors.	Low motivation, digital fatigue, and slow doubt-solving.	Build engagement loops (gamification, community) & use AI for instant support.
Outcome - Completes the course, gets a certificate, and seeks a job or score improvement.	Lack of career support and certificates with low industry value.	Provide robust career services and transparent outcome dashboards.

Macro Trends & Opportunities (2025)

- **Tier-2/3 Market Explosion:** India is the fastest-growing market, fueled by demand from tier-2/3 cities.
  - *Opportunity:* Build affordable, vernacular-first products that solve local needs and aspirations.
- **Profitability is King:** Focus is on sustainable unit economics (LTV/CAC > 3).
  - *Opportunity: Design products with strong organic growth loops.*
- **AI as a Co-pilot:** GenAI is being used for personalization and automation.
  - *Opportunity: Create AI-native learning tools.*
- **B2B Upskilling Booms:** Corporations are major EdTech spenders.
  - *Opportunity: Build enterprise platforms with clear ROI analytics.*
- **The Vernacular Wave:** The next 100M users need content in their language.
  - *Opportunity: Develop contextual, not just translated, vernacular products.*
- **Unbundling of Degrees & Micro-Credentials:** Shift from degrees to short, skill-focused certifications.
  - *Opportunity*: Offer flexible, affordable modules for learners to gain specific, job-ready skills.

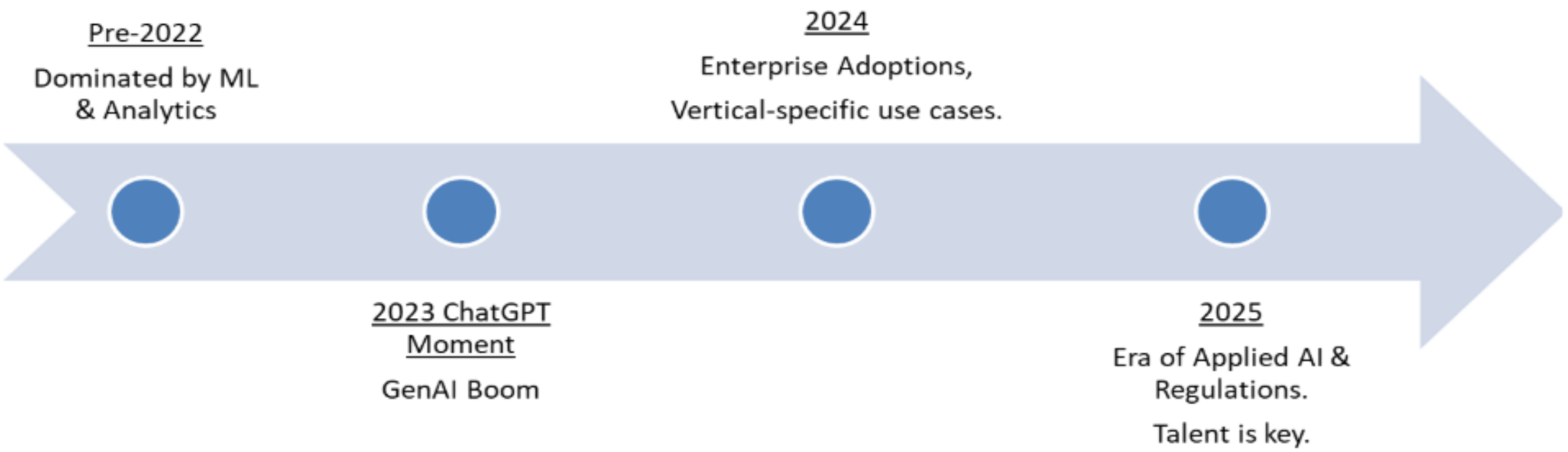
Common Product Pitfalls (with Examples)

- **Unsustainable CAC:** Burning cash on marketing without retention (e.g., Byju's aggressive sponsorships). Byju's case shows that overreliance on sales is also not a healthy path in the long term.
- **Ignoring Pedagogy:** Focusing on tech over actual learning outcomes (e.g., early MOOCs with <10% completion).
- **Scaling Operations Poorly:** Acquiring users faster than support can handle (e.g., Byju's refund backlash).
- **Missing Vernacular Needs:** Building English-first products and alienating the Tier-2/3 market.
- **Ignoring Learner Engagement:** Focusing on content acquisition while failing to solve for low motivation and high course drop-off rates.
- **Poor Hybrid Execution:** Failing to seamlessly integrate the online and offline user experience, creating operational chaos.
- **Breaking User Trust:** Replicating aggressive, opaque sales and support tactics that led to the downfall of older players.

Key Metrics

- **B2C EdTech KPIs:**
  - Engagement: DAU/MAU, Avg. Time Spent, Course Completion Rate.
  - Retention: D1/D7/D30 Retention, Cohort Completion %.
  - Monetization: Conversion Rate (Free to Paid), CAC / LTV Ratio (must be < 1:3), Avg. Revenue Per User (ARPU).
  - Advocacy: Net Promoter Score (NPS).
- **B2B EdTech KPIs:**
  - Adoption: % of Licensed Seats Activated, Learner Engagement Rate.
  - Retention: Net Revenue Retention (NRR), Renewal Rate.
  - Value: Customer Satisfaction (CSAT), ROI metrics for the client.
- **Sample Interview Frames:**
- "How would you increase course completion rates from 15% to 50%?"
- "Design an AI-powered personal tutor for JEE aspirants."

Sector Evolution Timeline, Drivers & Challenges



The AI Technology Stack & Key Players

L1: Infrastructure

Providers of core compute power, resources for AI to work.
PM Lens: Balance cost vs. performance.
Key Players:
Chipmakers - NVIDIA, AMD, Google, Intel
Cloud Providers - AWS, Azure, GCP, Shakti Cloud (Indian)

L2: Foundational Models

The general-purpose models.
PM Lens: Build vs. Model Fine-tune vs. Open-source.
Key Players:
Global - OpenAI, Gemini, Meta, Anthropic, Mistral
Indian - Sarvam, Krutrim (Ola), Project Indus (Tech Mahindra)

L3: Data & Middleware

PM Lens: Where trade-offs (latency, cost) are made.
Key Players:
Vector Databases: Pinecone, Milvus, Chroma DB.
Orchestration & Tooling: LangChain, LlamaIndex, HF.
MLOps & Data Platforms: Databricks, Snowflake.

L4: Applications

The end-user products (Haptik, Vertical SaaS).
PM Lens: Focus on UX and solving a specific user pain point.
Key Players:
Conversational AI: Haptik (a Jio company), Yellow.ai.
Enterprise AI & Analytics: Fractal Analytics, Mu Sigma.
AI-native SaaS (Global): Microsoft (Copilot), Adobe (Firefly), GitHub (Copilot), Notion.

L5: Integrations & Consulting

The service layer driving large-scale AI adoption.
PM Lens: Focus on GTM partnerships to scale enterprise sales.
Key Players:
Indian IT Services Giants: TCS, Infosys, Wipro, HCLTech.
Global Consulting Firms: Accenture, Deloitte, PwC, EY.

Common Features & Product Norms

- **Data and Computation Intensive:** AI models, especially deep learning models, require massive amounts of data and computation power for training.
- **Delivery:** APIs & SDKs are standard.
- **Interface:** "Co-pilot" assistants embedded in existing software.
- **Customization:** Fine-tuning & RAG (Retrieval-Augmented Generation) tools.
- **Enterprise-Grade:** Evals dashboards, audit logs, and red-teaming are now table stakes.
- **Pricing:** Usage-based (pay-per-token/API call) is the dominant model.

Business Model Canvas - Indian AI Sector

- **Customer Segments:** Large enterprises (seeking efficiency), tech startups (building AI-native products), SMEs (adopting AI tools), government, and individual developers/consumers.
- **Value Propositions:** Automation of complex tasks, hyper-personalization at scale, predictive insights for decision-making, and creation of novel content/media.
- **Channels:** Direct enterprise sales, product-led growth (PLG) for SaaS tools, APIs for developers, and cloud marketplaces (AWS, Google Cloud).
- **Revenue Streams:** API calls (pay-as-you-go), SaaS subscriptions (tiered), enterprise licensing, and consulting/implementation fees for custom solutions.
- **Key Resources:** AI/ML talent (engineers, researchers), high-quality proprietary **datasets**, massive compute infrastructure (GPUs), and strong research partnerships.
- **Key Partnerships:** Cloud providers (for infrastructure), data suppliers, academic institutions (for research), and industry-specific channel partners.
- **Cost Structure:** Dominated by **compute costs** (GPU clusters) and **talent costs** (high salaries for AI engineers), followed by R&D and data acquisition.

Use Cases & Pain Points

- **Developers:** High API costs, rate limits/quotas, model switching costs, poor documentation.
- **Opportunity:** Build better, cheaper developer tools.
- **Enterprises (B2B):** Data residency/DPDP compliance, integration complexity, TCO modeling & ROI proof.
- **Opportunity:** Offer enterprise-grade security and clear ROI dashboards.
- **SMEs:** Lack of expertise & high costs.
- **Opportunity:** Create simple, no-code AI tools for specific business functions.

Key Metrics

- **Model Performance: Groundedness/Hallucination Rate**, p95 Latency, Accuracy, F1 Score, Eval Coverage.
- **User Engagement:** Task Success Rate, Feature Adoption Rate.
- **Business Metrics:** Cost-Per-Resolved-Task, API Uptime, ROI for customers.

Macro Trends & Opportunities (2025)

- **AI Agents:** Moving from assistants to autonomous task completion.
  - *Opportunity:* Build agentic workflows for specific industries.
- **Edge AI:** Running smaller models on-device.
  - Opportunity: Build privacy-first, low-latency apps.
- **Multimodality:** Models understanding text, image, audio & video.
  - *Opportunity:* Create novel content generation tools.
- **Verticalization:** Fine-tuning models for specific domains.
  - *Opportunity:* Build a "GPT for Doctors" or "GPT for Lawyers".
- **Public Datasets:** IndiaAI datasets platform will unlock new product ideas for startups.
  - *Opportunity:* Build hyper-localized AI products for uniquely Indian problems.

Ethical & Regulatory Landscape

- **Core Challenge:** Balancing innovation with responsible development.
- **Key PM Concerns:** Model bias, hallucinations, data privacy, and job displacement.
- **Regulatory Outlook:** Governed by **DPDP Act 2023 + draft Rules**. India's **BHASHINI** program is a key case study for AI-driven inclusion. **Opportunity:** Build "Responsible AI" features from day one.

Common Product Pitfalls

- **Tech in Search of a Problem:** Building a cool feature that solves no real user need.
- **Ignoring the "Last Mile":** A powerful model with a poor, unusable UI.
- **Negative Unit Economics:** High inference costs leading to losses on every user action.
- **Shallow Evals:** AI models have probabilistic outputs, not testing for real-world failures, can lead to untrustworthy products.
- **Single-Vendor Lock-in:** Over-reliance on one model provider, risking high switching costs.
- **Scalability:** A model that works well may fail in a real-world production environment due to issues with latency, or cost.

AI PM Interview Tips

- **Think in Layers:** Structure your answer around the AI stack.
- **Anchor on User Problems:** Start with the "why," not the "what" (tech).
- **Address Ethics Proactively:** Discuss risks and mitigations before being asked.
- **Know Your Trade-offs:** Discuss RAG vs. Fine-tuning, compute budgets vs. latency targets.
- **Design for Trust:** Talk about your "eval harness design"—how you'd test for accuracy and safety.
- **Ensuring data quality/privacy -** Sourcing clean, unbiased data while adhering to DPDP Act.

Section 4

Tackling Cases



Guesstimate Tips

Introduction

In **Product Management interviews**, guesstimates are used to assess a candidate's skills in....

- **Problem Solving**
- **Reasoning**

This is vital because, **in real-world scenarios,** precise data on

- market sizes
- consumer preferences
- adoption rates

....is often unavailable.

What should a strong PM do then?

A strong PM must...

Make quick, logical estimates to seize opportunities

Allocate resources

Validate business cases



....Thus, enabling faster, data-driven decisions even amid uncertainty.

Guesstimate Tips

Basic Tips in Guesstimates

1. **Clarify Assumptions**
   Always confirm your assumptions with the interviewer before proceeding.

2. **Ask Relevant Questions**
   Gather as much clarity as possible on scope, timeframe, definitions, and constraints before starting calculations.

3. **Frame the Problem**
   Look at it from multiple angles (e.g., supply side vs. demand side) and pick the one which is simpler and with more reliable assumptions.

   This can include: ↓ **Top-down approach** ↑ **Bottom-up approach**

4. **Use Simple Numbers**
   Stick to whole numbers for ease of calculation and cleaner logic.

5. **Segment the Population**
   Break down the problem using filters such as Gender, Rural/Urban, Age Groups and Income.

6. **Keep Calculations Neat**
   Maintain clear step-by-step working for transparency and error reduction.

7. **Triangulate your Answer**
   Sanity-check the number using alternative methods or rough benchmarks to see if it makes sense.

Key Reference Data for Guesstimates (Specific to India only)



You can refer to the below figures that are updated rough estimates which are rounded for ease of solving calculations during guesstimates:

Population and Area

- Total Population – 1.4 billion
- Total Area – 3.3 Mn sq. km
- Avg. people/ household – 4

Gender Split

- Male – 50%
- Female – 50%

Rural Urban Split

- Rural – 60%
- Urban – 40%

Age Split

- 0-20 – 40%
- 21-40 – 30%
- 41-60 – 20%
- 60+ – 10%

Income Split

- Low – 50%
- Middle – 30%
- Upper Middle – 15%
- High – 5%

Major Cities (Population)

- Delhi – 2%
- Mumbai – 1.5%
- Bangalore – 0.9%
- Kolkata – 1%
- Hyderabad – 0.9%

Product Design: Do's & Don'ts

Approach Tips

Product Design interviews are generally asked in two ways....

- **Designing a new product**
- **Improving an existing product**

Both of these questions can be answered using the **CIRCLES framework.** CIRCLES is a very powerful way to structure your approach. **Do not mention it explicitly, but rather direct the interview accordingly.**

Importance of a Customer Journey

✓ Always ensure to **come up with the Customer Journey and verify with the interviewer.**	✓ This helps **structure the interview better,** but it requires some **user POV thinking.**	✓ **Be very minute in the steps** that the user is taking and then **come up with a Journey.**

The CIRCLES Framework

C — Context

- **The first should be to understand more about the product, the company, what their objectives are etc.**
- **Ask 3-4 questions to the interviewer.** Also, if the product is well known, give a brief about your understanding of the product and then ask for clarification from the interviewer.

I — Identify User Personas

- **Identify 2-3 user personas which could use your product.** You should try to make the persona as relatable as possible. Talk about the persona's demographics, their psychographics etc.
- **Give the persona a name as well.** For example, Aditya, a Sports enthusiast, looks for new sports gear online and wants the best quality products.
- **Select a persona** you would like to build your product for. Give your reasons for selecting this persona to the interviewer.

The CIRCLES Framework

R — Context

- **You need to identify the needs and pain points for your selected persona.**
- This is the step where you would **build the user journey** for the selected user persona. Talk about the exact points in the journey where we see maximum friction/issues for the customer.
- In the end, you should have **3 crisp problems** that you see for the user.

C — Identify User Personas

- You would likely find a lot of needs and pain points across the user journey.
- **Prioritize a few needs and pain points you would like to solve right now.**
- Give your reasoning to the interviewer. You can **evaluate the pain points** in the lens of impact (business or customer experience), resources required to build and the complexity of solution.

The CIRCLES Framework

L — List Solutions

❖ **Try to be detailed with your solution.** The more specific you are, the more relatable the solution becomes.

❖ Do not focus on the technical details too much. **Focus on the product story** and how the solution fixes issues for the customer. Clearly articulate each use case that would be solved.

❖ **Come up with 2-3 solutions for the selected problem.** Always specify the most obvious solution and then talk about fancier options. Do not over engineer at this stage.

E — Evaluate Tradeoffs

❖ Talk about **why a certain solution is better than all others.** You can use similar lens as Cut Through Prioritization to substantiate your solution.

S — Summarize using KPIs

❖ **Come up with metrics to measure your solution.** There should be one north star metric to measure the overall objective of the product and a few metrics to measure the success of your features. You can also evaluate trade-offs by thinking of metrics that might go down due to this and whether the new feature is still worth it.

The CIRCLES Framework

Summarizing the CIRCLES Framework

1. **C**omprehend situation (What? Who? Why? And How?)
2. **I**dentify customer (different personas)
3. **R**eport customer needs (as a customer, I want to..., so that I can ...)
4. **C**ut through prioritization
5. **L**ist solutions
6. **E**valuate trade-offs
7. **S**ummarize recommendations

The Do's in Product Design

Laying down the structure

Giving the interviewer a broad overview of how you plan to approach finding solutions to the given problem can be beneficial. It helps the interviewer follow your answer better and helps them understand the rationale behind each step in the approach.

Naming personas based on attributes

Giving your persona a proper name, e.g. Rahul, is common as it makes referring to the persona easier. However, a better approach would be to name the persona by their defining attribute (which is key to your solution), e.g. Rahul, our Sports Enthusiast.

Map the customer journey

A simple way to approach brainstorming user needs is to first think about the series of actions the intended customer would perform in the context of the question. For example, if the question is based on e-commerce, write down the list of actions a user performs to buy a product on an e-commerce platform. The needs and problems at each step can then be listed down.

Reporting needs as stories

✓ **User stories are a concise way of defining a user's needs.**

Format: *As a (user persona), I want to (need), so that (rationale)*.

Example 1: As a rider on Uber, I want to see prices before booking, so that I can decide if Uber is the best option for me.

Example 2: As a shopper, I want to filter products by delivery date, so that I can buy only items that arrive before my event

Product Design: Do's & Don'ts

The Don’ts in Product Design

Jumping to Solutions/Features

You will be judged more on the approach to the solution and the logic behind the various decisions you take to find the solution. Hence, it is critical to walk the interviewers through your approach.

Spelling out the framework

Frameworks are meant to structure your approach. **Avoid mentioning the name of the framework** or being too obvious in your application of it. A good practice would be to add some variations to the framework.

Reporting only the unique solution

One always feels the pressure to come up with a unique solution with advanced application of AI/ML/AR/VR, leaving out the most obvious solutions. While it is important to come up with creative solutions, **it is also necessary to show that you can identify the obvious solutions** that satisfy important customer needs.

Being married to your first solution

It is recommended to come up with multiple features to provide alternative strategies.

What is GTM?

- GTM strategies are action plans for a company to launch a new product/functionality in areas where the product is currently not present. These may be regions where the company is already operating with a different set of products, or it might be an untapped region for the company.
- GTM strategies entail that the product reaches the intended audience at the right time through the proper channels and solves the right use cases to fulfill the company expectations for that product.
- The aim of a GTM case is to ensure success for a product launched by a company. It requires devising plans to support customer awareness, acquisition and retention both before and after the product is launched.
- Customers should be made clear of the value proposition of the product, and the right marketing should be done to capture the intended target audience.

Why is GTM important?

- As Product Managers, you are responsible for the success of your product. This includes creating the strategy for product targeting and communication, ensuring that the intended target audience is attracted to your product and will be retained as users, ensuring product success.

- GTM cases may especially be asked in PMM (Product Marketing Manager) roles, where the job responsibility has a large overlap with the kind of strategic thinking required for GTM cases. GTM may also be asked as follow-ups to Product Design questions, testing whether you have developed a comprehensive outlook over your product functionalities and whether it will be useful to the target audience.

Approaching GTM Questions

Define the WHAT

Define the Product

❖ To start a GTM case, you can ask clarifying questions regarding the product to be launched. More specifically, gather information about...

1

Product functionalities

- What does the product do and how does it deliver value to customers?
- Any features that need to be highlighted?
- Is there a unique value proposition that separates us from competitors?

2

Product History

- Has the product been launched in other regions?
- Where does the product sit in the company portfolio?

Define the Company

❖ Get an overview of the company, what it specializes in and what are its aims from the product launch

- Are there other products launched in the same market?
- Are there any possible product synergies from other products?

Define the Market Landscape

- Are there other competitors operating in the same space we are trying to enter?
- What kind of a market is this? Is this a D2C market or a B2B space?
- What is the market size and distribution of market share by competitors?

Approaching GTM Questions

Define the WHY

Why is the company launching this product? What are their expectations from the product launch?

Define the WHO

Understand the customers

Understanding the customer persona is **extremely important** for product cases.

Spend a good amount of time clarifying the customer pain points, the market gaps and exploring potential opportunities to create a value differentiation for customers. **Customer persona** can be understood through exploring these dimensions.

- Who is our target segment that we want to sell the product to?
  - What are the dimensions of their demographic data (age, gender, working/students etc.)
  - What are the major pain points in the product marketplace that we can possibly solve?
- What value proposition is delivered to customers through the usage of our product
- What are the customer sentiments associated with the company's image and products?

Approaching GTM Questions

Define the HOW

Deliver a Clear Value Proposition for our Customers

On understanding the customer pain points and the features of the product, clearly strategize what product features you would like to highlight when promoting the product. These features can be:

- **Novel offerings not present in the market that solve a unique customer pain point**
- **An improvement over similar functionalities delivered by existing products in the market**

A common way of delivering a value proposition follows the below format:

"For (target customer) who (statement of need or opportunity), our (product name) is a (product category) that (key benefit, compelling reason to buy). Unlike (primary competitive alternative), our product (key differentiator)."

Approaching GTM Questions

Define the HOW

Develop a Pricing Strategy

Design a pricing model to attract customers and help the company turn a profit. Common pricing strategies include:

1. Value-Based Pricing

This is commonly used when offering a novel functionality or solving a previously unaddressed pain point. Price should be set close to the perceived value of the functionality for customers. Value-Based Pricing can be calculated as:

Value of the utility that customers gained by using the product + %age premium

OR

Cost paid by customers to solve the problem through alternate means + %age premium

Approaching GTM Questions

Define the HOW

Develop a Pricing Strategy

2. Competitor Based Pricing

Used in price-sensitive markets, competitive pricing is used to gain market share while balancing profitability for the company. This is used when our product is similar to other competitive products in the markets in the eyes of a customer.

To calculate a competitor-based price:

1. ***Calculate the average of price of similar competitors in the market***

2. ***Price a %age above or below the average price, depending on whether you want to prioritize market share or margins***

Approaching GTM Questions

Define the HOW

Develop a Pricing Strategy

3. Cost-Plus Pricing

Usually used when you are trying to offload a product in the market that is not selling well.

Cost-plus pricing should generally be avoided in GTM cases.

To calculate a cost-plus price:

cost-plus price = Total cost of product development*(1+markup %age)

Approaching GTM Questions

Define the HOW

Develop a Pricing Strategy

4. Dynamic Pricing

Dynamic Pricing is primarily used for services offered online where the prices are declared on customer request.

It balances customer demand and competitor pricing to maximize profit for the company.

It uses technologies like machine learning to find the optimal price for a given situation.

Approaching GTM Questions

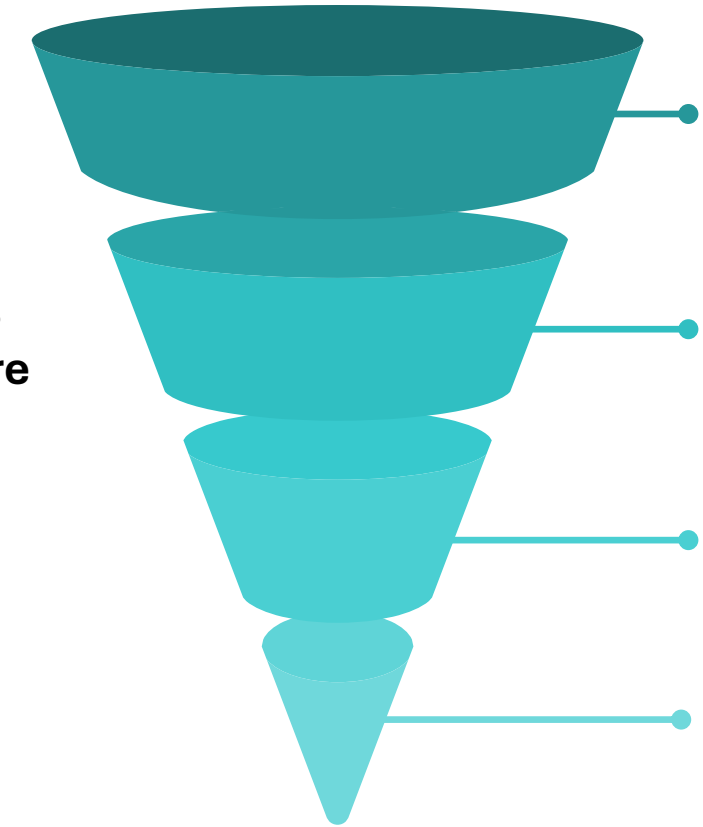
Define the HOW

Develop a Marketing Strategy

Discuss marketing tactics that expose the intended target audience to the appropriate product functionalities.

This involved choosing the marketing themes and choosing the appropriate channels **where the concentration of our target audience is high.**

Appropriate marketing strategies can be developed by choosing the right channels at each stage of the **marketing funnel.**



Awareness: Publicize the product to reach high customer awareness. **Eg:** paid ads, social media marketing etc

Interest: Develop strategies to build customer incentive in using the product and increasing first time users. **Eg:** e-mail campaigns

Decision: use insights to target likely buyers. **Eg:** sales enablement tools

Adoption: Develop after-sales service, onboarding tutorials etc to ensure customer satisfaction

Approaching GTM Questions

Define the HOW

Develop a Launch Strategy

Pre-Launch

- Develop a pre-launch strategy, choosing the appropriate pricing and marketing options to ensure customer exposure to the launched product.
- Your main goal here should be to increase customer awareness in the target audience at the lowest cost possible, by utilizing the right marketing channels

Post-Launch

- After customer adoption and product launch, focus on developing strategies to ensure customer retention and customer satisfaction.
- Develop the appropriate strategy to ensure customer redressals and reduce drop out rates.

Choose the right Metrics

Adopt the right set of metrics that are aligned with the product launch business goals. **For example:**

Customer satisfaction score	**Product Adoption Rates**	**Daily Active Users**	**Customer Acquisition Costs**

Evaluate Risk Factors

Address the risks that may be associated with the product launch and adoption. These risks may be:

External Risk Factors	Use PESTEL	**Product Related Risks**	Switching costs, Exit costs, etc.

"If you can't measure it, you can't improve it." - Peter Drucker

Why Product Metrics?

Metrics are the objective truth that transform opinions into facts, guiding every decision you make from feature prioritization to strategic pivots. From a recruiter's perspective, metrics test a candidate's skill at

- Connecting product changes to quantitative metrics
- Understanding the impact that metrics have on one another
- Prioritizing based on data.

❖ **Start here: You can watch this video (https://www.youtube.com/watch?v=wtDFgx41Rho) to get familiar with some common metrics.**

The Measurement Hierarchy: OKRs, KPIs, and Metrics

It's easy to get lost in the jargon. We can break these down, as if you were a taxi driver.

OKRs (Objectives & Key Results)

- **Objective:** Drive from Mumbai to Goa for a vacation.
- **Key Results:**
  - Arrive before 8 PM.
  - Keep fuel costs under ₹5,000.
  - Get a 5-star rating from your passengers.

KPIs (Key Performance Indicators):

These are the critical dials on your dashboard you must watch to ensure you reach your destination successfully.

Examples: Fuel Level, Average Speed

Metrics:

These are all the quantifiable, measurable data points available to you.
Examples: Current RPM, tire pressure, outside temperature, odometer reading. They provide context but aren't all *key* indicators for this specific trip's success.

As a PM, you set ambitious OKRs, track your most critical KPIs, and monitor a broader set of metrics to understand the full picture.

Finding Your North Star Metric (NSM)

A PM may track many metrics, but the North Star Metric is the single most important one. It best captures the **core value** your product delivers. An ideal North Star metric should:

- Lead to revenue
- Reflect customer value
- Measure progress.

Examples – Spotify ("Time Spent Listening"), Airbnb ("Nights Booked")

Must Reads

NSMs used by growth-stage companies – Real-world NSM examples across industries.

Shreyas Doshi on Basic Product Metrics – Fundamental metrics every PM should know.

How to Ace Any Product Metrics Interview Question

The chances of an interviewer asking a metrics-based question are low. However, knowing the process can help you to improve your product design, RCA skills. That being said, a structural way to tackle a product metric question could be:
When asked to define success metrics for a product (e.g., "What are the success metrics for Instagram Reels?")

1. Clarify the Goal & Context

First, ask clarifying questions.
"What is the primary objective of this product? Are we focused on growth, engagement, or revenue? Who are the target users?"

2. Map the User Journey

- Briefly outline the key user actions from discovery to becoming a power user.
- Use frameworks like AARRR here.

3. Analyze Metrics for Each Stage

List potential metrics for each step of the journey
(Acquisition, Activation, etc.).

4. Prioritize & Define the North Star

- Pick the 3-4 most critical KPIs.
- Commit to 1-2 North Star Metrics that best capture the core value of the product.
- Justify your choice.

5. Discuss Counter-Metrics: Trade-off Awareness

*E.g. "As a product manager at YouTube, you may want to increase the number of ads shown to drive **Revenue** (KPI), but you must monitor the Churn Rate (counter-metric) to ensure that it's not hurting the user experience and long-term **Retention**."*

As a PM, your goal is to move your thinking up the strategic ladder. The three core strategies represent an evolution in strategic thinking.

Level 1: Cost-Plus Pricing (The Foundation)	Level 2: Competitor-Based Pricing (The Market)	Level 3: Value-Based Pricing (The Goal)
• **The "How":** You calculate your costs (development, marketing, support) and add a markup. • **The PM's Reality:** This is your **floor**. • You must know your costs to ensure profitability. But pricing based only on your internal costs is lazy. It completely ignores the customer's perspective and the market dynamics. • **Never stop here.**	• **The "How":** You anchor your price to what your competitors are charging. • **The PM's Reality:** This is **table stakes.** • You must know your competitors' pricing to position your product effectively. However, this assumes your competitors have priced correctly and that your product is directly comparable. Relying on this is a reactive strategy that leads to "me-too" pricing and price wars.	• **The "How":** You price based on the **quantifiable economic value** your product delivers to a specific customer segment. • **The PM's Reality:** This is your **north star.** It forces you to answer : "How much better, faster, or cheaper does my product make my customer's life?" • It decouples your price from your costs and your competitors, anchoring it instead to the value you create. This is the hardest strategy to execute, but the most profitable and defensible. Hence, we see companies moving towards **Dynamic Pricing.**

Section	Key Questions	Why It Matters for a PM
Business Objective	Why are we pricing this? Are we aiming for market share, profit maximization, or user acquisition?	This sets the strategic direction for every subsequent decision.
Target Customer Segment	Who is the specific group of users we are creating the most value for? What are their characteristics?	You cannot determine value until you know who you're delivering it to. Be ruthlessly specific.
Quantifiable Value	What is the next best alternative for our customer? What is our unique differentiation? What is the **dollar value** of that differentiation?	This is the heart of value-based pricing. You must translate features into dollars (e.g., "$X saved per month," "Y% increase in revenue").
Willingness to Pay (WTP)	What is the customer's perceived value and ability to pay? How does this change based on their urgency or the opportunity cost of doing nothing?	This is the psychological ceiling. Your price must be less than or equal to the customer's WTP to drive a purchase.
Pricing Model & Tactic	How will we charge? (Subscription, Usage-Based, etc.) What tactics will we use? (Freemium, Tiered, etc.)	This is the implementation. The model should align with how your customer gets value over time.

Pricing Tactics

Model	When a PM Should Use This	Key PM Considerations & Risks
Freemium	When your product has low marginal costs and benefits from network effects or a "try-before-you-buy" experience.	**The Conversion Trap:** The free version must be good enough to attract users but "painful" enough to drive upgrades. You risk cannibalizing your paid product if the free offering is too generous.
Tiered (Feature Bundling)	When you have distinct customer segments with different needs and WTP (e.g., Individual, Team, Enterprise). This can also be used for decoy pricing. E.g. introduce a third option to make one of your other options look much more attractive.	**The "Right" Features:** The biggest challenge is correctly assigning features to tiers. Your goal is to make each tier a compelling upgrade, not a frustrating limitation. Too many choices can overwhelm users and lead to no decision at all.
Usage-Based	When a customer's consumption of your product directly correlates with the value they receive (e.g., cloud computing, SMS APIs).	**Predictability vs. Value:** Customers love the fairness but hate unpredictable bills. Consider hybrid models (a base subscription + overage charges) to offer the best of both worlds.

Pricing Tactics

Model	When a PM Should Use This	Key PM Considerations & Risks
Per-User	The standard SaaS model. It's simple to understand and forecast.	**The Collaboration Penalty:** This model can discourage adoption within a company (e.g., "I don't want to add my colleague because it will cost more"). Consider "active user" pricing or team-based bundles.
Market Skimming (Based on Product Lifecycle)	When launching a highly anticipated, innovative product with brand equity. The strategy is to set a high initial price to "skim" the maximum revenue from early adopters, and then gradually lower the price over time to attract more price-sensitive customer segments.	**Brand & Differentiation are Crucial:** This only works if your product is perceived as highly desirable and unique. Without strong brand loyalty, customers will simply wait for a cheaper alternative. **The Alienation Risk:** Early adopters who paid a premium can feel cheated if the price falls too quickly, potentially damaging brand loyalty for future launches.

Step 1: Frame the Problem & Validate the Data (Ask Clarifying Questions)

Before RCA, Know the Product: Understand **what the product does**

1. Reiterate & Define the Metric

- Clarify **what the metric means** and **how it's calculated**.
- Check if **calculation or reporting methods** have changed recently.
- Verify **data integrity** — confirm BI tools and data pipelines are working correctly.

2. Isolate the "Blast Radius"

- **Time Period**: Identify if it's a sudden drop or gradual decline, check for seasonality, and pinpoint when it started.
- **Geography**: Is the impact global or specific to certain regions?
- **User Segment**: Determine if all users are affected or specific personas — e.g., **new vs. existing users**, or by **platform** (Android vs. iOS).

Step 2: Investigate Internal Factors

1. Trace the User Journey

- Map the whole user flow in the app.
- Identify the **exact funnel stage** where the drop/change occurs (e.g., sign-up, checkout, key feature use).

2. Product & Technology Changes

- Check for recent **UI/UX releases** that could cause confusion or friction.
- Investigate **technical issues** — bugs, glitches, latency, crashes.

3. Business & Offering Changes

- **Demand-Side**: Review changes to pricing, subscription tiers, and free trials.
- **Supply-Side**: Check for salary shifts, service quality, or **cannibalizing products** pulling users away.

4. Qualitative User Feedback

- Analyze app store reviews, NPS scores, and support tickets.
- Look for patterns or complaints that align with the timing of the change.

Step 3: Investigate External Factors

(If internal causes are ruled out, widen the lens)

1. Competitive Landscape

- Identify **new entrants** or **competitive moves**.
- Check for **price cuts**, **promotions**, or **better offers** from competitors.
- Look for cheaper substitutes entering the market.

2. Market & Environment

- Review **regulatory or legal changes** affecting the product.
- Consider **environmental events** (pandemics, disasters) that could shift user behavior.
- Spot **shifts in consumer psychology** or buying patterns.

3. Public Perception

- Scan **PR coverage** and **social media sentiment**.
- Investigate any **trust or brand reputation issues**.

Tips for Aspiring PMs (Interview Execution)

1. Think Out Loud, Skip the Jargon

Walk through your logic step-by-step.

Example: say *"First, I'd clarify the metric..."* instead of naming the framework.

2. State Assumptions Clearly

Acknowledge data gaps.

Example: *"I'll assume this affects all platforms equally, but I'd verify with the data analyst."*

3. Be a Collaborative Partner

Frame your investigation as **questions + hypotheses**.

Let the interviewer guide where needed.

4. Link to Product Lifecycle

Adapt your approach based on stage (Launch, Growth, Maturity).

Example: *"Since we're in Maturity, I'd focus on competitor actions and retention."*

5. Prepare for the "What's Next?"

After RCA, propose a solution.

Outline **how to test it** and **how to measure success**.

Section 5

Practice Cases



Section 5.1

Guesstimates

Instagram Daily Uploads in India

Guesstimates

Estimate the number of Instagram uploads occurring in India.

I'd like to ask some clarifying questions. Do you want me to estimate the number of uploads per day or per year? And should these uploads include posts, stories or both?

Consider only posts and estimate uploads per day.

Should I consider both personal and business accounts?

Personal accounts only for now.

Can I assume this estimate is for a typical day, not a festival or a holiday?

You can assume a typical day and start working on your approach.

Thanks, I will approach this problem by breaking it down as:

of Instagram uploads per day = Active Personal Instagram Users × Average Posts per User per Day

First, I shall estimate the number of active personal users on Instagram in India. Then, I'll segment these users into groups based on their posting frequency to get the average posts. Can I proceed with this approach?

Yes, please go ahead.

First, I shall estimate the number of personal users as

[# of active users on Instagram] = [population of India] * [% population with smartphone and internet access] * [% of people on Instagram] * [% of monthly active users]

The Population of India is approximately 1.4 billion. Due to an all-time high penetration of mobile services, assuming 70% smartphone and internet penetration, we have around :

1.4B × 80% = 980M smartphone users with internet
Can I assume the Instagram penetration percentage is across smartphone users, or should I segment it according to app usage?

What do you think?

Instagram is one of the most popular apps on the Play Store with over a billion downloads, so I assume that 50% of these users have Instagram.

Sure, good point. For simplicity, assume 50% penetration among smartphone users and feel free to make reasonable assumptions.

Thanks, assuming 50% of these people use Instagram:
980M × 50% = 490M accounts

I'll assume that 80% of these accounts are active personal accounts:
490M × 80% = 400M
So, we have around 400 M active personal users.

Sounds reasonable.

Now, I'll estimate the average number of uploads per person per day. We can divide the monthly active users into three categories, with rough estimates for the percentage of users in each category from my own experience on the app as follows:

1. Light users: 60% of users, posting one post every month = 0.6 * 400 = 240 M users
2. Moderate users: 35% of users, posting one post every week = 0.35 * 400 = 140M users
3. Heavy users: 5% of users, posting one post every 2 days = 0.5 *400 = 20M users

Therefore, the number of posts per day = 240M × (1/30) + 140M × (1/10) + 20M × (1/2)
= 8M + 14M+ 10M
= 32 M

That sounds good. Anything else? I think Instagram has around 100 million daily posts.

Ok, let's see if my estimate makes sense. If Instagram globally has around 100 million posts per day, India contributing around 32 million posts (32%) seems plausible given our high population, and Instagram is banned in China.

Great, thank you. We can close this interview.

YouTube Daily Revenue

Guesstimates

Estimate the revenue that YouTube makes in a day.

Sure! I'll focus on estimating YouTube's daily ad revenue. YouTube has multiple revenue streams—Ads, YouTube Premium, Music, TV, and even SuperChats for livestreams—but since ads contribute the majority, I'll zoom into YouTube Ads as the core. Would you like me to estimate this globally or for a specific market?

Let's estimate YouTube's daily ad revenue specifically in the United States.

Perfect. Here's the approach I'll take:
YouTube Ad Revenue (Daily) = Ad Views × Revenue per Ad View, where:
Ad Views = Total Video Views × % Videos with Ads
Revenue per Ad View = CPM / 1000

For estimating the total number of ad views, there are two ways we can go about it:
1. Estimating from the demand side as to how many YouTube videos people watch daily.
2. Estimating the supply side by looking at YouTube's video views service capacity.

Since, in this case capacity will not be a problem as YouTube can easily scale up its infra to serve the demand, we should go via demand estimation. Does this sound fair?

Alright, that does sound like a plan. Let's go ahead.

Let's start with a US population of ~330M (2025 est.). I'll segment by age and assume an average daily YouTube watch behavior based on age group.

Age Group	% of Population	Viewership rate	Daily Users (Million)	Avg. Videos per day	Daily Views (Million)
0-5	10%	0%	0	0	0
6-12	10%	50%	15	5	75
13-24	20%	90%	54	20	1080
24-44	20%	85%	51	10	510
45-64	20%	60%	36	5	180
65+	20%	40%	24	3	72

Estimated total views – 2B

Not all views are monetized:
1. Some users have YouTube Premium (assume ~20% in US)
2. Some creators don't monetize
3. Some content has skippable or no ads

So, I'll assume 50% of views are monetized with ads.

Ad Views per Day ≈ 2B × 50% = 1B ad views

That sounds good, lets move forward.

YouTube charges advertisers via CPM (Cost per 1000 Impressions).

In the US, CPM varies by content and targeting. For this case, lets assume a CPM of $20/1000 impressions.

Revenue per ad view = $0.02/ad view.
Hence, daily revenue = 0.02 * 1B = $20M.

For the year, we have $20M * 365 = $7.3 B

Anything else you would like to add?

Google's annual ad revenue = $200B. Assuming YT gives 10-15% of that, we have global YouTube revenues to be around $20-30B. Hence, assuming USA gives $7.3B is a fair estimate.

Okay, we can close the interview.

Zoom Daily Server Usage

Guesstimates

Estimate Zoom's daily server usage

I'd like to ask some clarifying questions. My understanding is that Zoom's server usage would comprise the following:

1. Storing information about its users and their call logs
2. Streaming the data (audio, video and text) to and from participants in a zoom call
3. Miscellaneous processing tasks (backend)

Since most usage spikes are likely tied to live calls, especially video streams, I'll focus on streaming for now, the most significant driver of infra load. Shall I proceed with this assumption?

Yes, that sounds reasonable.

Do you want me to account for global usage or be constrained to a particular geography, such as India?

Consider the usage for India.

There are three broad categories of users for video conferencing.

1. Personal
2. Student
3. Business

Out of them, business accounts should be the most prominent users of Zoom due to higher average usage time, as many of the jobs after COVID have an important remote work component to them, leading to a higher proportion of their time in office being spent on video conferences.

You can go ahead with business accounts for now.

Among business accounts, should I account for event streaming, professional calls between employees or both? If we were to include webinars or events, the calculation would change, as the data pattern is one-to-many, which is a different kind of server load.

We can proceed with professional calls for now.

I will break down this problem as follows:

Zoom's daily server usage (on a weekday) = # of people using VC in India * Zoom's Market Share * Average time spent on VC (in mins) * Data requirements per minute

To estimate the number of people using video conferencing, I will take the share of the working population in India (predominantly aged between 20-60 years), which is 60%. I assume that VC is mostly used by white collared employees in India, whose share in the working population is around 10%.

To estimate the proportion of the white-collar workforce using VC, I will break down the jobs into client-facing and traditional roles:

1. About 40% are in tech, consulting, or client-facing roles where VC is essential (90% adoption).
2. The other 60% are in support of traditional roles with less need (let's say 40% adoption).

This gives us a blended average of (0.4*0.9) + (0.6*0.4) = 0.36 + 0.24 = 60%. The number of people using VC in India comes out as:

India's population * Share of working population * Share of white collared employees * Share of employees requiring VC services

= 140 Cr * 60% * 10% * 60%

= **5 Cr (approx.)**

Sounds reasonable. Please go ahead.

In the video conferencing market, there are three major players that I can think of: Zoom, Google Meet and Microsoft Teams. Exact market share might fluctuate by region, but I'll take 50% as a working estimate for India. To estimate the average time spent on VC services, I will break down the users as follows:

1. Heavy users (VC usage of more than 4 hours): 20% - This set will comprise executives, managers
2. Medium users (VC usage of around 2 hours): 40% - This set will comprise engineering managers, marketing and brand teams
3. Light users (VC usage of around 30 mins): 40% - This set will comprise core engineering teams, finance teams

The average usage comes out as: 20%*4hrs + 40%*2hrs + 40%*0.5hrs = 1.8hrs

Further, I will assume that the data requirement of a VC is 1Mbps, which makes it 3600 Mb per hour or 450 MB. Hence, Zoom's daily server usage comes out as 5 Cr * 1.8 * 450 MB = ~ 40,000 TB/day

Note: The data requirement can also be a weighted average based on video quality and usage. However, for simplicity it has been taken as 1 Mbps

Zoom Daily Server Usage

Before we end, I have one more question for you. If Zoom observes rising VC usage in Tier 2/3 cities or heavy mobile participation, what should it do?

In the event of higher participation from Tier 2/3 cities or mobiles, there is a high chance that the users coming in from this background will not have access to very high-quality, reliable internet. In this case, Zoom should consider launching a lite version wherein the data requirements are lowered from the current version, which requires a bandwidth of more than 1Mbps.

Okay, we can end here, thank you.

Number of Daily WhatsApp Chats

Guesstimates

Estimate the number of WhatsApp chats occurring in a day.

Sure. I'd like to understand the context a little better. Are we trying to gauge this for a particular geography?

Yes, let's focus on messages in India.

Okay. I'd further like to know which user types should we focus on: WhatsApp for business users or casual users?

Good point. Let's discuss about the casual users.

Thanks. To approach this problem, I'd like to first estimate the number of WhatsApp users in India. Then I'd like to segment this population and find out the average number of chats per segment to arrive at our larger question. Does this approach sound good?

Yes! Please go ahead.

Firstly, the number of WhatsApp users in India can be broken down as,

of WA users = Population of India * internet penetration % * % having smartphones * % having WhatsApp installed.

Sounds right, please go ahead.

Great! So I'll make a quick estimate of our multipliers. India's population is roughly 1.4 Billion people. About 50% of India has internet penetration with regular access. The majority of people use internet on smartphones these days, and hence, the percentage of people with internet access who have smartphones would be close to 70%.

Finally, as WhatsApp is one of the core apps on smartphones, I will assume that 80% of people have it and use it regularly.

This leads to number of WhatsApp users in India as:

1.4 Billion*0.5*0.7*0.8, or ~400 Million.

Perfect! That's a very good estimate. How will you go about the number of messages in a day?

Please spare me a moment to segment our users.
I'd like to firstly classify our users on their age into 15-29, 30-49 and 50+. The usage and affinity varies across these age groups. Then for each of the age group, I would break the users into heavy users, medium users and light users. Does this approach sound good?

Yes, I think this is fair.

Okay. I would distribute 60% of WhatsApp users in the first age category of 15-29, 30% in 30-49 and 10% in 50+.

Further, for the category 15-29, 60% would be heavy users with ~80 chats sent a day, 20% medium users with ~ 40 chats a day and the final 20% of light users with ~10 chats a day. This would lead to an average chats per day for the 15-29 age group as 60%*80 + 20%*40 + 20%*10, or ~60 chats per day.

I would similarly compute average number of chats for the other age groups. Should I continue with this approach and arrive at our final answer?
(Ensure to check-in with the interviewer in middle of large computations.)

The approach looks good to me. I have some data for you, the average number of chats for the middle and old age group are 25 chats and 5 chats a day.

Thank you! With all the data bits now ready, I can build-up to our final solution.

of chats per day in WA = Total WA Users* [(% of user 15-29 * avg. chats per day of 15-29) + (% of user 30-49 * avg. chats per day of 30-49) + (% of users older than 50 * avg. chats per day of users older than 50)]

This computes as:
of chats per day in WA = 400 Million * [(60% * 60)+(30% * 25)+(10% * 5)]
of chats per day in WA = 17.6 Billion or ~18 Billion

Hence, we have approximated the number of WhatsApp chats per day in India is close to 18 Billion.

Thank you! We can close this interview.

Fantasy Football Platform Users

Guesstimates

Estimate the users for a fantasy football platform

Sure, I have a couple of preliminary questions. What is the scope of the product that we are looking to design, in terms of geographical span, and the target users?

The platform is envisioned to be launched in a European country with a population of 200 million. What do you think can be the target users?

I think that depends on the current position of the platform in its lifecycle. However, assuming the platform has already been launched, it makes more sense that the target market would be the urban consumers which have a higher propensity to pay, than the rural customers.

Seems logical. You can proceed with this assumption.

Before I begin with the estimation, I would like to highlight some factors for segmentation that I intend to use. The first level of segmentation could be based on age and gender, as they have differing receptiveness to sports.

Then, the second level of segmentation could be done on the percentage of people who watch sports, which itself could be segmented into the percentage of sports viewers that watch football. Here, I assume that most of the platform users would be football viewers.

Further, since fantasy sports require financial transactions, it is necessary for the users to be comfortable with using digital payments. This again can form the third level of segmentation, i.e., the percentage of users who use digital payments. Finally, the last segmentation can be done based on the propensity to play fantasy sports.

Do let me know if the approach seems fair, and if there is any other factor that you would want me to consider? Also, do we have some data on the slicing thresholds for some of the industry-specific segmentation factors that I outlined above.

This seems fair. You can assume the slicing thresholds at different segmentation levels as you deem fit.

Understood. So out of the 400 M people, we can assume the urban population to form 70% chunk of the total. Further I would assume an equal split of Male and Female genders.

For incorporating age as a factor, I would bucket the age groups into 3: 18-25, 26-40 and 40+. I have purposefully kept under 18 users out of my analysis since it is not legal for them to engage in fantasy sports. The differing split points for the factors can be assumed as thus:

Age	18-25 (25%)		26-40 (25%)		40+ (25%)	
Gender	Male (50%)	Female (50%)	Male (50%)	Female (50%)	Male (50%)	Female (50%)
% who watch sports	60%	40%	50%	30%	40%	20%
% who watch football	70%	70%	70%	70%	70%	70%
% who use digital payment	80%	80%	70%	70%	60%	60%
% who play fantasy sports	20%	15%	15%	10%	5%	0%

Do the split-off percentages seem fair? I'll proceed with the actual calculation hereafter.

The percentages seem fair. Can you explain the trends in the segmentation basis for those who use digital payments and engagement in fantasy sports?

For the digital payments adoption factor, I have assumed decreasing trends with rising age as I understand that the digital literacy and comfort level with using digital solutions for activities such as financial transactions decreases. Furthermore, the trends in the engagement with fantasy sports is assumed to be decreasing with rising, again because fantasy sports in digital format has been a recent industry and may not find traction easily with older people.
Further, even within a certain age group, the adoption has been assumed to be lesser for females compared to males, which in fact mirrors the trends in the sports viewership factor.

Fair reasoning, proceed with the calculations.

Fantasy Football Platform Users

Guesstimates

Sure, assuming a 30 - 70 split for rural and urban population, the total count of urban population comes out to be 140 million. Applying the above thresholding factors, the final count of users comes out to be:

Age	18-25: 35 Mn		26-40: 35 Mn		40+: 35 Mn	
Gender	Male: 17.5 Mn	Female: 17.5 Mn	Male: 17.5 Mn	Female: 17.5 Mn	Male: 17.5 Mn	Female: 17.5 Mn
% who watch sports	10.5 Mn	7 Mn	8.8 Mn	5.3 Mn	7 Mn	3.5 Mn
% who watch football	7.4 Mn	4.9 Mn	6.1 Mn	3.7 Mn	4.9 Mn	2.5 Mn
% who use digital payment	5.9 Mn	3.9 Mn	4.3 Mn	2.6 Mn	2.9 Mn	1.5 Mn
% who play fantasy sports	1.2 Mn	0.6 Mn	0.6 Mn	0.3 Mn	0.1 Mn	0 Mn

Total number of users comes out to around 2.8 million.

The final figure seems plausible. We can close the discussion.

Food Delivery App Users

Guesstimates

Estimate the number of users for a food delivery app in a South American country.

Sure, I have a couple of clarifying questions before I start the estimation. What is the total population of the country, and should I assume this is a product in its early stage or already launched?

The country has a population of 100 million. You can assume the platform is already launched and is looking to scale. Go ahead with the assumptions you need.

Understood. I'll assume the app has achieved some market penetration and is targeting urban consumers first, as they have better access to infrastructure, smartphones, and restaurants. I'll also assume a significant proportion of users would be working professionals or students in urban areas, which drives demand for food delivery services.

Seems logical. You can proceed with this assumption.

Thank you. Before I begin with the calculation, I'd like to outline my segmentation framework:

1. **Urban vs Rural:** Given infrastructure and internet access challenges in rural areas, food delivery adoption will likely be concentrated in urban areas
2. **Age Segmentation:** Users aged 18–40 are the most likely to use food delivery services due to lifestyle and disposable income.
3. **Income & Digital Access:** I'll consider smartphone ownership, internet access, and digital payments as necessary enablers.
4. **Food Delivery Propensity:** Finally, I'll estimate what portion of this digitally-enabled group is likely to use food delivery services.

Are these factors exhaustive?

That approach works. Assume the slicing percentages as you see fit.

Great. I'll proceed with these assumptions:

1. **Urban population:** Assume 70% of 100M = 70 million
2. **Age Distribution** in urban areas:
    i. 0–17: 25%
    ii. 18–40: 50%
    iii. 41+: 25%

So, 18–40 urban population = 50% of 70M = **35 million**

3. **Income and Digital Access** (smartphone + internet + digital payments): Among 18–40 urban population: Assume 70% have sufficient income and access → 0.7 × 35M = 24.5 million
4. **Food Delivery Propensity:** Of the digitally active population, estimate 40% actively use food delivery platforms (based on competitive landscape and market maturity) → 0.4 × 24.5M = 9.8 million

Final Estimate: **~9.8 million active users**

That figure seems reasonable. Can you explain your rationale for the 40% adoption assumption?

Certainly. I based this on a few considerations:

1. **Market maturity:** Since the app is already launched and we're looking at a South American country with urban penetration, early adopters would already be on the platform.
2. **Competition:** Multiple food delivery platforms may exist, so active usage may be distributed across providers, reducing the app-specific share.
3. **User behavior:** While many users have access to the service, not all order frequently—some prefer cooking or dining out.

I chose 40% as a mid-point assumption considering these factors.

Makes sense. We can close the discussion.

Section 5.2

New Product Design

Design a mobile application for an office space sharing business for Airbnb.

PM Tip: The first step for the candidate is to clarify any ambiguities in the problem statement.

Before moving further, I would like to understand a bit more about the problem at hand. What is the exact business model of this office sharing app? For whom would we be designing the app, the owner or the renter, or both? And lastly, what is the objective of this app.

Good questions. The whole business idea is that people with extra office space can rent out to start-ups or freelancers, Airbnb will take the empty space and remodel it, but the owner will take care of the operations. We need to design the app for both owner and the renter. As of now, the objective is to get as many customers as possible on board.

That's helpful. I am assuming that the person who is renting the space is the owner, just to reduce the complexity of the design. Also, I believe we are not focusing on monetization for the business as of now, later for sure, please correct me if I am wrong. However, I will address the payment issues between the owner and the person renting it. Now that I understand the business, I will look for user segments, then find their needs, build user stories, search for solutions and create an MVP to fit these solutions.

PM Tip: After understanding the problem statement, the Interviewee clears all his/her assumptions beforehand. Also, he/she gives an overall flow of how he/she is attacking the problem. This helps to keep the interviewer intact with the broad view and aligned with the Interviewee's thoughts. It's always good to first give a high-level view and then go on to the details.

There are 3 users for this application:

1. Owner
2. The person renting it
3. Airbnb

Which one should I focus on?

You may decide that.

For any multi-sided platform, it is important to solve the chicken-egg problem. Taking the example of Uber who first focused on on-boarding the drivers to create supply, I will also focus on owners. Since we are focusing on the owner, we must first define their persona. Mostly, the owner will be a person or a company who have extra space in their building which can be monetized. They will be comfortable in sharing their space to a stranger. If it is a company, this will be an additional source of money.

PM Tip: The Interviewee chose a particular user to focus on, but he/she also gave a proper rationale. He/she backed it up with a business model similar to Uber's to choose the user correctly. It would have been better if the Interviewee had gone deeper about the user (the owner), like their demographics, their size, their location, etc.

Few of the needs of the owners would be:

1. Finding reliable people to rent it to, also the ability to approve or decline requests
2. Furnish and maintain the space with office supplies
3. Estimated rent and information on rental periods and owner's rules
4. Payment and customer support
5. Security of the place

The user needs can be analyzed by navigating through the steps the owner would need to go through the whole process.

So, the needs of the user should be identified at each stage of the process



PM Tip: It's always good to break down the whole process into intermediary stages and figure out the customer in these stages, this helps get the exhaustive list of user needs and avoids missing any important part.

Fair enough. But how will you prioritize among the different needs of the user?

Since we are focusing on MVP, I will look at the most needed features which are 2, 3, and 4. So my user stories as an owner are:

- Keep a check on the office supplies and other furniture, replenish them whenever necessary.
- I want to know the rent I will get so that I can decide whether it is meaningful for me or not.
- A way to interact with the system to confirm/cancel bookings and settlements

ProTip: Prioritizing the requirements is one of the key roles of a product manager. Since it's a boundary-spanning role, product managers are bombarded with multiple tasks of the product, from the business team, from the engineering team, from the marketing team, etc.. Still, the product managers must make hard calls to prioritize the requirements. Some use the effort-impact matrix to do this; however, multiple other frameworks exist.

For an MVP, I will address all the stories individually first.
Solution to story 1:

- Display the list of all the supplies and furniture available in the location.
- Provide the capability to the place orders through the app itself, Airbnb can provide support.
- An ability to display to the renters what all furniture and equipment is present in the office.

Solution to story 2:

- Option for the owner to send the rent to the person renting it
- Rent estimation based on
    - Space occupied
    - No. of days
    - Special requests/Customization
    - Number of people/employees

Solution to Story 3:

- A tab to manage request for renting.
- Option to confirm/cancel or chat with an interested person.
- Have a payment gateway to pay full rent in advance.
- Provide a weekly settlement into the owner's account.
- Cancellations/Refunds will be handled by Airbnb in between settlements

Would you go ahead with all of these?

No, I will evaluate the solutions in each story based on customer impact and its ease of implementation. After the evaluation, I will select 1,2 in S1, 2 in S2, and 1,2 in S3 for the MVP. Do you want me to go into detail about these?

No, this is good. Thanks for your time and all the best.

PM Tip: The Interviewee could have explained here a bit more about how they selected the features for the MVP; however, since the interviewer confirmed, it's fine. It's always better to back up your arguments with data and logic; interviewers are looking for the critical thinking of the Interviewee's ability.

Increasing Voter Turnout

Perugia's voting turnouts have stagnated at 55% in the last few years. The Government of Perugia has hired you to design an application to improve voter turnout. What are some of the features you would recommend for such an application?

Okay. Can I start with a few clarifying questions about the problem?

Sure, go ahead.

What has been attributed to as the reason behind the stagnation of the turnouts? Are the people skipping the vote from a specific demographic? What are the current methods of voting in Perugia? What is the specific aim of the government with the application, and is it supposed to serve as a method of voting as well?

The voters who have skipped have cited reasons of lack of knowledge of the candidates, lack of interest in the voting process, confusion about how the registration and voting process works, the perception that their vote doesn't count for much and that they were busy on the day of voting. The government runs awareness campaigns, but they have reached and affect only a few of the voters. The people skipping are primarily the new voters from the ages of 18-30 years of age. The current voting method is just a ballot paper system, and the government doesn't aim to change that. With the application, the government wishes to attract more voters to the process and engage them during the election season to get them interested and vote more. Does that help?

Yes, it does. From what I could gather, the government would like the application to make the voters more engaged with the voting process and show up to vote. As the most affected user base is the new voters, I'll keep the application focused on solving their needs. Is that okay?

Yes, that is the government's target segment as well.

Okay. From what you mentioned, it seems the new voters suffer from both the absence of information on multiple levels of the voting process and the lack of interest in the voting process This also shows that the currently accessible mediums are unable to solve this problem, be it conventional media or otherwise. Looking at the voting process itself, it could include information on the performance of the current/previous candidate, the profiles and activities of all contesting candidates, their manifestos, the current problems faced by the majority of the citizens, how to enter the voting process and where to vote. If these look fine, can I take a minute to think over the solution?

Sure, go ahead.

Alright. Keeping the needs in mind, I have come up with the following features in the app:
Addressing lack of information –

- Candidate Scorecard: This would be curated using the information from their previous work, their previous election performance, their headlines in the news, criminal and other charges against them. Since we are working with the government, some of this information can be directly sourced from the candidates, and the rest can be gathered from external sources.
- App-based registration: By digitizing the voter registration through the app, the voter would be able to register from anywhere using the app, reducing the time and money taken to set up and use an extensive offline registration system.
- News section: Capture the headlines to inform the user of the current issues in the news, which would be localized to their voting region.
- Timeline of the voting process: What is my booth and navigation to the voting booth.

Addressing lack of interest:

- Gamification of the voters: Running an app-based daily/weekly quiz to find the most informed voters. There may be perks associated with that, like preferential voting etc., depending on the rules and laws of the land.
- Notification center: Using notifications specifically to keep the voter engaged about the happenings of the process and the critical dates.
- Connect with the candidates: The candidates can leverage the app to communicate directly with their voters specific to the region. On the other side, the voters could ask questions and upvote the questions that they find relevant. It would be in the candidates' interest to answer more of the questions asked since it would only increase their reach, and the voters would be more engaged.
- Non-transferrable badges for voters: Perks like badges, QR to your exclusive photo at the booth etc., could be incorporated into the app to encourage voting.

Do these look fine, or should I explore more features?

No, these should do. Could you elaborate on one of the user workflows that you see the user having on the application?

Sure. Talking about a scenario a week before the election, when the app has the most information. The user would open the app and see their profile details with the gamified score (could be percentile of how much more informed they are than the others on the app), the next quiz they should attempt, and the latest news on their voting area. The user takes the quiz and moves to the candidate profile tab. The user can select the current/previous elected member who is re-contesting. The app gives the user a candidate scorecard, their recent work in the office, the candidate's latest headlines, and the recent legal charges against them (if any). The user selects the work tab, which then elaborates on their recent participation in the parliamentary debate, the bills they presented etc. The user reads the information and closes the app. That would be a flow where the candidate engages with the app and consumes some information. Does that work?

Yes, that is good enough. Can you quickly provide some metrics that you would track for this application?

Surely, we would want to a few things in this application. Engagement metrics like DAU/WAU, Session Duration, Notification Open Rate etc. would be important. We would also like to track the awareness metrics based on candidate profile views per user, number of articles read etc. Finally, we would want to track the number of users registered as voters through the app.

Thank you, that will be all.

Disney Theme Park Experience

New Product Design

You are a PM at Disney. You have been asked to design an application to improve the customer experience at Disney theme parks. What would you do?

I would like to ask a few questions before diving into the solution. I want to verify whether my understanding of the offerings of the theme park is correct. The parks generally offer amusement rides and shows for entertainment. Theme parks typically have eateries for customers to eat. Some theme parks have an in-premise hotel for customers to stay.

You can assume that the theme park you are developing for has all the offerings you mentioned except hotels.

Got it. I will start with identifying the primary user of the application and then listing down the actions they take while visiting the theme park.
The most common visitors of theme parks would be families where the adults are accompanying the children. The decision taker would be one of the parents, so they would potentially be the primary user of the application.

The users perform the following activities:
1. Book the entry ticket to the park
2. Decide on the attraction to engage with
3. Navigate to the selected attraction
4. Make the payment if the attraction has an associated fee
5. Visit eateries for meals or refreshments

Based on the activities I listed down, the users will have the following needs:
1. Ticket Booking: I would want to explore the different tiers of tickets available and their offerings to find the tier which best suits my budget and interests.
2. Picking the amusement ride/show: During this phase, the customers would have the following needs:
    i. understand the experience of the attraction I am planning to visit to ensure that the attraction would be interesting to me.
    ii. understand the restrictions (e.g., age, height) of the ride/show of my choice to validate that I am eligible to visit it.
3. I would be visiting the park for a fixed amount of time and want to make the best use of my time.
4. I want to visit an attraction when it is not too crowded so that I don't have to wait in line.
5. Navigate to selected attractions: I need help figuring out the way to the ride or the show I want to visit.
6. Eateries: I want to pick an eatery that would provide me with the food I like.

The pain point assessment seems fair. Assume that you had the budget to solve only one of these problems. Which one would you select, and what feature would you build for it?

I would prioritize the solutions based on the following parameters:
1. Effectiveness of Existing Solutions
2. Value customer would derive from the problem being solved

Pain Point	Existing Solutions	Solution Effectiveness	Value for customer
Ticket Booking	Third party booking platforms	High	Medium
Ride Picking	List of attractions on the park website or brochure	Low – solution is tedious	High
Navigation	Google maps, physical maps, directions in park	Medium	Medium
Eateries	Park Website, Brochures	Medium	Low

Based on the selected parameters, I would choose 'Ride Picking' as the pain point to be solved.

Solution:
The goal of the features to be built would be to help the customers optimize the time spent in the park. I propose the following features for the Ride Picking functionality:
1. Attraction Catalogue: The feature would help customers explore the rides and shows available in the park by filtering them based on their requirements.
2. The rides can be filtered based on the following parameters:
    i. Age, Weight, Height – To ensure eligibility
    ii. Attraction Type – Audio-Visual
    iii. Shows, Rollercoasters, Games, etc.

3. The feature would give the following information about the attraction:
    i. Rules & Regulations
    ii. Video of the ride
    iii. Timings with available slots (If it's a scheduled show)
    iv. Expected waiting time
    v. Customer Reviews
4. The feature would have an option to pre-book scheduled shows/rides.
    i. Itinerary Optimizer: The feature would assist the customers in building the itinerary for a visit.
    ii. The optimizer would be a chatbot that would take information such as Duration and Timings of the visit, Number of visitors, and Preferences of each visitor (type of rides/show they want to see)
    iii. The feature's output would be an itinerary that allows the customers to visit the most attractions, which are to their liking, in the duration they are present in the park. The feature would enable the customer to make changes to the itinerary manually.

Sounds good. We can close the case now

India's start-up ecosystem has grown immensely in the past decade. As a product manager at LinkedIn, suggest features to leverage the growth in the start-up ecosystem.

LinkedIn's primary mission is to connect professionals to make them more productive and successful. Will it be fair to assume that LinkedIn has successfully attracted the major stakeholders of start-ups, i.e., founders and investors, into registering on their platform (considering their massive user base of 1B+ users)?

Yeah, sure. Please go ahead.

With customer acquisition already done, the additional features of the platform should be focused on increasing customer engagement. The goal should be achieved by banking on LinkedIn's core strengths - its large user base, presence of experts from all domains, access to users' professional data such as past employers and education, and services like LinkedIn learning.
Moving ahead, I will focus on the needs of key stakeholders in the start-up ecosystem and list down their needs.

1. Founders: A person or group of people who come up with the fundamental idea that forms the core business of the start-up.
2. Investors: A high net-worth individual/group that provides capital to start-ups in exchange for ownership equity in the company.

Those are the two key stakeholders. Before moving to the needs, could you tell me which stakeholder you would prioritize?

As the current business goal is increasing customer engagement, I would prioritize the two stakeholders based on the potential for increased customer engagement. Investors are generally backed by a professional team or institution and prefer performing primary research for their needs. Thus, they have limited requirements for additional services and limited engagement potential. On the other hand, many founders have limited resources and are actively looking for avenues to create buzz and awareness around their company. This offers LinkedIn multiple potential avenues to engage with them. Will it be fair to focus on Startup founders for now?

Looks good to me.

A typical journey of a startup founder during the early stage can be broken down as follows:

1. Ideation and Validation: The founder develops a business idea and conceptualizes a product or service offering. The needs during the phase are –
    - Market Research
    - Guidance for conceptualization
    - Idea Validation
2. Building and Training: The founder moves from a concept to a company, putting their research into practice. The needs during the phase are
    - Prototyping/Simulating an idea
    - Building the core team
    - Training the team
    - Building an MVP product
3. Scaling the company: With the MVP in place, a founder needs to increase awareness and accessibility of the product, and access to more cash flows to increase the scale of the company. The needs during the phase are:
    - Marketing the product
    - Accumulate product feedback
    - Boost metrics based on consumer engagement
    - Open new opportunities to raise funds

The features are prioritized based on the compatibility with LinkedIn's mission and strengths, value to the customer and lack of existing solutions fulfilling the needs. Based on the above criteria, market research, mentorship, recruitment and training, and accumulation of product feedback can be prioritized.
I would like to suggest the following features to satisfy the prioritized needs:

1. Deploy a MentorConnect feature that makes it possible for founders to earn mentor credit hours and display a badge based on a tiered badging system based on engagement level on shared learnings and hours spent on reviewing ideas.
2. Enable In-Post Beta Surveys: Founders tag product posts with a "Beta" badge; commenters auto-receive a lightweight NPS poll after interaction.
3. Leverage LinkedIn Polls & Events to run micro-surveys in founder-centric groups; segment replies by firmographic filters.
4. Offer an AI Trend Miner that scrapes industry-specific conversations, hashtags, and job-post keywords to surface emerging pain points in real time for a particular industry.
5. Bundle LinkedIn Learning Team Pass: Three free licenses per founding team; unlock more seats via milestone achievements

Ola management has recognized the travel needs of school kids as a potential business opportunity. Design the experience on Ola to meet their needs.

Would the service be integrated into the existing Ola application or a new application? What would be the region of operation?

This service needs to be integrated into the existing Ola environment. The region of operation is India.

Alright. Also, the target audience for this service would primarily be children falling in the age group of 4-17 years. Before moving on to the features, I would like to identify the key stakeholders, their needs and the problems they face with the existing services.

Sure, go ahead.

The key stakeholders would be the **consumers** (school kids and their parents) and the **drivers**. We will be designing the service from the point of view of the consumers because the new service will be drastically different compared to their usual services from the user's point of view but similar from the driver's point of view.

We can segment the users based on the:

1. *Distance between the school and the student's residence*: A student residing very close to the school can either walk to the school, use a bicycle, or be dropped off by the parent but will probably not avail a bus/van service. On the other hand, a student who resides far away from school would look for a more comfortable mode of transport as he/she will have to travel for a longer time.
2. *Affordability*: The choice of service also depends on how much the parents can afford.

Sounds good. What segment should our service target?

Our service should target the middle-income segment. Higher-income segments prefer personal vehicles and school bus services and are less likely to look for cheaper alternatives. The lower-income segment is not an ideal segment to target at launch. The service could later be expanded at a future stage to cover all the remaining segments.

That sounds fair. What other factors will you consider while building the service.

Another decision-making point would be the kind of service Ola would be planning to offer. The standard services available are listed below:

Mode of Transport	**Features**
Bus	Cheaper, more secure, more timely and disciplined service
Auto	Unsafe, informal, multiple vendors, more options to choose from
Van	Flexible timings, multiple vendors, more options to choose from, safer option
Cab	Expensive, more suited for higher- income segments
Bike	Suited for travelling small distances, not feasible during monsoons

As the current Ola service already includes travel via autos, cabs, and bikes, we can target all the service segments excluding buses, as including buses could result in a huge establishment cost.

That sound alright. What are the relevant pain-points of the users which our service is trying to solve?

I will group the pain points into 3 buckets - Driver/Vehicle selection, Commute and Payments

1. Driver/Vehicle Selection

- Vehicle condition isn't good
- Lack of driver/vehicle options
- Prices are same irrespective of quality of service

2. Commute

- Lack of discipline shown by drivers
- Drivers taking uninformed/irregular leaves
- Information asymmetry regarding driving skills
- Many drivers don't have driving license/required paperwork
- Overcrowding in the vehicle

3. Payments

- Irregular price hikes
- No receipts or official paperwork

All the above-given problems are must- solves to engender trust in the parent's mind and ensure the safety and comfort of children. The incremental effort of adding features to solve the above pain points is also low. Therefore, the MVP should solve all the listed problems.

Okay, now let's move on to the solutions.

The solution is an end-to-end school drop and pick up service with the following features:

1. *Ola verified drivers*: The drivers on the platform will be tested and verified by Ola (driving tests and background checks) to allow parents to be sure of the driving standards of the driver. The driving quality will also be kept in check using GPS tracking on the vehicles at all times.
2. *Free trial rides*: To onboard new customers, a free trial ride will be offered so that the parents can experience the product features live, meet the driver and validate the vehicle quality before subscribing to the service. If the parents do not like the free trial ride, they can report the same on the app and provide feedback. A customer service rep will get in touch if needed.
3. *Real-time location updates*: Notification in the app for relevant locations – driver starts the ride, driver 10 mins away from your location, driver at your place, driver at the school, etc.
4. *Backup arrangements*: Ola will offer to arrange for a backup at no extra charges if the regular vehicle breaks down or the driver is unavailable.

Alright. Can you walk-me through the proposed MVP?

Sure. Below is the walk-through of the new service:

Step 1: Signing In

- Create a profile with name ,age, photograph
- Select school and home address

Step 2: Select Service

Type & the select the relevant driver for the chosen service. Look at price estimates for each service

- Select driver
- Look at driver profile & vehicle condition information
- Schedule test ride with the driver
- Finalize driver

Step 3: Payment and Service Agreement

- Select Subscription Model - Monthly/Quarterly/Yearly
- Sign the Online Agreement

Step 4: Avail the service

- Confirm Schedule for the week. (Driver to be notified for minor tweaks in schedule)
- See the live location of the driver while availing the service
- Select backup vehicle in case of driver/vehicle unavailability

That sums up the case nicely. Thank you for your time.

Airline Entertainment System

New Product Design

You are a PM for Microsoft Surface and Delta asked you to design the perfect airline entertainment system. You decide the features, scenarios, and users. What would you do?

I have some clarifying questions to ask before we proceed.

Sure, go ahead.

Since airline entertainment systems already exists, do I need to make improvements to the existing system or design one from scratch? What are their objectives with this new application? What is the problem with the current product that you are planning to solve for the user?

You would need to make improvements to the existing software system. The hardware, however, can be changed with the new corporate relationship between MS and Delta. MS wishes to become the leading provider of the entertainment system for airlines, and Delta wishes to improve the user experience. The current usage of the system remains low, especially on shorter flights

Yes, are there any constraints?

Yes, we are looking at a 6-8 months timeline of making the changes, and the budget is not a constraint.

Got it. Based on our discussion so far and my understanding,

On the user persona side, I think there would be the following user personas that we can look at:

1. The busy business traveler(on company time, often has personal devices and Wi-Fi) and has internet access
2. The bored economy traveler
3. The international leisure traveler

The needs of the users could look like this:

1. Casual Entertainment
2. Information on the flight
3. Other Information
4. Communication

Does this sound okay? Would you like me to focus on any specific persona?

We would ideally like to serve all of them, but let's look at the domestic bored economy traveler for the problem.

Sure. So, let's dig deeper into the user persona. These are travellers who are flying for personal use and are usually at their own expense. They are on a short flight and can't usually afford the in-flight Wi-Fi. The current system gives them minimal support unless they spend money to access the live tv, shows etc. and is virtually unusable.

The user is left to their own devices to spend the flight time. Addressing their requirements, with the needs in the background, I would like to propose a few features:

1. Interface with OTT platforms: Users can utilize their existing subscriptions during transit and continue their watchlist. The user can indicate preferences, and content will be preloaded before the flight.
2. Casual In-Flight Games and Trivia Challenges: Free-to-play casual titles like Tetris, word puzzles, Flight trivia and quizzes with rewards like drink coupons
3. Mood-Based and AI-Personalized Content powered by Copilot: Options like "Relax", "Laugh", "Focus", and "Explore India" to surface relevant content based on flight time and preferences.
4. Incorporate MS surface abilities like drawing, MS 365 ecosystem apps: The surface ability gives the user the apps to create drawings, documents, notes, etc. The user could also email their work to themselves once done and wouldn't need to log in to secure the user better.
5. The MS 365 ecosystem with a login option would give additional access to MS Teams and the user's past work for messaging and collaborating.
6. MSN available freely on the system: The user would have access to information and stay up to date with current happenings. Incorporating MSN as a free service would serve both companies better.
7. Detachable systems for drawing or reading mode
8. Making free and premium sections: to encourage users to rely on the entertainment system for the flight time rather than look for their own devices.
9. Quick Navigation to all important options

Would you like me to prioritize some of these features?

Yes. As stated, let think of an MVP that can be built within the stipulated timeline that we discussed earlier.

Sure. Since time is our only constraint. I will focus on the features with the highest impact and the least effort not only in development but also in the implementation. I've eliminated the physical changes to devices as they require a change in both the software and hardware. Therefore, the features we can focus on for the MVP could be:

1. Incorporate MS 365 ecosystem apps like OneNote, MS office without the login
2. Interface with the OTT platforms
3. MSN access on the system
4. Quick navigation, modern UI refresh
5. Casual games & trivia

These features enable the user with more entertainment options than currently available and make the experience more convenient to use. Moreover, it strengthens the corporate relationship between Delta and Microsoft, increasing their stickiness.

Assumptions:

1. The current system can take a software update
2. Revenue generation has not been a focus since we looked at only improving the user experience
3. Changes in content were also put out of the current scope of the problem
4. No complex real-time networking features required

Metrics:

1. Increase in the usage time of the entertainment system overall
2. Increase in the number of users accessing the entertainment system more a minimum threshold time
3. Increase in the customer satisfaction of the entertainment system showing a better perception of the system in the user's mind

Would you like me to elaborate further on any part?

No, this should do. Thank you

Design an Alexa skill for artists/designers that can be monetized.

CIRCLES – COMPREHEND THE SITUATION

Let me just clarify the problem statement. The goal is to design a monetizable, useful Alexa skill for a chosen artistic user segment.. In my understanding, Skills are like apps for Alexa. For example, controlling smart lights etc. through Alexa can be done through Alexa Skills. I am assuming that this means that the skill I am designing shall have a voice- interface? Monetization methods include offering the Alexa skill as a complementary feature to an existing product, providing access through a subscription model, or charging users on a pay-per-use basis.

Yes. It will primarily be a voice enabled product, but as a supplementary interface, UI/Video can be used to provide certain information.

C**I**RCLES – IDENTIFY THE CUSTOMER PERSONA

Okay. Are we targeting a particular segment of artists? sculptors/painters/graphic/music artists, etc.?

You can select a particular segment and think of a product around it.

I will first try to identify certain user personas that the Alexa skill can cater to. Among artists, the most attractive market that has a demonstrable monetization potential is that of musicians. They primarily work with audio through lyrics, rhythm and sounds.

Strength of Alexa lies in its audio recognition capabilities. I can think of two user personas in the realm of musicians:

a) Hobbyist learning instruments or singing in their homes
b) Professional musicians recording in studios but practicing at home.

Out of these two, I would pick the hobbyist as my target user. Professional musicians are set in their ways and usually use professional equipment. The hobbyist might be willing to invest in tools that speed up the learning process and help them become a seasoned musician from a beginner. Does that sound alright?

Yes, that sounds good.

Let's call our hobbyist musician Aditya. He spends 30 mins – 1 hr daily either singing or practicing playing instruments. Typically, he is either alone or is joined by some friends. Aditya plays Hindi, English and other contemporary music. I would like to focus on one type of user, someone who plays instruments or is practicing singing. Can you help me decide if I should pick a particular user?

You can choose the use case of playing instruments.

CI**R**CLES – REPORT USER NEEDS

Next, I would like to understand what problems does our instrumental hobbyist musician, Aditya face. Given that they practice alone, they need feedback on how they can improve. How their tempo, rhythm and chords are matching with the song. In case of guitar, they can get feedback if they are playing the right notes. In other cases, they might need vocals backing their playing and adjust according to the instruments pace. Finally, they might want to record their sessions and want to either use them or analyze, through tools or manually.

To summarize the user's problems – a) need for feedback, b) backing vocals and c) recording.

CIR**C**LES – CUT THROUGH PRIORITIZATION

We need to prioritise among these problem statements. Ideally, you would want to build a comprehensive solution which caters to multiple needs of the user but for our MVP, we will choose the most important feature and then build other features over time. We will use Revenue Potential, Impact and Ease of Development.

User Need	**Monetization**	**Impact**	**Development Need**
Need of feedback	High	High	High
Backing vocals	Medium	Medium	Medium
Recording	Low	Low	Low

Revenue potential is maximum for solving the need for feedback. The ability to partner with existing players like Guitar Tuna is huge.

I would like to start by talking about some of the features. After which, we will discuss some monetization strategies for the product and select the best way forward. Does that sound good?

Yes, that sounds good

CIRC**L**ES - LIST SOLUTIONS

Some core features of the application will be:

a) Ability to understand tone, rhythm, tempo and chords of the playing instrument beat
b) Ability to select a particular song and playback the general music so that the player can play their instrument on top of it
c) Spoken feedback using NLP & LLMs
d) Improvement of the LLM through targeted edits using data.

Monetizing the application can be through three methods:

a) Partnering with different music applications like Spotify, Guitar Tuna etc. to build it into their apps,
b) Pay per use model on the number of invocations
c) Building a standalone music subscription service, along with a mobile app to view analysis and track progress.

How would you evaluate which monetization strategy works best in this case?

CIRCL**E**S - EVALUATE TRADEOFFS

Building a standalone Alexa skill with only a voice mode would be very difficult to monetize, given the low monetization of Alexa skills in general. In most cases, like smart lights etc., the Alexa service is provided as a complementary service. Similarly, it makes sense to focus on either partnering with the right apps or building a standalone music app. Given the development effort required to develop a standalone app, it makes sense to initially partner with other music apps to build some credibility and eventually release as a separate premium service. The service can be offered as an add on subscription or can be paired with premium tiers of existing applications.

That sounds great. Let's summarize now.

CIRCLE**S** – SUMMARISE

We designed an Alexa skill targeted at hobbyist instrumental musicians like Aditya, who practice at home and seek structured feedback. Among user needs—real-time feedback, backing tracks, and recording—we prioritized feedback due to its high impact, monetization potential, and development feasibility.

The MVP includes features like tone and rhythm analysis, voice-based feedback using LLMs, and the ability to play along with chosen songs. We propose a phased monetization strategy: initially partner with music apps (e.g., Guitar Tuna) to integrate the skill, then evolve into a standalone subscription service with a companion mobile app. This approach balances development effort, user value, and long-term revenue potential.

Social Media app for Covid-19

New Product Design

Design a product for social media apps to fight Covid-19.

Is the main purpose of this product to access information easily and reduce misinformation going around and be a reliable source for data?

Yes, and along with this we would also use this product to build strong support communities in this difficult times.

Sure, We can also share safe practices to be followed.

Sure, go ahead.

I would like to segregate users based on the emergency and their impact on the situation. So, the main segments would be:

1. Healthcare workers
2. Covid -19 patients
3. Government
4. Others (Friends, family and everyone else)

I would like to first define the user needs of these segments and pick the most critical one to solve. Is that okay?

That sounds okay.

I will define the user needs of each category, following which I will discuss the prioritization and success metrics of our application:

Healthcare Workers:

- Need to communicate with other hospitals and government officials to confirm protocols to be followed
- Availability of kits, medicines etc.
- As this is a heavy stress period, concentration should be there on the mental health of healthcare workers.
  Express concerns and update the status of patients etc.

Covid-19 Patients:

- Confirmed patients: Steps to be followed next and actions to be taken
- Non - confirmed patients: Starting from testing procedure, what are the steps to be followed

Government:

- Send alert messages when new cases are detected
- General information and rules to be followed - in terms of conveying them to citizens or checking if they are being followed etc.,
- Track confirmed patients and returnees from abroad
- Act as single channel to address all the frequent concerns and questions that public might have
- Act as a medium to let public know of the new schemes by government

All the other users:

- Act as single trusted source for all the information
- Protocols to be followed as advised by WHO and other governments

For this interview I would like to address the issues faced by healthcare workers as they are the ones most impacted and must spend long hours at hospital, away from family and are really important for us to catch a hold of the situation.

To define these personas a little more, they are mostly doctors, nurses, hospital staff and other staff related to healthcare. The assumption here will be that most of these people are well connected with technology and are often 25 years or older.

Possible Solutions - Healthcare Workers (Especially Doctors And Nurses)

In the problems mentioned above for the healthcare workers I would suggest the following solutions:

Online counselling app to healthcare workers working directly with Covid patients: This will be available for both healthcare professionals and their families to help them get through this stressful situation.

Information portal: This will act as a source where doctors from different hospitals will be able to update different requirements they have (like blood, plasma, PPE kits etc.,) and update multiple things like number of patients etc., and because large amount of people already use different platforms this will be helpful

One of the other objectives in these times is morale building. If we enable users on Instagram to appreciate the real-world events happening around them, this will act as a motivator for health care workers in these tough times. This can probably include sending appreciation stories to their profile, tag them on appreciation posts or help in raising awareness of their requirements

PM Tip: While suggesting these you may face some questions like by whom, who will support the platform and funding etc., be careful to not quote something totally unrealistic.

Cut Through Prioritization:

To create the prioritization list let's consider the engineering costs and the number of people this will impact.

So, in my opinion the Information portal has the highest ROI amongst the suggested options because in these tough times, it is hard to get correct data and harder to fulfill the requirements needed. This might act as one of the ways in bridging that gap.

This can be followed by the online counselling app as keeping doctors mentally healthy is also critical. At the end we can have Instagram stories where others can appreciate the real word happenings to increase morale of doctors and nurses.

Success Metrics

1. Number of people using this service (in terms of regions, professions in a time frame like 1 months, 3 months etc.,) - helps us understand the attractiveness of our features.
2. Number of requests answered by government officials in terms of supplying kits, filling up vacancies, addressing concerns (converted) - on daily basis.

To summarize, data has a crucial role to play to understand the spread of disease, capacity of workforce we have, necessity of medical equipment etc., so the process of data sharing should be seamless, for this reason the idea of an information portal is prioritized.

Bicycle Renting App

New Product Design

Design a new bicycle renting app to be used in a college's campus.

So, we have to design a bicycle renting application for the people staying in large college campuses. I want to confirm that the goal of this product will be to provide an easy mode of commute for the college's residents? Also, will these bicycles be allowed to be driven outside campus?

Yes, this is the goal. The bicycle can be driven only inside the campus.

Then this can be a mobile application, thus, giving users the flexibility to plan the commute from anywhere. Moving forward, I will first identify the customers for the product, then list down and prioritize their needs. Then, I will brainstorm the solutions and prioritize among them for the MVP. Lastly, I will define the success metrics for the product. Does this plan look good?

Yes, please continue.

There can be following type of users:
Residents:

- College students: They would be young (generally, 18 yrs to 30 yrs) and tech savvy. However, they will have lower willingness to pay. They might use the bicycles for faster commuting between hostels and academic complexes, and for leisure campus rides.
- Faculty members: The younger section of faculty members might be interested in utilizing the bicycles for commuting within campus. Their willingness to pay would be higher than that of students.
- College staff: They would comprise guards, mess workers and workers in food outlets. Fewer of these will be tech savvy and have higher willingness to pay than students. Their main goal would be to travel from one location to another faster and carry goods.

Visitor: They may visit the campus during fests, conferences and competitions. Their willingness to pay might be higher when compared to resident students. Their goal would be to travel from dorms to event locations faster and sightseeing.
Whom shall I focus on for this interview?

Let us focus on students.

Sure. I will first list down the needs of a student as per their user journey.

1. A student would want to find the nearest available bicycle so that she is aware of where the bicycle can be picked from. Furthermore, she would block the bicycle for some time so that no one else takes it by the time she reaches there.
2. She might need directions to reach the nearest bicycle stand.
3. After reaching the location, she would have to identify her bicycle and unlock it to start the ride.
4. New students might want to be navigated to the destination. Students will also require an emergency contact number whom they can reach out in case the bicycle breaks in the route.
5. She will end the ride after reaching the destination so that she can stop the meter from charging her anymore.
6. She would want to make the payment through the apps she commonly uses so that she doesn't have to maintain a new wallet.
7. She would want to generate recurring bicycle booking requests so that she can have a mode to commute for each of her classes.
8. She might want to synchronize her academic calendar with bicycle booking so that a bicycle can be tentatively blocked for her.

For MVP, we would initially focus on fulfilling the following needs so that students rent the bicycle for easy commuting: 1, 2, 3, 5, 6.

Brainstorming few features for MVP of mobile application and also measuring their impact *(shown in the table at the end of the case)*. Based on this, I will prioritize the following features (from the table): 1, 2, 3, 8, 10, 11, 12, 13, 16.

To measure the success of the product, I would track the following metrics:

- No. of app downloads per month
- No. of new registrations per month
- No. of rides per month
- Average duration of a ride
- Average distance of a ride
- Percentage of pickups from each point
- Percentage of drops from each points

Would you like me to do something more?

This is good. Thank you!

S.No.	Features	User Impact/Value	Cost of building impact/Effort
1	GPS on college campus map for finding the nearest pickup point with available bicycle	High	Low
2	Maintaining state of bicycle: Available, Not Available, Soon Available	High	Low
3	Blocking the bicycle and making it unavailable for other users for a time duration	High	Low
4	Measuring estimated time to reach destination and making it soon available at that point	High	High
5	If a bicycle is not currently available, user should get to know an estimated availability time	High	High
6	Whenever a user leaves hostel, provide him with a push notification if a bicycle is available near them	Low	High
7	The user should be able to navigate to the available bicycle	Low	Low
8	The user can scan a barcode on bicycle through app to unlock it and start the ride	High	Medium
9	The user enters the destination and be navigated to the nearest drop point from there	Medium	Low
10	The user should be able to end his ride through the mobile app	High	Low
11	When the user ends the ride, the bicycle should be locked automatically	High	Medium
12	The user should be allowed to end a ride if he is not at a drop location	High	Low
13	The user should be able to link UPI and wallets to the app for ease of payments	High	Medium
14	The user should be able to see the ride duration and distance travelled for past rides	Low	Medium
15	Gamification can be implemented by maintaining a leaderboard on the distance travelled and also providing users with badges for motivation	Medium	High
16	An alert should be shown if user takes the bicycle outside the campus	High	Low

Table: Features for MVP and corresponding impact

Podcast App at Spotify

New Product Design

You are a PM at Spotify. You are asked to come with solutions to improve Spotify as a podcast app

As per my understanding, Spotify is a music and podcast streaming platform. It follows a freemium model. It gives listeners access to millions of songs and other content from creators all over the world.

That's correct.

Before delving deep into the problem, I have a few questions in mind:

- For whom, are we trying to improve the product i.e., Listeners or Creators?
- Are we have targeting any particular demography?
- Do we have any budget or timeline constraints?

Thank you for asking that. We are focusing only on the listeners at the moment. For the simplicity of the case we can assume there are no demography, budget or timeline constraints.

Thank you for clarifying that. I would now like to touch upon the various stages of user journey which involve: Acquisition, Activation, Engagement, Retention and Referral.
For the given problem statement, we need to focus on the Engagement and Retention stages. I'll prioritize Engagement, since a good engagement increases chances of retention, if that's fine with you?

Sure, Go ahead.

Now, I would like to list down the user-personas. Based on my understanding, there are three user personas for Spotify podcast users.

- School and College Students
- Working Professionals
- Retired Citizens

College students are primarily interested in music and podcasts. However, they struggle to find the right topics for podcasts, often losing interest and leaving halfway. Working professionals, who listen to both music and podcasts, are more focused on podcasts related to their professional work. Despite having a higher attention span, they also get distracted while listening to long podcasts. Retired professionals, who have more spare time, also struggle to find the right recommendations due to the various types of recommendations on the app's home page.

I would like to concentrate on the working professionals, as these are the people who tend to use podcasts more.

That's a good understanding. You can go ahead with your line of thought.

Thanks. Going forward, I see that the current pain points for the working professionals are:

- Users often struggle to find new podcasts aligned with their interests due to overwhelming volume and lack of engaging previews.
- There's no forum within the app where listeners can ask questions, comment, or discuss episodes. This pushes the community interactions to third-party platforms (e.g., Reddit, Discord).

That sounds fair. What product features you would like to introduce to mitigate the pain points?

- In order to diversify the content usage by the customers, we can introduce a function equivalent to "Podcast Snippets". With this feature, viewers may scroll through and view brief excerpts from different podcasts. Clicking on a clip may take a user to the original podcast channel, where they may listen to the entire episode, if they find it fascinating and would like to investigate more. This strategy could pique users' interest in podcast genres they never would have considered.
- To increase engagement among the users, we can have a community feed for podcasts after it attains specified number of followers where the creators can share notes and conduct Q&A sessions.

Those are some interesting suggestions. Can you also list down some success metrics for your suggestions?

Sure. We can use the following success metrics:

- Monthly increase in the no. of users who saved and listened to full episodes via podcast snippets.
- Total number of unique users saving podcast snippets.
- Average snippets saved per person.
- Increase in user engagement (likes, comments, Q&A participation) on podcast community feeds.
- Growth in followers for podcasts with active community feeds.

That sums up the case nicely. Thank you for your time.

Build a Product to buy and sell antiques

Sure, I have a couple of preliminary questions. What is the scope of the product that we are looking to design, in terms of users, the nature of antiques, and the orchestration of the solution?

Good question. We are looking at an online marketplace where users can buy and sell antiques. By antiques, we imply goods that are older than 100 years.

Understood. What is the geography that we are targeting, and does the client have experience in delivering similar products or marketplaces in the past?

Let's assume the geographical target to be pan-India. Also assume that it is a startup venture that the client is looking to enter, with no prior experience.

Right. Given this information, I would define the goal of such a product to be an online marketplace to enable sellers to list their antiques and sell them to prospects, and enable buyers to search, view and buy antiques. Is the goal sufficiently defined, or would you like me to consider any other aspects? Also, I take it that the antique-based market is presently unorganized in India - is this a fair assumption?

This is fine.

I'll like to structure the design into three parts. First, I'll define the user personas for buyers and sellers and outline some use cases for each of them. Second, against each of the use cases, I'll present some potential solutions. Lastly, I would define some success metrics that can be employed for these proposed solutions. Is it a fair structure to follow, or would you like me to look into a specific aspect of the solution.

Looks good. Let's go through each of the segments one by one.

Sure. given the goal of our product, we would primarily have two segments of users - the buyers and sellers.

Among the buyers, we would have two sub-categorizations:

1. Purchase for self: Users who want to purchase antiques for themselves
2. Purchase for re-sale: Users who want to purchase antiques for re-sale in another market or potential buyers.

For the sellers, we can again have two broad sub-categorizations:

1. Direct sellers: Users who directly sell their personal antiques
2. Merchants: Users who own a business to sell antiques.

Let us only consider direct customers for the buy-side.

Understood. I'll now proceed with defining some use-cases for each user.
For the seller, the following use-cases may be considered along with their priority order, which is defined in terms of usefulness to the customers for the MVP:

Use Case	Priority
I want to display and get maximum reach for my antiques	P1
I want to have a QnA platform to connect with the potential buyers and answer their queries	P2
I want to safely sell my product to the interested buyers	P1
I want to know about the customer feedback and ratings for the products sold in the past	P2

Use Cases for Direct Buyers (Self-purchase):

Use Case	Priority
I want to view the antiques that are available for sale	P1
I want to verify the authenticity of the antique	P1
I want to know the past reviews and ratings of the sellers that have been added by past buyers	P2
I want to safely carry out the purchase transaction	P1

In the initial phase, I would like to prioritize only the P1 features and propose solutions against them. Is this a fair approach? Do let me know in case there is any specific use-case that the client wants to incorporate or any re-prioritization that may be required by the client.

The use-cases are comprehensive. Let us proceed with some proposed solutions.

Understood. Before delving into the specific interface, I would also like to outline the interface for the product. Given the need for an online marketplace, I would propose the solution to be hosted as a reactive website at least in the initial phase, with further interfaces being added in the later stages of the product life cycle. The choice for web platform is based on two factors:

1. Lower cost of prototyping
2. Minimal issues with versioning and compatibility

The specific features for each of the P1 use cases are as under:

User	Use Case	Potential Solutions	Business Impact	Cost
Seller	I want to display and get maximum reach for my antiques	Standard form fields for listing of products for sale. Additional option for sponsored listing for bumping-up the listing on the listing page.	H	L
		Analytics dashboard to track product metrics such as impressions on temporal basis.	H	M
		Integration with social media APIs to enable direct sharing of products on them.	H	L
	I want to safely sell my product to the interested buyers	Have a 3rd-party payment gateway integration (eg Razorpay, PayU).	H	M
		Order management dashboard to update status of dispatch and delivery and track current orders.	H	M
Buyer	I want to view the antiques that are available for sale	Multifilter functionality basis location of seller, nature of antique, pricing, seller, etc.	H	L
		Statistical ML model to use past purchase history/profile details to bump up relevant listings on the search feed	H	M
Buyer	I want to verify the authenticity of the antique	Add mandatory attribute for "certificate of authenticity" for each listing.	M	L
	I want to safely carry out the purchase transaction	Have a 3rd-party payment gateway integration (eg Razorpay, PayU)	H	M
		Live tracking of order status (Payment->Dispatch->Delivery)	M	L

The proposed solutions are comprehensive. Why don't you quickly list down the solutions that should be prioritized first.

Sure. I would like to prioritize the solutions basis two factors - Business Impact and Cost. By Business Impact, I aim to incorporate the importance of the said solution in developing a MVP and the first level go-live, and through cost, I aim to take into consideration the cost of development on a financial level.

Using the above two factors, the key solutions that should be targeted first are as under.

1. Standard form fields for listing of products for sale
2. Integration with social media APIs to enable direct sharing of products on them.
3. Multifilter functionality basis location of seller, nature of antique, pricing, seller, etc.
4. Add mandatory attribute for "certificate of authenticity" for each listing.
5. Order management dashboard to update status of dispatch, delivery, and track current orders.
6. Have a 3rd-party payment gateway integration.

While the first four features are prioritized basis their high business impact and low cost, the last two are prioritized since payments and order status form a crucial part of the journey of a marketplace solution

Fair analysis. Can you quickly define some success criteria before we wrap up this case?

Sure, some of the key success metrics for this solution can be as thus:

1. **Adoption Metrics:**
   Number of new signups - both buyers and sellers - per month

2. Engagement Metrics:
    i. Average session duration
    ii. Number of listings viewed
    iii. Number of listings shared on social media
    iv. Purchase conversion rate
3. Retention Metrics:
    Churn rate (percentage of users who stop using the platform)

Good analysis, we can close the case now.

Can you walk me through how you'd improve a ride-hailing app like Uber or Ola?

Before diving into the solutions, I'd like to clarify a few things to better scope the problem. What is the state of our company? What is the goal of the company at this point? Are we focused on increasing bookings, improving retention, enhancing driver satisfaction, or expanding to new segments?

Let's assume the company is mature and has a significant urban user base. However, competition is intense, and we want to improve rider experience and increase loyalty.

That's helpful. I'll assume we are targeting urban Indian users, existing and new. One quick follow-up: Who is our primary user here – the rider or the driver? While both are critical, I'd like to prioritize based on the product goals.

Good question. Let's prioritize riders for now. Improving their experience is our current focus.

Understood. For the rider segment in urban India, common pain points include:

1. Long wait times during peak hours
2. Price surges
3. Safety concerns (especially for female users at night)
4. Lack of visibility on driver location or ETA accuracy
5. Unreliable driver cancellations

So, with retention and satisfaction as our north stars, I'll try to solve for reducing friction and building more rider trust.

Sounds fair. Go ahead—what would you build to address this?

1. Smart Pickup Suggestions - Use real-time location data, traffic patterns, and geofencing to recommend better pickup points nearby. This helps avoid confusion and failed pickups, especially in areas with traffic restrictions or confusing road layouts.
2. Trust Mode for Night Rides - An opt-in safety feature, especially for women. Riders get matched with high-rated drivers (or even only female drivers, where available), and the ride comes with enhanced live tracking and optional emergency sharing.
3. Priority Matching for Loyal Users - Offer faster ride assignments and reduced cancellations to users who frequently use the app. This can also include a loyalty badge and early access to new features.
4. Predictive Surge Alerts - Notify users ahead of time if surge pricing is likely to occur soon in their area. This gives them a chance to book early or adjust their plans.

How would you prioritize amongst these features?

As a mature ride-hailing company, our product goals are:

- Retention
- User satisfaction
- Competitive differentiation

	Retention	User satisfaction	Competitive differentiation
Smart Pickup Suggestions	Reduces frustration, improves reliability	Lower pickup delays, better experience	Industry-standard
Trust Mode for Night Rides	Builds long-term trust with a key segment	Directly addresses safety concerns	Strong
Priority Matching for Loyal Users	Rewards frequent use	Speeds up access, builds habit	Encourages staying over switching
Predictive Surge Alerts	Neutral to low	Manages expectations	Easy to copy, not sticky

I'd start with Smart Pickup Suggestions—it hits two of our top goals, is relatively simple to implement, and solves a daily pain point.
Next, I'd prioritize Trust Mode. While it may not apply to everyone, it significantly improves the experience for a key user segment, and it positions us as a more thoughtful, safety-first brand.
Priority Matching would follow as a loyalty feature that rewards repeat usage—something we'll want once core functionality is improved.

Sounds good. Let's say you launch Smart Pickup Suggestions. How would you measure success?

- Failed Pickup Rate – Did we reduce cancellations or missed pickups?
- Pickup Time Accuracy – Is the gap between ETA and actual arrival smaller now?
- User Feedback – Specifically comments or CSAT scores related to pickup experience

Nice. Would you roll this out all at once?

I'd suggest a soft launch in a few zones of high-density cities like Bangalore or Delhi, where pickup issues are common due to traffic and infrastructure.

Great answer. We can end the discussion here.

Health and Fitness App

New Product Design

Develop a health and fitness application for smartphone users.

Alright. Do we have any specifics in terms of the geography we are targeting or any particular aims of the company?

For now, lets focus on the urban Indian market.

Ok. Do we have a target set of customers in mind we want to sell our product to?

How about you recommend the same? What would be the best target for our application?

Ok. One preliminary question. How does our company look like? Are we a startup building a single application, an enterprise with a suite of apps or something else? Do we have prior experience in building related products?

Let's say you are a startup that is building this app for the first time.

Cool. Since we are a startup, we will want to focus on growth and adoption to get an initial set of users to try our application. Since experimentation rate is higher among young populations and since the young urban population is more geared towards fitness these days, I would suggest that we target the younger urban demographic as our initial set of customers. More specifically, people in the age of 18-30 who are both willing to try out new things and also have enough disposable income to spend on new products.

Sounds good. We can move forward.

Great. I want to discuss more about the target customers' needs. Since these are urban young consumers, with a majority of them working full time jobs, they do not have much time to dedicate toward health and fitness. Most of them frequent gyms on a regular basis or occasionally play sports like cricket and badminton or run marathons.

Ok, are there any other aspects of health and fitness that urban consumers might try to focus on?

Yes, diet control is gaining popularity these days. Hence, providing features that aid in the same can be a helpful feature in such applications.

Cool. How should we prioritize what to build first?

The following solutions come to my mind

1. A tracking feature for tracking health analytics - this includes features like tracking workouts, calories burnt, heart rate etc. We can build a profile for the user and give them recommendations based on their workout history.
2. A scheduling feature to plan workouts for the week - this will help users track which body part they are training when and with what intensity.
3. Workout plans for customers - We can create personalized workout plans for users based on their health history and aims. We can also onboard famous trainers who can then sell their workouts on our application which will help us with engagement.
4. A diet tracking plan - we can implement a calorie counter that can help users track their calorie and their nutritional intake. We can then provide them recommendations on food plans and what to eat based on what their goals are.

These sound like nice solutions. Although, how do we plan on building them?

Since we are a startup, we should focus on the following factors when prioritizing building features

- Reach - how many users are likely to be interested in this feature
- Cost - How much time and money will be required to build out these features
- Revenue - How can we likely earn revenues from these features.

Since we are in early stage, I would suggest prioritizing reach and cost over revenues.

Alright, then based on these factors, what should we build first?

I will prioritize the listed features in the following order

1. Scheduling feature
2. Health Analytics tracking
3. Workout Plans
4. Diet tracking

We should build scheduling first since it will be relatively cost effective to implement, and given the busy lifestyle of urban Indians, will be a helpful solution that can provide them value in planning out their workouts. This will help us with Reach and Cost.

We can then build health analytics. The reason I put it second is that there are already lots of applications that build this and have dedicated accessories for tracking. However, it is an integral part of fitness apps and hence should be built out.

We can then focus on Workout Plans. For this, we will need a repository of workout plans and onboard trainers, which will require a lot of marketing. However, this can help us with revenue as we can take a percentage share in sales. However, I will suggest building this in later stages and not in the MVP.

Diet tracking can be kept last. As per my personal experience, only hardcore gym enthusiasts actively update their charts and track diet effectively. Plus, we will have to build knowledge in the nutritional content of a variety of Indian meals, which will take time.

Sounds good. Any other recommendations you would like to make?

Yes. I guess we can focus on metrics to track success. Since our priority is growth and adoption, we can focus on two metrics:

1. New User Adoption Rate – this is the percentage of users gained on a periodic basis. This will help us track how popular our application is and with some geographical analytics can help us focus on which regions we can prioritize for growth
2. DAU(Daily Active Users) – this can help us figure how actively our existing users are using the app, which can help us approximate how much value we are delivering to customers and how useful our built features are to them.

We can use these 2 metrics to define success for our product and help grow in an effective way.

That sounds great. We can close the case now. Thank you for your time!

Internal Knowledge Repository

New Product Design

You are a Product manager of XYZ company. Design a web portal that will serve as an internal knowledge base of this company.

Sure, I have a couple of preliminary questions. What is the scope of the product we are looking to design - in terms of users, the types of knowledge shared, and whether the portal should focus on just consumption or include contribution and collaboration as well?

Good question. We are looking at a company-wide internal portal. It will serve as a single source for resources like SOPs, playbooks, and internal decks (e.g., Stage 0 decks). Currently, there's no formal structure; documents are scattered. You may decide whether collaboration and contribution are in scope for the MVP.

Understood. One more question - do we have any constraints in terms of geography, scale of organization, or existing tools in use (e.g., SharePoint, OneDrive)? And is this a revamp of an existing tool or a fresh build?

Assume it's a fresh build for a mid-to-large organization, spread across functions and locations in India. Knowledge is currently scattered in drives and email threads.

Got it. Based on this, I would define the goal of the product as: To create a centralized, searchable, and structured portal for employees to access and engage with key organizational knowledge such as SOPs, playbooks, and internal decks. Would you like me to proceed with structuring the problem now? I'll define the user personas, outline their key use cases, and then propose MVP features and success metrics.

That sounds good. Let's go through each of the sections one by one.

Great. Let me start with the user personas. We would primarily have two segments of users:

1. Consumers (Knowledge Seekers): Employees who are looking for information - SOPs, playbooks, strategy decks, etc.
2. Contributors (Knowledge Owners): SMEs or team leads responsible for uploading and maintaining documents.

Alright. Now walk me through the use-cases for each of the user personas.

Yes, I'll now proceed with defining some use-cases for each user.

For the consumers, the following use-cases may be considered along with their priority order, which is defined in terms of usefulness to the customers for the MVP.

Use Case	Priority
I want to search for and view relevant documents (SOPs, Playbooks, etc.)	P1
I want to see who created a document and when it was last updated	P1
I want to download documents for offline access	P1
I want to ask questions or leave comments on a document	P1
I want to bookmark frequently used documents	P2

For the contributors, the following use-cases may be considered along with their priority order.

Use Case	Priority
I want to upload knowledge documents with proper tags and metadata	P1
I want to receive and respond to comments on my uploaded resources	P1
I want to be reminded or able to update outdated documents	P2

Would you like me to move on to the proposed solutions for the P1 use cases?

Yes, let's proceed with the proposed solutions

Before diving into the specific features, I recommend building this as a responsive web portal for the MVP. This allows:

- Easy access across desktops and mobile browsers
- Lower prototyping cost
- No app store dependencies or versioning issues

Now, based on the P1 use cases, I will proceed with the solutions targeted towards the Consumers and I have used the RICE framework for feature prioritization:

Feature	Reach	Impact	Confidence	Effort	Priority
Global search with filters to view relevant documents (SOPs, Playbooks, etc.)	High	High	Medium	Low	High
Display metadata on each document page	Medium	High	Medium	Low	High
Download button with version timestamp and file type shown	High	High	High	Low	High
Threaded comment section under each document to ask questions or leave comments on a document	Medium	Medium	Medium	Medium	Medium

I will now proceed with the solutions targeted towards the Contributors in the similar manner:

Feature	Reach	Impact	Confidence	Effort	Priority
Upload form with fields (title, tags, category, contact info) to upload knowledge documents with proper tags and metadata	High	High	High	Low	High
Email notifications + comment response dashboard	Medium	High	Medium	Medium	Medium

The proposed solutions are comprehensive. Why don't you quickly list down the solutions that should be prioritized first for the MVP?

Sure. Based on the above analysis, I would propose the features in the following priority order for the MVP for both Consumers and the Contributors:

Consumers:

- Global Search with Filters - It is central to discovery and while slightly more complex, it delivers high value to most users.
- Metadata Display (Author, Timestamps, etc.) - It enhances document credibility and transparency and is easy to implement with immediate benefit.
- Download Button with Version Info - It has high reach, impact, and confidence with low effort which enables offline access and builds trust with clear versioning.
- Threaded Commenting on Documents - This introduces collaboration and clarifications at a moderate cost and is useful for feedback loops.

Contributors:

- Upload Form with Metadata Fields - This is a critical enabler for structured knowledge contribution which is low effort and high impact for long-term content growth.
- Email Notifications to Contributors - This is useful for engagement and can be added after core workflows are functional.

Fair analysis. Can you quickly define some success metrics before we wrap up the case?

Sure. I'd group the success metrics across three dimensions:

1. Adoption Metrics
    - % of employees using the portal weekly/monthly
    - No. of unique documents uploaded per month
2. Engagement Metrics
    - Avg. time spent per session
    - No. of downloads made
    - No. of comments or feedback submitted
    - No. of searches performed
3. Retention Metrics:
    - Churn rate (percentage of users who stop using the platform).

These metrics will help gauge both the utility and sustainability of the portal.

Good analysis. We can close the case now.

Book Recommendation App

New Product Design

You are a Product manager of ABC company. Design a book recommendation platform that helps users discover and find the best book recommendations suited to their interests.

Sure, I have a couple of preliminary questions. What is the scope of the product we are looking to design, in terms of target users, the types of books, and recommendation flow?

Good question. We're building a mobile-first book recommendation platform powered by an AI assistant where users can get book suggestions, compare prices across marketplaces, and write or read reviews.

Understood. What geography are we targeting? Also, is this a new initiative or part of an existing business?

We're targeting India initially, and this is a fresh startup with no existing product in the books space.

Great. Based on that, I'd define the goal as creating a mobile-first platform that helps users discover, evaluate, and buy books using an intuitive AI interface. Would you like to focus only on the discovery-to-purchase journey or include post-purchase engagement as well?

Let's keep the focus on discovery, exploration, and purchase for now.

Sounds good. I'll structure the solution into three parts:

1. Defining user personas and their key use cases
2. Proposing mobile-first solutions
3. Outlining success metrics

Does that work?

Yes, please go ahead.

Great. Let's start with the user personas. We identified 3 primary user personas for this platform:

- Engaged Reader: A student or a working professional who enjoys reading, either casually or with intent. Looks for relevant, high-quality recommendations and wants a quick way to browse, evaluate, and purchase books.
- Review Contributor: Enjoys reviewing books and influencing the reading community.
- Deal Seeker: Looks for the best prices and fastest delivery before buying.

Based on these personas, here are the core use cases prioritized for MVP:

Use Cases for Engaged Reader:

Use Case	Priority
Ask the AI assistant for book suggestions based on genre, mood, or specific themes.	P1
Ask targeted queries like "best books on productivity" or "top thrillers of 2023".	P1
Scroll through suggestions and reviews easily on mobile	P1
Save or revisit previously viewed books through History	P2
Bookmark books for later consideration	P2

Use Cases for Review Contributor:

Use Case	Priority
Write and submit detailed book reviews with star ratings and genre tags	P2
Read community reviews and respond to comments	P2
Receive notifications when someone reacts to their reviews	P3
View and manage a list of books they've reviewed	P2

Use Cases for Deal Seeker:

Use Case	Priority
View price comparisons across Amazon, Flipkart, and other online marketplaces.	P1
Receive alerts or notifications on price drops or new offers	P2
Filter books by price, format (paperback/hardcover), and delivery speed	P2
Quickly access external purchase links with minimal steps	P1

Would you like me to move on to the proposed solutions for the P1 use cases?

Yes, let's proceed with the proposed solutions.

Given the mobile-first nature of the product, I propose an intuitive, responsive app design with minimal typing, high discoverability, and quick access flows. We can include the following features:

- A sidebar view (AI Assistant, History, Notifications) - with AI Assistant being the default view
- Swipe between book info and reviews
- Tappable prompts for common AI queries
- Lazy loading of prices and review sections to ensure performance

Now, based on the P1 use cases, I will proceed with the solutions and use the RICE framework for feature prioritization:

Feature	Reach	Impact	Confidence	Effort	Priority
Conversational AI Assistant for book discovery	High	High	Medium	High	Medium
Book Detail Screen with scrollable tabs (summary, reviews, metadata)	High	High	High	Low	High
Price Comparison View and external buy links	High	High	Medium	Medium	Medium
Side Bar with quick access to AI Assistant, History, and Notifications to comments	Medium	Medium	Medium	Low	High

The proposed solutions are comprehensive. Why don't you quickly list down the solutions that should be prioritized first for the MVP?

Based on the above analysis, I would propose this priority order for the features in MVP:

- **Book Detail Screen:** This has high reach and impact with low effort, making it the first step for enabling discovery and evaluation.
- **Sidebar Navigation:** It is quick to implement and critical for user accessibility and flow continuity.
- **Price Comparison View:** This would drive user purchase decisions and monetization, and should follow after book discovery core features.
- **Conversational AI Assistant:** This involves high impact and reach but it also involves higher development efforts and can be rolled out iteratively after validating core user journeys.

Fair analysis. Can you quickly define some success metrics before we wrap up the case?

Sure. I'd group the success metrics across three dimensions:

Adoption Metrics:

- Percentage of users who initiate an AI query within their first session
- Daily active users (DAU) growth over time

Engagement Metrics:

- Average number of books viewed per session
- Click-through rate on marketplace links

Retention Metrics:

- Percentage of users who return within 7 days
- Number of reviews posted by users

Good analysis. We can close the case now.

Samsung Galaxy Buds Experience

New Product Design

Thanks for joining! Imagine you're working at Samsung in the product team for Galaxy Buds. Using listening data, Samsung wants to investigate new approaches to improve the listening experience for Galaxy Buds owners. **In particular, there's interest in exploring how this data might be made social.** Let's brainstorm how you would approach building something valuable here.

Before jumping into solutions, I'd like to ask a few questions to better understand the problem.

1. Is the goal primarily to improve user engagement, or are we also trying to build brand loyalty or monetization opportunities?
2. Do Galaxy Buds users typically use an app for post-purchase interaction, and is the app usage limited to setup and occasional settings?
3. Are we open to building a new app from scratch or would this ideally live within the existing ecosystem?

Great clarifying questions.

1. Engagement is the priority here.
2. You're correct with the assumption. Current Wearable app usage often drops off after setup.
3. We're open to both approaches. I am interested in what you think would make the most sense from a product standpoint.

Understood. Given that context and considering Samsung already has an app installed on the majority of Galaxy Buds users (the Wearable app), I would like to recommend that we can leverage and enhance the existing app rather than build a new one.

Sure. I am with you on the approach. Please go ahead.

Great! Firstly, I would like to identify and prioritize the user segments most likely to benefit from a social listening feature. I would like to segment the customers based on age, location, likelihood of social sharing and the underlying potential for engagement.

Segment	Age	Location	Description	Engagement Potential	Social	Product Affinity	Priority
Casual Urban Listeners	25-35	Urban	Use earbuds for commute, work focus, light music	High	Medium	High	1
Students/ Young Adults	16-25	Mostly Urban	Stream heavily, love social sharing (listen during studies / lectures)	High	High	Medium-High	1
Semi-urban Music Enthusiasts	25-35	Semi-Urban	Use offline + streaming apps, interested in community-based sharing	Medium	Medium	Medium	2
Fitness-first users	20-40	Urban/Rural Mix	Music is secondary to fitness tracking, low social intent	Medium	Low	Medium	3
Rural Listeners	20-40	Rural	Prefer local/offline music, low app engagement and limited digital sharing	Low	Low	Low	4

According to the filtering parameters, defined above I would like to prioritize the following customer segments:

1. Casual Listeners – large base, high engagement potential, and use buds frequently.
2. Students/Young Adults – inherently social, love personalization and expression through music.

That's a nice way to segment and break-up. However, I would like you to think if you have missed any user segment in the above classification?

I think I have covered an extensive spread of the user base. However, I feel one segment is particularly missing from there which are Audiophiles / Tinkerers who are highly engaged in sound customization but are less into social features. But I would not like to prioritize them as compared to the current top users as they will have medium engagement potential and low-medium social sharing likelihood.

That's a great catch. Please proceed with the analysis.

Based on the top two segments, I would like to jot down the key user needs to understand the underlying the use-cases:

Understand and track listening behavior over time	Users want to consider their music and audio habits, trends, duration, genres to develop self-awareness and feel more linked to their routines. It turns inert listening into a more conscious, customized experience.
Share music preferences and listening highlights with others	Users want to consider their music and audio habits, trends, duration, genres to develop self-awareness and feel more linked to their routines. It turns inert listening into a more conscious, customized experience.
Discover new content or tuning setups through peers	Users want to consider their music and audio habits, trends, duration, genres to develop self-awareness and feel more linked to their routines. It turns inert listening into a more conscious, customized experience.
Feel part of a like-minded audio community	Being part of communities with like tastes gives the product emotional worth. It encourages repeated participation and transforms a personal hobby into a group, communal one.

Before proceeding, I would like to know if there are any inputs you might have over the needs?

No, I think you defined the needs in detail. Please proceed to mention the solutions.

Great! Now, let me outline three potential features/solutions we could build into the Galaxy Wearable app based on the customer segments and their needs defined above:

Feature	What It Is
Weekly Listening Digest (Shareable)	A personalized, visual summary that shows your weekly listening highlights, top songs, favorite genres, total listening time, mood tags (e.g., "Chill", "Energetic"). Users can share a summary with friends/family or can post it to community groups as well.
Genre clubs	In-app social spaces organized around music genres or moods (e.g., "Chill Lo-fi," "90s Bollywood," "Focus Beats"). Users can join, share their digests, see others' stats, discuss music, and discover new tracks.
Peer Sound Preset Exchange	A simple tool for users to create and share their own sound/EQ settings. Others can try, save, and upvote presets like "Bass Boost for EDM" or "Crisp Vocals for Podcasts." Presets can also be recommended within Genre Clubs.

I would like to prioritize Genre Clubs feature in this scenario as it is:

1. Most aligned with the core goal of building social interaction around Galaxy Buds usage.
2. Encourages ongoing participation, not just one-off sharing.
3. Unlocks opportunities for community-based features (polls, recommendations, badges, etc.).

RICE Framework (optional, scoring out of 5)

Feature	Reach	Impact	Confidence	Effort	Priority (R*I*C/E)
Genre Clubs	4	4	4	4	1
Weekly Listening Digest	4	3	3	3	2
Preset Sharing	3	3	3	3	3

That was a thoughtful walk-through. I liked how you explored the full landscape, asked clarifying questions, and grounded your prioritization in both user behavior and feasibility. Now, as a last step, can you also mention a few of the success metric we could track for the feature you plan on introducing?

Sure! I would like to segregate the metrics based on discovery, adoption, engagement, retention and revenue.

Stage	Metric
Discovery	% of app users who tap to explore club previews (genre tiles, trending clubs, etc.)
Adoption	1. % of users who join at least one club within 7 days of discovery 2. % who complete onboarding
Engagement	1. Average club visits per user per week 2. Average posts/replies per user per week (including shared digests, comments) 3. % of users who react/like content
Retention	1. % of users who return to a club within 7 and 30 days of first visit 2. % of users who post/comment in a club more than 3 times in a month

Great! Genre Clubs feels like a natural extension of the Galaxy Buds experience. That wraps up the interview. Thanks for joining!

Section 5.3

GTM Strategies



Social Fitness Product

GTM

Our company wants to launch a social fitness product. How would you develop its go-to-market strategy? *(Ask clarifying questions about the product, what it does, what problem it solves and how, target market and the goal of the launch)*

What kind of a product is this? Is it a tangible product like smartwatches or a smartphone app?

It is a smartphone app where users can connect with other fitness enthusiasts. The users can share updates about their workouts, diet plans, progress etc. with each other. They can work together through video conferencing on the app itself and compete against other users in groups or individually in regular challenges hosted on the app. The aim is to encourage users about health and fitness through peer motivation.

Okay. Are you partnering with other third-party apps?

Yes, users can also share their updates on other social networking accounts like Facebook and Instagram. It can also connect to smartwatches and wearables like Apple Watch, Fitbit, and Garmin to enable automatic workout and health data syncing.

So, who are the target users for this app? I assume it would be fitness enthusiasts who are also tech-savvy.

Yes, you are right. We are targeting people in the age group of 20-60 years, living in metropolitan and urban areas of India.

Is this a new product or an existing one being launched now in India? And, what is the goal of this launch?

This is a new product, and India is the first country where it would be launched. The MVP launch and beta testing were successful and now we want to launch the full version in the market. The goal, therefore, is to build awareness, drive adoption and generate revenue.

Okay, thank you for answering my questions.
(Analyze the market situation – take company, product, customer, competitor, and market landscape into consideration)

The company is a sports equipment company that manufactures footwear, sports and casual apparel. They are looking to diversify their offerings by entering the smartphone app space.

There are some synergies on both sides of the business. Major competitors would be other fitness apps currently in the market like MyFitnessPal, cult.fit, HealthifyMe, FTTR etc. This product covers both physical activities and social interactions. Moreover, people are becoming aware of the importance of good health, and the fitness industry is in the growing phase in India.

Now that we have analyzed different components, we can go to the launch strategy. I will divide the activity into 3 stages: pre-launch, launch and post-launch.

Pre-Launch:

- Pricing – The product could be a free app with ad-based revenue or follow a freemium model or could be a mixture of both (depends on the discussion with the interviewer)
- Create buzz – through marketing activities like releasing teasers on social media channels and promotion by fitness influencers
- Early Community Building: We can select 50–100 fitness influencers and personal trainers for an invite-only beta; collect feedback and build anticipation

Launch:

- Place – Since this is an app, it will be released via app stores(android and iOS)
- Promotion - start marketing the product actively by employing digital and social media marketing tactics. We should also advertise the product in gyms and wellness centres. And we can also use SEO to help gain traction for the application on the web as well as on the app stores. We can also look at exclusive launch offers like fitness merchandise giveaways for early adopters.

Post -Launch:

Now, we will measure the performance by tracking relevant metrics. Some examples could be:

- Acquisition - conversion rate, cost of customer acquisition, # of downloads
- Activation: % users completing onboarding, first post/workout log within Day 1
- Retention - churn rate
- Engagement – avg. session time, what features are being used, # of updates/posts posted
- Customer satisfaction – feedback in the form of ratings and reviews, NPS scores
- Revenue and market share – are we meeting our revenue and market share goals

I will use the feedback information and engagement information to plan for v2 potentially and add features like group leaderboards, corporate wellness tie-ups, and cashback rewards.

Sounds good. We can close the case now.

Table Reservation

GTM

Swiggy has decided to enter the table reservation category in restaurants. Formulate a launch strategy for the same.

Can you tell me more about the product, like if you want to create a new app for this requirement, or would this be a new feature in the existing app?

So the product is still in the idea conceptualization phase. I was hoping you could help me with that.

All right, for table reservation, there would be three major stakeholders, the customers, restaurants and Swiggy. I would suggest creating a new feature in the existing app as Swiggy already has a large customer base, therefore creating awareness and growing customer adoption about this product would be easier. Next, I want to look at the pain points of the stakeholders involved to come up with the product features.

1. Customers –When going out to eat, a customer would like to know the various restaurants available around them, price range and the cuisine. Under cuisine, they would like to know the variety of dishes served. Customers do not like waiting, so they would want to know the waiting time beforehand. They would also like to know about the availability of parking space, timings etc. And most importantly, they would like to reserve tables so that they don't have to join a queue when they reach the restaurant.
2. Restaurants – Restaurants are aware that if they cannot handle demand effectively in their restaurants, they would lose out on customers who had to wait too long. On the other hand, if a restaurant reserves a table for a customer, then they can't give it to another customer. They must keep the table vacant until the designated customer arrives. Due to this, there are chances of them losing out on prospective customers.

But don't you think Zomato already provides these features?

Yes, so one differentiating feature of this product would be that customers can also pre-order their dishes while reserving a table. This way, they wouldn't have to wait for their orders to be prepared in the restaurant. Swiggy can provide estimated arrival times of customers to the restaurants so that they can start preparing their orders accordingly.

Okay, I have one more question. If the customer has booked a table for a particular time and is late, what happens then? Should it continue to be blocked at the cost of losing another customer?

We can provide a buffer time beyond a customer's reservation time till which the restaurant will reserve the table for them. If the customer arrives later than the allotted buffer time then he/she will have to wait if the restaurant is fully occupied.

Okay. Considering this to be what the application does, how would you go about launching it?

I would like to divide the marketing activities into three buckets. Pre-launch, Launch, and Post-launch. I would like to go through this one by one.

Pre-launch:
Swiggy can release this product in big cities like Mumbai, Bangalore, Delhi etc. first, where the dining culture is big to observe customer adoption. They can create hype around the new feature through marketing activities like releasing teasers on social media channels, putting up billboards. This will help us in attracting new customer segments who previously did not use Swiggy.

Launch:
We can start marketing and promotional activities to build customer awareness: traditional and modern channels. We can also introduce smart nudges integrated into the app to push the existing users to use the newly introduced feature. We can also add features to allow them to share the new feature with their friends, providing the existing users with a discount after a certain number of new users join through their link.

Post Launch:
Now, we will measure the performance by tracking relevant metrics. Some examples could be:

- Acquisition - conversion rate, cost of customer acquisition.
- Engagement – # of reservations, bounce rate, % of reservations where customer visited the restaurant out of total reservations, % of reservations where customer arrived before the end of buffer period.
- Customer satisfaction – feedback in the form of ratings and reviews, NPS scores.
- Revenue and market share – Are we meeting our revenue and market share goals?

I will use the feedback and engagement information to expand it to other cities potentially.

Sounds good. We can close the case now

Facebook Dating App

GTM

You are a PM at Facebook India. You're tasked to come up with a Go to Market strategy for a dating application that Facebook has come up with.

Before I move on, I want to clarify the marketing goal. Are we planning to launch the application in India?

Yes, you need to come up with a strategy to launch this application in India.

Sure. I would like to start off by analyzing the current dating application market. Dating is among the top-grossing categories on the Google PlayStore in India. People are willing to pay for such products. The top applications which I can recall are Tinder & Bumble. They both are global products. There are quite a few local products as well. But all these are dating applications first. None of them have the capabilities that Facebook has, in terms of Social graph, user preferences, etc. Most of the users are between the ages of 18 - 27. They are usually in college or early stages of their professional careers. Most of the dating platforms follow a freemium model. They also use ads to generate revenue. One of the main reasons for a freemium model is to get the users to use the product and realize the value before asking them to pay. There is also the aspect of network effects. Getting more users on the platform is very important for its success.

So how does Facebook fit in this market? What can Facebook do differently?

Talking about Facebook, I think the goal of Facebook is to bring the world closer together and help build communities. A dating platform that enables building new connections fits right into its vision. The key strength of Facebook is its data. It can leverage this data to find an almost perfect match. However, dating experiences vary from culture to culture. Facebook has to keep in mind that dating expectations are starkly different in India compared to western countries. So the data about users and their networks can be leveraged to build a product that people can trust. Dating platforms in the market struggle to keep the platform clean of bots and fake accounts. This is a huge problem, which is hard to solve without taxing the actual users to give out personal information. However, the current platforms have the advantage of understanding the user behavior. They also have brand equity. This would take quite some time to build. Facebook can also include features like fraud detection and a better recommendation system to win over the customers' trust. So Facebook should focus on providing a platform for building lasting relationships as they have enough data about the users and their behavior.

Okay, this is good. Considering these are the main features of the application, how would you go about launching it?

Sure. I would like to divide the marketing activities into three buckets. Pre-launch, Launch, and Post- launch. I would like to go through this one by one.

Sounds great. Go on

Okay.

- **Pre-launch** - Facebook should ideally start by testing the application in a few regions within India. They can start with metro cities like Mumbai & Bangalore where the dating culture is more progressive. Facebook should create hype around the product. They could release teasers & small clips showing what the product could do.
- **Launch** - They should launch the product by creating a virtual or a live event. All the competitors in the market are somewhat established, so Facebook should seek to make an impression on its users.
- **Post-launch** - Facebook should actively market the product and educate the users about how this platform is different from its competitors. They can use existing products like their Facebook & Instagram to market this product.

Now, talking about distribution; Facebook can launch this as a separate application. Facebook has access to 300 million Indian users and they can use this user base to market the product to.

Now let's move on to user retention.

Sure. When it comes to dating platforms, retention is dependent on the value people derive in terms of connections made. Facebook should not focus on whether the said match goes on a date or meet up in real life as there are a lot of other factors beyond the control of Facebook. The primary focus should be on how happy a user is with the kind of people he/she meets on the platform. We can measure this by seeing how long they interact with each other on the platform and also if they go one step further and connect on Facebook as well. We can also ask them for their feedback a week after they start interacting with each other. We need to craft the feedback in such a way that the users realize why the feedback is being taken.

Sounds good. How would you go about pricing this product?

The product should be free for all users. New features can be added later on which can be charged. We can also tap into live experiences with virtual parties, events, etc. Facebook can also

create a pro version for the user who unlocks several features to find better matches or even find matches quicker.

This is good. Could you please summarize the whole plan?

Sure. The goal for Facebook would be to stand out and differentiate itself from the competition. The focus should be on building long term relationships, be it friendship or dating. Moving on to marketing, Facebook should focus on creating hype towards the product. This will also set the right expectations for the users. They should back that up with a launch that involves its users. However, most platforms don't have enough data to find the perfect match for a person and hence, not enough value can be derived from them. Also, these platforms struggle to identify bots or fake accounts on their platform. Facebook can differentiate itself by solving these very problems.

That sums up the case very well. Thank you.

Lyft's Indian Market Entry

GTM

Lyft wants to enter Indian market, what should be its go to market strategy?

Lyft has both ride sharing and Food delivery. Is there anything specific that I should be focusing on?

Yes, you need to come up with a strategy to launch this application in India.

Let's focus on ride sharing.

So how does Facebook fit in this market? What can Facebook do differently?

In ride sharing should we focus on only cars or should we also look at Autos and bikes?

We'll just keep it to cars for now.

For this lets first analyze the situation in the market.

Company

Lyft is a ride sharing app like OLA and Uber. It connects riders and drivers and allows an easier mode of transportation. Lyft has many verticals in other countries like Uber does in India (Autos, Shared, normal etc.,). We need to decide in which segments we are going to launch initially and the future plan.

Opportunity

Lets now explore the opportunity on the segments. One of the main things to keep in mind is the traffic in India, along with this we also need to keep in mind demand and supply for these segments. My suggestion in here is to take use of high number of cars existing already and give good benefits to drivers so they use Lyft more than other competitors

Customers

Coming to our customers most of them might not know about the brand itself and most of them might be regular customers to other competitors we have. We need to give them a compelling reason to try out our app, can be discounts, more safety etc.

As we know we have 2 major competitors currently and many upcoming ones. Uber expanded into this market with a lot of competition and was successful, if we have a compelling differentiating factor we can also be successful as customers look for best options for them. We can probably tie up with local car dealers to create localization feelings among customers which would probably encourage drivers to join us. We should also invest heavily into marketing initially and probably be a loss making firm like Amazon initially to get customers to try our product. In addition to this our pricing strategy is unique and will help us in remaining competitive.

Market Conditions

If we divide India into rural, suburban and Metropolitan we need to decide on where to launch first and what would make sense to achieve our goal. Because major customers are there in Metropolitans that would be the first place to launch according to me.

Regulatory Issues

Along with these we also need to consider regulatory issues and rules that the country has so that we won't get into trouble later which might spoil the brand reputation.

Sounds fair. How should Lyft go about the launch strategies ?

Now that we have analyzed different components we should go to the launching strategies. Let's divide the activities into 3 stages: Pre launch, Launch itself and Post launch.

Pre-launch

- Marketing Activities - Billboard ads, newspaper ads, pamphlets, experience videos from other countries etc.
- Enough supply according to the demand in the area
- Offer monetary benefits to drivers better than competitors or experimenting with new revenue sharing methods like subscription etc.
- Referral programs
- Tie ups with local dealers - training and adhering to rules
- Compulsory training for drivers to improve customer experience as they are the face of our brand

Launch

- Promotions and discounts for riders, customers
- Advertisements in local newspapers, pamphlets Tie ups with corporates Blogs with influencers and celebrities to create positive images etc., can be done.

Post Launch

This is where we measure the metrics. I think these are the metrics which we make more sense in this case.

Acquisition

- # customers in time frame like 1/3/6 months
- # drivers in time frame like 1/3/6 months
- # app downloads in time frame like 1/3/6 months

Revenue

- Expenditure per ride - 3 months Vs 6 months
- Revenue per ride - 3 months Vs 6 months

Customer Experience

- # repeat customers in a week
- Wait time (average) in a day
- # cancellations in a day
- Ratings given by customers on average

In addition to all this we should proactively reach out to customers with complaints to resolve them soon for better experience and also reach out to people for feedback for continuous improvement.

Okay, we can close the interview.

Tata Group Loyalty Program

GTM

The tata group wants to incentivize their customers and want to increase their loyalty towards the brand. We need your help to prepare a loyalty program for the conglomerate.

To make sure I understand the situation correctly, Tata group wants to design a conglomerate wide incentive program which would cover all the brands falling under the Tata brand including Taj, Titan, Tanishq, automobile etc.

Yes, that is correct.

As I understand, many brands under Tata group already has their standalone loyalty program. May I please know what will happen to these programs.

You can assume that all the existing loyalty programs will be replaced by the new loyalty program that you would suggest.

I am not sure that I understand the problem statement. Do the Tata want to make the existing benefits (reward points) that customers have earned null and void? I believe that would make the customers angry.

Good that you mentioned it. In your solution, ensure that there is a smooth transition for the existing reward system.

Okay. One last question before I lay out my solution, will the brands have a say in incentivizing their customers or the complete control will be with the of Tata group?

We would like you to think how this Loyalty program should be structured. Also, comment on its technical feasibility.

Okay, can I take 2 minutes to structure my thoughts and provide a solution?

Sure.

So according to me, different stakeholders like the Tata Group, brand owners and store owners should have the option to award reward points to the customers. Also, the system should have the capability to award different points to consumers using different brands. For example, different reward should be awarded to customer of Starbucks and customer of Titan.

Makes sense, carry on.

So, the structure should be as follows:

- Tata Group points: Under this, the Tata group can award points to all the customers irrespective of the brands.
- Brand points: Under this, any independent brand can award points to their customers. They can define rules for the loyalty program.
- Store owners: Every store owner should have the ability to award points to the local customers.

Another thing I need to add is that a customer can be eligible for any points, depending on the rules defined. A customer can get Tata group points, brand promotions and even points by the store owner.

Usually when the campaign runs, the points are provided in a particular city. How will you ensure that?

Fair enough. I believe if all the store owners are asked to update the offers, it might be quite ineffective. We should probably give this functionality in brand points and Tata group points. Generally, the configuration should be done on store id which acts as an identifier for the store.

That seems like a good plan. Can you explain the implementation plan?

Okay. I know that we need a unique identifier to identify each person uniquely. It is hard to provide the unique number to the customer as they would tend to forget. I suggest we use the mobile number of the consumer as a unique identifier. However, with this, we will have one limitation. The identity of the person will be saved at the Tata group level and not the brand level. To counter this, we need to create a tag where the customer registers.

Can you suggest other things to consider for a good loyalty program?

Sure, please let me think about it for a minute.
The whole idea of accruing the points is to have the ability to redeem the points. Usually, the points can be redeemed at the same brand where they were accrued. I believe Tata group can look to allow some cross- brand redemptions. For example, the points accrued at Titans can be used to redeem at Starbucks.

Interesting, anything else you'd like to add?

One last point which we discussed in the beginning about the current reward points of the users. I believe if we are going with the new Loyalty program, all the existing points should be mapped to the new customer identifier so that customers can use those points in future. I believe the personal data of all the existing customer is also important to import. Therefore, there needs to be a one-time data migration activity to get the old data in the new system. That will be all, sir.

Thank you for the insights.

New Geographies

GTM

WinZO has seen a great success in the Indian market and now is looking to expand to other geographical areas. After thorough research, we have found that Brazil would be a suitable market to expand. Can you chalk out a Go-to-market strategy for the same?

Sure. Can you tell me more about the product and its features? Also, do you want to create a new app with new games for this particular requirement, or would this just be a new version of the existing app?

Well, WinZO is a social gaming platform with over 100 games. Currently it has 20 Cr+ registered players who compete against each other for actual monetary benefits. Answering to your second question, I was hoping that you would help me with the same.

Alright. Before I move on, I want to understand two things:

- You have described WinZO as a gaming "platform". Does it mean that not all games are owned by WinZO?
- And, are all the games paid on the app? And how are the games priced?

You are right; the games on the app are a mixture of owned and hosted games. WinZO collaborates with game developers to host their games on the app either on a fixed fee model or a revenue sharing model. Most of the games on the app are paid. Users can use the practice mode for free if they want to. Pricing of the games is different across different formats of the games. There are single player games, two-player live games, tournaments, etc. However, the ticket value for each game is small (ranging from INR 2, 5, 10, 20 and 50 to 100), considering our target audience which is typically from Tier-2,3 and 4 cities.

Thank you for providing the details. I would first analyze the market situation - taking the company, product, users, collaborator and competitors into consideration. Then, I shall explore the pricing and launch strategy.

In the interest of time, I would like you to focus on the product, users and collaborators. We can assume that the pricing strategy would be similar to that in India for different formats.

Alright. When it comes to the product and its users, I feel that there is a need to look at the games, which are the key value proposition offered to the users. There will be a significant difference between games liked by people in Brazil. WinZO has games like Ludo and Snakes & Ladders in India which hold a traditional significance. However, these might not be performing in Brazil on the same scale as that in India. For other casual games, I feel Brazil as a country that is more inclined towards games related to football compared to India which has more cricket buffs.

That's right. So, how do you go about choosing which games to add in the platform? What all factors would you consider?

For choosing the games, I would be looking at these factors:

- Player preferences: Looking at the general trends, we can focus on popular game genres like sports, FPS, MOBA, etc.
- Cultural Context: We can focus on games that align with cultural elements or themes of the country that might have a greater appeal (Like Ludo in India).
- Existing Platforms: By closely watching our competitors, we can analyze the kind of games they offer and their top grossing games.
- Collaborators: We can look at possible partnerships with games and game developers to host them onto the platform, along with top grossing games in Apple Store, Google Play Store, etc.
- Tech & Infra: We need to loop in the device preferences of the users (i.e., mobile vs tabs vs console), internet stability etc. that will help decide whether to host cloud-based lightweight games, or client-based games.
- Integration: We should also look at technological feasibility of integration (Unity, Android, etc.) when considering possible partnerships,

This should give us a good starting point to decide on the games. After the launch, we can also take inputs from the users about the games they would like to see on the platform.

This looks good. What all metrics do you think would be important to track when we decide to release the application?

We could these useful metrics to track the progress of the app after the release:

- Acquisition - Installs, Conversion Rate, Cost of Customer Acquisition.
- Engagement – # of games played, time spent on the app, DAU, MAU
- Customer satisfaction – Feedback in the form of ratings and reviews on Google Play Store/App Store, NPS scores.
- Revenue – Amount of real money spent, Amount of money withdrawn, CLTV, etc.

I would use the valuable feedback from users and the above metrics to strengthen the product.

Sounds good. Thank you for your time.

Section 5.4

Metrics

Instagram Creator Marketplace

Metrics

You are a product manager at Instagram working on the Creator Commerce product tagging feature. How would you go about measuring its success?

I would like to start by shedding light on my understanding of Instagram shops to check whether it's accurate. Please correct any inaccuracies that I might state. Creator Commerce Checkout is a feature where Instagram creators can tag products from partnered brands in their posts, reels, and stories. When users view this content, they can tap on the product tag, see product details, and complete their purchase by visiting an external website.

The description is accurate. Please proceed with the rest of the solution.

Great. To define meaningful metrics, I'd like to first map out the journeys of the three primary stakeholders involved in making this feature work: the creator, the business/brand partner that lists and fulfils the products, and the buyer who makes the purchase. For defining the metrics, I would like to proceed with defining the user journey of the stakeholders. Should I proceed with this approach?

The approach seems good. Let's keep the focus on the creators and buyers. You can skip the brand-side metrics for now. Please go on with the journeys.

Sure, Thanks. I'll start by mapping out the journey for creators. It would broadly involve these steps:

- Awareness of the Creator Commerce program through in-app notifications, creator community announcements, or outreach from Instagram's team.
- Confirming eligibility by linking a Business or Creator account and passing Instagram's commerce policy review.
- Activating product tagging by enabling the feature within account settings.
- Tagging products from the catalogue in their posts, reels, and stories.
- Tracking performance and earnings via Instagram's insights, including metrics like product views, taps, and conversions.
- Iterating and optimizing by tagging new products and participating in brand campaigns based on past performance.

For buyers, the journey would be:

- Discovering a tagged product while consuming a creator's content through feed, reels, stories, or the explore page.
- Tapping on the product tag to open the product detail page and viewing product information, price, and images.
- Clicking through to the merchant's website for checkout.
- Completing the purchase externally and optionally returning to Instagram for feedback or future engagement via comments, posts.

You've captured the journeys accurately. Please go ahead with the metrics.

I'll start with the creator side and follow the awareness, engagement, and conversion funnel to define the metrics.

- **Awareness:** At this stage, the focus is on whether eligible creators know about the Creator Commerce program.
  For example:
  - What percentage of eligible creators have seen the program announcement via in-app notifications or email campaigns?
  - How many creators have clicked through those invites to learn more? Click-through rate (CTR) on program invite notifications/emails
- **Engagement:** At this stage, we check if creators are adopting and actively using the product.
  Metrics I would track here include:
  - % of creators who enabled product tagging
  - Average number of product-tagged posts, stories, or reels per creator per week/month
  - % of creators actively tagging products over consecutive weeks
- **Conversion:** To measure whether creators are successfully driving transactions through their content, the key metrics could be:
  - Average number of outbound clicks to merchant websites per creator
  - GMV attributed to creator-tagged product referrals
  - % of creators whose tagged products generated outbound clicks

That's a solid set. What about metrics for buyers?

I'd approach it in a similar way

- **Awareness:** Whether buyers are noticing product tags in creator content.
  For instance:
  - % of users who encountered product-tagged content in their feed, reels, or stories
  - % of users who tapped on product tags
- **Engagement:** Here, I'd measure how meaningfully users interact with the product listings.
  Here, the examples would be:
  - Average number of product detail views per user session originating from creator content
  - Average time spent on product detail pages within Instagram
- **Conversion:** To capture how effectively we're converting those views into purchases.
  The metrics would be:
  - Click-through rate (CTR) from product detail views to merchant website
  - External conversion rate (purchases made per outbound click
  - Repeat outbound clicks per user

Do you think we can consider any other metrics to track for this program?

Yes, beyond these metrics, we can also monitor operational and counter metrics to keep a check on the health of the program.

For operational health, I'd track:
- Outbound link click success rate (any click failures or dead links)
- Time-to-load metrics for merchant landing pages (to avoid drop-offs)
- Partner merchant conversion attribution accuracy

And for counter metrics:
- Creator dissatisfaction rate — via periodic in-app surveys
- Fraud reports or abuse of product tagging by creators
- Number of buyer complaints about poor merchant checkout or product quality

Makes sense. How would you go about defining a North Star and tracking all these metrics?

Since Instagram does not control the checkout, our goal is to drive qualified outbound traffic that converts on partner sites. The North Star metric would be –

- Outbound Purchase Conversion GMV driven by Creator-tagged content

To track these metrics, we'll need to set up event tracking in the internal analytics platform for all major actions, like product tag views, product detail page views, UTM-tagged URLs for outbound links, capturing content ID, creator ID, and campaign info and completed purchases. Operational data such as payment success rates and order fulfilment times would come from our backend systems integrated with other brands' apps/websites. And we could capture buyer and creator satisfaction scores using lightweight in-app surveys triggered post-purchase or post-campaign.

Thanks for the great discussion. We can end the interview.

Promotions in Google Pay

Metrics

You are a product manager in the Google Pay team. You have been given the task of defining the metrics for tie-in promotions on Google Pay. How will you go about it?

I believe I can start by describing how tie-in promotions work on Google Pay to get my context right. Please correct if my description is inaccurate at any point. Google Pay is used for making payments. People use it for transferring money to their friends or making online payments mostly via UPI or cards. Once the user makes a payment, they get a scratch card as a reward. On scratching the card, users get rewards which can be cashback or coupons of various companies. I believe these coupons of various companies like Lenskart, Mamaearth, Zomato & others are tie-in promotions.

Your description of tie-in promotions is accurate. Essentially tie-in promotions involve a company partnership with another company for promotions. You can proceed further with your analysis.

The metrics need to be aligned with the business value. The tie-in promotions are an important factor for users to repeat payments using Google Pay. Users will keep using Google Pay if they get relevant offers on some occasions. Similarly, promoters will partner with Google Pay if they benefit out of the partnership by gaining more customers & revenue.

The business goals seem accurate. Please proceed with the metrics.

I will cover the metrics from aspects of both users and promotional partners.

From the user's perspective, the metrics will be covering the frequency of promotions, its utilisation, and its impact on user engagement on the platform. We would ideally expect the promotions to be useful for the users.
No. of successful uses of promotions = No. of payments * % of tie-in promotion scratch cards * % of tie-in promotions used.
The user success metrics are:

- No. of payments or payment rates every month
- Percentage of scratch cards which are tie-in promotions (Scratch cards be cashback, promotions, or better luck next time)
- Percentage of promotions getting expired
- Average of no. of different brands/types of promotions given to the user
- Percentage of repeat promotions of a specific type or brand.

From a promotion partner's perspective, the metrics will cover how Google Pay, and promotion partners can benefit from the exercise.
No. of successful use of a specific promotion = No. of payments * % of promotional coupons for a given partner * % of a specific coupon * % of users successfully using the promotional coupon
The partner metrics are:

- No. of tie-in promotion partners
- Retention rate of promotion partners
- Churn rate of promotion partners
- Average number of promotional coupons shown for every partner
- Percentage of promotions which get expired without use, both overall and for a given partner
- Average number of recipients of a given promotional coupon
- Percentage of users successfully using a given promotional coupon (this needs to be shared by the partner)

Overall, these metrics will help track the success of tie-in promotions and help identify some issues. For example, if a promotional coupon by Lenskart has very low conversion rates, it can be discontinued or be modified to increase the usage.

I think this covers most of the important metrics for tie-in promotions.

Facebook Marketplace

Metrics

You are the PM for Facebook Marketplace. What metrics would you track to measure the health and success of the feature?

I would first like to clarify my understanding of the feature and the stakeholders involved. Facebook marketplace is a platform within Facebook to discover, buy and sell items with people in your community. The functionality caters to two stakeholders –the buyer and the seller. The feature allows sellers to list both new and second-hand items they want to sell that would then show up on the marketplace tab. Buyers can then explore the listed products in both the marketplace tab and groups in which product details are shared.
I'll assume that Facebook's goal with the feature is to increase customer engagement on the platform.

You are right. Please go ahead.

Before I list down the metrics, I would like to map out the journey of both the stakeholders to understand the goal of each action the feature asks the user to take.

Seller's Journey:
1. Visit the marketplace tab
2. Select the option to list an item
3. Add item description and price
4. Receive an offer to buy from the buyer
5. Accept or reject the offer
6. Receive the payment
7. Handover the item

The last two steps happen outside Facebook's ecosystem

Buyers Journey
1. Visit the marketplace
2. Search for a product
3. Contact the seller
4. Make the payment
5. Receive the product

An alternate customer journey would start with buyers discovering a product they like in groups

Your understanding of the journey of the stakeholders is fair. Let's move ahead to the metrics

I would divide the metrics into Awareness, Activation, Conversion and Retention metrics.

1. **Awareness:** The goal would be to measure whether users can discover Facebook Marketplace. The metrics would include
    a) #Users who have visited Facebook Marketplace
    b) #Clicks on Facebook Marketplace tab per day
2. **Activation:** The goal of the phase is to measure the level of active engagement with the feature. Relevant metrics will be
    a) #Products listed per week
    b) #Bounce Rate
    c) #Clicks on product listings per week
3. **Conversion:** The goal of this phase is to measure how effective is the feature in getting users to perform the primary action, i.e. buying a product.
    a) #Unique Interactions per week
    b) #Purchases made per week
4. **Retention:** Retention indicates the users' propensity to come back to the platform and use the feature again.
    a) #Repeat Purchase Rate
    b) #Repeat Product Listing Rate

Alright, out of all these metrics mentioned, which one should be prioritized by us as the top metric?

The Activation and Conversion metrics should be tracked for individual categories to understand differences in buying and selling process for different product categories. With the goal of increasing customer engagement in mind, the North Star Metric would be - #Unique Interactions per week. #Purchases per week would be used as the counter metric to ensure that product quality on the platform is not compromised for increasing user engagement.

You are the Digital Product Manager of Big Basket. What would be the metrics you would like to follow?

Okay, first I want to confirm that my understanding of Big Basket is correct.

Big Basket is an Indian food and grocery E-Retail store. A user can order a wide range of products like fruits, vegetables, packaged foods, personal care items, and household supplies. Moreover, the users can use both a mobile app and website for placing an order.

Yes, you may move forward.

Could you please share the business goal which we would like to report through the metrics.

Assume that the goal of the business is to increase the value of orders placed via the BigBasket mobile app.

I will create a user journey map, identify which actions indicate success and failure towards placing order and then define the actionable lagging indicators. Should I do it for both, website and mobile app?

You can define the metrics for only the mobile app for now.

Sure. The value of orders can be formulated as follows:

- Total Value = # of orders * average value per order
- # of orders = # of visitors * % of visitors who viewed/added items to cart * % visitors who bought the items
- # of visitors = # of new visitors + # of retained visitors.

I would then map out the user journey:

- The user installs the app via Play Store or App Store.
- The user opens the app and lands on the home screen.
- The user browses products using the homepage cards, app navigation, or in-app search.
- The user adds items to the basket or saves them to a custom list (requires login).
- The user logs in or registers when prompted either to save items, place an order, or view personalized offers.
- The user reviews the cart, selects a delivery address and slot, applies vouchers, and may be nudged to opt for BBstar membership for better value.
- The user proceeds to the payment gateway, completes the transaction, and is shown an order confirmation screen.
- Confirmation is also sent via push notification, SMS, and email.
- Users may contact customer support via the in-app help section to modify, cancel, or track orders.
- The user receives the delivery at the doorstep and can rate the experience via the app.

In this journey, the user goes through 5 stages:

Awareness: The users can discover BigBasket app via Play Store, App Store, referrals, paid marketing or even through BigBasket's own website. Hence, we can have following metrics:

- Number of app installs per month
- Percentage of installs from organic sources (search/website)
- Percentage of installs from paid campaigns (ads, influencer links)
- App ranking in relevant categories (e.g., Shopping, Grocery)
- Click-to-install conversion rate
- App uninstall rate per month
- Average app rating

Engagement: The user browses the app to find and shortlist products. The following can be measured to improve user engagement:

- Bounce rate per product or category page per month
- Average session duration per user per month
- Average number of products viewed per session
- Average quantity per product added to the basket
- Number of custom lists created per month
- Percentage of users using in-app search per month
- Percentage of logged-in users per month
- Percentage of new registrations per month
- Daily active users (DAU)
- Monthly active users (MAU)
- Value of returns per month

Conversion: After adding the products to the basket, the user reviews the basket and checks out. The following can be measured to improve conversion:

- Average number of items in the basket per month
- Average value of items in the basket per month
- Basket abandonment rate per month
- Order cancellation rate per month
- Average order value per user per month
- Percentage of users who select similar products during checkout per month
- Percentage of users applying vouchers per month
- Value of vouchers applied per month

Retention: The user should return to the Big Basket app for further purchases within 2 weeks. The following can be measured to improve retention:

- Percentage of users opting for BBstar membership per month
- Percentage of users buying again per month
- Repeat order frequency (orders/user/month)
- Average time between two app purchases

Referral: The users can also be a marketer for our product through word of mouth and sharing referral code.

- Percentage of users sharing referral codes per month
- Conversion rate of shared referral codes
- Revenue generated through referrals per month
- Net Promoter Score (NPS) collected via in-app feedback

This is great. Thank you!

WhatsApp In-App Video Playing

Metrics

How will you measure the success of WhatsApp's in-app video playing feature?

First, I would like to describe the feature and my understanding of its objective. WhatsApp has introduced an in-app video playing feature, whereby users can play YouTube videos from within the chat window itself. I think the objective of the feature is to keep the users within the app context, so that the time spent on WhatsApp can be increased. There are three types of interactions that can be done when this feature is in use:

1. Expand the video to full screen
2. Close the video
3. Watch the video on YouTube by clicking on the YouTube icon

That sounds good. Please proceed.

The user's journey for using this feature would look like this:

- Activation: The user receives a link to a YouTube video on WhatsApp
- Engagement:
    - They click on the link, and the video starts playing within the app
    - They can expand the video to full screen or stop viewing by clicking on the close button.
- Alternatively, they can click on the YouTube button to be taken to the YouTube app. If they choose this option, they can either continue browsing on YouTube (Leakage) or return to WhatsApp.
- Post-Engagement: After video interaction, the user may resume chatting, share the video, or initiate a call within WhatsApp

Since this feature would be available only on phones with an Android version that supports in-app windows and an updated version of the app, all the metrics should account for that.

Activation:

- % of users who play a link they are sent
- Relative activation rate: % of users with this feature available that play a video v/s % of users without the feature who play a video (which opens the YouTube app instead)

Engagement:

- % of users who played the link and watched the video (either partly or wholly)
- Relative return rate: % of users who stayed in WhatsApp after they stopped watching the video v/s % of users without the feature who return to WhatsApp after watching the video
- % of users who share the video link within WhatsApp after watching
- Time spent on WhatsApp by users with this feature v/s without this feature

Leakage:

- % of users with the feature opting to view the video on YouTube
- % of users who opted to view the video on YouTube that returned to WhatsApp

Okay, and how would you use these metrics to evaluate the feature's success?

The engagement metrics will show whether the feature has boosted time spent on WhatsApp and whether it enables users to engage with YouTube content more frequently. The leakage metrics will help identify if the feature successfully limits app-switching or if significant drop-offs still occur.

Great, can you please summarize in 30 seconds?

To summarize, the in-app video playing feature is an effort by WhatsApp to boost time spent by users within WhatsApp and enable more media consumption for users. Primarily, the relative activation and retention rates will help evaluate these feature value propositions.

Instagram Story Highlights

Metrics

How would you measure the success of the Instagram story highlights features?

I would like to clarify my understanding of the Instagram story highlight feature and its objective. It is a feature to describe a particular account in crisp selected stories. The intention of the highlights could be multiple.
For instance, if we talk of a photography page, the highlight might include its best photo works to indicate the level of quality to its clients. If the Instagram page is about a tech product, its highlight may include how to use the product in stepwise, or maybe the answer to the FAQs about the product.
The objective of the feature is to reduce viewer's time to understand/know about the Instagram profile/product in a short span of time and thus increase the conversion rate for these accounts. Conversion might mean different things for different accounts, for instance, for the personal account it could mean increased followers/likes, for business accounts, it could mean increased product sales etc. Also, Instagram has introduced a story archive feature which is the backbone of highlights feature. So, even after 24 hours of posting a story, you can get those stories in the archive.

Bang on. You are right, please go ahead.

Thanks. So, users can make any old stories a highlight, which will be present on the Instagram account until and unless it is removed manually by the account owner. It just acts like key highlighted posts for the account. These highlights are like small circular icons, similar to story icons present on top of all of the posts on the account page. So, there are two stakeholders in this system:

Account owner: The account owner follows the below steps:
- Click on the new button on account page
- Select the Instagram stories to be highlighted
- Confirm the stories to make them the highlight

Account visitor: Once a user visits the Instagram account, they can see all the highlighted stories in circular icons on the account page. Then they can click on the highlight of their choice, and they will be able to get the story like any other Instagram story.

Yes correct, you have got the product correctly. Now tell me, how will we measure its success.

Let's go through the customer journey in phases. I have already explained the detailed steps for both the account owner and visitor. Now, let's get into the customer journey stages.

Awareness: The account owner should be made aware that such a feature exists. It is currently being done at two places, on the account page and on the story page.

- Average # of times highlight button clicks per story on the story page
- Average # of times highlight button clicks on the account page per account page visit per user

Engagement: The engagement here indicates whether the account owner is able to engage with this feature or not, basically the highlights they created. For account visitors, the engagement depends on various other factors like time spent by visitors on other's account seeing these highlights and so on.
- Percentage of number highlights created to the number of stories posted per user per month.
- Percentage of number highlights created, from account page to the number of stories posted per user per month.
- Percentage of number highlights created, from story page to the number of stories posted per user per month.
- Average time spent on watching other account's
- Highlights per user per month.

Conversion: The whole purpose of the feature is to improve the conversion for users. So, once a user has engaged with other's highlights, he/she should be able to quickly understand about the other's account and make decisions whether to follow or not, whether to buy their product or not etc.
- Average time spent on another account's page per user per month.
- Percentage of highlight viewers following other's page per user per month.
- Percentage of highlight viewers ordering services from other's account per user per month
- Average time spend on other's account before following them per user per month
- Average time spent on other's account before placing the order for their products/services per user per month.

PM Tip: Always quantify metrics (e.g., per day, per month). Align metrics with the product's stage by first defining the stage's objectives (e.g., Awareness, Conversion) and then matching metrics to those goals.

Section 5.5

Pricing

Amazon Prime Pricing Projection

Assume you are the new product manager in our Amazon Prime business and are deciding pricing. The vice president would like to lower the price from $99.99 per year to $89.99 per year. Making your own assumptions, develop the financial projections for this decision.

With lower prices, customer demand will increase. The new Prime customers might also purchase more things. However, increased demand will reduce membership revenues from existing Prime customers and increase shipping costs for existing non-Prime customers. Let's assume there are 5 million Prime customers.

1. **Existing Prime customers**

With a $10 price reduction that reduces annual revenues by $2 billion ($100 * 200M Prime customers = $20B annual revenue)

2. **New Prime customers**

We need to estimate how many new Prime customers will be gained from the changes. Finally, let's consider additional revenue from the membership, along with incremental sales. Let's say Prime membership jumps 20 percent due to the price cut. That's an additional 40 million Prime customers. The incremental membership revenue is roughly **$3.6 billion** (40m * $90).

Average shipping costs per customer is around $40 per year. Let's say that moving these customers into Prime increases shipping costs by $20 per year per new Prime customer. That's an additional **$0.8 billion** cost (40m * $20).

Lastly, we can anticipate more frequent purchases due to free 2-day shipping. Let's say there's an additional $40 worth of purchases per new Prime customer per year. Amazon's gross margins are 25%, so that's an extra $10 per year per new Prime customer, in total **$0.4 billion** (40m * $10).

To recap on the net annual impact:

Existing Prime customers: -$2 billion net

New Prime customers: +$3.2 billion net (+$3.6 billion from member revenue - $0.8 billion incremental shipping costs + $0.4 billion incremental gross margin)

So, what's the overall change in revenue, and what's your recommendation?

Overall, we're looking at a $1.2 billion annual increase from reducing price by $10. **I recommend we reduce the price**.

JioSaavn Family Plan

Pricing

JioSaavn is a music streaming service that wants to launch a family music plan for its users. How would you go about it?

Sure, from what I understand JioSaavn wants to launch a new subscription plan aimed at families. So, I need to come up with a pricing strategy for this plan?

Yes, go ahead.

Okay, and I assume the goal of the pricing strategy is to attract more users. Especially, those who are non-earning members or dependents within a household (e.g., children or elderly).

Fair enough. How do you go about pricing such a plan?

From what I know music streaming is a very competitive market in India and I would like to concentrate on competitor-based pricing. However, various players cater to different customer segments. To identify appropriate benchmarks, I'd first look at which segment JioSaavn targets which I believe is regional music consumers, especially in tier-2 and tier-3 cities.

Okay, go ahead.

Also, I think Spotify and YouTube Music are the appropriate competitors given their popularity in India. I would look at their pricing strategies and the offerings they provide to their customers. Could you provide the latest pricing tiers for us and our key competitors?

Of course. Here are the current approximate monthly plans:

- JioSaavn: Individual: ₹99/month, Duo Plan (2 users): ₹129/month
- Spotify: ₹179/month for 6 users (Family Plan)
- YouTube Music: ₹189/month for 6 users (Family Plan; part of YouTube Premium)

Most music streaming platforms offer quite similar features, but JioSaavn stands out because of its strong regional music collection and how well it can tie into Jio's telecom ecosystem. That's why I'd suggest pricing the family plan at ₹149/month for up to 6 users as it would be a sweet spot that would give families great value and fit well with users in Tier 2 and Tier 3 cities and even in rural areas.

We can probably run experiments to identify user perception regarding this new feature. I'd also look at how many paid users we currently have versus the competition. If we're behind, then this aggressive pricing could help us catch up without sparking a price war. Overall, the idea is to keep things affordable, play to our strengths, and build a family plan that makes sense for our audience.

Great! This sounds good.

Netflix Pricing Strategy

Pricing

The Indian OTT market is intensely competitive and price-sensitive. Netflix has already implemented price cuts and a mobile-only plan. Now, with the global rollout of its 'paid sharing' policy to curb password sharing, you, as a Product Manager at Netflix, need to refine the pricing and product strategy for India to continue growing the subscriber base and revenue.

I understand that the objective here is twofold: continue to grow our subscriber base while also increasing revenue, specifically by converting password-sharers into paying customers.

Yes, that sounds fair. Go ahead.

To structure my approach, I'll use the 5 Cs framework: Company, Customers, Competitors, Collaborators, and Context. I'll begin with what I know and then ask for a few details.

Sure, you may take your time and go ahead.

1. **Company (Netflix)**: Our goals are shifting from pure subscriber growth to profitable growth, meaning Average Revenue Per User (ARPU) is now a key metric. The "paid sharing" feature is the critical global initiative to support this. Our core competencies remain our content engine and personalization algorithm.
2. **Customers**: The Indian market isn't monolithic. I see three key segments:
    - **Mobile-First Youth**: Highly price-sensitive, needs a low-cost, mobile-only option.
    - **Family Household**: Values multiple screens and 4K quality for Smart TVs. They are less price-sensitive and prioritize convenience.
    - **Converted Sharer**: This is a new, important segment. They are being forced into a purchase decision and need an affordable, low-friction way to keep their profile and viewing history.
3. **Competitors**: I know the landscape is defined by bundling and low prices from major players like Amazon Prime Video, Jio Hotstar. To ground my recommendations in the current market, could you provide the latest pricing tiers for us and our key competitors?

Of course. Here are the current approximate monthly plans:

- Netflix: Mobile (₹149), Basic (₹199), Standard (₹499), Premium (₹649). Netflix doesn't offer annual plans.
- Amazon Prime: Included with Prime membership at ₹299/month (or ₹1499/year), which also includes shopping and music.
- Jio Hotstar: Has multiple tiers, with its premium ad-free plan at ₹299/month (or ₹1499/year).

This data confirms my hypothesis: our premium plans are priced significantly higher than competitors, who leverage bundles (Amazon) and live sports (Jio Hotstar) as key value propositions. Our core value is our original content library. Competing with them on prices will neither justify our costs nor our value proposition. Moving on to cost-based pricing, it doesn't make sense to analyze it from a cost perspective, as the variable cost involved in serving every additional subscriber is close to null. This industry primarily comprises high fixed costs of obtaining content licenses, content production, IT infrastructure, etc., and hence shouldn't be looked at from a cost-plus pricing angle.

4. **Collaborators**: Telco operators like Jio and Airtel are key acquisition channels through bundling. OEM partnerships for pre-installation on devices also reduce friction.
5. **Context**: The password-sharing crackdown is the most significant trend. Additionally, the 5G rollout makes HD/4K plans more appealing, and the ubiquity of UPI simplifies payments.

That's a thorough analysis, especially highlighting the new "Converted Sharer" segment and the competitive pressures from bundled services. Based on this, what are your specific recommendations on the "paid sharing feature"?

Based on that analysis, my recommendations are focused on successfully implementing paid sharing while refining our offerings for key segments.

1. **Optimize the "Paid Sharing" Feature for India**: This is our biggest opportunity to boost ARPU. The "Profile Transfer" feature must be flawless. I propose pricing the "add an extra member" slot at ₹99/month.
    a. *Rationale*: Our ad-free Basic plan is ₹199. At ₹99, the fee is less than half the cost of a new plan, making it a low-friction "yes" for someone forced to pay for the first time.

2. **Introduce an Ad-Supported Tier**: The crackdown will cause some user loss. This plan is our safety net to increase ARPU. It would offer most of our catalogue with ad breaks, capped at 720p on a single screen. I suggest pricing it at ₹149/month.
    a. *Rationale*: This creates a clear value ladder. It's cheaper than the ad-free Basic plan (₹199), giving a clear choice to our most price-sensitive users and keeping them in our ecosystem. Note: This would effectively replace the current Mobile plan by offering more device flexibility at the same price point, but with ads.

Any other recommendations going forward with the strategy?

1. **Refine Acquisition with "Freemium" Pilots**: Since 30-day free trials are gone, we need a new hook. I recommend making the first episode of our top 5 Indian and top 5 global originals free to watch for all non-subscribers.
    a. *Rationale*: This provides a risk-free sample of our best content, acting as a powerful acquisition tool to drive sign-ups.
2. **Deepen Telco Bundling Partnerships**: We should create custom plans for our telco partners, like a "Netflix Ad-Supported + 15GB data" bundle. This makes Netflix a value-add for the telco and an easy acquisition channel for us.
3. **Annual subscriptions**: This serves as a commitment discount, which can achieve two goals: it lowers the perceived monthly cost for price-conscious users and, more importantly, it will help lock in subscribers and reduce monthly churn.

These are very concrete recommendations. How would you measure the success of this new strategy if we were to implement it?

I would track a hierarchy of metrics:

Primary Metrics (The North Stars):

- Subscriber Growth Rate: Are we still adding net new subscribers?
- Average Revenue Per User (ARPU): Is the revenue per user increasing, driven by paid sharing and upsells?

Secondary Metrics (The Diagnostic Metrics):

- Conversion Rate of Shared Accounts: What percentage of accounts that receive the sharing notification convert to a new plan or an "extra member" slot? This is the direct measure of Recommendation #1's success.
- Churn Rate or otherwise Customer Retention: How does our overall churn rate change? And specifically, is the ad-supported plan successfully catching users who would have otherwise churned?
- Ad-Supported Tier Engagement: How many hours are users on this new tier watching? This is a leading indicator for our ad revenue potential.

Excellent. You've covered the strategy from analysis to execution and measurement. We can close the case here.

Vendor Price Negotiation

Pricing

Excerpt: While most pricing questions cater to setting the appropriate price for new products or services, negotiating vendor contracts is another kind of pricing problem that a manager might encounter

Hi, you are a product manager at a popular cloud services company. Your main product is a platform that provides on-demand GPU instances for AI startups and researchers to train their models. You source your capacity from a third-party GPU server vendor. This vendor is dissatisfied with the current pricing contract and is asking for the payment of significant overage charges. How would you proceed?

That's an interesting question. To begin with I would like to clarify some aspects of the problem.

Go ahead.

I would like to understand what kind of pricing structure is currently in place, and what is the reason for which the vendor is asking for the renegotiation?

Currently, we have a pay-per-use payment model. Here, the amount to be paid per user depends on the band in which the lump sum requirements of the GPU server's capacity lie. Recently, a major new open-source AI model was released, and there has been a massive, unexpected surge in demand from our customers wanting to use it.

Alright. So, this surge in usage could be from more customers signing up to use the new model, or our existing customers running much larger or more frequent training jobs. Or both. Which one has been the case with us?

Good question. So, we have had a spike in the number of users and in addition, the average time for which a user engages with the platform has increased too.

Also, to understand the business requirements better, I would like to know more about the business model in place. Typically, such platforms would either have a pure pay-as-you-go model or a subscription-based model with a certain number of credits. If our model is purely pay-as-you-go, we should be in a better position, because the increase in usage would translate directly to higher revenue, offsetting the vendor's overage charges. If we have fixed-price subscription customers, the existing fee would have been decided considering the break-even point with the ongoing GPU costs - hence, the loss of profits would be higher.

Fair analysis. Assume for simplicity that there are no pay-as-you-go revenues, we are simply using a subscription fee-based model.

Great, so we can approach the issue in the following way. There are two counter forces to shape the response to the situation - Server costing & Product pricing.

GPUs can either be our own or from a third-party. The third-party server additionally can be existing or new. The existing one can have the ongoing costing model, or it can be renegotiated.

Since the ongoing costing model is clearly not working, we would exclude that from our analysis.

On the pricing end. We can either continue with the current pricing strategy or shift to a new one. The new strategy can, in turn, be subscription- based or usage-based. Is there something else I should consider?

No, this looks perfect. What parameters would you evaluate these options on?

We could evaluate the options on the revenues (a proxy of pricing success), cost (a proxy of costing success), and risk (a sanity check on whether the option makes sense for the business).

Great. Now you can present your analysis and recommendations?

Vendor Price Negotiation

Pricing

H=High, M=Medium, L=Low.

S. No.	*Alternative*	*Rev. Impact*	*Cost Impact*	*Business Risk*	*Evaluation*
1	Build Own Data Center	L	H	H	Although this option would reduce reliance on third-party vendors, it involves extremely high capital costs and a long timeline. It carries risks of migration downtime and, critically, the surge in demand for this specific AI model might cool off, leading to unused capacity. Since the process is long, revenue realization might be lower than the potential.
2	Diversify (Add a New GPU Server Vendor)	M	L	M	Costs are potentially low due to competitive pricing. However, there is a medium risk due to the time and effort required for technical integration with our platform, quality assurance, and the added operational complexity of managing a second vendor relationship.
3	Renegotiate with Existing Vendor	H	M	M	The business risk is moderate as this is the least disruptive option, but our vendor has significant bargaining power due to our dependency. Therefore, the impact on cost may only be moderate. Revenue is not directly impacted due to no transition losses. Also, no capital or fixed payments would be required.
4	Revise Subscription Pricing (New Customers Only)	M	H	L	This would lead to higher revenue from new customers, improving future profitability. However, it doesn't solve the immediate cost problem, as existing subscribers would continue to cause losses. The risk of churn is low since it doesn't affect current customers.
5	Introduce usage-based tiers for all new subscribers	M	H	L	Improves future profitability without upsetting the existing customer base. Doesn't solve the immediate loss from current customers.
6	Launch usage-based pricing tiers for all users (fair-use)	H	L	H	This would lead to the highest revenue by fully monetizing the usage surge. However, the risk is extremely high. Forcing existing fixed-price subscribers into a usage-based model would likely cause significant customer churn, negative PR, and brand damage. Also, technical changes might be required in the platform.

Based on this analysis, I would like to recommend opting for option 5 immediately. The reason is that it's least disruptive to our customer base and addresses both our cost structure and future revenue without taking on excessive risk.

On the other hand, we need to explore option 2 as a strategic initiative. Building relationships with a second vendor will protect us from being in this position again and will give us bargaining power for future negotiations.

We could evaluate the options on the revenues (a proxy of pricing success), cost (a proxy of costing success), and risk (a sanity check on whether the option makes sense for the business).

That'd be all, thanks!

Section 5.6

RCA

Drop in Engagement

RCA

We are seeing a drop in Engagement for Duolingo over the last month. How would you go about analyzing this issue?

Thank you for this interesting problem. To make sure I analyze it thoroughly, I'd like to proceed in a structured manner. First, I'll clarify the problem and the metrics. Second, I'll state my framework for investigating the root cause. Third, I'll systematically drill down to isolate the cause. Finally, I'll propose solutions and a validation plan.
To begin, I'd like to confirm my understanding of Duolingo and the problem. Duolingo is a freemium language-learning app that gamifies the experience through points, levels, and streaks to drive user retention. Is that correct?

Yes, your description of Duolingo is accurate.

Great. Now, clarifying the core metric: "Engagement." You mentioned it has dropped over the last month. Could you define what specific metric we use for engagement? Is it DAU/MAU (Daily Active Users / Monthly Active Users), average session time, or perhaps the number of lessons completed per user per week?

Excellent question. For this discussion, let's define Engagement as the weekly average number of users who complete at least one lesson. We've seen this metric decline by about 15% over the past month.

Thank you, that's a very clear definition. A 15% drop is certainly significant.

Now, I'll outline my approach. I will structure my analysis by first looking for External Factors and then drilling down into Internal Factors.

1. External Factors: Things outside our direct control, like seasonality, competitor actions, or major news events.
2. Internal Factors: Things within our control, like product changes, technical bugs, or marketing campaigns.

Does this structured approach sound good to you?

Yes, that sounds logical. Please proceed.

[Phase 1: Investigating External Factors]

To start with external factors, has this 15% drop been sudden or gradual over the last month? And to understand if this is a pattern, I'd like to check our year-over-year data. Have we seen a similar dip around this time in previous years? This would help me identify any seasonality.

The drop has been gradual over the last 4 weeks. And yes, while we often see a slight dip around this time of year, this 15% drop is significantly steeper than the 3-5% seasonal variation we've seen historically.

That's a key piece of information. It suggests that while seasonality might be a small contributing factor, it doesn't explain the majority of the drop. There's likely another, more dominant cause. Have there been any major actions by our competitors, like a big marketing push or a viral new feature from an app? Or any negative PR affecting Duolingo?

Our market intelligence shows no unusual competitor activity or any significant negative press.

Okay. It seems unlikely the root cause is purely external. So, I'll now focus my investigation on Internal Factors.

[Phase 2: Investigating Internal Factors]

I'll break down the internal factors into a few buckets:

- User Segments: Is the drop uniform across all users, or is it concentrated in a specific segment?
- User Journey Funnel: Where in the user's path is the drop-off happening?
- Technical Issues & Product Changes: Have we introduced bugs or made a product change that's causing this?

First, let's segment the data. I'd want to analyze the engagement drop across different dimensions:

- Geography: Are we seeing this drop globally or is it localized to a specific country, say the US or India?
- Platform: Is there a difference between iOS and Android users?
- User Type: Are new users or returning users more affected?
- Language: Is this drop seen across all languages or specific ones like Spanish or French?

That's a great line of questioning. After looking at the data, we've found the drop is most pronounced among returning users on iOS in the United States, across all the major languages they learn.

Excellent. That narrows down the problem space considerably. We're looking for something that specifically affected returning iOS users in the US in the last month.
Now, let's map out the user journey for a returning user to see where they are dropping off. A simplified funnel would look like this:

1. User receives a notification or opens the app.
2. User lands on the home screen/learning path.
3. User starts a lesson.
4. User progresses through the lesson's various exercises (e.g., matching pairs, typing translations, speaking phrases).
5. User completes the lesson.

The metric you mentioned was "users who complete at least one lesson." This implies the problem lies somewhere in this funnel. My hypothesis is that we'll see a significant drop-off at one specific step. For instance, are users failing to start a lesson, or are they starting but not completing it?

You're on the right track. The conversion from Step 2 (Home Screen) to Step 3 (Start Lesson) is stable. However, we are seeing a major drop in the completion rate of lessons. That is, users start a lesson but don't finish it.

Got it. So, the problem is within the lesson itself. Now we need to figure out what changed inside our lessons in the last month for returning iOS users in the US. I can think of a few possibilities:

1. Technical Bugs: A new iOS update might be causing our app to crash during lessons, or maybe lesson assets (images, audio) aren't loading correctly.
2. Performance Issues: The lesson could have become slower or laggier.
3. Product/Content Change: We may have rolled out a change to the structure, content, or type of exercises within the lessons. I'd want to review our release logs and feature flag dashboard.

Let's assume we've checked for major bugs and performance issues and they aren't the primary cause. You mentioned product changes. We did recently roll out a change via a feature flag to a segment of our iOS users in the US. We increased the proportion of speaking exercises in lessons to improve language retention.

That is a very strong lead. My primary hypothesis now is: The increased frequency of mandatory speaking exercises is causing users to drop off mid-lesson because they are often in environments where they cannot or are uncomfortable speaking aloud (e.g., on public transport, in a quiet office, or at night).
[Phase 3: Validation and Solutions]

To validate this hypothesis, I would propose a few quick data analyses:

- Correlate Drop-off with Exercise Type: Can we see which specific exercise screen users are on when they exit the lesson? If there's a spike in drop-offs on speaking exercises, that would strongly support the hypothesis.
- Analyze Time of Day: Are drop-offs higher during commute times or typical work hours, when users are more likely to be in public?

Let's say the data confirms your hypothesis. We see a huge number of users dropping off the moment a speaking exercise appears. What would you propose as a solution?

This is a classic case of a well-intentioned feature creating a poor user experience in certain contexts. I would brainstorm a few solutions and prioritize them based on user impact and engineering effort.

1. Solution 1 (Quick Fix): "Can't Speak Now" Button. Introduce a button on speaking exercises that allows users to skip it or replace it with a different type of exercise (e.g., typing or listening). This directly addresses the user's contextual problem without disabling the feature entirely.
2. Solution 2 (More Control): Exercise Preferences. In the settings, allow users to set their preference for how frequently they receive speaking exercises, or to disable them entirely. This gives power users more control.
3. Solution 3 (Smarter System): Context-Aware Suggestions. Use device signals (e.g., headphones connected, microphone access denied, time of day) to intelligently decide whether to serve a speaking exercise. This is more complex but would provide the best experience.

For an immediate rollout, I would champion Solution 1. It's the most direct and likely lowest-effort solution to stop the bleeding.

Drop in Engagement

We could roll this out as an A/B test.

- Control Group: The current experience with increased speaking exercises.
- Variant Group: The experience with the "Can't Speak Now" button.

Success Metric: Our primary success metric would be the lesson completion rate. I would look for a statistically significant increase in this metric for the variant group, bringing it closer to the pre-drop baseline. A secondary metric would be to ensure that long-term retention isn't negatively impacted.
If the A/B test is successful, we can roll this feature out to 100% of the affected users and then explore Solution 2 or 3 as part of our long-term product roadmap.

That's an excellent and thorough analysis. You've clearly identified the problem, drilled down to a plausible root cause, and proposed a concrete, data-driven plan to validate and fix it. Thank you.

Low Adoption Rate

RCA

Instagram recently launched a new feature allowing users to purchase merchandise of clothes shown in Reels. However, the initial adoption rate for this feature is very low. Can you walk me through how you would start investigating this issue?

Let me know if I understood correctly about the product in question. Instagram is a popular social media platform that revolves around visual content sharing. Its core value proposition lies in enabling users to express themselves through photos, videos, short form content, connect with a global community, and discover inspiring content.

Yes, you are right about the product.

Firstly, I will narrow down the scope of the problem. Is this specific to a geography? Or Is it specific to app users? Or any such specific demographic.

The low adoption rate has been low for all geographies.

Is this a recent issue, or has it been gradually declining since the feature was launched?

It's a recent issue.

Have there been any technical issues or performance problems reported with this feature?

No technical issues have been reported.

Is the low adoption seen across all types of Reels or concentrated in some categories?

It has been observed across the content.

Thank you for the information. Now, I'd like to analyze both internal and external factors. Have there been any external factors such as changes in the social media landscape or competitors launching similar features or government regulations?

No, there haven't been any significant external changes.

Okay, let's move on to analyzing the internal factors. Let me chart out the user journey of user interacting with reels. Instagram users open the app's homepage. Then, they open a reel on their timeline. They, then, do vertical scrolling to watch new recommended reels. They can like, comment or share a reel in the feed when they are watching it or they can follow the content creator. Users would open the profile page of content creator to check old content of the content creator. Let me know where the new feature got introduced and has users been dropping at any part of the above user flow.

Good analysis of user flow. The new feature was introduced on the profile page of content creator. Where user gets option to visit the page to buy merchandise of content creator by clicking on the profile image of content creator. No, user metrics have not changed in the old user flow.

Based on this new information, the user flow would be modified as follows. Instagram users open the app's homepage. Then, they open a reel on their timeline. They, then, do vertical scrolling to watch new recommended reels. They can like, comment or share a reel in the feed when they are watching it. If they like the reel and content creator, they open the profile page of the content creator and click on the profile image of the content creator to buy their merchandise.

That's right. That would be the new user flow.

Since, problem is observed across the content and user metrics have not changed at any part of the old user flow. What has been the rate of users clicking on the profile image and clicking on the merchandise link?

There was no change in rate of users clicking on the profile image. However, rate at which users click on the merchandise link has been minimal to say the least.

Since, the whole user flow metrics have been normal till the last point. The problem could be awareness about the feature.

Yes, you are right. Users did not know that the feature existed. Only users who read or heard about the feature release interacted with it. How can we improve awareness about the feature?

Following could be done so as to increase adoption rate. Send notifications to all users or targeted user segments when they open the Instagram app informing them about the new shopping capability. Have a "Shop from Reels" spotlight as the first post users see in their feed for a period of time to announce the feature. Show reels that are ads to users on how to use the feature. This way interventions can be made on current the user flow to nudge user behavior.

Thank you. We can close the interview.

Drop in Watch Time

RCA

Let's say you're a Product Manager at YouTube, and you've noticed a 10% drop in overall watch time on the platform this week. How would you go about analyzing this issue?

Let me know if I understood correctly about the product in question. YouTube is a leading video-sharing platform that allows users to upload, share, and view a wide variety of videos. With its vast and diverse content library, including user-generated content, official music videos, and educational resources, YouTube offers a one-stop destination for entertainment, information, and creative expression.

Yes, you are right about the product.

Firstly, I would like to seek more context about the problem. Could you provide insights into whether the drop is consistent across all content categories, or if specific types are more affected?

The drop seems to be consistent across most content types.

Understood. Next, I'd like to understand if this drop is observed on all devices or if it's more pronounced on mobile or desktop.

It appears both have the same drop.

Great. I'd also like to know if specific regions are experiencing a more significant drop in watch time compared to others.

It seems to be a global issue without any significant regional variations.

Lastly, have there been any recent changes in how watch time is calculated or measured on the platform?

No, there haven't been any changes in the calculation method recently.

When did this drop in watch time start, and has it been consistent over the past week?

It has been consistent.

Have there been any external factors such as changes in the social media landscape or competitors launching similar features or government regulations?

No, there haven't been any significant external changes.

With that context, I would proceed to internal factors analysis. Do we have any indications that data accuracy might be an issue? Are there discrepancies in our metrics or reports?

The data appears to be accurate and consistent with other data sources.

Have there been technical problems, such as increased downtime, latency, or errors in the mobile app? Could these technical issues be discouraging users from engaging with the platform?

No, technical issues observed.

Let's understand the user journey and see where the problem is occurring for users. Users open the YouTube app. Then, they search for a video based on keywords or click on a video recommended to them. They watch the video. They click on search for a new video, or a new video gets auto played after the video is played.

This user journey is appropriate and relevant.

Let's pinpoint the point in user journey where the watch time is dropping. are fewer people launching the YouTube app?

No.

Are people finding less content that intrigues them to watch? Or not receiving relevant search results? or are fewer titles and thumbnails enticing users to click videos open?

Relevant keyword results are shown. Recommendation algorithm is working well.

Are people watching videos for shorter durations once they start?

The watch time per video has been normal.

Has there been change in drop offs after watching the video? If yes, it could be due to disabling of default auto-play because search and recommendation are working normally.

Yes. This was the reason for drop off of watching times. Thank You.

Spike in Ride Cancellations

RCA

Thank you for joining us today. We are facing a significant issue with our Uber app. There has been a sudden spike in cancellation rates, and our customers are experiencing poor service as a result. I'd like you to walk me through how you would approach identifying the root cause of this issue.

Thank you for the opportunity. To start, can you clarify what the Uber app does in this context and which specific services are being impacted by these cancellations?

The Uber app connects riders with drivers for transportation services. The issue is primarily with ride cancellations.

Understood. Could you specify the time period during which this increase in cancellations has been observed? Is it a recent development, or has it been a gradual increase over time?

It's a recent development, primarily observed in the last couple of weeks.

That's helpful to know. Is there a particular geographic region or city where this problem is more pronounced, or is it widespread across all areas where Uber operates?

The issue seems to be more prevalent in metropolitan areas, but it's not limited to a specific city.

Interesting. What kind of customers are being affected the most by these cancellations? Are they new users, existing users, or a mix of both?

Both new and existing users are affected, but we've noticed more complaints from our frequent users.

Thank you. On which platform are these cancellations happening? Is it more on iOS, Android, or both equally?

The cancellations are happening across both platforms, so it doesn't seem to be platform-specific.

Got it. Have there been any recent changes to how the cancellation metric is calculated or reported? Any updates to the BI tool that tracks these metrics?

No, the way we calculate and report cancellations hasn't changed recently.

Understood. Now, let's look at the customer journey on the app. Typically, a customer would open the app, enter their destination, request a ride, and wait for a driver to accept the ride. The customer is then notified of the driver's details, price and ETA.

Yes, that's correct.

That's very helpful. Moving on to internal factors, has there been any recent update or major release to the Uber app, particularly anything that might affect the user experience or the functionality of the driver's side of the app?

Yes, we recently rolled out an update that allows drivers to see the drop-off location before they accept a ride.

Interesting. Let's go back to the user journey once. Can you specify at which point in this journey the cancellations are occurring?

The drivers are usually cancelling the rides after accepting them but before reaching the user's pickup location.

That provides valuable insight. Have there been any technical issues or glitches reported since this new feature was released?

No technical glitches have been reported with this feature.

Aside from the new feature, have there been any other recent changes in the offerings on the app, such as pricing, internal salaries, service quality, or any new product offerings that might influence driver behavior?

No, the main change has been the new feature allowing drivers to see the drop-off location.

Let's consider external factors as well. Are there any new regulations or laws affecting our service? Any environmental factors like recent natural disasters?

No new regulations or environmental factors that we're aware of.

How about competitor actions? Have there been any significant changes in competitor offerings or promotions that could be impacting driver behavior?

Not that we've noticed. Our competitors seem to be operating as usual.

Have there been any PR issues or negative sentiment on social media that could be influencing driver or customer behavior?

Spike in Ride Cancellations

RCA

We haven't noticed any significant PR issues or social media backlash recently.

Based on our discussion, it seems the root cause might be related to the recent update allowing drivers to see the drop-off location. Drivers could be cancelling more frequently if they find the drop-off location undesirable. This matches the feedback you've received from the drivers. Addressing this feature, perhaps by implementing incentives for completing rides to less desirable locations or refining the feature to balance driver preferences with customer needs, might help reduce the cancellation rates.

That makes sense. Thank you for walking through the analysis so thoroughly.

Decline in Subscriptions

RCA

Great to have you here today. Let's dive into the case. Netflix subscriptions have reduced on a month-to-month basis by 8%. How would you approach identifying the root cause of this decline?

Thank you for the opportunity. Let me know if I understood correctly about the product in question. Netflix is a streaming service that offers a wide variety of TV shows, movies, documentaries, and more, available to subscribers for a monthly fee.

Yes, that's correct.

Now, regarding the time period, is the 8% decline observed just in the last month, or has this been a trend over several months?

The decline has been observed over the last month.

Understood. Is this decline observed only in India, or is it a global phenomenon?

It's specific to India.

Okay. Are we seeing this decline across all the customer segments, or is it more pronounced in a particular segment, such as new subscribers or long-term users?

It seems to affect both new and existing subscribers.

Noted. Is this change in subscription rates sudden, or has it been gradual over the month?

It's been quite sudden.

That's helpful to know. Could you clarify how the subscription metric is calculated? Is it the number of active subscriptions, revenue, or something else?

It's based on the number of active subscriptions.

Thanks. Has there been any recent change in how this metric is calculated or reported?

No, the calculation method has remained consistent.

Lastly, is the BI tool that reports this metric functioning correctly?

It seems to be working fine.

Great, thank you. Now, I would look into internal factors. To do this, I'll map out the user journey on the Netflix app: users typically log in, navigate to the payment section to adjust any payment preferences and pay, set preferences of language and genre and then proceed to search for or watch recommended content. Is that correct?

Yes, it sounds perfect.

Moving on, have there been any recent tech-related issues or updates or major UI releases that could have contributed to this decline? Any glitches, non-intuitive UX changes, or performance issues?

Yes. There was an update related to instant messaging service to customers reminding them of scheduled auto debit subscription setting. Furthermore, a new readily available and accessible auto debit cancellation button has been introduced in the payment section of the websites and app.

Okay, that's really insightful. Have there been any changes in the offerings on the app, such as pricing adjustments, changes in service quality, or new product features that might be cannibalizing existing offerings?

There haven't been any significant changes in offerings.

Understood. Are there any notable trends in user reviews that could give us insight into their satisfaction or dissatisfaction?

We had noticed some negative reviews mentioning the ease of cancellation due to the new regulations.

That's insightful. Now, let's consider external factors. Has there been any new government regulation or law, other than the one you've mentioned, that might affect Netflix's performance?

Yes, the Government of India has recently introduced other policy changes aimed at increasing consumer protection. These include stricter data privacy laws and more stringent rules around auto-renewal of subscriptions across various digital services.

That's important to know. Can you confirm if the mentioned updates were due to these factors?

Yes, that's correct.

Decline in Subscriptions

RCA

How about competitor analysis? Have any new entrants or existing competitors launched more attractive options or promotions recently?

Some competitors have been offering discounts, but nothing drastically different from usual.

Noted. Have there been any recent PR issues or negative sentiment on social media or in the news about Netflix?

No major PR issues, but the new regulation has been a topic of discussion.

Finally, are there any observable changes in consumer behavior or psychology that could be influencing this decline?

The users have been opting for cancellation of auto debit feature in significant numbers.

That makes sense. The government regulation requiring text reminders and an easily accessible cancellation button seems to have made users more aware and cautious about automatic payments, leading to the observed decline in subscriptions. We can address this by enhancing the value proposition to retain users and perhaps offer reminders or loyalty programs of the benefits of staying subscribed.

Sounds good, thank you!

Drop in Daily Active Users

RCA

Welcome! You're the Product Manager for our mobile banking app. Recently, we observed a concerning drop: Android Daily Active Users (DAU) decreased by 40% in the last two weeks, whereas iOS DAU remained steady. Can you walk me through how you'd diagnose and identify the root cause?

Absolutely. Let me first ensure I clearly understand the issue. Android DAU dropped by 40% over two weeks, while iOS DAU remained flat. Is that accurate?

Yes, that's correct.

Great. I'd start by asking a few clarifying questions: (Metrics Definition) Are we measuring DAU as total unique users who opened the app at least once per day? (User Segments) Is this drop consistent across new and existing users? (Geographic Scope) Does this issue appear globally, or is it localized in certain markets?

(Metrics Definition) Yes, it's unique users per day. (User Segments) Both new and existing users are impacted. (Geographic Scope) It's a global issue but more pronounced in North America.

Understood. First, I'd like to verify the reliability of our metrics before digging deeper: (Metrics Reliability) Have we recently changed our analytics system, or any event definitions? Is there a possibility of data processing delays or incorrect aggregation?

No changes to analytics tools, or events definitions. Also, no delays or data aggregation errors detected.

Given metrics are accurate, I'd next explore external market factors: (Industry Trends) Has the industry witnessed any recent shifts or disruptions impacting Android specifically? (Competitor Influence) Are there competitors who recently launched any significant promotions, discounts, or features targeting Android users specifically?

No significant external events or competitor promotions recently.

Okay, ruling out external factors. Let's segment the users more deeply to see where precisely this drop is happening: (Device Analysis) Is this issue consistent across all Android OS versions, or specific to a particular Android version? Does this issue affect all Android devices equally, or is it specific to certain manufacturers (e.g., Samsung, Google Pixel)?

Good question. Upon checking, the drop is primarily among users with Android 12 and above, regardless of manufacturer.

That's insightful. Let's investigate internal changes next. Typically, sudden drops coincide with product changes: (Recent Product Updates) Did we deploy a new Android app version around the time this DAU decline started? Did the update include major UX/UI, backend, security, or permissions changes?

Yes. We pushed a major Android update two weeks ago, specifically targeting improved security and permissions management. No equivalent iOS update was rolled out.

This aligns well with the timeline of our issue. Let's drill down deeper here: (Release Quality) Did we observe a spike in crash reports or user complaints post-release? Has our app rating on the Google Play Store dropped significantly after the update?

Yes, crash reports have surged dramatically, specifically around app startup. Reviews mentioning crashes and negative ratings have sharply increased since the update.

That's a critical clue. I'd next examine the crash reports closely: (Crash Diagnosis) Do crash logs indicate a specific error or exception consistently across devices? Do crashes correlate to the new permissions or security logic we introduced?

The logs indicate repeated crashes tied directly to permission checks during app launch, specifically on Android 12+.

Given this, it strongly suggests our recent security update has introduced a permission-handling bug that specifically impacts users on Android 12 and above. To summarize our diagnostic path so far clearly: Verified metrics accuracy (no issue found). Ruled out external market factors and competitors (no impact). Identified the affected segment (Android 12+ devices globally, strongly in North America). Narrowed the problem to our recent internal app update with permission or security logic changes causing app startup crashes. Do you agree with this conclusion so far?

Yes, I agree with your analysis and conclusion.

Excellent. Now, with root cause identified, let's discuss solutions and next steps:
Short-term fixes: Immediately roll back the problematic update to the last stable version. If rollback isn't feasible, urgently issue a hot-fix correcting the permissions logic and deploy quickly.

Drop in Daily Active Users

RCA

User Communication: Inform affected users about the issue and provide clear instructions to update the app promptly once fixed. Consider compensatory gestures (e.g., promotional offers) to regain user trust.
Long-term preventive measures: Strengthen QA testing, especially for Android OS upgrades, permissions, and startup routines. Implement automated alerting systems to catch anomalies in user metrics and crash rates earlier.

Thank you. We can close the interview.

Decline in Cart Additions

RCA

There is a decline of about 10% in cart additions (Add to Cart) in the last 3 days in Myntra. Can you help us investigate the problem?

Sure, to make sure we are on the same page, I would clarify the problem statement. The cart addition refers to adding an item through the 'Add to Bag' button on Myntra and I have to find the root cause behind the decline in the cart additions.

Yes, that is correct.

Myntra has both mobile application (android and IOS) and website. Are we looking at a generic 10% decline on all platforms or is it specific to any one platform i.e. app or website?

Good question, we are noticing the decline in the Myntra android app.

Is the decline seen across all categories of products or in any specific category of product? Also, I am assuming that this is an abrupt decline and not associated with month-end/start seasonalities.

The decline is seen throughout all the categories. Yes, you may assume that.

Is this specific to certain user segments as in new users, premium users or based on any kind of demography?

No, the decline is broadly similar.

Okay. I will take just 2 mins to prepare the root cause analysis framework and explain it further.

Sure, take your time.

Okay, so I am assuming that all the analytical tools are working fine and we do not need to check the tool that are providing us with the data.

Yes, you can assume this. We have checked the logs multiple times.

Thank You. I will divide the framework in 2 parts: External factors and internal factors.

1. External factors: What competitors have done in the last 3 days. This can include any new product launch or new announcement by the competitor. This can take away some of the existing customers of the Myntra or their purchase decision can be delayed. We can also look at macro-economic changes: is it the seasonality impact or has been there any particular news in the market?
2. Internal factors: Is there any new announcement or campaigns run by the Myntra? Has Myntra made any design changes in the app or released any updates in the android app? Apart from these factors, we can look at the customer journey to see whether any step before the cart addition has seen any glitch or bugs. If none of this is the reason, then probably we can look at some of the factors that impact the purchase decision of the consumer. Does the approach look good, or do you want me to make any changes?

Yes, that looks great. Please continue.

Has there been any sale or product announcement by the online or offline competitors of Myntra?

No, nothing unusual. To the best of our knowledge, they are running their regular campaigns.

Is it safe to assume that there have been no macroeconomic changes that can influence customer behavior? Personally, I did not come across any such news.

Yes, it is safe to assume.

Okay, so I will move on to the internal factors. Firstly, has there been any new announcement or campaigns run by the Myntra? The information can help me identify whether customers are waiting for any special event to shop or has there been decline in any one product category and increase in another.

The decline is spread over all the categories and not specific to any product or customer base. Also, we are running regular campaigns that we run throughout the year.

Interesting. My next question would be whether Myntra app has released any new updates?

Decline in Cart Additions

RCA

Yes, we have released few updates to improve the user experience of the customer.

Are these updates related to the customer journey of the user, and does it involve changes in Add to Bag feature?

Yes, it involves design changes related to 'Add to Bag' feature. We want you to analyze this thing further. What other factors do you think can be there that can lead to decline?

Okay.

- The first thing I can think of is whether this update released for the android app was a force update or not. If it was not a force update, then some consumers might be using the older version, and the backend services might not support the older version. Further, we need to focus on whether a rollback for such an update is possible to mitigate this impact.
- Second, we can look at the success rate of the 'Add to Bag' API in the last 3 days to analyze if the API is working fine or not.
- Thirdly, I hope consumers are aware of the new changes in the app and they might not consider the changes fishy and abort the transaction.

Actually, when the consumer clicks on the 'Add to bag' button, that action is not being captured by the API. Looks like there was a mistake in the deployment. Thank You for resolving the issue.

Do you want me to analyze it further by going through the customer journey?

No, I think that will be all.

Section 5.7

App Improvement



Facebook Reactions

App Improvement

Let us say that Facebook wants to improve its Reactions feature. How would you go about suggesting some actions that they can take?

I'd like to clarify a few things before I deep dive into this question. Firstly, when we say 'Reactions', are we talking about the feature through which users react to the posts made by them or other users through emojis etc.? Second, do we have any specific user persona in mind for which we should suggest some focal areas - for example, based on their engagement? Third and most importantly, is there any specific reason behind the call to action for improvement such as negative user feedback or any other thing?

You are correct in your understanding of the feature. We have multiple user personas that use Facebook, however for the sake of this discussion, let's restrict ourselves to individual user personas only. You can assume that Facebook wants to drive higher engagement on its platform and wants to explore how the Reactions feature can be improved to achieve this.

Understood. I'd like to broadly outline my high-level structure before I start. First, I'll start by identifying the different user personas that use this feature. Then: identifying the pain points, I'll formulate some solutions that can address them and evaluate the tradeoffs and risks that may be associated with them. Is this a fair approach, or would you like me to consider any additional analyses?

This seems comprehensive. We can proceed.

Thanks, since our focus is on individual users, I would like to bucket them into three segments:

1. **Passive viewers**: Users who do not react to any posts
2. **Nominal engagers**: Users who react to posts using emojis and/or like button
3. **Active engagers**: Users who are deeply engage with posts through emojis and comments

Now, to prioritize the user personas that I would take into consideration two factors - degree of unmet needs and TAM. Based on this, I feel that 'Active engagers' are the segment of users that we should prioritize as they have the highest unmet needs for expression and they act as the pull for Passive and Nominal engagers to increase their current engagement levels, thus expanding the current base of active engagers. Are there any inputs, or should I proceed with identification of pain points as the next step?

Yes we can proceed.

For the active engagers, since their objective is to have a deep expression of their opinions and feelings, I believe that that they would have the following major pain points with the current implementation:

1. Not enough emojis to express themselves, and since the emojis usually represent a single expression, there is a lack of expression avenues to express more complex expressions.
2. Emojis lack any dynamic visual characteristics, hence limiting the ability of the users to be more engaging and expressive in their reactions.
3. Reactions are limited to only one emoji per post, which may act as an inhibitor to express multiple reactions.
4. There are limited counter-reaction opportunities to the existing reactions, which leads to premature termination of the exchange.

Good observations. Let us say you want to prioritize any one of these pain points. Which one will you pick, and what solutions do you think can address them?

I think since the core objective of this analysis is to drive more engagement, offering greater expression avenues beyond emojis with the prioritization of point 2 - having audio-visual reactions - would be the way forward. To that end, some of the solutions can be:

1. Have avatar-based reactions through integration with facial-recognition solutions to make the reactions much more personalized.
2. Offer functionality to users to record or choose from a standard gallery, short videos that can be used for visually-expressive reactions
3. Include visual cues at the UX level to direct users towards the reactions discussed above. This can be in the form of a catchy, highlighted icon(s) that would drive attention of the users towards it.

However, some key challenges for these solutions can be:

1. Censorship challenges
2. Higher storage requirements

Good points. I would like you to think now of some success factors that can be used to measure the success of your solution, and some mitigation measures for the challenges that you have identified.

Some of the KPIs for the proposed feature can be:

1. The number of users engaging with video reactions
2. The percentage of reactions comprising of video reactions

For the challenges, the following mitigants can be adopted:

1. Use an AI model that screens the video reactions for risky content, and based on detection of uncensored content, flag the content to the user and prevent it from getting posted.
2. Limit the duration of the video to a maximum value, say 5 seconds, to reduce the storage requirement. Additionally, the resolution of the video can be adjusted to a set value to prevent storage overruns from high resolution content.

Nicely elaborated. We can close the discussion now.

BookMyShow User Engagement

App Improvement

Let us say that BookMyShow wants to improve engagement and user satisfaction through new experiential features. How would you go about suggesting some actions they can take?

I'd like to clarify a few things before I dive deeper. Firstly, are we targeting the end-user flow from discovery to post-booking, or are we focusing on a particular phase (e.g., event discovery, booking, post-attendance)? Second, are we looking at improving engagement across all users, or is there a priority persona we want to focus on? Lastly, is there any trigger behind this need for improvement - such as a drop in repeat usage or a need to increase content creation on the platform?

You can assume that we are looking at increasing user engagement and stickiness across the lifecycle. Let's focus on individual users who use BookMyShow regularly to book movies and events.

Understood. I'll start by outlining my high-level structure. First, I'll define relevant user personas based on engagement behavior. Then, I'll identify key pain points across their journey, followed by proposed solutions for the highest-impact areas. I'll also touch on trade-offs and mitigation strategies. Does that sound good?

That works. Go ahead.

Based on BookMyShow's use cases, I'd classify users into the following segments:

1. **Occasional Users**: Use the app only for major releases or popular events
2. **Transactional Users**: Regularly book tickets but don't engage beyond the core booking flow
3. **Engaged Users**: Book often and also explore events, write reviews, and refer others

To drive sustainable engagement, I would prioritize the Transactional Users segment. These users already trust and transact with the platform but are under-leveraged from an engagement and retention perspective. By increasing their post-booking or mid-journey interaction, we can move them closer to the Engaged User segment and improve overall stickiness.

Should I move on to identifying pain points next?

Yes, please proceed.

For Transactional Users, the key pain points include:

1. Static location settings: Users traveling to other cities often don't get relevant suggestions
2. Event discovery is passive: There is no intelligent or voice-based assistance for finding events
3. No meaningful rewards for writing reviews: Users contribute ratings but receive no incentive
4. Review contribution doesn't drive social visibility: Users aren't acknowledged for being consistent reviewers

From these, I would prioritize point 2 and 3, as they are directly linked to habit formation and engagement loops.

Makes sense. What solutions would you propose to address those?

I'd suggest the following:

1. **AI Assistant with Prompt-Based Search** (Voice Supported): Introduce a smart assistant on the homepage with prompts like "top movies near me", "sports this weekend", or "comedy shows in Mumbai". The user can type or speak a query, and it redirects them to a pre-filtered result set. This reduces cognitive effort and encourages proactive exploration.

2. **Reviewer Rewards Program**: Gamify the review submission process. Verified reviewers earn points for each review they write after attending an event or movie. These points can be redeemed for wallet credits, partner discounts, or early access offers. This can help improve community content while also driving post-transaction re-engagement.

3. **Dynamic Location Updating**: Enable automatic location detection so users don't have to manually change cities. This is useful for travelers and ensures that recommendations are always contextually relevant. A fallback toggle should be provided for users to override auto-location for privacy reasons.

These three solutions focus on reducing friction in discovery, increasing meaningful user contributions, and personalizing content.

The are thoughtful. What challenges or trade-offs do you foresee?

Some potential trade-offs and risks include:

1. The AI assistant might require strong natural language understanding to avoid frustration from misinterpretation. Needs a fallback experience (e.g., pre-filtered cards).
2. Risk of low-quality or spam reviews just to earn points. This can dilute trust and create moderation overhead.
3. Location Auto-Detection can raise privacy concerns among users who don't want location tracking.

Good points. I would like you to think now of some success factors that can be used to measure the success of your solution, and some mitigation measures for the challenges that you have identified.

Here are some relevant KPIs:

1. Percentage of users using the AI assistant (text or voice) in the discovery journey
2. Percentage of first-time reviewers converting to repeat reviewers
3. Average reviews per 1000 bookings
4. Uplift in session-to-booking conversion due to faster discovery
5. Opt-in rate for location-based recommendations

For the challenges, the following mitigations can be adopted:

1. Use pre-built AI intents (e.g., Google Dialogflow) for assistant implementation and limit scope to predefined genres/events to start.
2. Implement content filters and quality thresholds before awarding points (e.g., minimum word count, flagged words).
3. Make auto-location opt-in and clearly communicate its benefit to the user upfront.

Well structured and nicely articulated. We can close the discussion now.

Section 6

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Thank You